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MODULATORS OF DIACYGLYCEROL ACYLTRANSFERASE 2 (DGAT2)

Abstract

The present embodiments provide methods, compounds, and compositions useful for inhibiting DGAT2 expression, which may be useful for treating, preventing, or ameliorating a disease associated with DGAT2.

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Background/Summary

SEQUENCE LISTING
[0001] The present application is being filed along with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled BIOL0178SEQ.xml created Apr. 4, 2024, which is 4,151 KB in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety.
FIELD
[0002] The present embodiments provide methods, compounds, and compositions for treating, preventing, or ameliorating a disease associated with nonalcoholic fatty liver disease, including non-alcoholic steatohepatitis (NASH) and hepatic steatosis, by administering a diacylglycerol acyltransferase 2 (DGAT2) specific inhibitor to an individual.
BACKGROUND
[0003] Nonalcoholic fatty liver diseases (NAFLDs) including NASH (nonalcoholic steatohepatitis) are considered to be hepatic manifestations of the metabolic syndrome (Marchesini G, et al. Hepatology 2003; 37:917-923) and are characterized by the accumulation of triglycerides in the liver of patients without a history of excessive alcohol consumption. The majority of patients with NAFLD are obese or morbidly obese and have accompanying insulin resistance (Byrne C D and Targher G. J Hepatol 2015 April; 62 (1S): S47-S64). The incidence of NAFLD/NASH has been rapidly increasing worldwide consistent with the increased prevalence of obesity, and is currently the most common chronic liver disease. Recently, the incidence of NAFLD and NASH was reported to be 46% and 12%, respectively, in a largely middle-aged population (Williams C D, et al. Gastroenterology 2011; 140:124-131).
[0004] NAFLD is classified into simple steatosis, in which only hepatic steatosis is observed, and NASH, in which intralobular inflammation and ballooning degeneration of hepatocytes is observed along with hepatic steatosis. The proportion of patients with NAFLD who have NASH is still not clear but might range from 20-40%. NASH is a progressive disease and may lead to liver cirrhosis and hepatocellular carcinoma (Farrell G C and Larter C Z. Hepatology 2006; 43: S99-S112; Cohen J C, et al. Science 2011; 332:1519-1523). Twenty percent of NASH patients are reported to develop cirrhosis, and 30-40% of patients with NASH cirrhosis experience liver-related death (McCullough A J. J Clin Gastroenterol 2006; 40 Suppl 1: S17-S29). Recently, NASH has become the third most common indication for liver transplantation in the United States (Charlton M R, et al. Gastroenterology 2011; 141:1249-1253).
[0005] Currently, the principal treatment for NAFLD/NASH is lifestyle modification by diet and exercise. However, pharmacological therapy is indispensable because obese patients with NAFLD often have difficulty maintaining improved lifestyles.
[0006] Lipodystrophy syndromes are a group of rare metabolic diseases characterized by selective loss of adipose tissue that leads to ectopic fat deposition in liver and muscle and the development of insulin resistance, diabetes, dyslipidemia and fatty liver disease (Shulman G I. N Engl J Med 2014; 371:1131-1141; Garg A. J Clin Endocrinol Metab 2011; 96:3313-3325; Chan J L and Oral E A. Endocr Pract 2010; 16:310-323; Garg A. N Engl J Med 2004; 350:1220-1234). These

syndromes constitute a significant medical unmet need as these patients are refractory to current therapies, mainly used to treat diabetes and elevated TG levels, in an attempt to reduce the risk of serious associated complications (coronary artery disease, diabetic nephropathy, cirrhosis and pancreatitis).

[0007] Partial Lipodystrophy has a higher prevalence (estimated ˜2-3 in one (1) million) than generalized lipodystrophy, but the extent of the prevalence is unknown because these patients are greatly under-diagnosed (Chan J L and Oral E A. Endocr Pract 2010; 16:310-323; Garg A. J Clin Endocrinol Metab 2011; 96:3313-3325). Partial lipodystrophy is further divided into acquired partial lipodystrophy (APL) or Familial partial lipodystrophy (FPL). The diagnosis of PL is mainly clinical and needs to be considered in patients presenting with the triad of insulin resistance (with or without overt diabetes), significant dyslipidemia in the form of hypertriglyceridemia, and fatty liver (Huang-Doran I, et al. J Endocrinol 2010; 207:245-255). Patients often present with diabetes and severe insulin resistance requiring high doses of insulin.

[0008] Current treatment includes lifestyle modification such as reducing caloric intake and increasing energy expenditure via exercise. Conventional therapies used to treat severe insulin resistance (metformin, thiazolidinediones, GLP-1s, insulin), and/or high TGs (dietary fat restriction, fibrates, fish oils) are not very efficacious in these patients (Chan J L and Oral E A. Endocr Pract 2010; 16:310-323; Garg A. J Clin Endocrinol Metab 2011). Partial Lipodystrophy is an ultra-orphan indication for which there is a significant unmet medical need. Diabetes, NASH, and/or hypertriglyceridemia associated with this condition can lead to serious complications (Handelsman Y, et al. Endocr Pract 2013; 19:107-116).

[0009] Diacylglycerol O-acyltransferase (DGAT) catalyzes the final step in triglyceride (TG) synthesis by facilitating the linkage of sn-1,2-diacylglycerol (DAG) with an acyl-CoA. There are two isoforms of DGATs (DGAT1 and DGAT2), and studies indicate that both DGAT1 and DGAT2 play important roles in TG synthesis. DGAT1 is most highly expressed in small intestine and white adipose tissue (WAT), whereas DGAT2 is primarily expressed in liver and WAT (Cases S, et al. Proc Natl Acad Sci USA 1998; 95, 13018-13023; Cases S, et al. J Biol Chem 2001; 276, 38870-38876). Although both DGAT1 and DGAT2 catalyze the same reactions in TG synthesis with DAG or monoacylglycerol (MAG) and acyl-CoA as substrates, they are functionally distinguished not only by their tissue expression, but by their differences in catalytic properties (Cao J, et al. J Lipid Res 2007; 48:583-591; Cheng D, et al. J Biol Chem 2008; 283:29802-29811), subcellular localization (Stone S J, et al. J Biol Chem 2004; 279:11767-11776), and physiological regulation (Meegalla R L, et al. Biochem Biophys Res Commun 2002, 298, 317-323). For example, suppression of DGAT2, but not of DGAT1, by antisense oligonucleotide treatment improved hepatic steatosis and blood lipid levels independent of adiposity in rodent models of obesity and the data indicated that these effects were related to decreased hepatic lipid synthesis (Yu X X, et al. Hepatology 2005; 42:362-371).

[0010] These studies demonstrate that DGAT2 inhibition may improve NAFLD/NASH, as well as the metabolic profile of patients with lipodystrophy syndromes by reducing triglycerides, improving insulin sensitivity, and decreasing hepatic steatosis.

SUMMARY

[0011] The present embodiments provided herein are directed to potent and/or tolerable compounds and compositions useful for treating, preventing, ameliorating, or slowing progression of NAFLD, such as NASH, as well as lipodystrophy syndromes, such as partial lipodystrophy.

[0012] Several embodiments provided herein are directed to several antisense compounds that are more potent and efficacious than antisense oligonucleotides from an earlier published application, WO 2005/019418. Several embodiments provided herein are directed to compounds and compositions that are potent and tolerable.

Description

DETAILED DESCRIPTION

[0013] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed. Herein, the use of the singular includes the plural unless specifically stated otherwise. As used herein, the use of "or" means "and/or" unless stated otherwise. Furthermore, the use of the term "including" as well as other forms, such as "includes" and "included", is not limiting. Also, terms such as "element" or "component" encompass both elements and components comprising one unit and elements and components that comprise more than one subunit, unless specifically stated otherwise.

[0014] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described. All documents, or portions of documents, cited in this application, including, but not limited to, patents, patent applications, articles, books, treatises, and GenBank and NCBI reference sequence records are hereby expressly incorporated by reference for the portions of the document discussed herein, as well as in their entirety.

[0015] It is understood that the sequence set forth in each SEQ ID NO in the examples contained herein is independent of any modification to a sugar moiety, an internucleoside linkage, or a nucleobase. As such, antisense compounds defined by a SEQ ID NO may comprise, independently, one or more modifications to a sugar moiety, an internucleoside linkage, or a nucleobase. Antisense compounds described by ISIS number (ISIS #) indicate a combination of nucleobase sequence, chemical modification, and motif.

[0016] Unless otherwise indicated, the following terms have the following meanings: [0017] "2′-deoxynucleoside" means a nucleoside comprising 2′-H(H) furanosyl sugar moiety, as found in naturally occurring deoxyribonucleic acids (DNA). In certain embodiments, a 2′-deoxynucleoside may comprise a modified nucleobase or may comprise an RNA nucleobase (e.g., uracil). [0018] "2′-O-methoxyethyl" (also 2′-MOE and 2′-O(CH.sub.2).sub.2—OCH.sub.3) refers to an O-methoxy-ethyl modification at the 2′ position of a sugar ring, e.g. a furanose ring. A 2′-O-methoxyethyl modified sugar is a modified sugar. [0019] "2′-MOE nucleoside" (also 2′-O-methoxyethyl nucleoside) means a nucleoside comprising a 2′-MOE modified sugar moiety. [0020] "2′-substituted nucleoside" or "2-modified nucleoside" means a nucleoside comprising a 2′-substituted or 2′-modified sugar moiety. As used herein, "2′-substituted" or "2-modified" in reference to a sugar moiety means a furanosyl sugar moiety comprising a 2′-substituent group other than H or OH. [0021] "3′ target site" refers to the nucleotide of a target nucleic acid which is complementary to the 3′-most nucleotide of a particular antisense compound. [0022] "5′ target site" refers to the nucleotide of a target nucleic acid which is complementary to the 5′-most nucleotide of a particular antisense compound. [0023] "5-methylcytosine" means a cytosine modified with a methyl group attached to the 5 position. A 5-methylcytosine is a modified nucleobase.

[0024] "About" means within ±10% of a value. For example, if it is stated, "the compounds affected at least about 70% inhibition of DGAT2", it is implied that DGAT2 levels are inhibited within a range of 60% and 80%.

[0025] "Administration" or "administering" refers to routes of introducing a compound or composition provided herein to an individual to perform its intended function. An example of a route of administration that can be used includes, but is not limited to parenteral administration, such as subcutaneous, intravenous, or intramuscular injection or infusion.

[0026] "Administered concomitantly" or "co-administration" means administration of two or more compounds in any manner in which the pharmacological effects of both are manifest in the patient. Concomitant administration does not require that both agents be administered in a single pharmaceutical composition, in the same dosage form, by the same route of administration, or at the same time. The effects of both agents need not manifest themselves at the same time. The effects need only be overlapping for a period of time and need not be coextensive. Concomitant administration or co-administration encompasses administration in parallel or sequentially.

[0027] “Amelioration” refers to a lessening of at least one indicator, sign, or symptom of an associated disease, disorder, or condition. In certain embodiments, amelioration includes a delay or slowing in the progression of one or more indicators of a condition or disease. The severity of indicators may be determined by subjective or objective measures, which are known to those skilled in the art.

[0028] “Animal” refers to a human or non-human animal, including, but not limited to, mice, rats, rabbits, dogs, cats, pigs, and non-human primates, including, but not limited to, monkeys and chimpanzees.

[0029] “Antisense activity” means any detectable or measurable activity attributable to the hybridization of an antisense compound to its target nucleic acid. In certain embodiments, antisense activity is a decrease in the amount or expression of a target nucleic acid or protein encoded by such target nucleic acid compared to target nucleic acid levels or target protein levels in the absence of the antisense compound to the target.

[0030] “Antisense compound” means a compound comprising an antisense oligonucleotide and optionally one or more additional features, such as a conjugate group or terminal group. Examples of antisense compounds include single-stranded and double-stranded compounds. Examples are antisense oligonucleotides, ribozymes, siRNAs, shRNAs, ssRNAs, and occupancy-based compounds.

[0031] “Antisense inhibition” means reduction of target nucleic acid levels in the presence of an antisense compound complementary to a target nucleic acid compared to target nucleic acid levels in the absence of the antisense compound.

[0032] “Antisense mechanisms” are all those mechanisms involving hybridization of a compound with target nucleic acid, wherein the outcome or effect of the hybridization is either target degradation or target occupancy with concomitant stalling of the cellular machinery involving, for example, transcription or splicing.

[0033] “Antisense oligonucleotide” means an oligonucleotide having a nucleobase sequence that is complementary to a target nucleic acid or a region or segment thereof. In certain embodiments, an antisense oligonucleotide is specifically hybridizable to a target nucleic acid or a region or segment thereof.

[0034] “Bicyclic nucleoside” or “BNA” means a nucleoside comprising a bicyclic sugar moiety. As used herein, “bicyclic sugar” or “bicyclic sugar moiety” means a modified sugar moiety comprising two rings, wherein the second ring is formed via a bridge connecting two of the atoms in the first ring thereby forming a bicyclic structure. In certain embodiments, the first ring of the bicyclic sugar moiety is a furanosyl moiety. In certain embodiments, the bicyclic sugar moiety does not comprise a furanosyl moiety.

[0035] “Branching group” means a group of atoms having at least 3 positions that are capable of forming covalent linkages to at least 3 groups. In certain embodiments, a branching group provides a plurality of reactive sites for connecting tethered ligands to an oligonucleotide via a conjugate linker and/or a cleavable moiety.

[0036] “Cell-targeting moiety” means a conjugate group or portion of a conjugate group that is capable of binding to a particular cell type or particular cell types.

[0037] “Cleavable moiety” means a bond or group of atoms that is cleaved under physiological conditions, for example, inside a cell, an animal, or a human.

[0038] “cEt” or “constrained ethyl” means a bicyclic sugar moiety comprising a bridge connecting the 4′-carbon and the 2′-carbon, wherein the bridge has the formula: 4′-CH(CH.sub.3)—O-2′.

[0039] “Chemical modification” means a chemical difference in a compound when compared to a naturally occurring counterpart. Chemical modifications of oligonucleotides include nucleoside modifications (including sugar moiety modifications and nucleobase modifications) and internucleoside linkage modifications. In reference to an oligonucleotide, chemical modification does not include differences only in nucleobase sequence.

[0040] "Chemically distinct region" refers to a region of an antisense compound that is in some way chemically different than another region of the same antisense compound. For example, a region having 2′-O-methoxyethyl nucleotides is chemically distinct from a region having nucleotides without 2′-O-methoxyethyl modifications.

[0041] "Chimeric antisense compounds" means antisense compounds that have at least 2 chemically distinct regions, each position having a plurality of subunits.

[0042] "Cleavable bond" means any chemical bond capable of being split. In certain embodiments, a cleavable bond is selected from among: an amide, a polyamide, an ester, an ether, one or both esters of a phosphodiester, a phosphate ester, a carbamate, a di-sulfide, or a peptide.

[0043] "Cleavable moiety" means a bond or group of atoms that is cleaved under physiological conditions, for example, inside a cell, an animal, or a human.

[0044] "Complementary" in reference to an oligonucleotide means the nucleobase sequence of such oligonucleotide or one or more regions thereof matches the nucleobase sequence of another oligonucleotide or nucleic acid or one or more regions thereof when the two nucleobase sequences are aligned in opposing directions. Nucleobase matches or complementary nucleobases, as described herein, are limited to adenine (A) and thymine (T), adenine (A) and uracil (U), cytosine (C) and guanine (G), and 5-methyl cytosine (mC) and guanine (G) unless otherwise specified. Complementary oligonucleotides and/or nucleic acids need not have nucleobase complementarity at each nucleoside and may include one or more nucleobase mismatches. By contrast, "fully complementary" or "100% complementary" in reference to oligonucleotides means that such oligonucleotides have nucleobase matches at each nucleoside without any nucleobase mismatches.

[0045] "Conjugate group" means a group of atoms that is directly or indirectly attached to a parent compound, e.g., an oligonucleotide.

[0046] "Conjugate linker" means a group of atoms that connects a conjugate group to a parent compound, e.g., an oligonucleotide.

[0047] "Constrained ethyl nucleoside" (also cEt nucleoside) means a nucleoside comprising a bicyclic sugar moiety comprising a 4′-CH(CH.sub.3)—O-2′ bridge.

[0048] "Contiguous" in the context of an oligonucleotide refers to nucleosides, nucleobases, sugar moieties, or internucleoside linkages that are immediately adjacent to each other. For example, "contiguous nucleobases" means nucleobases that are immediately adjacent to each other.

[0049] "Designing" or "Designed to" refer to the process of designing an oligomeric compound that specifically hybridizes with a selected nucleic acid molecule.

[0050] "DGAT2" means any nucleic acid or protein of DGAT2. "DGAT2 nucleic acid" means any nucleic acid encoding DGAT2. For example, in certain embodiments, a DGAT2 nucleic acid includes a DNA sequence encoding DGAT2, an RNA sequence transcribed from DNA encoding DGAT2 (including genomic DNA comprising introns and exons), including a non-protein encoding (i.e. non-coding) RNA sequence, and an mRNA sequence encoding DGAT2. "DGAT2 mRNA" means an mRNA encoding a DGAT2 protein.

[0051] "DGAT2 specific inhibitor" refers to any agent capable of specifically inhibiting DGAT2 RNA and/or DGAT2 protein expression or activity at the molecular level. For example, DGAT2 specific inhibitors include nucleic acids (including antisense compounds), peptides, antibodies, small molecules, and other agents capable of inhibiting the expression of DGAT2 RNA and/or DGAT2 protein.

[0052] "Dose" means a specified quantity of a pharmaceutical agent provided in a single administration, or in a specified time period. In certain embodiments, a dose may be administered in two or more boluses, tablets, or injections. For example, in certain embodiments, where subcutaneous administration is desired, the desired dose may require a volume not easily accommodated by a single injection. In such embodiments, two or more injections may be used to achieve the desired dose. In certain embodiments, a dose may be administered in two or more injections to minimize injection site reaction in an individual. In other embodiments, the pharmaceutical agent is administered by infusion over an extended period of time or continuously. Doses may be stated as the amount of pharmaceutical agent per hour, day, week or month.

[0053] “Dosing regimen” is a combination of doses designed to achieve one or more desired effects.

[0054] “Double-stranded antisense compound” means an antisense compound comprising two oligomeric compounds that are complementary to each other and form a duplex, and wherein one of the two said oligomeric compounds comprises an antisense oligonucleotide.

[0055] “Effective amount” means the amount of compound sufficient to effectuate a desired physiological outcome in an individual in need of the agent. The effective amount may vary among individuals depending on the health and physical condition of the individual to be treated, the taxonomic group of the individuals to be treated, the formulation of the composition, assessment of the individual's medical condition, and other relevant factors.

[0056] “Efficacy” means the ability to produce a desired effect.

[0057] “Expression” includes all the functions by which a gene's coded information is converted into structures present and operating in a cell. Such structures include, but are not limited to the products of transcription and translation.

[0058] “Fully modified” in reference to an oligonucleotide means a modified oligonucleotide in which each nucleoside is modified. “Uniformly modified” in reference to an oligonucleotide means a fully modified oligonucleotide in which at least one modification of each nucleoside is the same. For example, the nucleosides of a uniformly modified oligonucleotide can each have a 2′-MOE modification but different nucleobase modifications, and the internucleoside linkages may be different.

[0059] “Gapmer” means a chimeric antisense compound in which an internal region having a plurality of nucleosides that is positioned between external regions having one or more nucleosides, wherein the nucleosides comprising the internal region are chemically distinct from the nucleoside or nucleosides comprising the external regions. The internal region may be referred to as the “gap” and the external regions may be referred to as the “wings.” In certain embodiments, the structure of a gapmer may support RNase H cleavage.

[0060] “Hybridization” means the pairing or annealing of complementary oligonucleotides and/or nucleic acid molecules. While not limited to a particular mechanism, the most common mechanism of hybridization involves hydrogen bonding, which may be Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding, between complementary nucleobases. In certain embodiments, complementary nucleic acid molecules include, but are not limited to, an antisense compound and a nucleic acid target. In certain embodiments, complementary nucleic acid molecules include, but are not limited to, an antisense oligonucleotide and a nucleic acid target.

[0061] “Identifying an animal having, or at risk for having, a disease, disorder and/or condition” means identifying an animal having been diagnosed with the disease, disorder and/or condition or identifying an animal predisposed to develop the disease, disorder and/or condition. Such identification may be accomplished by any method including evaluating an individual's medical history and standard clinical tests or assessments.

[0062] “Immediately adjacent” means there are no intervening elements between the immediately adjacent elements of the same kind (e.g. no intervening nucleobases between adjacent nucleobases).

[0063] “Individual” means a human or non-human animal selected for treatment or therapy.

[0064] “Inhibiting the expression or activity” refers to a reduction, blockade of the expression or activity relative to the expression or activity in an untreated or control sample, and does not necessarily indicate a total elimination of expression or activity.

[0065] “Internucleoside linkage” means a group or bond that forms a covalent linkage between adjacent nucleosides in an oligonucleotide. As used herein “modified internucleoside linkage” means any internucleoside linkage other than a naturally occurring, phosphate internucleoside linkage. Naturally occurring, non-phosphate linkages are referred to herein as modified internucleoside linkages. “Phosphorothioate linkage” means a linkage between nucleosides wherein the phosphodiester bond of a phosphate linkage is modified by replacing one of the non-bridging oxygen atoms with a sulfur atom. A phosphorothioate linkage is a modified internucleoside linkage.
[0066] “Lengthened” antisense oligonucleotides are those that have one or more additional nucleosides relative to an antisense oligonucleotide disclosed herein; e.g. a parent oligonucleotide.
[0067] “Linearly modified sugar” or “linearly modified sugar moiety” means a modified sugar moiety that comprises an acyclic or non-bridging modification. Such linear modifications are distinct from bicyclic sugar modifications.
[0068] “Linked deoxynucleoside” means a nucleic acid base (A, G, C, T, U) substituted by deoxyribose linked by a phosphate ester to form a nucleotide.
[0069] “Linked nucleosides” are nucleosides that are connected in a continuous sequence (i.e. no additional nucleosides are present between those that are linked).
[0070] As used herein, “mismatch” or “non-complementary” means a nucleobase of a first oligonucleotide that is not complementary to the corresponding nucleobase of a second oligonucleotide or target nucleic acid when the first and second oligonucleotides are aligned. For example, a universal nucleobase, inosine, and hypoxanthine, are capable of hybridizing with at least one nucleobase but are still mismatched or non-complementary with respect to nucleobase to which it hybridized. As another example, a nucleobase of a first oligonucleotide that is not capable of hybridizing to the corresponding nucleobase of a second oligonucleotide or target nucleic acid when the first and second oligonucleotides are aligned is a mismatch or non-complementary nucleobase.
[0071] “Modified nucleobase” means any nucleobase other than adenine, cytosine, guanine, thymidine, or uracil. An “unmodified nucleobase” means the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C) and uracil (U). A “universal base” is a nucleobase that can pair with any one of the five unmodified nucleobases.
[0072] “Modified nucleoside” means a nucleoside having, independently, a modified sugar moiety and/or modified nucleobase.
[0073] “Modified nucleotide” means a nucleotide having, independently, a modified sugar moiety, modified internucleoside linkage, or modified nucleobase.
[0074] “Modified oligonucleotide” means an oligonucleotide comprising at least one modified internucleoside linkage, a modified sugar, and/or a modified nucleobase.
[0075] “Modulating” refers to changing or adjusting a feature in a cell, tissue, organ or organism. For example, modulating DGAT2 RNA can mean to increase or decrease the level of DGAT2 RNA and/or DGAT2 protein in a cell, tissue, organ or organism. A “modulator” effects the change in the cell, tissue, organ or organism. For example, a DGAT2 antisense compound can be a modulator that decreases the amount of DGAT2 RNA and/or DGAT2 protein in a cell, tissue, organ or organism.
[0076] “Monomer” refers to a single unit of an oligomer. Monomers include, but are not limited to, nucleosides and nucleotides, whether naturally occurring or modified.
[0077] “Motif” means the pattern of unmodified and/or modified sugar moieties, nucleobases, and/or internucleoside linkages, in an oligonucleotide.
[0078] “Natural” or “Naturally occurring” means found in nature. “Naturally occurring internucleoside linkage” means a 3′ to 5′ phosphodiester linkage. “Natural sugar moiety” means a sugar moiety found in DNA (2′-H) or RNA (2′-OH). “Naturally occurring nucleobase” means the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C) and uracil (U). “Non-complementary nucleobase” refers to a pair of nucleobases that do not form hydrogen bonds with one another or otherwise support hybridization.

[0079] “Nucleic acid” refers to molecules composed of monomeric nucleotides. A nucleic acid includes, but is not limited to, ribonucleic acids (RNA), deoxyribonucleic acids (DNA), single-stranded nucleic acids, and double-stranded nucleic acids.
[0080] “Nucleobase” means a heterocyclic moiety capable of pairing with a base of another nucleic acid.
[0081] “Nucleobase sequence” means the order of contiguous nucleobases independent of any sugar, linkage, and/or nucleobase modification.
[0082] “Nucleoside” means a compound comprising a nucleobase and a sugar moiety. The nucleobase and sugar moiety are each, independently, unmodified or modified.
[0083] “Nucleotide” means a nucleoside having a phosphate group covalently linked to the sugar portion of the nucleoside.
[0084] “Oligomeric compound” means a compound comprising an oligonucleotide and optionally one or more additional features, such as a conjugate group or terminal group. Examples of oligomeric compounds include single-stranded and double-stranded compounds, such as, antisense compounds, antisense oligonucleotides, ribozymes, siRNAs, shRNAs, ssRNAs, and occupancy-based compounds.
[0085] “Oligonucleoside” means an oligonucleotide in which the internucleoside linkages do not contain a phosphorus atom.
[0086] “Oligonucleotide” means a polymer of linked nucleosides each of which can be modified or unmodified, independent one from another.
[0087] “Parent oligonucleotide” means an oligonucleotide whose sequence is used as the basis of design for more oligonucleotides of similar sequence but with different lengths, motifs, and chemistries. The newly designed oligonucleotides may have the same or overlapping sequence as the parent oligonucleotide.
[0088] “Parenteral administration” means administration through injection or infusion. Parenteral administration includes subcutaneous administration, intravenous administration, intramuscular administration, intraarterial administration, intraperitoneal administration, or intracranial administration, e.g. intrathecal or intracerebroventricular administration.
[0089] “Pharmaceutically acceptable carrier or diluent” means a medium or diluent suitable for use in administering to an animal. For example, a pharmaceutically acceptable carrier can be a sterile aqueous solution, such as PBS or water-for-injection.
[0090] “Pharmaceutically acceptable salts” means physiologically and pharmaceutically acceptable salts of compounds, such as oligomeric compounds or antisense compounds, i.e., salts that retain the desired biological activity of the parent compound and do not impart undesired toxicological effects thereto.
[0091] “Pharmaceutical agent” means a compound that provides a therapeutic benefit when administered to an individual.
[0092] “Pharmaceutical composition” means a mixture of compounds suitable for administering to an individual. For example, a pharmaceutical composition may comprise one or more compounds or salts thereof and a sterile aqueous solution.
[0093] “Phosphorothioate linkage” means a modified internucleoside linkage between nucleosides where the phosphodiester bond is modified by replacing one of the non-bridging oxygen atoms with a sulfur atom.
[0094] “Phosphorus moiety” means a group of atoms comprising a phosphorus atom. In certain embodiments, a phosphorus moiety comprises a mono-, di-, or tri-phosphate, or phosphorothioate.
[0095] “Portion” means a defined number of contiguous (i.e., linked) nucleobases of a nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucleobases of a target nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucleobases of an oligomeric compound
[0096] “Prevent” refers to delaying or forestalling the onset, development or progression of a

disease, disorder, or condition for a period of time from minutes to indefinitely. Prevent may also mean reducing the risk of developing a disease, disorder, or condition.

[0097] “Prodrug” means a form of a compound which, when administered to an individual, is metabolized to another form. In certain embodiments, the metabolized form is the active, or more active, form of the compound (e.g., drug).

[0098] “Prophylactically effective amount” refers to an amount of a pharmaceutical agent that provides a prophylactic or preventative benefit to an animal.

[0099] “Ref Seq No.” is a unique combination of letters and numbers assigned to a sequence to indicate the sequence is for a particular target transcript (e.g., target gene). Such sequence and information about the target gene (collectively, the gene record) can be found in a genetic sequence database. Genetic sequence databases include the NCBI Reference Sequence database, GenBank, the European Nucleotide Archive, and the DNA Data Bank of Japan (the latter three forming the International Nucleotide Sequence Database Collaboration or INSDC).

[0100] “Region” is defined as a portion of the target nucleic acid having at least one identifiable structure, function, or characteristic.

[0101] “Ribonucleotide” means a nucleotide having a hydroxy at the 2′ position of the sugar portion of the nucleotide.

[0102] “RNAi compound” means an oligomeric compound that acts, at least in part, through RISC or Ago2 to modulate a target nucleic acid and/or protein encoded by a target nucleic acid. RNAi compounds include, but are not limited to double-stranded siRNA, single-stranded RNA (ssRNA), and microRNA, including microRNA mimics. The term RNAi compound excludes antisense oligonucleotides that act through RNase H.

[0103] “Segment” is defined as a smaller or sub-portion of region within an antisense compound, an oligonucleotide, or a target nucleic acid.

[0104] “Side effects” means physiological disease and/or conditions attributable to a treatment other than the desired effects. In certain embodiments, side effects include injection site reactions, liver function test abnormalities, renal function abnormalities, liver toxicity, renal toxicity, central nervous system abnormalities, myopathies, and malaise. For example, increased aminotransferase levels in serum may indicate liver toxicity or liver function abnormality. For example, increased bilirubin may indicate liver toxicity or liver function abnormality.

[0105] “Single-stranded” in reference to an antisense compound or oligomeric compound means there is one oligonucleotide in the compound. “Self-complementary” in reference to an antisense compound or oligomeric compound means a compound that at least partially hybridizes to itself. A compound consisting of one antisense or oligomeric compound, wherein the oligonucleotide of the compound is self-complementary, is a single-stranded compound. A single-stranded antisense or oligomeric compound may be capable of binding to a complementary compound to form a duplex.

[0106] “Sites,” as used herein, are defined as unique nucleobase positions within a target nucleic acid.

[0107] “Slows progression” means decrease in the development of the said disease.

[0108] “Specifically hybridizable” refers to an antisense compound having a sufficient degree of complementarity between an antisense oligonucleotide and a target nucleic acid to induce a desired effect, while exhibiting minimal or no effects on non-target nucleic acids.

[0109] “Specifically inhibit” a target nucleic acid means to reduce or block expression of the target nucleic acid while exhibiting fewer, minimal, or no effects on non-target nucleic acids reduction and does not necessarily indicate a total elimination of the target nucleic acid's expression.

[0110] “Sugar moiety” means a group of atoms that can link a nucleobase to another group, such as an internucleoside linkage, conjugate group, or terminal group. In certain embodiments, a sugar moiety is attached to a nucleobase to form a nucleoside. As used herein, “unmodified sugar moiety” or “unmodified sugar” means a 2′-OH(H) furanosyl moiety, as found in RNA, or a 2′-H(H) moiety, as found in DNA. Unmodified sugar moieties have one hydrogen at each of the 1′, 3′, and

4′ positions, an oxygen at the 3′ position, and two hydrogens at the 5′ position. As used herein, “modified sugar moiety” or “modified sugar” means a furanosyl moiety comprising a non-hydrogen substituent in place of at least one hydrogen of an unmodified sugar moiety, or a sugar surrogate. In certain embodiments, a modified sugar moiety is a 2′-substituted sugar moiety. Such modified sugar moieties include bicyclic sugars and linearly modified sugars.

[0111] “Sugar surrogate” means a modified sugar moiety having other than a furanosyl moiety that can link a nucleobase to another group, such as an internucleoside linkage, conjugate group, or terminal group. Modified nucleosides comprising sugar surrogates can be incorporated into one or more positions within an oligonucleotide. In certain embodiments, such oligonucleotides are capable of hybridizing to complementary oligomeric compounds or nucleic acids.

[0112] “Synergy” or “synergize” refers to an effect of a combination that is greater than additive of the effects of each component alone.

[0113] “Target gene” refers to a gene encoding a target.

[0114] “Target nucleic acid,” “target RNA,” “target RNA transcript” and “nucleic acid target” mean a nucleic acid capable of being targeted by antisense compounds.

[0115] “Targeting” means the process of design and selection of an antisense compound that will specifically hybridize to a target nucleic acid and induce a desired effect.

[0116] “Target region” means a portion of a target nucleic acid to which one or more antisense compounds is targeted.

[0117] “Target segment” means the sequence of nucleotides of a target nucleic acid to which an antisense compound is targeted. “5′ target site” refers to the 5′-most nucleotide of a target segment. “3′ target site” refers to the 3′-most nucleotide of a target segment.

[0118] “Terminal group” means a chemical group or group of atoms that is covalently linked to a terminus of an oligonucleotide.

[0119] “Therapeutically effective amount” means an amount of a compound, pharmaceutical agent, or composition that provides a therapeutic benefit to an individual.

[0120] “Treat” refers to administering a compound or pharmaceutical composition to an animal in order to effect an alteration or improvement of a disease, disorder, or condition in the animal.

[0121] “Unmodified” nucleobases mean the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C) and uracil (U).

[0122] “Unmodified nucleotide” means a nucleotide composed of naturally occurring nucleobases, sugar moieties, and internucleoside linkages. In certain embodiments, an unmodified nucleotide is an RNA nucleotide (i.e. β-D-ribonucleotides) or a DNA nucleotide (i.e. β-D-deoxyribonucleotide).

Certain Embodiments

[0123] Certain embodiments provide methods, compounds and compositions for inhibiting DGAT2 (DGAT2) expression.

[0124] Certain embodiments provide antisense compounds targeted to a DGAT2 nucleic acid. In certain embodiments, the DGAT2 nucleic acid has the sequence set forth in Ref Seq No. NM_032564.3 (incorporated by reference, disclosed herein as SEQ ID NO: 1) or nucleotides 5669186 to 5712008 of Ref Seq No. NT_033927.5 (incorporated by reference, disclosed herein as SEQ ID NO: 2). In certain embodiments, the antisense compound is a single-stranded oligonucleotide.

[0125] Certain embodiments provide an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the antisense compound is a single-stranded oligonucleotide.

[0126] Certain embodiments provide an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 9, at least 10, at least 11, or at least 12 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the antisense compound

is a single-stranded oligonucleotide.
[0127] Certain embodiments provide an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the antisense compound is a single-stranded oligonucleotide.
[0128] Certain embodiments provide an antisense compound comprising a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the antisense compound is a single-stranded oligonucleotide.
[0129] In certain embodiments, antisense compounds or antisense oligonucleotides target the 26711-26802 of a DGAT2 nucleic acid. In certain aspects, antisense compounds or antisense oligonucleotides target within nucleotides 26711-26802 of a DGAT2 nucleic acid having the nucleobase sequence of SEQ ID NO: 2 (nucleotides 5669186 to 5712008 of Ref Seq No. NT_033927.5). In certain aspects, antisense compounds or antisense oligonucleotides have at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion complementary to an equal length portion within nucleotides 26711-26802 of a DGAT2 nucleic acid having the nucleobase sequence of SEQ ID NO: 2 (nucleotides 5669186 to 5712008 of Ref Seq No. NT_033927.5).
[0130] In certain embodiments, antisense compounds or antisense oligonucleotides target a region of a DGAT2 nucleic acid having the nucleobase sequence of SEQ ID NO: 2 within nucleobases 26711-26799, 26711-26730, 26721-26740, 26755-26744, 26778-26797, 26779-26798, 26755-26798, and 26780-26799. In certain aspects, antisense compounds or antisense oligonucleotides target at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobases within the aforementioned nucleobase regions.
[0131] In certain embodiments, the following nucleotide regions of SEQ ID NO: 2, when targeted by antisense compounds or oligonucleotides, display at least 60% inhibition: 9930-9949, 9953-9973, 9959-9978, 9961-9981, 9970-9992, 9975-9994, 9984-10003, 10040-10059, 10043-10063, 10045-10064, 10046-10065, 10049-10068, 10054-10073, 10160-10179, 10209-10229, 10212-10231, 10214-10235, 10217-10236, 10218-10238, 10327-10346, 10389-10413, 10490-10511, 10522-10548, 10530-10549, 10531-10551, 10533-10553, 10645-10669, 10651-10670, 10653-10673, 10655-10674, 10656-10675, 10657-10676, 10658-10680, 10662-10681, 10663-10684, 10666-10685, 10667-10684, 10670-10689, 10702-10721, 10727-10747, 10729-10748, 10730-10750, 10733-10755, 10737-10756, 10742-10761, 10763-10782, 10814-10835, 10817-10836, 10818-10837, 10940-10964, 11008-11027, 11074-11097, 11123-11150, 11158-11182, 11168-11187, 11170-11190, 11171-11197, 11199-11218, 11200-11220, 11209-11228, 11318-11342, 11413-11432, 11559-11578, 11594-11618, 11707-11731, 11872-11891, 11920-11939, 12247-12266, 12285-12304, 12549-12573, 12671-12695, 12806-12826, 12839-12863, 12890-12909, 12927-12961, 13098-13117, 13212-13231, 13239-13258, 13441-13461, 13921-13945, 13962-13996, 14177-14201, 14194-14218, 14239-14258, 14385-14404, 14457-14522, 14504-14527, 14537-14556, 14648-14697, 14715-14790, 14715-14739, 14735-14764, 14746-14765, 14749-14774, 14756-14779, 14765-14789, 14771-14790, 14807-14831, 14852-14871, 14882-14901, 14953-15015, 14997-15065, 15041-15107, 15041-15092, 15083-15107, 15178-15197, 15214-15243, 15219-15243, 15248-15276, 15255-15276, 15292-15311, 15318-15338, 15321-15342, 15353-15372, 15368-15392, 15514-15533, 15573-15665, 15573-15597, 15729-15753, 15804-15828, 15814-15833, 15876-15900, 15876-15896, 15933-15962, 15933-15954, 15991-16010, 16046-16073, 16082-16124, 16237-16256, 16288-16314, 16288-16478, 16308-16329, 16428-16478, 16510-16529, 16808-16855, 16808-16828, 16811-16855, 16898-16921, 17007-17031, 17038-17057, 17207-17229, 17269-17298, 17273-17294, 17328-17357, 17420-17444, 17450-17471, 17530-17559, 17328-17559, 17960-17989, 18078-18097, 18155-18176, 18155-18198, 18177-18200, 18225-18305, 18306-18327, 18309-18330, 18312-18330, 18312-18332, 18315-18344, 18329-18355, 18379-18398, 18417-18446, 18430-18451, 1818450-18473, 18470-18494, 18581-18601, 18583-18605, 18581-18611, 18588-18611, 18821-18850, 18821-18847, 18836-

18899, 18890-18914, 18898-18919, 19013-19037, 19013-19040, 19360-19411, 19363-19383, 19804-19823, 19909-19932, 19987-20009, 20232-20256, 20296-20325, 20462-20481, 20541-20560, 20668-20692, 20775-20797, 20779-20797, 20817-20841, 20828-20863, 20878-20897, 20956-20979, 21046-21120, 21091-21120, 21164-21203, 21164-21188, 21374-21403, 21659-21678, 21708-21727, 21765-21784, 21933-21955, 22153-22191, 22153-22175, 22158-22179, 22167-22191, 22439-22463, 22547-22567, 22770-22790, 22770-22795, 22775-22795, 22835-22870, 22841-22865, 22881-22901, 22885-22905, 22889-22910, 23095-23115, 23098-23119, 23099-23119, 23128-23157, 23179-23251, 23238-23258, 23240-23260, 23242-23262, 23429-23449, 23480-23566, 23547-23581, 23553-23573, 23555-23576, 23562-23586, 23597-23616, 23641-23670, 24123-24144, 24123-24147, 24202-24223, 24202-24241, 24799-25039, 24799-24847, 25018-2509, 25023-25053, 25026-25053, 25069-25092, 25069-25114, 25077-25104, 25088-25114, 25183-25205, 25183-25217, 25198-25217, 25226-25250, 25226-25265, 25236-25260, 25244-25291, 25312-25394, 25380-25400, 25717-25736, 25782-25860, 25845-25866, 25866-25935, 25919-25939, 25950-25971, 25954-25973, 25981-26030, 26102-26126, 26162-26220, 26300-26324, 26331-26372, 26363-26405, 26389-26409, 26396-26420, 26479-26504, 26516-26550, 26534-26555, 26534-26560, 26540-26560, 26610-26634, 26620-26640, 26626-26648, 26620-26664, 26633-26655, 26640-26664, 26716-26740, 26716-26799, 26755-26799, 26786-26807, 26789-26809, 26786-26809, 26811-26831, 26849-26869, 26869-26898, 26883-26903, 26926-26943, 27046-27075, 27106-27130, 27174-27230, 27221-27241, 27226-27258, 27241-27265, 27249-27269, 27367-27406, 27367-27402, 27386-27406, 27638-27664, 27769-27791, 27775-27810, 27852-27877, 28026-28049, 28128-28154, 28462-28492, 28475-28508, 28516-28542, 28584-28605, 28804-28826, 29010-29035, 29143-29167, 29151-29180, 29196-29218, 29244-29268, 29253-29282, 29361-29383, 29369-29391, 29450-29495, 29480-29504, 29538-29654, 29673-29697, 29681-29702, 29781-29801, 29804-29828, 29827-29847, 29832-29852, 29838-29858, 29844-29878, 30484-30504, 30524-30548, 30554-30588, 30669-30731, 30972-30997, 30984-31005, 31551-31571, 31554-31584, 31794-31818, 31932-31952, 31936-31956, 31939-31963, 31946-31971, 32115-32136, 32119-32140, 32130-32159, 32175-32201, 32428-32454, 32436-32467, 32459-32479, 32463-32489, 32474-32495, 32480-32529, 32586-32638, 32621-32641, 32756-32776, 32801-32825, 32816-32840, 33070-33180, 33176-33199, 33182-33202, 33411-33433, 33572-33601, 33729-33759, 33846-33876, 33860-33882, 33866-33886, 34063-34117, 34211-34232, 34686-34710, 34739-34887, 34901-34950, 35263-35292, 35277-35302, 35322-35346, 35408-35433, 35421-35441, 35436-35457, 35665-35685, 36246-36267, 36250-36270, 36253-36274, 36258-36281, 36268-36531, 36516-36537, 36573-36597, 36581-36603, 36632-36666, 36677-36698, 36682-36702, 36689-36757, 36742-36772, 36841-36874, 36846-36874, 36865-37041, 37054-37073, 37838-37858, 39668-39694, 39684-39705, 39820-39840, 39830-39852, and 40909-40933.

[0132] In certain embodiments, the following nucleotide regions of SEQ ID NO: 2, when targeted by antisense compounds or oligonucleotides, display at least 70% inhibition: 9971-9991, 9975-9994, 9984-10003, 10040-10059, 10044-10063, 10045-10064, 10049-10068, 10054-10179, 10210-10229, 10214-10234, 10216-10235, 10217-10236, 10218-10237, 10327-10346, 10490-10509, 10491-10511, 10522-10541, 10528-10548, 10530-10549, 10531-10550, 10532-10551, 10533-10553, 10536-10555, 10537-10556, 10645-10664, 10647-10666, 10649-10669, 10653-10673, 10655-10674, 10656-10675, 10657-10676, 10658-10677, 10659-10679, 10661-10680, 10662-10681, 10663-10683, 10665-10684, 10666-10685, 10667-10686, 10668-10687, 10670-10689, 10672-10691, 10702-10721, 10727-10747, 10729-10748, 10730-10750, 10732-10751, 10733-10752, 10734-10753, 10735-10754, 10737-10756, 10742-10756, 10742-10761, 10763-10782, 10815-10835, 10817-10836, 10818-10837, 10819-10840, 10940-10959, 10941-10960, 10942-10961, 10944-10963, 11008-11027, 11074-11093, 11075-11094, 11076-11095, 11078-11097, 11124-11146, 11128-11147, 11129-11148, 11130-11149, 11131-11177, 11159-11178, 11160-11179, 11161-11180, 11168-11187, 11170-11190, 11172-11191, 11173-11192, 11174-11196, 11180-11218, 11200-11219, 11201-11221, 11203-11222, 11204-11220, 11209-11228, 11323-11342, 11413-11432, 11559-11578, 11599-11618, 11872-11891, 112285-12304, 12549-12568, 12806-12826, 12809-12828, 12927-12946, 12942-12961, 13441-13461, 13926-13945, 14182-14201, 14194-14213, 14196-14216, 14239-14258, 14385-14404, 14462-14522, 14504-14523, 14505-14524, 14506-14527, 14537-14556, 14648-14667, 14649-14668, 14650-14669, 14651-14670, 14658-14682, 14664-14683, 14665-14684, 14666-14686, 14668-14687, 14673-14739, 14741-14764, 14746-14765, 14749-14774, 14756-14775, 14757-14779, 14765-14784, 14767-14786, 14769-14789, 14771-14790, 14772-14793, 14807-14831, 14852-15005, 14992-15012, 14995-15015, 14997-15016, 14998-15017, 14999-15020, 15011-15040, 15073-15092, 15214-15233, 15219-15243, 15248-15269, 15248-15276, 15251-15270, 15252-15272, 15254-15273, 15255-15274, 15256-15276, 15321-15342, 15573-15595, 15578-15665, 15729-15748, 15809-15828, 15876-15895, 15881-15952, 15991-16066, 16051-16070, 16053-16102, 16289-16309, 16292-16312, 16310-16329, 16248-16448, 16432-16451, 16438-16457, 16808-16828, 16811-16830, 16813-16832, 16815-16835, 16828-16852, 16836-16855, 16898-16917, 16900-16921, 17007-17027, 17207-17227, 17210-17229, 17274-17294, 17328-17347, 17330-17352, 17450-17549, 17960-17979, 18155-18175, 18157-18176, 18160-15180, 18162-18181, 18163-18182, 18164-18183, 18165-18184, 18166-18185, 18172-18193, 18175-18194, 18176-18194, 18176-18195, 18178-18199, 18181-18200, 18225-18244, 18235-18305, 18307-18327, 18309-18328, 18310-18330, 18312-18332, 18315-18337, 18320-18340, 18322-18341, 18323-18342, 18324-18343, 18325-18347, 18330-18355, 18379-18436, 18418-18437, 18419-18438, 18420-18439, 18422-18441, 18431-18451, 18450-18469, 18451-18470, 18452-18471, 18470-18489, 18581-18600, 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25193-25212, 25198-25250, 25236-25255, 25245-25291, 25312-25394, 25380-25399, 25429-25448, 25717-25808, 25841-25860, 25846-25865, 25900-25928, 25911-25930, 25913-25935, 25920-25971, 25981-26026, 26010-26030, 26102-26121, 26165-26186, 26169-26219, 26300-26324, 26353-26372, 26358-26377, 26363-26404, 26389-26409, 26397-26416, 26479-26498, 26482-26503, 26516-26535, 26517-26536, 26518-26537, 26519-26538, 26520-26540, 26526-26550, 26535-26555, 26537-26556, 26538-26557, 26539-26558, 26615-26634, 26620-26640, 26626-26646, 26628-26647, 26629-26648, 26630-26649, 26631-26650, 26632-26651, 26633-26652, 26635-26654, 26640-26659, 26711-26730, 26721-26799, 26786-26807, 26811-26830, 26852-26871, 26854-26873, 26869-26898, 26883-26903, 26927-26946, 27050-27075, 27106-27125, 27221-27240, 27228-27247, 27231-27250, 27235-27257, 27241-27265, 27249-27268, 27367-27386, 27377-27396, 27379-27398, 27381-27401, 27383-27402, 27387-27406, 27438-27457, 27639-27658, 27642-27661, 27644-27664, 27769-27790, 27775-27799, 27782-27809, 27853-27877, 28026-28048, 28128-28154, 28463-28484, 28468-28489, 28475-28500, 28516-28542, 28806-28826, 29010-28033, 29140-29159, 29143-29165, 29148-29167, 29154-29177, 29160-29180, 29193-29212, 29196-29218, 29201-29220, 29245-29265, 29253-29272, 29255-29274, 29257-29278, 29262-29282, 26361-29382, 29369-29388, 29371-29390, 29485-29504, 29538-29557, 29548-29594, 29635-29654, 29673-29692, 29675-29695, 29678-29697, 29682-29702, 29781-29801, 29809-29828, 29827-59847, 29829-29848, 29830-29849, 29831-29850, 29832-29852, 29838-29858, 29840-29859, 29842-29861, 29843-29862, 29844-29873, 30483-30502, 30484-30503, 30514-30533, 30529-30548, 30559-30578, 30669-30693, 30972-30992, 30977-30996, 30982-31001, 30986-31005, 31554-31573, 31768-31787, 31797-31817, 31933-31952, 31936-31956, 31942-31961, 32116-32135, 32120-32140, 32131-32151, 32135-32158, 32177-32198, 32182-32201, 32336-32355, 32428-32447, 32429-32448, 32430-32449, 32431-32450, 32432-32451, 32433-32452, 32434-32450, 32435-32454, 32436-32455, 32446-32466, 32449-32468, 32450-32469, 32451-32470, 32452-32471, 32453-32472, 32454-32473, 32455-32474, 32456-32475, 32457-32476, 32459-32479, 32461-32480, 32462-32481, 32463-32483, 32465-32486, 32468-32489, 32475-32495, 32477-32496, 32478-32497, 32480-32500, 32510-32529, 32619-32638, 32621-32640, 32622-32641, 32623-32642, 32645-32664, 32670-32689, 32711-32730, 32752-32771, 32753-32772, 32755-32774, 32756-32776, 32816-32840, 33096-33125, 33161-33180, 33179-33199, 33181-33200, 33182-33201, 33412-33433, 33729-33750, 33861-33881, 33866-33885, 33918-33937, 34017-34036, 34211-34231, 34216-34235, 34350-34369, 34460-34479, 34739-34887, 34901-34950, 35278-35298, 35322-35346, 35362-35381, 35410-35430, 35414-35433, 35421-35440, 35436-35457, 35665-36685, 36246-36267, 36249-36268, 36250-36269, 36251-36270, 36252-36271, 36253-36272, 36258-36277, 36262-36281, 36268-36287, 36354-36428, 36512-36531, 36517-36536, 36573-36597, 36582-36602, 36617-36636, 36677-36698, 36680-36699, 36681-36700, 36682-36702, 36691-36754, 36738-36757, 36743-36772, 36758-36777, 36841-36860, 36846-36865, 36865-36942, 37054-37138, 37389-37408, 37497-37516, 37499-37518, 37501-37520, 37745-37764, 39670-39689, 39685-39704, 39820-39839, 39830-39849, 39832-39852, and 40333-40352.

[0133] In certain embodiments, the following nucleotide regions of SEQ ID NO: 2, when targeted by antisense compounds or oligonucleotides, display at least 80% inhibition: 10040-10059, 10044-10063, 10045-10064, 10054-10073, 10210-10229, 10214-10234, 10217-10236, 10490-10509, 10531-10550, 10532-10551, 10533-10552, 10645-10664, 10650-10669, 10655-10674, 10656-10675, 10657-10676, 10659-10678, 10660-10679, 10661-10680, 10663-10682, 10664-10683, 10665-10684, 10666-10685, 10667-10686, 10670-10689, 10727-10746, 10730-10749, 10731-10750, 10732-10751, 10734-10753, 10735-10754, 10737-10756, 10816-10835, 10817-10836, 10818-10837, 10819-10838, 10820-10840, 10940-10959, 10941-10960, 10942-10961, 10944-10963, 11074-11093, 11075-11094, 11127-11146, 11128-11147, 11129-11148, 11130-11149, 11131-11177, 11159-11178, 11160-11179, 11170-11190, 11172-11191, 11173-11192, 11174-11193, 11175-11196, 11178-11197, 11199-11218, 11200-11219, 11201-11220, 11202-11221, 11203-11222, 11559-11578, 11599-11618, 12285-12304, 13442-13461, 14462-14481, 14503-14522, 14504-14523, 14505-14524, 14506-14527, 14648-14667, 14649-14668, 14650-14669, 14651-14670, 14658-14682, 14664-14683, 14666-14686, 14673-14697, 14741-14760, 14743-14764, 14746-14765, 14750-04769, 14752-14774, 14756-14775, 14757-14775, 14758-14777, 14759-14778, 14769-14789, 14771-14790, 14772-14791, 14773-14792, 14774-14793, 14807-14826, 14882-14901, 14986-15005, 14993-15012, 14995-15015, 14997-15016, 14998-15017, 14999-15018, 15000-15019, 15001-15020, 15016-15035, 15248-15268, 15250-15269, 15251-15270, 15252-15272, 15254-15273, 15319-15338, 15321-15342, 15574-15593, 15646-15665, 15881-15952, 16310-16329, 16813-16832, 16830-16852, 16900-16921, 17007-17026, 17208-17227, 17450-17470, 18156-18175, 18157-18176, 18161-18180, 18164-18183, 18166-18185, 18173-18193, 18175-18194, 18176-18195, 18180-18199, 18308-18327, 18310-18329, 18311-18330, 18312-18331, 18321-18340, 18323-18342, 18324-18343, 18325-18344, 18326-18345, 18327-18346, 18328-18347, 18331-18350, 18379-18436, 18432-18451, 18450-18469, 18581-18600, 18585-18604, 18828-18847, 18829-18848, 18900-18919, 19911-19930, 20300-20321, 20668-20687, 20830-20849, 20837-20857, 20958-20977, 22444-22463, 22548-22567, 22771-22790, 22772-22791, 22773-22792, 22886-22905, 23096-23115, 23100-23119, 23239-23258, 23242-23261, 23243-23262, 23244-23263, 23245-23264, 23246-23262, 23427-23446, 23551-23570, 23552-

23571, 23557-23576, 24123-24142, 24878-24897, 25023-25042, 25028-25047, 25033-25088, 25073-25092, 25075-25094, 25078-25101, 25085-25104, 25183-25202, 25193-25212, 25246-25291, 25375-25394, 25380-25399, 25782-25804, 25787-25806, 25841-25860, 25846-25865, 25900-25919, 25905-25928, 25911-25930, 25915-25934, 25950-25970, 26005-26025, 26010-26029, 26162-26185, 26169-26191, 26174-26194, 26179-26219, 26300-26319, 26302-26321, 26304-26324, 26353-26372, 26358-26377, 26482-26501, 26517-26536, 26519-26538, 26520-26539, 26521-26540, 26536-26550, 26537-26556, 26538-26557, 26615-26634, 26620-26639, 26627-26646, 26629-26648, 26630-26649, 26631-26650, 26632-26651, 26635-26654, 26640-26659, 26711-26730, 26778-26798, 26788-26807, 26811-26830, 26884-26903, 27050-27070, 27053-27072, 27055-27075, 27221-27240, 27231-27250, 27236-27256, 27241-27260, 27382-27401, 27383-27402, 27387-27406, 27770-27789, 27775-27794, 27777-27796, 27779-27799, 27782-27804, 27787-27809, 27853-27877, 28026-28046, 28029-28048, 28128-28149, 28133-28152, 28463-28482, 28468-28488, 28476-28497, 28480-28500, 29011-29030, 29143-29163, 29160-29179, 29193-29212, 29245-29265, 26253-26272, 29369-29388, 29548-29589, 29678-29697, 29683-29702, 29782-29801, 29827-29847, 29829-29848, 29830-29849, 29838-29858, 29840-29859, 29841-29860, 29842-29861, 29844-29863, 29854-29873, 30483-30502, 30484-30503, 30529-30548, 30674-30693, 30982-31001, 31554-31573, 32131-32151, 32135-32154, 32182-32201, 32431-32450, 32432-32451, 32433-32452, 32434-32453, 32446-32466, 32448-32467, 32449-32468, 32450-32469, 32451-32470, 32452-32471, 32453-32472, 32454-32473, 32455-32474, 32456-32475, 32459-32479, 32461-32480, 32462-32481, 32463-32482, 32464-32483, 32465-32484, 32466-32485, 32467-32486, 32468-32487, 32469-32488, 32470-32489, 32476-32495, 32477-32496, 32480-32499, 32619-32638, 32620-32639, 32621-32640, 32623-32642, 32645-32664, 32752-32771, 32753-32772, 32757-32776, 32816-32835, 33179-33199, 33181-33200, 33182-33201, 33412-33433, 33729-33749, 33866-33885, 34211-34231, 34739-34758, 34931-34950, 35436-35456, 35666-35685, 36247-36266, 36249-36268, 36250-36269, 36251-36270, 36252-36271, 36253-36272, 36258-36277, 36354-36373, 36409-36428, 36512-36531, 36517-36536, 36583-36602, 36680-36699, 36681-36700, 36841-36860, 36846-36865, 39670-39689, and 39832-39851.

[0134] In certain embodiments, the following nucleotide regions of SEQ ID NO: 2, when targeted by antisense compounds or oligonucleotides, display at least 90% inhibition: 32447-32466, 32452-32471, 32460-32479, 32462-32481, 32464-32483, 32465-32484, 32466-32485, 32467-32486, 32468-32487, 32619-32638, 32620-32639, 32753-32772, 36250-36269, 36517-36536, 32432-32451, 32431-32450, 30483-30502, 29828-29847, 29011-29030, 28026-28045, 29011-29030, 29828-29847, 30483-30502, 32431-32450, 32432-32451, 32447-32466, 32452-32471, 32460-32479, 32462-32481, 32464-32483, 32465-32484, 32466-32485, 32467-32486, 32468-32487, 32619-32638, 32620-32639, 32753-32772, 36250-36269, 36252-36271, 36409-36428, and 36517-36536.

[0135] In certain embodiments, the following antisense compounds or antisense oligonucleotides target a region of a DGAT2 nucleic acid and effect at least a 60% inhibition of a DGAT2 mRNA, ISIS NOs: 413236, 413284, 413391, 413399, 413413, 413422, 413433, 413433, 413441, 413446, 423460-423466, 423520-423527, 423601-423604, 472182, 472188, 472189, 472194-472196, 472203-472206, 472208, 472347, 472349-472352, 472398, 483803, 483811-483814, 483816-483818, 483821-483823, 483825-483835, 483838-483842, 483846-483848, 483852, 483853, 483862, 483866, 483868-483875, 483879, 483886-483890, 483892, 483895, 483897-483903, 483906-483913, 483916-483929, 483932-483934, 483936, 483942, 483948-483950, 483952, 483954, 483956, 483961, 483968-483973, 483975-483979, 483981, 483983-483989, 483992-483997, 484001, 484002, 484004-484006, 484008-484013, 484015-484017, 484019, 484020, 484028, 484030, 484031, 484041, 484049, 484050, 484054-484057, 484063-484065, 484069-484073, 484081-484083, 484085, 484089, 484094, 484097-484107, 484110, 484111, 484114-484119, 484123, 484125-484127, 484129-484137, 484139-484142, 484145-484149, 484152, 484154, 484156-484159, 484161, 484162, 484164, 484165, 484167-484172, 484178-484182, 484184, 484185, 484188, 484190-484198, 484202-484204, 484209-484211, 484213, 484215, 484217-484221, 484227, 484231, 484232, 484234, 484235, 484237, 484240-484243, 484248-484250, 484257, 484263, 484265-484271, 484273-484276, 484281-484285, 484290, 484292, 484293, 484298, 484299, 484301, 484302, 484310, 484319-484325, 484327, 484336, 484338, 484342-484344, 484346-484357, 484359, 484362-484365, 484368, 484370-484387, 484390, 484404, 484411, 495425, 495428, 495429, 495430, 495436, 495438, 495440-495459, 495461-495473, 495475, 495476, 495479-495492, 495495-495511, 495514-495545, 495548-495580, 495584, 495585, 495587-495594, 495596-495626, 495630-495639, 495641-495644, 495648-495651, 495656, 495657, 495660-495662, 495664-495667, 495669-495671, 495676, 495677, 495684-495689, 495692-495707, 495711, 495713, 495714, 495717-495732, 495734, 495736-495739, 495742-495757, 495762-495766, 495772-495774, 495781-495786, 495788, 495789, 495791, 495793-495797, 495799, 495800, 495804, 495808-495813, 495817-495844, 495847-495860, 495862-495870, 495873-495883, 495886, 495899-495904, 495909, 495911-495914, 495918, 495920-495923, 495955, 495957, 495958, 495983, 495992, 496004, 496009, 496010, 496012, 496033, 500841, 500843, 500844, 500852, 500859, 500866, 500867, 500870, 500871, 500892, 500895, 500912, 500913, 500928, 500929, 500934, 500935, 500942, 500944-500946, 500950, 500953, 500957, 500960, 500961, 500966, 500974, 500975, 500979, 500982, 500985, 500986, 500989, 500990, 500994, 500995, 500997, 500998, 501004, 501007, 501008, 501016, 501018-501020, 501025, 501026, 501029-501031, 501033-501035, 501037-501041, 501043, 501055, 501061, 501062, 501064, 501067-501070, 501076, 501093-501098, 501100, 501103, 501104, 501106, 501108, 501111, 501117-501119, 501122-501130, 501140, 501154, 501155, 501158, 501165, 501171, 501176, 501177, 501182-501184, 501193, 501194, 501199, 501200, 501209, 501210, 501212-501214, 501217, 501224, 501227-501230, 501240, 501241, 501244, 501247-501249, 501254, 501256, 501264, 501270, 501287, 501289, 501290, 501297, 501319, 501322, 501326, 501332, 501335, 501342, 501345, 501353, 501357, 501370, 501382-501406, 501410-501412, 501414, 501426-501431, 501434-501443, 501445-501457, 501831, 501835, 501837, 501838, 501849-501853, 501855, 501861, 501867, 501871, 501880, 501883, 501884, 501886-501890, 501900, 501903, 501916, 501932-501934, 501944, 501946, 501947, 501950, 501951, 501957, 501959, 501960, 501966, 501968, 501976, 501977, 501979, 501981-501983, 501991, 501996, 502011, 502013-502015, 502019, 502024-502028, 502031, 502034, 502036-502042, 502045, 502046, 502050, 502053, 502055, 502056, 502062, 502065, 502068-502070, 502083-502085, 502095-502100, 502102, 502106, 502110, 502113, 502114, 502117, 502119, 502120, 502122, 502124, 502127, 502131, 502135, 502144, 502146, 502149, 502150, 502154-502156, 502158, 502163, 502164, 502167, 502172-502176, 502179, 502186, 502189, 502191, 502192, 502194, 502204, 502206, 502215, 502220, 502223, 502228, 502236, 502314, 502319, 502320, 502322, 502334, 502336, 507663, 507665, 507667, 507669, 507675, 507677, 507679, 507684, 507689-507696, 507705, 507709-507719, 507724, 507726, 522363, 522365, 522366, 522368-522375, 522383-522386, 522389, 522391, 522395, 522396, 522398-522413, 522416-522422, 522424-522440, 522442-522452, 522454, 522456, 522457, 522461-522467, 522469-522474, 522478-522490, 522492-522513, 522516-522518, 522521, 522522, 522525, 522529, 522530, 522533-522535, 522540-522550, 522552-522558, 522561-522565, 522573, 522577-522582, 522586-522593, 522597, 522598, 522602-522614, 522617-522625, 522627-522639, 522642-522645, 522657-522663, 522666-522668, 522671-522684, 522686-522702, 522704-522728, 522732, 522733, 522738-522740, 522744-522749, 522754, 522757-522761, 522763-522771, 522776-522778, 522780, 522782-522785, 522788-522797, 522801-522803, 522805, 522807-522809, 522813, 522821-522824, 522829-522831, 522835-522839, 522842, 522844, 522846-522850, 522852-522854, 522856, 522857, 522865-522873, 522875-522880, 522887-522897-522899, 522901, 522904-522906, 522909, 522913, 522914, 522917, 522924, 522925, 522927, 522931-522935, 522939-522943, 522945-522950, 522956, 522964, 522965, 522970, 522977-522982, 522986, 522987, 522996, 522997, 523000-523002, 523015, 523016, 523022, 523025-523027, 525388, 525391, 525394, 525395, 525399, 525401-525405, 525414-525416, 525419, 525420, 525423, 525424, 525431, 525434-525436, 525441-525444, 525450, 525461, 525468-525474, 525477, 525479, 525480, 525484, 525485, 525488-525491, 525498-525501, 525504-525507, 525511-525518, 525520, 525528, 525535-525537, 525539, 525541, 525542, 525544, 525547-525558, 525565, 525577-525579, 525606, 525609-525612, 525619, 525631, 525632, 525649, 525650, 525658, 525683, 525687, 525688, 525690, 525705-525711, 525733, 525737, 525740, 525741, and 525754.

[0136] In certain embodiments, the following antisense compounds or antisense oligonucleotides target a region of a DGAT2 nucleic acid and effect at least a 60% inhibition of a DGAT2 mRNA, SEQ ID NOs: 20, 38, 39, 41, 46-48, 61-64, 68, 69, 74, 75, 76, 103, 120, 121, 122, 131, 134, 144, 145, 150, 152, 153, 154, 181, 182, 183, 202-204, 221, 225, 226, 229, 236, 237, 246, 247, 251, 259, 262, 272, 277-280, 283, 383, 391, 405, 414, 425, 433, 438, 450, 452, 453, 455-479, 482-496, 504-507, 509-511, 514-516, 518-528, 531-535, 539-541, 545, 546, 555, 559, 561-568, 574, 575, 577, 578, 583, 584, 589, 594-596, 597, 601, 607, 608, 613-615, 622-625, 627, 652, 653, 660, 674, 706, 709-711, 718-720, 726, 738, 740-745, 749-752, 754-756, 758-769, 774, 789, 793, 794, 807-812, 815, 819, 820, 839, 841-846, 848-852, 878-880, 884, 885, 889, 890, 892-895, 897, 899, 900, 902, 904, 907, 910, 913-916, 923, 935-940, 960-964, 968, 972-977, 979-981, 983-985, 988, 991-1010, 1014, 1015, 1017, 1018, 1020, 1023-1025, 1032-1034, 1036-1038, 1040, 1043-1049, 1051-1054, 1057-1061, 1065, 1071-1073, 1078-1080, 1083, 1084, 1092-1098, 1103, 1107, 1108, 1117, 1118, 1136, 1141, 1142, 1148, 1149, 1153, 1155, 1159, 1165, 1172-1176, 1178, 1181, 1183-1189, 1192-1199, 1202-1215, 1218-1220, 1222, 1228, 1234-1236, 1238, 1240, 1242, 1247, 1254-1259, 1261-1265, 1267, 1269-1275, 1278-1283, 1287, 1288, 1290-1292, 1294-1299, 1301-1303, 1305, 1306, 1314, 1316, 1317, 1327, 1335, 1336, 1340-1343, 1349-1351, 1355-1359, 1367-1369, 1371, 1375, 1380, 1383-1393, 1396, 1397, 1400-1405, 1409, 1411-1413, 1415-1423, 1425-1428, 1431-1435, 1438, 1440, 1442-1445, 1447, 1448, 1450, 1451, 1453-1458, 1464-1468, 1470, 1471, 1474, 1476-1484, 1488-1490, 1495-1497, 1499, 1501, 1503-1507, 1513, 1517, 1518, 1520, 1521, 1523, 1526-1529, 1534-1536, 1543, 1549, 1551-1557, 1559-1562, 1567-1571, 1576, 1578, 1579, 1584, 1585, 1587, 1588, 1596, 1605-1611, 1613, 1622, 1624, 1628-1630, 1632-1643, 1645, 1648-1651, 1654, 1656-1673, 1676, 1677, 1690, 1697, 1698, 1701-1703, 1709, 1711, 1713-1746, 1748, 1749, 1752-1765, 1768-1784, 1787-1818, 1821-1853, 1857, 1858, 1860-1867, 1869-1899, 1903-1912, 1914-1917, 1921-1924, 1929, 1930, 1933-1935, 1937-1940, 1942-1944, 1949, 1950, 1957-1962, 1965-1980, 1984, 1986, 1987, 1990-2005, 2007, 2009-2012, 2015-2030, 2035-2039, 2045-2047, 2054-2059, 2061, 2062, 2064, 2066-2070, 2072, 2073, 2077, 2081-2086, 2090-2117, 2120-2133, 2135-2143, 2146-2156, 2159, 2172-2177, 2182, 2184-2187, 2191, 2193-2196, 2228, 2230, 2231, 2256, 2265, 2277, 2282, 2283, 2285, 2306, 2308, 2310-2334, 2338-2340, 2342, 2350, 2351, 2354-2359, 2362-2371, 2373-2385, 2402, 2405, 2409, 2415, 2418, 2425, 2428, 2436, 2440, 2453, 2465-2468, 2478, 2479, 2482, 2485-2487, 2492, 2494, 2502, 2508, 2525, 2527, 2528, 2535, 2543, 2544, 2547, 2554, 2560, 2565, 2566, 2571-2573, 2582, 2583, 2588, 2589, 2598, 2599, 2601-2603, 2606, 2613, 2615, 2632-2637, 2639, 2642, 2643, 2645, 2647, 2650, 2656-2658, 2661-2669, 2679, 2691, 2692, 2694, 2695, 2701, 2704, 2705, 2713, 2715-2717, 2722, 2723, 2726, 2727, 2728, 2730-2732, 2734-2739, 2748, 2754, 2755, 2757, 2760-2763, 2779, 2780, 2785, 2786, 2793, 2795-2797, 2801, 2804, 2808, 2811, 2812, 2817, 2825, 2826, 2830, 2833, 2836, 2837, 2840, 2841, 2846, 2848, 2849, 2857, 2864, 2871, 2872, 2875, 2876, 2897, 2900, 2917, 2918, 2929, 2933, 2935, 2936-2951, 2953, 2959, 2965, 2969, 2978, 2981, 2982, 2984-2988, 2998, 3001, 3014, 3030-3032, 3042, 3044, 3045, 3048, 3049, 3055, 3057, 3058, 3064, 3066, 3074, 3075, 3077, 3079, 3080, 3081, 3089, 3094, 3109, 3111-3113, 3117, 3122-3126, 3129, 3132, 3134-3140, 3143, 3144, 3148, 3151, 3153, 3154, 3160, 3163, 3166-3168, 3172-3174, 3181-3183, 3193-3198, 3200, 3204, 3208, 3211, 3212, 3215, 3217, 3218, 3220, 3222, 3225, 3229, 3233, 3242, 3244, 3247, 3248, 3252-3254, 3256, 3261, 3262, 3265, 3270-3274, 3277, 3284, 3287, 3289, 3290, 3292, 3302, 3304, 3313, 3318, 3321, 3326, 3334, 3388, 3390, 3392, 3393, 3396, 3398, 3404, 3408, 3410, 3412, 3417, 3418, 3420, 3422-3429, 3432, 3434, 3438, 3442-3452, 3464, 3466, 3468, 3470, 3471, 3473-3480, 3488-3491, 3494, 3496, 3500, 3501, 3503-3518, 3521-3527, 3529-3545, 3547-3557, 3559, 3561, 3562, 3566-3572, 3574-3579, 3583-3595, 3597-3618, 3621-3623, 3626, 3627, 3630, 3634, 3635, 3638-3640, 3645-3655, 3657-3663, 3666-3670, 3678, 3682-3687, 3691-3698, 3702, 3703, 3707-3719, 3722-3730, 3732-3744, 3747-3750, 3762-3768, 3771-3773, 3776-3789, 3791-3807, 3809-3833, 3837, 3838, 3843-3845, 3849-3854, 3859, 3862-3866, 3868-3876, 3881-3883, 3885, 3887-3890, 3893-3902, 3906-3908, 3910, 3912-3914, 3918, 3926-3929, 3934-3936, 3940-3944, 3947, 3949, 3951-3955, 3957-3959, 3961, 3962, 3970-3978, 3980-3985, 3992-3995, 3997-4000, 4002-4004, 4006, 4009-4011, 4014, 4018, 4019, 4022, 4029, 4030, 4032, 4036-4040, 4044-4048, 4050-4055, 4061, 4069, 4070, 4075, 4082, 4083, 4085, 4086, 4092, 4095-4097, 4105-4109, 4113, 4114, 4123, 4124, 4127-4129, 4143, 4146, 4149, 4150, 4154, 4156-4160, 4169-4171, 4174, 4175, 4178, 4179, 4186, 4189-4191, 4196-4199, 4205, 4222, 4229-4235, 4238, 4240, 4241, 4245, 4246, 4249-4252, 4259-4262, 4265-4268, 4272-4279, 4281, 4289, 4296-4298, 4300, 4302, 4303, 4305, 4308-4319, 4326, 4338-4340, 4364, 4367, 4368, 4372, 4373, 4380, 4392, 4393, 4410, 4411, 4419, 4444, 4448, 4449, 4451, 4466-4472, 4494, 4498, 4501, 4502, 4515, 4526-4530, 4532-4535, 4537, 4538-4541, 4544-4546, 4556-4573, 4575-4578, 4580-4583, 4588, 4589, 4592-4594, 4596, 4597, 4599-4601, 4604, 4605, 4612-4614, 4616, 4620, 4622-4625, 4634-4636, 4644, 4652, 4655, 4656, 4667-4670, and 4672.

[0137] In certain embodiments, the following antisense compounds or antisense oligonucleotides target a region of a DGAT2 nucleic acid and effect at least a 70% inhibition of a DGAT2 mRNA, ISIS NOs: 413236, 413433, 413446, 423437, 423440-423445, 423447, 423449, 423450, 423452-423454, 423460-423465, 423522-423528, 423601-423604, 472204, 472205, 472208, 472349-472351, 483803, 483811-483814, 483816-483818, 483825, 483826, 483828-483835, 483838-483842, 483846, 483848, 483852, 483862, 483866, 483869, 483870, 483872-483875, 483879, 483887-483890, 483892, 483895, 483897-483903, 483908-483913, 483916-483921, 483923, 483924, 483926-483928, 483934, 483948-483950, 483952, 483954, 483968-483970, 483972, 483973, 483975, 483977-483979, 483984-483989, 483992-483994, 483996, 483997, 484001, 484004, 484006, 484010, 484012, 484017, 484019, 484020, 484030, 484031, 484041, 484049, 484056, 484064, 484069, 484070, 484071, 484073, 484082, 484083, 484085, 484089, 484094, 484099-484102, 484104, 484105, 484107, 484110, 484111, 484114-484118, 484126, 484127, 484129-484131, 484133, 484135-484137, 484139-484142, 484145-484148, 484156-484158, 484161, 484167, 484169-484172, 484178, 484180-484182, 484185, 484190, 484192-484194, 484198, 484202-484204, 484209, 484211, 484215, 484217-484220, 484231, 484232, 484235, 484241, 484248, 484249, 484263, 484265, 484267-484271, 484273, 484274, 484276, 484283-484285, 484290, 484292, 484293, 484301, 484302, 484320-484322, 484324, 484327, 484336, 484342-484344, 484346, 484348-484350, 484352, 484353, 484355-484357, 484359, 484368, 484370-484378, 484381-484383, 484390, 495425, 495429, 495430, 495440-495443, 495445-495454, 495456, 495458, 495461-495473, 495476, 495479-495485, 495487-495492, 495495-495499, 495501-495511, 495514-495531, 495533-495545, 495548-495556, 495558, 495560-495566, 495568-495580, 495584, 495585, 495588-495592, 495597-495600, 495602-495610, 495612-495614, 495616-495626, 495631-495633, 495636-495639, 495641-495643, 495649, 495650, 495656, 495657, 495661, 495664, 495666, 495669-495671, 495677, 495685, 495686, 495694, 495695, 495697, 495699-495702, 495705-495707, 495711, 495714, 495718-495720, 495724, 495730, 495731, 495736-495739, 495743-495746, 495748-495754, 495756, 495757, 495762, 495763, 495766, 495772, 495782, 495784, 495785, 495789, 495794, 495795, 495800, 495804, 495808-495813, 495817-495822, 495825-495833, 495835-495837, 495838-495844, 495848-495850, 495852-495857, 495859, 495860, 495867-495869, 495873-495878, 495882, 495899-495902, 495904, 495911, 495912, 495955, 495957, 495958, 495992, 500841, 500844, 500852, 500859, 500867, 500892, 500913, 500928, 500942, 500950, 500953, 500966, 500975, 500986, 500989, 500990, 500994, 500998, 501018-501020, 501025, 501030, 501033-501035, 501038-501041, 501062, 501064, 501067, 501069, 501093, 501094, 501097, 501098, 501100, 501103, 501111, 501118, 501122, 501123, 501127, 501128, 501154, 501165, 501171, 501176, 501183, 501184, 501199, 501200, 501210, 501212-501214, 501224, 501240, 501249, 501270, 501287, 501382-501385, 501387-501396, 501398-501402, 501404-501406, 501410-501412, 501414, 501426-501430, 501435, 501436, 501438-501443, 501445, 501447-501452, 501454-501457, 501849, 501850, 501852, 501855, 501861, 501871, 501883, 501884, 501886, 501890, 501916, 501932, 501944, 501946, 501950, 501951, 501959, 501960, 501966, 501968, 501981-501983, 502013, 502015, 502025, 502026, 502037, 502040, 502045, 502046, 502056, 502083, 502098, 502099, 502106, 502110, 502119, 502120, 502131, 502135, 502154-502156, 502163, 502164, 502175, 502179, 502189, 502191, 502194, 502220, 502319, 502322, 502334, 507663, 507679, 507684, 507692-507696, 507710, 507716-507718, 507724, 522363, 522366, 522368, 522370, 522373-522375, 522383-522385, 522399, 522404-522406, 522408, 522409, 522411, 522418, 522421, 522422, 522424, 522427, 522428, 522434-522437, 522440, 522442, 522444-522446, 522450, 522452, 522457, 522462-522467, 522470, 522473, 522474, 522478, 522479, 522482, 522484-522490, 522492, 522494-522496, 522501, 522503-522505, 522509, 522510, 522518, 522529, 522534, 522540, 522542-522550, 522552-522556, 522561, 522562, 522565, 522578-522582, 522587-522592, 522602-522610, 522612-522614, 522618, 522619, 522621, 522622, 522625, 522627, 522630-522638, 522642-522645, 522657, 522658, 522661-522663, 522666, 522667, 522671-522683, 522687-522694, 522696-522702, 522705, 522706, 522709, 522710, 522714-522719, 522722-522728, 522740, 522745-522748, 522754, 522757-522759, 522761, 522766-522771, 522776-522778, 522780, 522783, 522784, 522789, 522791, 522793, 522794, 522797, 522801, 522802, 522807, 522821, 522831, 522838, 522842, 522844, 522853, 522857, 522865, 522866, 522869-522872, 522877, 522878, 522888, 522889, 522894, 522897, 522898, 522905, 522913, 522917, 522924, 522925, 522932, 522939, 522940-522943, 522947, 522950, 522964, 522965, 522996, 523000, 523002, 523026, 525395, 525401, 525402, 525415, 525431, 525442, 525443, 525468-525472, 525474, 525477, 525479, 525480, 525499-525501, 525505, 525506, 525512, 525513, 525515, 525536, 525544, 525551-525554, 525578, 525612, 525631, 525688, 525705, 525708, 525711, 525733, and 525754.

[0138] In certain embodiments, the following antisense compounds or antisense oligonucleotides target a region of a DGAT2 nucleic acid and effect at least a 70% inhibition of a DGAT2 mRNA, SEQ ID NOs: 20, 38, 39, 46, 47, 62, 63, 103, 120, 121, 131, 144, 145, 150, 152, 153, 154, 182, 183, 202, 203, 204, 221, 225, 229, 246, 259, 283, 425, 438, 452, 453, 455-458, 460, 462-477, 479, 482-485, 487-496, 504-507, 509-511, 518, 519, 521-528, 531-535, 539, 541, 545, 555, 559, 562, 563, 565-568, 574, 575, 577, 594, 595, 597, 623, 624, 627, 653, 706, 710, 740, 742-744, 749-752, 754-756, 758, 760-766, 768, 769, 789, 808-811, 841-843, 845, 846, 848, 850-852, 880, 890, 893, 895, 897, 914-916, 939, 940, 960, 961, 973, 974, 981, 991-997, 1000, 1001, 1003-1008, 1015, 1020, 1024, 1032, 1034, 1036, 1043-1045, 1047-1049, 1052, 1059-1061, 1079, 1080, 1083, 1084, 1092, 1093, 1096, 1098, 1155, 1165, 1173-1176, 1178, 1181, 1183-1189, 1194-1199, 1202-1207, 1209, 1210, 1212-1214, 1220, 1234-1236, 1238, 1240, 1254-1256, 1258, 1259, 1261, 1263-1265, 1270-1275, 1278-1280, 1282, 1283, 1287, 1290, 1292, 1296, 1298, 1303, 1305, 1306, 1316, 1317, 1327, 1335, 1342, 1350, 1355-1357, 1359, 1368, 1369, 1371, 1375, 1380, 1385-1388, 1390, 1391, 1393, 1396, 1397, 1400-1404, 1412, 1413, 1415-1417, 1419, 1421-1423, 1425-1428, 1431-1434, 1442-1444, 1447, 1453, 1455-1458, 1464, 1466-1468, 1471, 1476, 1478-1480, 1484, 1488-1490, 1495, 1497, 1501, 1503-1506, 1517, 1518, 1521, 1527, 1534, 1535, 1549, 1551, 1553-1557, 1559, 1560, 1562, 1569-1571, 1576, 1578, 1579, 1587, 1588, 1606-1608, 1610, 1613, 1622, 1628-1630, 1632, 1634-1636, 1638, 1639, 1641-1643, 1645, 1654, 1656-1664, 1667-1669, 1676, 1677, 1698, 1702, 1703, 1713-1716, 1718-1727, 1729, 1731, 1734-1746, 1749, 1752-1758, 1760-1765, 1768-1772, 1774-1784, 1787-1804, 1806-1818, 1821-1829, 1831, 1833-1839, 1841-1853, 1857, 1858, 1861-1865, 1870-1873, 1875-1883, 1885-1887, 1889-1899, 1904-1906, 1909-1912, 1914-1916, 1922, 1923, 1929, 1930, 1934, 1937, 1939, 1942-1944, 1950, 1958, 1959, 1967, 1968, 1970, 1972-

1975, 1978-1980, 1984, 1987, 1991-1993, 1997, 2003, 2004, 2009-2012, 2016-2019, 2021-2027, 2029, 2030, 2035, 2036, 2039, 2045, 2055, 2057, 2058, 2062, 2067, 2068, 2073, 2077, 2081-2086, 2090-2095, 2098-2106, 2108-2117, 2121-2123, 2125-2130, 2132, 2133, 2140-2142, 2146-2151, 2155, 2172-2175, 2177, 2184, 2185, 2228, 2230, 2231, 2265, 2308, 2310-2313, 2315-2324, 2326-2330, 2332-2334, 2338-2340, 2342, 2350, 2351, 2354-2358, 2363, 2364, 2366-2371, 2373, 2375-2380, 2382-2385, 2478, 2487, 2508, 2525, 2543, 2554, 2560, 2565, 2572, 2573, 2588, 2589, 2599, 2601-2603, 2613, 2632, 2633, 2636, 2637, 2639, 2642, 2650, 2657, 2661, 2662, 2666, 2667, 2691, 2695, 2715-2717, 2722, 2727, 2730-2732, 2735-2738, 2755, 2757, 2760, 2762, 2779, 2793, 2801, 2804, 2817, 2826, 2837, 2840, 2841, 2846, 2849, 2857, 2864, 2872, 2897, 2918, 2947, 2948, 2950, 2953, 2959, 2969, 2981, 2982, 2984, 2988, 3014, 3030, 3042, 3044, 3048, 3049, 3057, 3058, 3064, 3066, 3079-3081, 3111, 3113, 3123, 3124, 3135, 3138, 3143, 3144, 3154, 3173, 3181, 3196, 3197, 3204, 3208, 3217, 3218, 3229, 3233, 3252-3254, 3261, 3262, 3273, 3277, 3287, 3289, 3292, 3318, 3388, 3393, 3404, 3412, 3417, 3420, 3425-3429, 3432, 3443, 3449-3451, 3464, 3468, 3471, 3473, 3475, 3478-3480, 3488-3490, 3504, 3509-3511, 3513-3516, 3523, 3526, 3527, 3529, 3532, 3533, 3539-3542, 3545, 3547, 3549-3551, 3555, 3557, 3562, 3567-3572, 3575, 3578, 3579, 3583, 3584, 3587, 3589-3595, 3597, 3599-3601, 3606, 3608-3610, 3614, 3615, 3623, 3634, 3639, 3645, 3647-3655, 3657-3661, 3666, 3667, 3670, 3683-3687, 3692-3697, 3707-3715, 3717-3719, 3723, 3724, 3726, 3727, 3730, 3732, 3735-3743, 3747-3750, 3762, 3763, 3766-3768, 3771, 3772, 3776-3788, 3792-3799, 3801-3807, 3810, 3811, 3814, 3815, 3819-3824, 3827-3833, 3845, 3850-3853, 3859, 3862-3864, 3866, 3871-3876, 3881-3883, 3885, 3888, 3889, 3894, 3896, 3898, 3899, 3902, 3906, 3907, 3912, 3926, 3936, 3943, 3947, 3949, 3958, 3962, 3970, 3971, 3974-3977, 3982, 3983, 3993, 3994, 3999, 4002, 4003, 4010, 4018, 4022, 4029, 4030, 4037, 4044-4048, 4052, 4055, 4069, 4070, 4096, 4123, 4127, 4129, 4150, 4156, 4157, 4170, 4186, 4197, 4198, 4229-4233, 4235, 4238, 4240, 4241, 4260-4262, 4266, 4267, 4273, 4274, 4276, 4297, 4305, 4312-4315, 4339, 4373, 4392, 4449, 4466, 4469, 4472, 4494, 4515, 4526-4530, 4532, 4533, 4535, 4537-4540, 4545, 4546, 4558-4560, 4562-4570, 4573, 4575, 4576, 4578, 4580, 4582, 4588, 4589, 4592, 4596, 4600, 4601, 4604, 4612-4614, 4616, 4622-4625, 4634, 4635, 4652, 4667, and 4668.

[0139] In certain embodiments, the following antisense compounds or antisense oligonucleotides target a region of a DGAT2 nucleic acid and effect at least an 80% inhibition of a DGAT2 mRNA, ISIS NOs: 413433, 423463, 423464, 423523, 423524, 423526, 472351, 483817, 483825, 483826, 483828, 483830-483835, 483839-483841, 483848, 483852, 483866, 483869, 483870, 483873-483875, 483887, 483889, 483890, 483895, 483897, 483898, 483900, 483901, 483908-483910, 483913, 483916, 483919, 483921, 483923, 483924, 483927, 483952, 483968-483970, 483972, 483984, 483986-483988, 483992, 483993, 483996, 483997, 484004, 484017, 484031, 484041, 484049, 484064, 484070, 484085, 484094, 484099, 484100, 484115-484117, 484126, 484127, 484129-484131, 484133, 484137, 484139-484141, 484148, 484156-484158, 484167, 484169-484171, 484181, 484182, 484192, 484193, 484203, 484204, 484209, 484215, 484217, 484218, 484220, 484231, 484235, 484249, 484267-484271, 484273, 484283, 484284, 484292, 484293, 484301, 484327, 484336, 484343, 484344, 484348, 484350, 484353, 484357, 484368, 484377, 484378, 495425, 495429, 495440, 495442, 495446, 495449-495451, 495453, 495454, 495463, 495464, 495466, 495467, 495469-495472, 495481, 495482, 495484, 495485, 495488-495492, 495495-495498, 495505-495510, 495514-495531, 495533-495537, 495539-495541, 495543-495545, 495550-495556, 495560, 495561-495564, 495566, 495568-495578, 495585, 495591, 495598-495600, 495604, 495606, 495608, 495609, 495618-495622, 495639, 495649, 495650, 495685, 495686, 495697, 495702, 495705-495707, 495718, 495730, 495736, 495738, 495739, 495744, 495745, 495749, 495751-495753, 495756, 495785, 495808-495810, 495817-495820, 495822, 495825-495832, 495835-495844, 495849, 495852-495854, 495856, 495857, 495867-495869, 495874-495878, 495902, 495992, 500859, 500867, 500913, 500998, 501020, 501034, 501035, 501064, 501094, 501098, 501100, 501103, 501127, 501183, 501199, 501213, 501385, 501387-501393, 501396, 501398, 501404-501406, 501411, 501412, 501427-501430, 501435, 501436, 501438, 501440, 501442, 501443, 501447-501452, 501454-501855, 501861, 501884, 501944, 501950, 501983, 502040, 502046, 502056, 502083, 502099, 502119, 502154, 502164, 502175, 502179, 502189, 502191, 502319, 507692, 507694, 507696, 507717, 522366, 522373-522375, 522383, 522435-522437, 522444, 522445, 522450, 522465, 522473, 522484, 522485, 522495, 522501, 522509, 522529, 522546, 522550, 522553-522556, 522579-522582, 522587-522589, 522606-522610, 522618, 522621, 522627, 522630-522632, 522634, 522638, 522643, 522645, 522667, 522671, 522672, 522674-522683, 522687-522692, 522694, 522696-522700, 522705, 522709, 522715-522719, 522745, 522757, 522770, 522783, 522784, 522807, 522865, 522866, 522888, 522889, 522897, 522913, 522942, 522964, 523002, 525401, 525469, 525470, 525501, and 525612.

[0140] In certain embodiments, the following antisense compounds or antisense oligonucleotides target a region of a DGAT2 nucleic acid and effect at least a 80% inhibition of a DGAT2 mRNA, SEQ ID NOs: 103, 120, 121, 131, 144, 145, 150, 152, 153, 182, 204, 246, 259, 283, 425, 456, 457, 462, 463, 467-469, 487-490, 504, 506, 509, 510, 518, 519, 521, 523, 524-528, 532-534, 541, 545, 559, 562, 563, 566-568, 594, 742, 750, 751, 762, 808-810, 843, 845, 893, 915, 939, 973, 993-995, 997, 1000, 1001, 1005, 1006, 1008, 1034, 1060, 1061, 1079, 1084, 1173, 1175, 1176, 1181, 1183, 1184, 1186, 1187, 1194-1196, 1199, 1202, 1205, 1207, 1209, 1210, 1213, 1238, 1254-1256, 1258, 1270, 1272-1274, 1278, 1279, 1282, 1283, 1290, 1303, 1317, 1327, 1335, 1350, 1356, 1371, 1380, 1385, 1386, 1401-1403, 1412, 1413, 1415-1417, 1419, 1423, 1425-1427, 1434, 1442-1444, 1453, 1455-1457, 1467, 1468, 1478, 1479, 1489, 1490, 1495, 1501, 1503, 1504, 1506, 1517, 1521, 1535, 1553-1557, 1559, 1569, 1570, 1578, 1579, 1587, 1613, 1622, 1629, 1630, 1634, 1636, 1639, 1643, 1654, 1663, 1664, 1677, 1698, 1702, 1713, 1715, 1719, 1722-1724, 1726, 1727, 1736, 1737, 1739, 1740, 1742-1745, 1754, 1755, 1757, 1758, 1761-1765, 1768-1771, 1778-1783, 1787-1804, 1806-1810, 1812-1814, 1816-1818, 1823-1829, 1833-1837, 1839, 1841-1851, 1858, 1864, 1871-1873, 1877, 1879, 1881, 1882, 1891-1895, 1912, 1922, 1923, 1958, 1959, 1970, 1975, 1978-1980, 1991, 2003, 2009, 2011, 2012, 2017, 2018, 2022, 2024-2026, 2029, 2058, 2081-2083, 2090-2093, 2095, 2098-2105, 2108-2117, 2122, 2125-2127, 2129, 2130, 2140-2142, 2147-2151, 2175, 2265, 2313, 2315-2321, 2324, 2326, 2332-2334, 2339, 2340, 2355-2358, 2363, 2364, 2366, 2368, 2370, 2371, 2375-2380, 2382, 2383, 2572, 2588, 2602, 2633, 2637, 2639, 2642, 2666, 2695, 2717, 2731, 2732, 2757, 2864, 2872, 2918, 2953, 2959, 2982, 3042, 3048, 3081, 3138, 3144, 3154, 3181, 3197, 3217, 3252, 3262, 3273, 3277, 3287, 3289, 3404, 3417, 3425, 3427, 3429, 3450, 3471, 3478-3480, 3488, 3540-3542, 3549, 3550, 3555, 3570, 3578, 3589, 3590, 3600, 3606, 3614, 3634, 3651, 3655, 3658-3661, 3684-3687, 3692-3694, 3711-3715, 3723, 3726, 3732, 3735-3737, 3739, 3743, 3748, 3750, 3772, 3776, 3777, 3779-3788, 3792-3797, 3799, 3801-3803, 3805, 3810, 3814, 3820-3824, 3850, 3862, 3875, 3888, 3889, 3912, 3970, 3971, 3993, 3994, 4002, 4018, 4047, 4069, 4129, 4156, 4230, 4231, 4262, 4373, 4526, 4529, 4532, 4538, 4540, 4560, 4562-4564, 4566-4570, 4578, 4623, 4625, and 4667.

[0141] In certain embodiments, the following antisense compounds or antisense oligonucleotides target a region of a DGAT2 nucleic acid and effect at least a 90% inhibition of a DGAT2 mRNA, ISIS NOs: 413433, 423463, 483817, 483825, 483834, 483848, 483866, 483869, 483873, 483874, 483895, 483898, 483908, 483910, 483913, 483952, 484085, 484148, 484157, 484170, 484181, 484231, 484271, 484283, 484350, 484353, 495449, 495450, 495495, 495498, 495505, 495506, 495516-495520, 495522-495524, 495526, 495527, 495535, 495553-495555, 495562, 495563, 495570, 495571, 495575-495577, 495620, 495752, 495809, 495825, 495829, 495836, 495837, 495839-495842, 495853, 495857, 495876, 495878, 501385, 501387, 501404, 501412, 501427, 501430, 501442, 501443, 501447, 501448, 501450, 501452, 501861, 522632, 522688, and 522745.

[0142] In certain embodiments, the following antisense compounds or antisense oligonucleotides target a region of a DGAT2 nucleic acid and effect at least a 90% inhibition of a DGAT2 mRNA, SEQ ID NOs: 283, 425, 457, 467, 510, 518, 527, 541, 559, 562, 566, 567, 1181, 1184, 1194, 1196, 1199, 1238, 1371, 1434, 1443, 1456, 1467, 1517, 1557, 1569, 1636, 1639, 1722, 1723, 1768, 1771, 1778, 1779, 1789-1793, 1795-1797, 1799, 1800, 1808, 1826-1828, 1835, 1836, 1843, 1844, 1848-1850, 1893, 2025, 2082, 2098, 2102, 2109, 2110, 2112-2115, 2126, 2130, 2149, 2151, 2313, 2315, 2332, 2340, 2355, 2358, 2370, 2371, 2375, 2376, 2378, 2380, 2959, 3737, 3793, 3850, 4562, 4564, 4566, 4568, and 4569.

[0143] In certain embodiments, a compound comprises a modified oligonucleotide consisting of 8 to 80 linked nucleosides having at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion complementary to an equal length portion within nucleotides 26778-26797, 23242-23261, 26630-26649,15251-15270, 28026-28045, 35436-35455, 10820-10836, 23246-23262 of SEQ ID NO: 2

[0144] In certain embodiments, a compound comprises a modified oligonucleotide consisting of 10 to 30 linked nucleosides complementary within nucleotides 26778-26797, 23242-23261, 26630-26649,15251-15270, 28026-28045, 35436-35455, 10820-10836, 23246-23262 of SEQ ID NO: 2.

[0145] In certain embodiments, a compound comprises a modified oligonucleotide consisting of 8 to 80 linked nucleosides having a nucleobase sequence comprising at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373.

[0146] In certain embodiments, a compound comprises a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373.

[0147] In certain embodiments, a compound comprises a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, a modified oligonucleotide targeted to DGAT2 is ISIS 484137, 484085, 484129, 495576, 501861, 502194, 525443, and 525612.

[0148] In certain embodiments, a modified oligonucleotide targeted to DGAT2 is ISIS 484137, 484085, 484129, 495576, 501861, 502194, 525443, and 525612. Out of about 5,000 antisense oligonucleotides that were screened as described in the Examples section below, ISIS 484137, 484085, 484129, 495576, 501861, 502194, 525443, and 525612 emerged as the top lead compounds. In particular, ISIS 484137 exhibited the best combination of properties in terms of potency and tolerability out of about 5,000 antisense oligonucleotides.

[0149] In certain embodiments, any of the foregoing compounds or oligonucleotides comprises at least one modified internucleoside linkage, at least one modified sugar, and/or at least one modified nucleobase.

[0150] In certain aspects, any of the foregoing compounds or oligonucleotides comprises at least one modified sugar. In certain aspects, at least one modified sugar comprises a 2′-O-methoxyethyl group. In certain aspects, at least one modified sugar is a bicyclic sugar, such as a 4′-CH(CH.sub.3)—O-2′ group, a 4′-CH.sub.2—O-2′ group, or a 4′—(CH.sub.2).sub.2—O-2′ group.

[0151] In certain aspects, the modified oligonucleotide comprises at least one modified internucleoside linkage, such as a phosphorothioate internucleoside linkage.

[0152] In certain embodiments, any of the foregoing compounds or oligonucleotides comprises at least one modified nucleobase, such as 5-methylcytosine.

[0153] In certain embodiments, any of the foregoing compounds or oligonucleotides comprises: [0154] a gap segment consisting of linked 2′-deoxynucleosides; [0155] a 5′ wing segment consisting of linked nucleosides; and [0156] a 3′ wing segment consisting of linked nucleosides; [0157] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. In certain embodiments, the oligonucleotide consists of 8 to 80 linked nucleosides having a nucleobase sequence comprising the sequence recited in SEQ ID NO: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain aspects, the oligonucleotide consists of 10 to 30 linked nucleosides having a nucleobase sequence comprising the sequence recited in SEQ ID NO: 1423, 1371, 1415,

1849, 2959, 3292, 4198, and 4373.
[0158] In certain aspects, the modified oligonucleotide has a nucleobase sequence comprising the sequence recited in SEQ ID NO: 1423, 1371, 1415, 1849, 2959, or 3292, wherein the modified oligonucleotide comprises [0159] a gap segment consisting of ten linked 2′-deoxynucleosides; [0160] a 5′ wing segment consisting of five linked nucleosides; and [0161] a 3′ wing segment consisting of five linked nucleosides; [0162] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment, wherein each nucleoside of each wing segment comprises a 2′-O-methoxyethyl sugar; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. In certain embodiments, the modified oligonucleotide consists of 20-80 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 20-30 linked nucleosides.
[0163] In certain aspects, the modified oligonucleotide has a nucleobase sequence comprising the sequence recited in SEQ ID NO: 4198 or 4373, wherein the modified oligonucleotide comprises a gap segment consisting of ten linked 2′-deoxynucleosides; [0164] a 5′ wing segment consisting of three linked nucleosides; and [0165] a 3′ wing segment consisting of four linked nucleosides; [0166] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment, wherein each nucleoside of each wing segment comprises a 2′-O-methoxyethyl sugar; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. In certain embodiments, the modified oligonucleotide consists of 17-80 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 17-30 linked nucleosides.
[0167] In certain embodiments, a compound comprises or consists of ISIS 484137 and has the following chemical structure or a salt thereof:
##STR00001##
[0168] In certain embodiments, a compound comprises or consists of ISIS 484137, named by accepted oligonucleotide nomenclature, showing each 3′-O to 5′-O-linked phosphorothioate diester internucleotide linkage as follows:
[0169] 2′-O-(2-methoxyethyl)-5-methyl-P-thiouridylyl-(3′-O.fwdarw.5′-O)—2′-O-(2-methoxyethyl)-P-thioguanylyl-(3′-O.fwdarw.5′-O)—2′-O-(2-methoxyethyl)-5-methyl-P-thiocytidylyl-(3′-O.fwdarw.5′-O)—2′-O-(2-methoxyethyl)-5-methyl-P-thiocytidylyl-(3′-O.fwdarw.5′-O)—2′-O-(2-methoxyethyl)-P-thioadenylyl-(3′-O.fwdarw.5′-O)—P-thiothymidylyl-(3′-O.fwdarw.5′-O)—P-thiothymidylyl-(3′-O.fwdarw.5′-O)—P-thiothymidylyl-(3′-O.fwdarw.5′-O)—2′-deoxy-P-thioadenylyl-(3′-O.fwdarw.5′-O)—2′-deoxy-P-thioadenylyl-(3′-O.fwdarw.5′-O)—P-thiothymidylyl-(3′-O.fwdarw.5′-O)—2′-deoxy-P-thioguanylyl-(3′-O.fwdarw.5′-O)—2′-deoxy-P-thioadenylyl-(3′-O.fwdarw.5′-O)—2′-deoxy-P-thioguanylyl-(3′-O.fwdarw.5′-O)—2′-deoxy-5-methyl-P-thiocytidylyl-(3′-O.fwdarw.5′-O)—2′-O-(2-methoxyethyl)-5-methyl-P-thiouridylyl-(3′-O.fwdarw.5′-O)—2′-O-(2-methoxyethyl)-5-methyl-P-thiouridylyl-(3′-O.fwdarw.5′-O)—2′-O-(2-methoxyethyl)-5-methyl-P-thiocytidylyl-(3′-O.fwdarw.5′-O)—2′-O-(2-methoxyethyl)-P-thioadenylyl-(3′-O.fwdarw.5′-O)—2′-O-(2-methoxyethyl)-5-methyl-cytidine, 19 sodium salt.
[0170] In any of the foregoing embodiments, the compound or oligonucleotide can be at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% complementary to a nucleic acid encoding DGAT2.
[0171] In any of the foregoing embodiments, the antisense compound can be a single-stranded oligonucleotide.
[0172] In any of the foregoing embodiments, the oligonucleotide can consist of 8 to 80, 10 to 30, 12 to 50, 13 to 30, 13 to 50, 14 to 30, 14 to 50, 15 to 30, 15 to 50, 16 to 30, 16 to 50, 17 to 30, 17 to 50, 18 to 22, 18 to 24, 18 to 30, 18 to 50, 19 to 22, 19 to 30, 19 to 50, or 20 to 30 linked nucleosides.
[0173] In certain embodiments, compounds or compositions provided herein comprise a salt of the modified oligonucleotide. In certain embodiments, the salt is a sodium salt. In certain

embodiments, the salt is a potassium salt.
[0174] In certain embodiments, a compound comprises a modified oligonucleotide described herein and a conjugate group. In certain embodiments, the conjugate group is linked to the modified oligonucleotide at the 5′ end of the modified oligonucleotide. In certain embodiments, the conjugate group is linked to the modified oligonucleotide at the 3′ end of the modified oligonucleotide. In certain embodiments, the conjugate group comprises at least one N-Acetylgalactosamine (GalNAc), at least two N-Acetylgalactosamines (GalNAcs), or at least three N-Acetylgalactosamines (GalNAcs).
[0175] In certain embodiments, a compound having the following chemical structure comprises or consists of ISIS 484137 or salt thereof with a conjugate group comprising GalNAc as described herein:
##STR00002##
[0176] wherein for each pair of R.sub.1 and R.sub.2 on the same ring, independently for each ring, either R.sub.1 is —OCH.sub.2CH.sub.2OCH.sub.3 (MOE) and R.sub.2 is H; or R.sub.1 and R.sub.2 together form a bridge, wherein R.sup.1 is —O— and R.sub.2 is —CH.sub.2—, —CH(CH.sub.3)—, or —CH.sub.2CH.sub.2—, and R.sup.1 and R.sub.2 are directly connected such that the resulting bridge is selected from: —O—CH.sub.2—, —O—CH(CH.sub.3)—, and —O—CH.sub.2CH.sub.2—; and for each pair of R.sub.3 and R.sub.4 on the same ring, independently for each ring: either R.sub.3 is selected from H and —OCH.sub.2CH.sub.2OCH.sub.3 and R.sub.4 is H; or R.sub.3 and R.sub.4 together form a bridge, wherein R.sup.3 is —O—, and R.sub.4 is —CH.sub.2—, —CH(CH.sub.3)—, or —CH.sub.2CH.sub.2—, and R.sub.3 and R.sub.4 are directly connected such that the resulting bridge is selected from: —O—CH.sub.2—, —O—CH(CH.sub.3)—, and —O—CH.sub.2CH.sub.2—; and each R.sup.5 is independently selected from H and —CH.sub.3; and each Z is independently selected from S— and O—.
[0177] In certain embodiments, a compound comprises or consists of SEQ ID NO: 1423, 5′-GalNAc, and chemical modifications as represented by the following chemical structure:
##STR00003## [0178] wherein for each pair of R.sub.1 and R.sub.2 on the same ring, independently for each ring, either R.sub.1 is —OCH.sub.2CH.sub.2OCH.sub.3 (MOE) and R.sub.2 is H; or R.sub.1 and R.sub.2 together form a bridge, wherein R.sup.1 is —O— and R.sub.2 is —CH.sub.2—, —CH(CH.sub.3)—, or —CH.sub.2CH.sub.2—, and R.sup.1 and R.sub.2 are directly connected such that the resulting bridge is selected from: —O—CH.sub.2—, —O—CH(CH.sub.3)—, and —O—CH.sub.2CH.sub.2—; and for each pair of R.sub.3 and R.sub.4 on the same ring, independently for each ring: either R.sub.3 is selected from H and —OCH.sub.2CH.sub.2OCH.sub.3 and R.sub.4 is H; or R.sub.3 and R.sub.4 together form a bridge, wherein R.sup.3 is —O—, and R.sub.4 is —CH.sub.2—, —CH(CH.sub.3)—, or —CH.sub.2CH.sub.2—, and R.sub.3 and R.sub.4 are directly connected such that the resulting bridge is selected from: —O—CH.sub.2—, —O—CH(CH.sub.3)—, and —O—CH.sub.2CH.sub.2—; and each R.sup.5 is independently selected from H and —CH.sub.3; and each Z is independently selected from S— and O—; and X is H or a conjugate group that optionally comprises a conjugate linker and/or cleavable moiety.
[0179] In certain embodiments, a compound comprises ISIS 769355 or salt thereof. In certain embodiments, a compound consists of ISIS 769355 or salt thereof. In certain embodiments, ISIS 769355 has the following chemical structure:
##STR00004##
[0180] In certain embodiments, a compound comprises ISIS 769356 or salt thereof. In certain embodiments, a compound consists of ISIS 769356 or salt thereof. In certain embodiments, ISIS 769356 has the following chemical structure:
##STR00005##
[0181] In certain embodiments, a compound comprises ISIS 769357 or salt thereof. In certain embodiments, a compound consists of ISIS 769357 or salt thereof. In certain embodiments, ISIS

769357 has the following chemical structure:
##STR00006##
[0182] Certain embodiments provide compositions comprising any of the compounds comprising or consisting of a modified oligonucleotide targeted to DGAT2 or salt thereof and a conjugate group, and at least one of a pharmaceutically acceptable carrier or diluent.
[0183] In certain embodiments, the compounds or compositions as described herein are efficacious by virtue of having at least one of an in vitro IC.sub.50 of less than 2 nM, less than 3 nM, less than 4 nM, less than 5 nM, less than 6 nM, less than 7 nM, less than 8 nM, less than 9 nM, less than 10 nM, less than 20 nM, less than 30 nM, less than 35 nM, less than 40 nM, less than 45 nM, less than 50 nM, less than 60 nM, less than 70 nM, less than 80 nM, less than 90 nM, less than 100 nM, less than 110 nM, less than 300 nM, less than 400 nM, less than 500 nM, less than 600 nM, less than 700 nM, less than 800 nM, less than 900 nM, less than 1 μM, less than 1.1 μM, less than 1.2 μM, less than 1.3 μM, less than 1.4 μM, less than 1.5 μM, less than 1.6 μM, less than 1.7 μM, less than 1.8 μM, less than 1.9 μM, less than 2 μM, less than 2.5 μM, less than 3 μM, less than 3.5 μM, less than 4 μM, less than 4.5 μM, less than 5 μM, less than 5.5 μM, less than 6 μM, less than 6.5 μM, or less than 10 μM.
[0184] In certain embodiments, the compounds or compositions as described herein are highly tolerable as demonstrated by having at least one of an increase an ALT or AST value of no more than 4 fold, 3 fold, or 2 fold over saline treated animals or an increase in liver, spleen, or kidney weight of no more than 30%, 20%, 15%, 12%, 10%, 5%, or 2%. In certain embodiments, the compounds or compositions as described herein are highly tolerable as demonstrated by having no increase of ALT or AST over saline treated animals. In certain embodiments, the compounds or compositions as described herein are highly tolerable as demonstrated by having no increase in liver, spleen, or kidney weight over saline treated animals.
Certain Indications
[0185] Certain embodiments provided herein relate to methods of treating, preventing, or ameliorating a disease associated with DGAT2 in an individual by administration of a DGAT2 specific inhibitor, such as an antisense compound targeted to DGAT2.
[0186] Examples of diseases associated with DGAT2 treatable, preventable, and/or ameliorable with the methods provided herein include nonalcoholic fatty liver diseases (NAFLD), nonalcoholic steatohepatitis (NASH), hepatic steatosis, fatty liver diseases, lipodystrophy syndromes, including congenital generalized lipodystrophy (CGL), acquired generalized lipodystrophy (AGL), familial partial lipodystrophy (FPL), and acquired partial lipodystrophy (PL), metabolic syndrome and cardiovascular diseases.
[0187] In certain embodiments, a method of treating, preventing, or ameliorating a disease associated with NAFLD in an individual comprises administering to the individual a specific inhibitor of DGAT2, thereby treating, preventing, or ameliorating the disease. In certain embodiments, a method of treating, preventing, or ameliorating a disease associated with lipodystrophy syndrome in an individual comprises administering to the individual a specific inhibitor of DGAT2, thereby treating, preventing, or ameliorating the disease. In certain embodiments, the lipodystrophy syndrome is partial lipodystrophy. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound targeted to DGAT2, such as an antisense oligonucleotide targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is a compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides and has a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified

oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is ISIS 484137, ISIS 484085, ISIS 484129, ISIS 495576, ISIS 501861, ISIS 502194, ISIS 525443, and ISIS 525612. In certain embodiments the DGAT2 specific inhibitor is ISIS 769355, ISIS 769356, and ISIS 769357. In any of the foregoing embodiments, the antisense compound can be a single-stranded oligonucleotide.

[0188] In certain embodiments, a method of treating, preventing, or ameliorating NAFLD/NASH comprises administering to the individual a DGAT2 specific inhibitor, thereby treating, preventing, or ameliorating NALFD/NASH. In certain embodiments, a method of treating, preventing, or ameliorating lipodystrophy comprises administering to the individual a DGAT2 specific inhibitor, thereby treating, preventing, or ameliorating lipodystrophy syndrome. In certain embodiments, the lipodystrophy syndrome is partial lipodystrophy. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound targeted to DGAT2, such as an antisense oligonucleotide targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is a compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides and has a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is ISIS 484137, ISIS 484085, ISIS 484129, ISIS 495576, ISIS 501861, ISIS 502194, ISIS 525443, and ISIS 525612. In certain embodiments the DGAT2 specific inhibitor is ISIS 769355, ISIS 769356, and ISIS 769357. In any of the foregoing embodiments, the antisense compound can be a single-stranded oligonucleotide. In certain aspects, the antisense compound is administered to the individual parenterally. In certain aspects, administering the antisense compound results in specific reduction of DGAT2 expression, reduction in the rate of triglyceride synthesis, improvement of hepatic steatosis and blood lipid levels, reduction of hepatic lipids, reversal of diet-induced hepatic insulin resistance, and improvement in hepatic insulin sensitivity. In certain aspects, administering the antisense compound results in reduction in liver steatosis and fibrosis. In certain aspects, administering the antisense compound results in improvements in ALT levels. In certain aspects, administering the antisense compound results in decrease in NAFLD Activity Score (NAS). In certain aspects, administering the antisense compound results in reduction in serum total cholesterol and triglycerides. In certain aspects, administering the antisense compound results in neutral insulin sensitivity and glycemic control. In certain aspects, administering the antisense compound results in improved insulin sensitivity and glycemic control. In certain aspects, the individual is identified as having or at risk of having a disease associated with NAFLD. In certain aspects, the individual is identified as having or at risk of having a disease associated with lipodystrophy syndrome.

[0189] In certain embodiments, a method of inhibiting expression of DGAT2 in an individual having, or at risk of having, a disease associated with NAFLD comprises administering a DGAT2 specific inhibitor to the individual, thereby inhibiting expression of DGAT2 in the individual. In certain embodiments, a method of inhibiting expression of DGAT2 in an individual having, or at risk of having, a disease associated with lipodystrophy syndrome comprises administering a DGAT2 specific inhibitor to the individual, thereby inhibiting expression of DGAT2 in the individual. In certain aspects, administering the inhibitor inhibits expression of DGAT2 in the liver. In certain aspects, administering the inhibitor inhibits expression of DGAT2 in adipose tissue. In certain aspects, the individual has, or is at risk of having lipodystrophy syndrome, partial lipodystrophy, liver steatosis, NAFLD, or NASH. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound targeted to DGAT2, such as an antisense oligonucleotide targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is a compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides and has a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is ISIS 484137, ISIS 484085, ISIS 484129, ISIS 495576, ISIS 501861, ISIS 502194, ISIS 525443, and ISIS 525612. In certain embodiments the DGAT2 specific inhibitor is ISIS 769355, ISIS 769356, and ISIS 769357. In any of the foregoing embodiments, the antisense compound can be a single-stranded oligonucleotide.

[0190] In certain embodiments, a method of reducing or inhibiting triglyceride synthesis, hepatic lipid synthesis and insulin resistance in the liver of an individual having, or at risk of having, a disease associated with DGAT2 comprises administering a DGAT2 specific inhibitor to the individual, thereby reducing or inhibiting triglyceride synthesis, hepatic lipid synthesis and insulin resistance in the liver of the individual. In certain embodiments, a method of reducing or inhibiting triglyceride synthesis, lipid synthesis and insulin resistance in the adipose tissue of an individual having, or at risk of having, a disease associated with DGAT2 comprises administering a DGAT2 specific inhibitor to the individual, thereby reducing or inhibiting triglyceride synthesis, lipid synthesis and insulin resistance in the adipose tissue of the individual. In certain aspects, administering the DGAT2 inhibitor leads to improvement in hepatic insulin signaling, hepatic insulin sensitivity and cardiovascular risk profile. In certain aspects, administering the DGAT2 specific inhibitor results in reduction in liver steatosis and fibrosis. In certain aspects, administering the DGAT2 specific inhibitor results in improvements in ALT levels. In certain aspects, administering the DGAT2 specific inhibitor results in decrease in NAFLD Activity Score (NAS). In certain aspects, administering the DGAT2 specific inhibitor results in reduction in serum total cholesterol and triglycerides. In certain aspects, administering the DGAT2 specific inhibitor results in neutral insulin sensitivity and glycemic control. In certain aspects, administering the antisense compound results in improved insulin sensitivity and glycemic control. In certain aspects, the individual has, or is at risk of having, NAFLD or NASH. In certain aspects, the individual has, or is at risk of having, lipodystrophy syndrome or partial lipodystrophy. In certain aspects, the inhibitor is an antisense compound targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound targeted to DGAT2, such as an antisense oligonucleotide targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is a compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides and has a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is ISIS 484137, ISIS 484085, ISIS 484129, ISIS 495576, ISIS 501861, ISIS 502194, ISIS 525443, and ISIS 525612. In certain embodiments the DGAT2 specific inhibitor is ISIS 769355, ISIS 769356, and ISIS 769357. In any of the foregoing embodiments, the antisense compound can be a single-stranded oligonucleotide. In certain aspects, the antisense compound is administered to the individual parenterally.

[0191] Certain embodiments are drawn to a DGAT2 specific inhibitor for use in treating a disease associated with NAFLD. In certain aspects, the disease is NASH. In certain aspects, the disease is hepatic steatosis. In certain aspects, the disease is liver cirrhosis. In certain aspects, the disease is hepatocellular carcinoma. Certain embodiments are drawn to a DGAT2 specific inhibitor for use in treating a disease associated with lipodystrophy syndromes. In certain aspects, the disease is partial lipodystrophy. In certain aspects, the inhibitor is an antisense compound targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound targeted to DGAT2, such as an antisense oligonucleotide targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is a compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides and has a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is ISIS 484137, ISIS 484085, ISIS 484129, ISIS 495576, ISIS 501861, ISIS 502194, ISIS 525443, and ISIS 525612. In certain embodiments the DGAT2 specific inhibitor is ISIS 769355, ISIS 769356, and ISIS 769357. In any of the foregoing embodiments, the antisense compound can be a single-stranded oligonucleotide. In certain aspects, the antisense compound is administered to the individual parenterally.

[0192] Certain embodiments are drawn to a DGAT2 specific inhibitor for reducing or inhibiting triglyceride synthesis, hepatic lipid synthesis and insulin resistance in the liver of an individual having, or at risk of having, a disease associated with NAFLD or lipodystrophy syndromes comprises administering a DGAT2 specific inhibitor to the individual, thereby reducing or inhibiting triglyceride synthesis, hepatic lipid synthesis and insulin resistance in the liver of the individual. Certain embodiments are drawn to a DGAT2 specific inhibitor for reducing or inhibiting triglyceride synthesis, lipid synthesis and insulin resistance in the adipose tissue of an individual having, or at risk of having, a disease associated with NAFLD or lipodystrophy syndromes comprises administering a DGAT2 specific inhibitor to the individual, thereby reducing or inhibiting triglyceride synthesis, lipid synthesis and insulin resistance in the adipose tissue of the individual. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound targeted to DGAT2, such as an antisense oligonucleotide targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is a compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides and has a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is ISIS 484137, ISIS 484085, ISIS 484129, ISIS 495576, ISIS 501861, ISIS 502194, ISIS 525443, and ISIS 525612. In certain embodiments the DGAT2 specific inhibitor is ISIS 769355, ISIS 769356, and ISIS 769357. In any of the foregoing embodiments, the antisense compound can be a single-stranded oligonucleotide.

[0193] Certain embodiments are drawn to use of a DGAT2 specific inhibitor for the manufacture of a medicament for treating a disease associated with NAFLD. In certain aspects, the disease is NASH. In certain aspects, the disease is hepatic steatosis. In certain aspects, the disease is liver cirrhosis. In certain aspects, the disease is hepatocellular carcinoma. Certain embodiments are drawn to use of a DGAT2 specific inhibitor for the manufacture of a medicament for treating a disease associated with lipodystrophy syndromes. In certain aspects, the disease is partial lipodystrophy. In certain aspects, the inhibitor is an antisense compound targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound targeted to DGAT2, such as an antisense oligonucleotide targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is a compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides and has a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is ISIS 484137, ISIS 484085, ISIS 484129, ISIS 495576, ISIS 501861, ISIS 502194, ISIS 525443, and ISIS 525612. In certain embodiments the DGAT2 specific inhibitor is ISIS 769355, ISIS

769356, and ISIS 769357. In any of the foregoing embodiments, the antisense compound can be a single-stranded oligonucleotide. In certain aspects, the antisense compound is administered to the individual parenterally.

[0194] Certain embodiments are drawn to use of a DGAT2 specific inhibitor for the manufacture of a medicament for reducing or inhibiting triglyceride synthesis, hepatic lipid synthesis and insulin resistance in the liver of an individual having, or at risk of having, a disease associated with NAFLD or lipodystrophy syndromes. Certain embodiments are drawn to use of a DGAT2 specific inhibitor for the manufacture of a medicament for reducing or inhibiting triglyceride synthesis, hepatic lipid synthesis and insulin resistance in the liver of an individual having, or at risk of having, a disease associated with NAFLD or lipodystrophy syndromes. Certain embodiments are drawn to use of a DGAT2 specific inhibitor for the manufacture of a medicament for reducing or inhibiting triglyceride synthesis, lipid synthesis and insulin resistance in the adipose tissue of an individual having, or at risk of having, a disease associated with NAFLD or lipodystrophy syndromes. Certain embodiments are drawn to use of a DGAT2 specific inhibitor for the manufacture of a medicament for reducing or inhibiting triglyceride synthesis, lipid synthesis and insulin resistance in the adipose tissue of an individual having, or at risk of having, a disease associated with NAFLD or lipodystrophy syndromes. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound targeted to DGAT2, such as an antisense oligonucleotide targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is a compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides and has a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is ISIS 484137, ISIS 484085, ISIS 484129, ISIS 495576, ISIS 501861, ISIS 502194, ISIS 525443, and ISIS 525612. In certain embodiments the DGAT2 specific inhibitor is ISIS 769355, ISIS 769356, and ISIS 769357. In any of the foregoing embodiments, the antisense compound can be a single-stranded oligonucleotide.

[0195] In certain embodiments, a method of inhibiting expression of DGAT2 in a cell comprises contacting the cell with a DGAT2 specific inhibitor. In certain embodiments, the cell is a hepatocyte. In certain embodiments, the cell is in the liver of an individual. In certain embodiments, the cell is an adipose tissue cell of an individual. In certain embodiments, the cell is in the adipose tissue of an individual. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound targeted to DGAT2, such as an antisense oligonucleotide targeted to DGAT2. In certain embodiments, the DGAT2 specific inhibitor is a compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides and has a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising or consisting of a modified oligonucleotide and a conjugate group, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is an antisense compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, and 4373. In certain embodiments, the DGAT2 specific inhibitor is ISIS 484137, ISIS 484085, ISIS 484129, ISIS 495576, ISIS 501861, ISIS 502194, ISIS 525443, and ISIS 525612. In certain embodiments the DGAT2 specific inhibitor is ISIS 769355, ISIS 769356, and ISIS 769357. In any of the foregoing embodiments, the antisense compound can be a single-stranded oligonucleotide. In certain aspects, the antisense compound is administered to the individual parenterally.

[0196] In any of the foregoing embodiments, the DGAT2 specific inhibitor can be an antisense compound targeted to DGAT2. In certain aspects, the antisense compound is an antisense oligonucleotide, for example an antisense oligonucleotide consisting of 8 to 80 linked nucleosides, 12 to 30 linked nucleosides, or 20 linked nucleosides. In certain aspects, the antisense oligonucleotide is at least 80%, 85%, 90%, 95% or 100% complementary to any of the nucleobase sequences recited in SEQ ID NOs: 1-4. In certain aspects, the antisense oligonucleotide comprises at least one modified internucleoside linkage, at least one modified sugar and/or at least one modified nucleobase. In certain aspects, the modified internucleoside linkage is a phosphorothioate internucleoside linkage, the modified sugar is a bicyclic sugar or a 2′-O-methoxyethyl, and the modified nucleobase is a 5-methylcytosine. In certain aspects, the modified oligonucleotide comprises a gap segment consisting of linked 2′-deoxynucleosides; a 5′ wing segment consisting of linked nucleosides; and a 3′ wing segment consisting of linked nucleosides, wherein the gap segment is positioned immediately adjacent to and between the 5′ wing segment and the 3′ wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

[0197] In any of the foregoing embodiments, the antisense oligonucleotide consists of 12 to 30, 15 to 30, 15 to 25, 15 to 24, 16 to 24, 17 to 24, 18 to 24, 19 to 24, 20 to 24, 19 to 22, 20 to 22, 16 to 20, or 17 or 20 linked nucleosides. In certain aspects, the antisense oligonucleotide is at least 80%, 85%, 90%, 95% or 100% complementary to any of the nucleobase sequences recited in SEQ ID NOs: 1-4. In certain aspects, the antisense oligonucleotide comprises at least one modified internucleoside linkage, at least one modified sugar and/or at least one modified nucleobase. In certain aspects, the modified internucleoside linkage is a phosphorothioate internucleoside linkage, the modified sugar is a bicyclic sugar or a 2′-O-methoxyethyl, and the modified nucleobase is a 5-methylcytosine. In certain aspects, the modified oligonucleotide comprises a gap segment consisting of linked 2′-deoxynucleosides; a 5′ wing segment consisting of linked nucleosides; and a 3′ wing segment consisting of linked nucleosides, wherein the gap segment is positioned

immediately adjacent to and between the 5′ wing segment and the 3′ wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.
[0198] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 16-4679, wherein the modified oligonucleotide comprises: [0199] a gap segment consisting of linked 2′-deoxynucleosides; [0200] a 5′ wing segment consisting of linked nucleosides; and [0201] a 3′ wing segment consisting of linked nucleosides; [0202] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.
[0203] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising a conjugate group and a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 16-4679, wherein the modified oligonucleotide comprises: [0204] a gap segment consisting of linked 2′-deoxynucleosides; [0205] a 5′ wing segment consisting of linked nucleosides; and [0206] a 3′ wing segment consisting of linked nucleosides; [0207] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.
[0208] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 10 to 30 linked nucleosides having a nucleobase sequence comprising any one of SEQ ID NOs: 1423, 1371, 1415, 1849, 2959, 3292, 4198, or 4373, wherein the modified oligonucleotide comprises: [0209] a gap segment consisting of linked 2′-deoxynucleosides; [0210] a 5′ wing segment consisting of linked nucleosides; and [0211] a 3′ wing segment consisting of linked nucleosides; [0212] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.
[0213] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 16 to 30 linked nucleosides having a nucleobase sequence comprising the sequence recited in SEQ ID NO: 1423, wherein the modified oligonucleotide comprises: [0214] a gap segment consisting of ten linked 2′-deoxynucleosides; [0215] a 5′ wing segment consisting of five linked nucleosides; and [0216] a 3′ wing segment consisting of five linked nucleosides; [0217] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; [0218] wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0219] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 20 linked nucleosides having a nucleobase sequence consisting of the sequence recited in SEQ ID NO: 1423, wherein the modified oligonucleotide comprises: [0220] a gap segment consisting of ten linked 2′-deoxynucleosides; [0221] a 5′ wing segment consisting of five linked nucleosides; and [0222] a 3′ wing segment consisting of five linked nucleosides; [0223] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0224] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising a conjugate group and a modified oligonucleotide consisting of 20 linked nucleosides having a nucleobase sequence consisting of the sequence recited in SEQ ID NO: 1423, wherein the modified oligonucleotide comprises: [0225] a gap segment consisting of ten linked 2′-deoxynucleosides; [0226] a 5′ wing segment consisting of five linked nucleosides; and [0227] a 3′ wing segment consisting of five linked nucleosides; [0228] wherein the gap segment is positioned

between the 5′ wing segment and the 3′ wing segment; wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.

[0229] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 16 to 30 linked nucleosides having a nucleobase sequence comprising the sequence recited in SEQ ID NO: 1371, wherein the modified oligonucleotide comprises: [0230] a gap segment consisting of ten linked 2′-deoxynucleosides; [0231] a 5′ wing segment consisting of five linked nucleosides; and [0232] a 3′ wing segment consisting of five linked nucleosides; [0233] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; [0234] wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.

[0235] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 20 linked nucleosides having a nucleobase sequence consisting of the sequence recited in SEQ ID NO: 1371, wherein the modified oligonucleotide comprises: [0236] a gap segment consisting of ten linked 2′-deoxynucleosides; [0237] a 5′ wing segment consisting of five linked nucleosides; and [0238] a 3′ wing segment consisting of five linked nucleosides; [0239] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.

[0240] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 16 to 30 linked nucleosides having a nucleobase sequence comprising the sequence recited in SEQ ID NO: 1415, wherein the modified oligonucleotide comprises: [0241] a gap segment consisting of ten linked 2′-deoxynucleosides; [0242] a 5′ wing segment consisting of five linked nucleosides; and [0243] a 3′ wing segment consisting of five linked nucleosides; [0244] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; [0245] wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.

[0246] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 20 linked nucleosides having a nucleobase sequence consisting of the sequence recited in SEQ ID NO: 1415, wherein the modified oligonucleotide comprises: [0247] a gap segment consisting of ten linked 2′-deoxynucleosides; [0248] a 5′ wing segment consisting of five linked nucleosides; and [0249] a 3′ wing segment consisting of five linked nucleosides; [0250] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.

[0251] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 16 to 30 linked nucleosides having a nucleobase sequence comprising the sequence recited in SEQ ID NO: 1849, wherein the modified oligonucleotide comprises: [0252] a gap segment consisting of ten linked 2′-deoxynucleosides; [0253] a 5′ wing segment consisting of five linked nucleosides; and [0254] a 3′ wing segment consisting of five linked nucleosides; [0255] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; [0256] wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.

[0257] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 20 linked nucleosides having

a nucleobase sequence consisting of the sequence recited in SEQ ID NO: 1849, wherein the modified oligonucleotide comprises: [0258] a gap segment consisting of ten linked 2′-deoxynucleosides; [0259] a 5′ wing segment consisting of five linked nucleosides; and [0260] a 3′ wing segment consisting of five linked nucleosides; [0261] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0262] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 16 to 30 linked nucleosides having a nucleobase sequence comprising the sequence recited in SEQ ID NO: 2959, wherein the modified oligonucleotide comprises: [0263] a gap segment consisting of ten linked 2′-deoxynucleosides; [0264] a 5′ wing segment consisting of five linked nucleosides; and [0265] a 3′ wing segment consisting of five linked nucleosides; [0266] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; [0267] wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0268] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 20 linked nucleosides having a nucleobase sequence consisting of the sequence recited in SEQ ID NO: 2959, wherein the modified oligonucleotide comprises: [0269] a gap segment consisting of ten linked 2′-deoxynucleosides; [0270] a 5′ wing segment consisting of five linked nucleosides; and [0271] a 3′ wing segment consisting of five linked nucleosides; [0272] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0273] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 16 to 30 linked nucleosides having a nucleobase sequence comprising the sequence recited in SEQ ID NO: 3292, wherein the modified oligonucleotide comprises: [0274] a gap segment consisting of ten linked 2′-deoxynucleosides; [0275] a 5′ wing segment consisting of five linked nucleosides; and a 3′ wing segment consisting of five linked nucleosides; [0276] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; [0277] wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0278] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 20 linked nucleosides having a nucleobase sequence consisting of the sequence recited in SEQ ID NO: 3292, wherein the modified oligonucleotide comprises: [0279] a gap segment consisting of ten linked 2′-deoxynucleosides; [0280] a 5′ wing segment consisting of five linked nucleosides; and [0281] a 3′ wing segment consisting of five linked nucleosides; [0282] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0283] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 14 to 30 linked nucleosides having a nucleobase sequence comprising the sequence recited in SEQ ID NO: 4198, wherein the modified oligonucleotide comprises: [0284] a gap segment consisting of ten linked 2′-deoxynucleosides; [0285] a 5′ wing segment consisting of three linked nucleosides; and [0286] a 3′ wing segment consisting of four linked nucleosides; [0287] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; [0288] wherein each nucleoside of each

wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0289] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 17 linked nucleosides having a nucleobase sequence consisting of the sequence recited in SEQ ID NO: 4198, wherein the modified oligonucleotide comprises: [0290] a gap segment consisting of ten linked 2′-deoxynucleosides; [0291] a 5′ wing segment consisting of three linked nucleosides; and [0292] a 3′ wing segment consisting of four linked nucleosides; [0293] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0294] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 14 to 30 linked nucleosides having a nucleobase sequence comprising the sequence recited in SEQ ID NO: 4373, wherein the modified oligonucleotide comprises: [0295] a gap segment consisting of ten linked 2′-deoxynucleosides; [0296] a 5′ wing segment consisting of three linked nucleosides; and [0297] a 3′ wing segment consisting of four linked nucleosides; [0298] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; [0299] wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0300] In any of the foregoing methods or uses, the DGAT2 specific inhibitor can be a compound comprising or consisting of a modified oligonucleotide consisting of 17 linked nucleosides having a nucleobase sequence consisting of the sequence recited in SEQ ID NO: 4373, wherein the modified oligonucleotide comprises: [0301] a gap segment consisting of ten linked 2′-deoxynucleosides; [0302] a 5′ wing segment consisting of three linked nucleosides; and [0303] a 3′ wing segment consisting of four linked nucleosides; [0304] wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment; wherein each nucleoside of each wing segment comprises a 2′MOE sugar; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
[0305] In any of the foregoing methods or uses, the DGAT2 specific inhibitor is administered subcutaneously, such as by subcutaneous injection.

Antisense Compounds

[0306] Oligomeric compounds include, but are not limited to, oligonucleotides, oligonucleosides, oligonucleotide analogs, antisense compounds, antisense oligonucleotides, and siRNAs. An oligomeric compound may be “antisense” to a target nucleic acid, meaning that is capable of undergoing hybridization to a target nucleic acid through hydrogen bonding.
[0307] In certain embodiments, an antisense compound has a nucleobase sequence that, when written in the 5′ to 3′ direction, comprises the reverse complement of the target segment of a target nucleic acid to which it is targeted.
[0308] In certain embodiments, an antisense compound is 10 to 30 subunits in length. In certain embodiments, an antisense compound is 12 to 30 subunits in length. In certain embodiments, an antisense compound is 12 to 22 subunits in length. In certain embodiments, an antisense compound is 14 to 30 subunits in length. In certain embodiments, an antisense compound is 14 to 20 subunits in length. In certain embodiments, an antisense compound is 15 to 30 subunits in length. In certain embodiments, an antisense compound is 15 to 20 subunits in length. In certain embodiments, an antisense compound is 16 to 30 subunits in length. In certain embodiments, an antisense compound is 16 to 20 subunits in length. In certain embodiments, an antisense compound is 17 to 30 subunits in length. In certain embodiments, an antisense compound is 17 to 20 subunits in length. In certain embodiments, an antisense compound is 18 to 30 subunits in length. In certain embodiments, an antisense compound is 18 to 21 subunits in length. In certain embodiments, an antisense compound

is 18 to 20 subunits in length. In certain embodiments, an antisense compound is 20 to 30 subunits in length. In other words, such antisense compounds are from 12 to 30 linked subunits, 14 to 30 linked subunits, 14 to 20 subunits, 15 to 30 subunits, 15 to 20 subunits, 16 to 30 subunits, 16 to 20 subunits, 17 to 30 subunits, 17 to 20 subunits, 18 to 30 subunits, 18 to 20 subunits, 18 to 21 subunits, 20 to 30 subunits, or 12 to 22 linked subunits, respectively. In certain embodiments, an antisense compound is 14 subunits in length. In certain embodiments, an antisense compound is 16 subunits in length. In certain embodiments, an antisense compound is 17 subunits in length. In certain embodiments, an antisense compound is 18 subunits in length. In certain embodiments, an antisense compound is 19 subunits in length. In certain embodiments, an antisense compound is 20 subunits in length. In other embodiments, the antisense compound is 8 to 80, 12 to 50, 13 to 30, 13 to 50, 14 to 30, 14 to 50, 15 to 30, 15 to 50, 16 to 30, 16 to 50, 17 to 30, 17 to 50, 18 to 22, 18 to 24, 18 to 30, 18 to 50, 19 to 22, 19 to 30, 19 to 50, or 20 to 30 linked subunits. In certain such embodiments, the antisense compounds are 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, or 80 linked subunits in length, or a range defined by any two of the above values. In some embodiments the antisense compound is an antisense oligonucleotide, and the linked subunits are nucleotides, nucleosides, or nucleobases.

[0309] In certain embodiments, the antisense compound or oligomeric compound may further comprise additional features or elements, such as a conjugate group, that are attached to the oligonucleotide. In embodiments where a conjugate group comprises a nucleoside (i.e. a nucleoside that links the conjugate group to the oligonucleotide), the nucleoside of the conjugate group is not counted in the length of the oligonucleotide.

[0310] In certain embodiments antisense oligonucleotides may be shortened or truncated. For example, a single nucleoside may be deleted from the 5′ end (5′ truncation), or alternatively from the 3′ end (3′ truncation). A shortened or truncated antisense compound targeted to an DGAT2 nucleic acid may have two subunits deleted from the 5′ end, or alternatively may have two subunits deleted from the 3′ end, of the antisense compound. Alternatively, the deleted nucleosides may be dispersed throughout the antisense compound, for example, in an antisense compound having one nucleoside deleted from the 5′ end and one nucleoside deleted from the 3′ end.

[0311] When a single additional subunit is present in a lengthened antisense compound, the additional subunit may be located at the 5′ or 3′ end of the antisense compound. When two or more additional subunits are present, the added subunits may be adjacent to each other, for example, in an antisense compound having two subunits added to the 5′ end (5′ addition), or alternatively to the 3′ end (3′ addition), of the antisense compound. Alternatively, the added subunits may be dispersed throughout the antisense compound, for example, in an antisense compound having one subunit added to the 5′ end and one subunit added to the 3′ end.

[0312] It is possible to increase or decrease the length of an antisense compound, such as an antisense oligonucleotide, and/or introduce mismatch bases without eliminating activity (Woolf et al. (Proc. Natl. Acad. Sci. USA 89:7305-7309, 1992; Gautschi et al. *J. Natl. Cancer Inst.* 93:463-471, March 2001; Maher and Dolnick *Nuc. Acid. Res.* 16:3341-3358,1988). However, seemingly small changes in oligonucleotide sequence, chemistry and motif can make large differences in one or more of the many properties required for clinical development (Seth et al. J. Med. Chem. 2009, 52, 10; Egli et al. *J. Am. Chem. Soc.* 2011, 133, 16642).

[0313] In certain embodiments, antisense compounds are single-stranded, consisting of one oligomeric compound. The oligonucleotide of such single-stranded antisense compounds is an antisense oligonucleotide. In certain embodiments, the antisense oligonucleotide of a single-stranded antisense compound is modified. In certain embodiments, the oligonucleotide of a single-stranded antisense compound or oligomeric compound comprises a self-complementary nucleobase sequence. In certain embodiments, antisense compounds are double-stranded, comprising two oligomeric compounds that form a duplex. In certain such embodiments, one oligomeric compound of a double-stranded antisense compound comprises one or more conjugate groups. In certain embodiments, each oligomeric compound of a double-stranded antisense compound comprises one or more conjugate groups. In certain embodiments, each oligonucleotide of a double-stranded antisense compound is a modified oligonucleotide. In certain embodiments, one oligonucleotide of a double-stranded antisense compound is a modified oligonucleotide. In certain embodiments, one oligonucleotide of a double-stranded antisense compound is an antisense oligonucleotide. In certain such embodiments, the antisense oligonucleotide is a modified oligonucleotide. Examples of single-stranded and double-stranded antisense compounds include but are not limited to antisense oligonucleotides, siRNAs, microRNA targeting oligonucleotides, and single-stranded RNAi compounds, such as small hairpin RNAs (shRNAs), single-stranded siRNAs (ssRNAs), and microRNA mimics.

[0314] In certain embodiments, antisense compounds are interfering RNA compounds (RNAi), which include double-stranded RNA compounds (also referred to as short-interfering RNA or siRNA) and single-stranded RNAi compounds (or ssRNA). Such compounds work at least in part through the RISC pathway to degrade and/or sequester a target nucleic acid (thus, include microRNA/microRNA-mimic compounds). As used herein, the term siRNA is meant to be equivalent to other terms used to describe nucleic acid molecules that are capable of mediating sequence specific RNAi, for example short interfering RNA (siRNA), double-stranded RNA (dsRNA), micro-RNA (miRNA), short hairpin RNA (shRNA), short interfering oligonucleotide, short interfering nucleic acid, short interfering modified oligonucleotide, chemically modified siRNA, post-transcriptional gene silencing RNA (ptgsRNA), and others. In addition, as used herein, the term RNAi is meant to be equivalent to other terms used to describe sequence specific RNA interference, such as post transcriptional gene silencing, translational inhibition, or epigenetics.

[0315] In certain embodiments, a double-stranded compound can comprise any of the oligonucleotide sequences targeted to DGAT2 described herein. In certain embodiments, a double-stranded compound comprises a first strand comprising at least an 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleobase portion of any one of SEQ ID NOs: 16-4679 and a second strand. In certain embodiments, a double-stranded compound comprises a first strand comprising the nucleobase sequence of any one of SEQ ID NOs: 16-4679 and a second strand. In certain embodiments, the double-stranded compound comprises ribonucleotides in which the first strand has uracil (U) in place of thymine (T) in any one of SEQ ID NOs: 16-4679. In certain embodiments, a double-stranded compound comprises (i) a first strand comprising a nucleobase sequence complementary to the site on DGAT2 to which any of SEQ ID NOs: 16-4679 is targeted, and (ii) a second strand. In certain embodiments, the double-stranded compound comprises one or more modified nucleotides in which the 2′ position in the sugar contains a halogen (such as fluorine group; 2′-F) or contains an alkoxy group (such as a methoxy group; 2′-OMe). In certain embodiments, the double-stranded compound comprises at least one 2′-F sugar modification and at least one 2′-OMe sugar modification. In certain embodiments, the at least one 2′-F sugar modification and at least one 2′-OMe sugar modification are arranged in an alternating pattern for at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleobases along a strand of the dsRNA compound. In certain embodiments, the double-stranded compound comprises one or more linkages between adjacent nucleotides other than a naturally-occurring phosphodiester linkage. Examples of such linkages include phosphoramide, phosphorothioate, and phosphorodithioate linkages. The double-stranded compounds may also be chemically modified nucleic acid molecules as taught in U.S. Pat. No. 6,673,661. In other embodiments, the dsRNA contains one or two capped strands, as disclosed, for example, by WO 00/63364, filed Apr. 19, 2000. In certain embodiments, the first strand of the double-stranded compound is an siRNA guide strand and the second strand of the double-stranded compound is an siRNA passenger strand. In certain embodiments, the second strand of the double-stranded compound is complementary to the first strand. In certain embodiments, each strand of the double-stranded compound consists of 16, 17, 18, 19, 20, 21, 22, or 23 linked nucleosides. In certain embodiments, the first or second strand of the double-stranded compound can comprise a conjugate group.

[0316] In certain embodiments, a single-stranded RNAi (ssRNAi) compound can comprise any of the oligonucleotide sequences targeted to DGAT2 described herein. In certain embodiments, an ssRNAi compound comprises at least an 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleobase portion of any one of SEQ ID NOs: 16-4679. In certain embodiments, an ssRNAi compound comprises the nucleobase sequence of any one of SEQ ID NOs: 16-4679. In certain embodiments, the ssRNAi compound comprises ribonucleotides in which uracil (U) is in place of thymine (T) in any one of SEQ ID NOs: 16-4679. In certain embodiments, an ssRNAi compound comprises a nucleobase sequence complementary to the site on DGAT2 to which any of SEQ ID NOs: 16-4679 is targeted. In certain embodiments, an ssRNAi compound comprises one or more modified nucleotides in which the 2′ position in the sugar contains a halogen (such as fluorine group; 2′-F) or contains an alkoxy group (such as a methoxy group; 2′-OMe). In certain embodiments, an ssRNAi compound comprises at least one 2′-F sugar modification and at least one 2′-OMe sugar modification. In certain embodiments, the at least one 2′-F sugar modification and at least one 2′-OMe sugar modification are arranged in an alternating pattern for at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleobases along a strand of the ssRNAi compound. In certain embodiments, the ssRNAi compound comprises one or more linkages between adjacent nucleotides other than a naturally-occurring phosphodiester linkage. Examples of such linkages include phosphoramide, phosphorothioate, and phosphorodithioate linkages. The ssRNAi compounds may also be chemically modified nucleic acid molecules as taught in U.S. Pat. No. 6,673,661. In other embodiments, the ssRNAi contains a capped strand, as disclosed, for example, by WO 00/63364, filed Apr. 19, 2000. In certain embodiments, the ssRNAi compound consists of 16, 17, 18, 19, 20, 21, 22, or 23 linked nucleosides. In certain embodiments, the ssRNAi compound can comprise a conjugate group.

[0317] In certain embodiments, antisense compounds comprise modified oligonucleotides. Certain modified oligonucleotides have one or more asymmetric center and thus give rise to enantiomers, diastereomers, and other stereoisomeric configurations that may be defined, in terms of absolute stereochemistry, as (R) or(S), as α or β such as for sugar anomers, or as (D) or (L) such as for amino acids etc. Included in the modified oligonucleotides provided herein are all such possible isomers, including their racemic and optically pure forms, unless specified otherwise. Likewise, all cis- and trans-isomers and tautomeric forms are also included.

[0318] In certain embodiments, antisense compounds and oligomeric compounds described herein comprise modified oligonucleotides. Such modified oligonucleotides may contain any combination of the modified sugar moieties, modified nucleobases, modified internucleoside linkages, motifs, and/or lengths described herein.

Certain Antisense Compound Mechanisms

[0319] In certain embodiments, antisense compounds are capable of hybridizing to a target nucleic acid, resulting in at least one antisense activity. In certain embodiments, antisense compounds specifically affect one or more target nucleic acid. Such specific antisense compounds comprises a nucleobase sequence that hybridizes to one or more target nucleic acid, resulting in one or more desired antisense activity and does not hybridize to one or more non-target nucleic acid or does not hybridize to one or more non-target nucleic acid in such a way that results in an undesired antisense activity.

[0320] In certain antisense activities, hybridization of an antisense compound to a target nucleic acid results in recruitment of a protein that cleaves the target nucleic acid. For example, certain antisense compounds result in RNase H mediated cleavage of the target nucleic acid. RNase His a cellular endonuclease that cleaves the RNA strand of an RNA:DNA duplex. The DNA in such an

RNA:DNA duplex need not be unmodified DNA. In certain embodiments, the invention provides antisense compounds that are sufficiently “DNA-like” to elicit RNase H activity. Further, in certain embodiments, one or more non-DNA-like nucleoside in the gap of a gapmer is tolerated.
[0321] In certain antisense activities, an antisense compound or a portion of an antisense compound is loaded into an RNA-induced silencing complex (RISC), ultimately resulting in cleavage of the target nucleic acid. For example, certain antisense compounds result in cleavage of the target nucleic acid by Argonaute. In certain embodiments, antisense compounds that are loaded into RISC are RNAi compounds.
[0322] In certain embodiments, hybridization of an antisense compound to a target nucleic acid dose not result in recruitment of a protein that cleaves that target nucleic acid. In certain such embodiments, hybridization of the antisense compound to the target nucleic acid results in alteration of splicing of the target nucleic acid. In certain embodiments, hybridization of an antisense compound to a target nucleic acid results in inhibition of a binding interaction between the target nucleic acid and a protein or other nucleic acid. In certain such embodiments, hybridization of an antisense compound to a target nucleic acid results in alteration of translation of the target nucleic acid.
[0323] Antisense activities may be observed directly or indirectly. In certain embodiments, observation or detection of an antisense activity involves observation or detection of a change in an amount of a target nucleic acid or protein encoded by such target nucleic acid, a change in the ratio of splice variants of a nucleic acid or protein, and/or a phenotypic change in a cell or animal.
[0324] In certain embodiments, modified oligonucleotides having a gapmer sugar motif described herein have desirable properties compared to non-gapmer oligonucleotides or to gapmers having other sugar motifs. In certain circumstances, it is desirable to identify motifs resulting in a favorable combination of potent antisense activity and relatively low toxicity. In certain embodiments, compounds of the present invention have a favorable therapeutic index (measure of activity divided by measure of toxicity).
Target Nucleic Acids, Target Regions and Nucleotide Sequences
[0325] In certain embodiments, antisense compounds comprise or consist of an oligonucleotide comprising a region that is complementary to a target nucleic acid. In certain embodiments, the target nucleic acid is an endogenous RNA molecule. In certain embodiments, the target nucleic acid encodes a protein. In certain such embodiments, the target nucleic acid is selected from: an mRNA and a pre-mRNA, including intronic, exonic and untranslated regions. In certain embodiments, the target RNA is an mRNA. In certain embodiments, the target nucleic acid is a pre-mRNA. In certain such embodiments, the target region is entirely within an intron. In certain embodiments, the target region spans an intron/exon junction. In certain embodiments, the target region is at least 50% within an intron.
[0326] Nucleotide sequences that encode DGAT2 include, without limitation, the following: Ref Seq No. NM_032564.3 (incorporated herein by reference; designated herein as SEQ ID NO: 1), nucleotides 5669186 to 5712008 of Ref Seq No. NT_033927.5 (incorporated herein by reference; designated herein as SEQ ID NO: 2), nucleotides 20780400 to 20823450 of Ref Seq No. NT_167190.1 (incorporated herein by reference; designated herein as SEQ ID NO: 4878), Ref Seq No. AK091870.1 (incorporated herein by reference; designated herein as SEQ ID NO: 4879), and nucleotides 1232000 to 1268000 of Ref Seq No. NW_001100387.1 (incorporated herein by reference; designated herein as SEQ ID NO: 3).
Hybridization
[0327] In some embodiments, hybridization occurs between an antisense compound disclosed herein and a DGAT2 nucleic acid. The most common mechanism of hybridization involves hydrogen bonding (e.g., Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding) between complementary nucleobases of the nucleic acid molecules.
[0328] Hybridization can occur under varying conditions. Hybridization conditions are sequence-

dependent and are determined by the nature and composition of the nucleic acid molecules to be hybridized.
[0329] Methods of determining whether a sequence is specifically hybridizable to a target nucleic acid are well known in the art. In certain embodiments, the antisense compounds provided herein are specifically hybridizable with a DGAT2 nucleic acid.
Complementarity
[0330] An oligonucleotide is said to be complementary to another nucleic acid when the nucleobase sequence of such oligonucleotide or one or more regions thereof matches the nucleobase sequence of another oligonucleotide or nucleic acid or one or more regions thereof when the two nucleobase sequences are aligned in opposing directions. Nucleobase matches or complementary nucleobases, as described herein, are limited to adenine (A) and thymine (T), adenine (A) and uracil (U), cytosine (C) and guanine (G), and 5-methyl cytosine (mC) and guanine (G) unless otherwise specified. Complementary oligonucleotides and/or nucleic acids need not have nucleobase complementarity at each nucleoside and may include one or more nucleobase mismatches. An oligonucleotide is fully complementary or 100% complementary when such oligonucleotides have nucleobase matches at each nucleoside without any nucleobase mismatches.
[0331] Non-complementary nucleobases between an antisense compound and a DGAT2 nucleic acid may be tolerated provided that the antisense compound remains able to specifically hybridize to a target nucleic acid. Moreover, an antisense compound may hybridize over one or more segments of a DGAT2 nucleic acid such that intervening or adjacent segments are not involved in the hybridization event (e.g., a loop structure, mismatch or hairpin structure).
[0332] In certain embodiments, the antisense compounds provided herein, or a specified portion thereof, are, or are at least, 70%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% complementary to a DGAT2 nucleic acid, a target region, target segment, or specified portion thereof. In certain such embodiments, the region of full complementarity is from 6 to 20 nucleobases in length. In certain such embodiments, the region of full complementarity is from 10 to 18 nucleobases in length. In certain such embodiments, the region of full complementarity is from 18 to 20 nucleobases in length. Percent complementarity of an antisense compound with a target nucleic acid can be determined using routine methods.
[0333] For example, an antisense compound in which 18 of 20 nucleobases of the antisense compound are complementary to a target region, and would therefore specifically hybridize, would represent 90 percent complementarity. In this example, the remaining noncomplementary nucleobases may be clustered or interspersed with complementary nucleobases and need not be contiguous to each other or to complementary nucleobases. As such, an antisense compound which is 18 nucleobases in length having four noncomplementary nucleobases which are flanked by two regions of complete complementarity with the target nucleic acid would have 77.8% overall complementarity with the target nucleic acid and would thus fall within the scope of the present invention. Percent complementarity of an antisense compound with a region of a target nucleic acid can be determined routinely using BLAST programs (basic local alignment search tools) and PowerBLAST programs known in the art (Altschul et al., *J. Mol. Biol.,* 1990, 215, 403 410; Zhang and Madden, Genome Res., 1997, 7, 649 656). Percent homology, sequence identity or complementarity, can be determined by, for example, the Gap program (Wisconsin Sequence Analysis Package, Version 8 for Unix, Genetics Computer Group, University Research Park, Madison Wis.), using default settings, which uses the algorithm of Smith and Waterman (Adv. Appl. Math., 1981, 2, 482 489).
[0334] In certain embodiments, the antisense compounds provided herein, or specified portions thereof, are fully complementary (i.e. 100% complementary) to a target nucleic acid, or specified portion thereof. For example, an antisense compound may be fully complementary to a DGAT2 nucleic acid, or a target region, or a target segment or target sequence thereof. As used herein, “fully complementary” means each nucleobase of an antisense compound is capable of precise base

pairing with the corresponding nucleobases of a target nucleic acid. For example, a 20 nucleobase antisense compound is fully complementary to a target sequence that is 400 nucleobases long, so long as there is a corresponding 20 nucleobase portion of the target nucleic acid that is fully complementary to the antisense compound. Fully complementary can also be used in reference to a specified portion of the first and/or the second nucleic acid. For example, a 20 nucleobase portion of a 30 nucleobase antisense compound can be "fully complementary" to a target sequence that is 400 nucleobases long. The 20 nucleobase portion of the 30 nucleobase oligonucleotide is fully complementary to the target sequence if the target sequence has a corresponding 20 nucleobase portion wherein each nucleobase is complementary to the 20 nucleobase portion of the antisense compound. At the same time, the entire 30 nucleobase antisense compound may or may not be fully complementary to the target sequence, depending on whether the remaining 10 nucleobases of the antisense compound are also complementary to the target sequence.

[0335] In certain embodiments, antisense compounds comprise one or more mismatched nucleobases relative to the target nucleic acid. In certain such embodiments, antisense activity against the target is reduced by such mismatch, but activity against a non-target is reduced by a greater amount. Thus, in certain such embodiments selectivity of the antisense compound is improved. In certain embodiments, the mismatch is specifically positioned within an oligonucleotide having a gapmer motif. In certain such embodiments, the mismatch is at position 1, 2, 3, 4, 5, 6, 7, or 8 from the 5′-end of the gap region. In certain such embodiments, the mismatch is at position 9, 8, 7, 6, 5, 4, 3, 2, 1 from the 3′-end of the gap region. In certain such embodiments, the mismatch is at position 1, 2, 3, or 4 from the 5′-end of the wing region. In certain such embodiments, the mismatch is at position 4, 3, 2, or 1 from the 3′-end of the wing region.

[0336] The location of a non-complementary nucleobase may be at the 5′ end or 3′ end of the antisense compound. Alternatively, the non-complementary nucleobase or nucleobases may be at an internal position of the antisense compound. When two or more non-complementary nucleobases are present, they may be contiguous (i.e. linked) or non-contiguous. In one embodiment, a non-complementary nucleobase is located in the wing segment of a gapmer antisense oligonucleotide.

[0337] In certain embodiments, antisense compounds comprise one or more mismatched nucleobases relative to the target nucleic acid. In certain such embodiments, antisense activity against the target is reduced by such mismatch, but activity against a non-target is reduced by a greater amount. Thus, in certain such embodiments selectivity of the antisense compound is improved. In certain embodiments, the mismatch is specifically positioned within an oligonucleotide having a gapmer motif. In certain such embodiments, the mismatch is at position 1, 2, 3, 4, 5, 6, 7, or 8 from the 5′-end of the gap region. In certain such embodiments, the mismatch is at position 9, 8, 7, 6, 5, 4, 3, 2, 1 from the 3′-end of the gap region. In certain such embodiments, the mismatch is at position 1, 2, 3, or 4 from the 5′-end of the wing region. In certain such embodiments, the mismatch is at position 4, 3, 2, or 1 from the 3′-end of the wing region.

[0338] In certain embodiments, antisense compounds that are, or are up to 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 nucleobases in length comprise no more than 4, no more than 3, no more than 2, or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as a DGAT2 nucleic acid, or specified portion thereof.

[0339] In certain embodiments, antisense compounds that are, or are up to 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleobases in length comprise no more than 6, no more than 5, no more than 4, no more than 3, no more than 2, or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as a DGAT2 nucleic acid, or specified portion thereof.

[0340] The antisense compounds provided also include those which are complementary to a portion of a target nucleic acid. As used herein, "portion" refers to a defined number of contiguous (i.e. linked) nucleobases within a region or segment of a target nucleic acid. A "portion" can also refer to a defined number of contiguous nucleobases of an antisense compound. In certain embodiments, the antisense compounds, are complementary to at least an 8 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least a 9 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least a 10 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least an 11 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least a 12 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least a 13 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least a 14 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least a 15 nucleobase portion of a target segment. Also contemplated are antisense compounds that are complementary to at least a 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more nucleobase portion of a target segment, or a range defined by any two of these values.

Identity

[0341] The antisense compounds provided herein may also have a defined percent identity to a particular nucleotide sequence, SEQ ID NO, or compound represented by a specific Isis number, or portion thereof. As used herein, an antisense compound is identical to the sequence disclosed herein if it has the same nucleobase pairing ability. For example, a RNA which contains uracil in place of thymidine in a disclosed DNA sequence would be considered identical to the DNA sequence since both uracil and thymidine pair with adenine. Shortened and lengthened versions of the antisense compounds described herein as well as compounds having non-identical bases relative to the antisense compounds provided herein also are contemplated. The non-identical bases may be adjacent to each other or dispersed throughout the antisense compound. Percent identity of an antisense compound is calculated according to the number of bases that have identical base pairing relative to the sequence to which it is being compared.

[0342] In certain embodiments, the antisense compounds, or portions thereof, are, or are at least, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% identical to one or more of the antisense compounds or SEQ ID NOs, or a portion thereof, disclosed herein.

[0343] In certain embodiments, a portion of the antisense compound is compared to an equal length portion of the target nucleic acid. In certain embodiments, an 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 nucleobase portion is compared to an equal length portion of the target nucleic acid.

[0344] In certain embodiments, a portion of the antisense oligonucleotide is compared to an equal length portion of the target nucleic acid. In certain embodiments, an 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 nucleobase portion is compared to an equal length portion of the target nucleic acid.

Modifications

[0345] Modifications to antisense compounds encompass substitutions or changes to internucleoside linkages, sugar moieties, or nucleobases. Modified antisense compounds are often preferred over native forms because of desirable properties such as, for example, enhanced cellular uptake, enhanced affinity for nucleic acid target, increased stability in the presence of nucleases, or increased inhibitory activity.

[0346] Chemically modified nucleosides may also be employed to increase the binding affinity of a shortened or truncated antisense oligonucleotide for its target nucleic acid. Consequently, comparable results can often be obtained with shorter antisense compounds that have such chemically modified nucleosides.

Modified Internucleoside Linkages

[0347] The naturally occurring internucleoside linkage of RNA and DNA is a 3′ to 5′ phosphodiester linkage. Antisense compounds having one or more modified, i.e. non-naturally occurring, internucleoside linkages are often selected over antisense compounds having naturally occurring internucleoside linkages because of desirable properties such as, for example, enhanced cellular uptake, enhanced affinity for target nucleic acids, and increased stability in the presence of nucleases.

[0348] Oligonucleotides having modified internucleoside linkages include internucleoside linkages that retain a phosphorus atom as well as internucleoside linkages that do not have a phosphorus atom. Representative phosphorus containing internucleoside linkages include, but are not limited to, phosphodiesters, phosphotriesters, methylphosphonates, phosphoramidate, and phosphorothioates. Methods of preparation of phosphorous-containing and non-phosphorous-containing linkages are well known.

[0349] In certain embodiments, nucleosides of modified oligonucleotides may be linked together using any internucleoside linkage. The two main classes of internucleoside linking groups are defined by the presence or absence of a phosphorus atom. Representative phosphorus-containing internucleoside linkages include but are not limited to phosphates, which contain a phosphodiester bond ("P═O")(also referred to as unmodified or naturally occurring linkages), phosphotriesters, methylphosphonates, phosphoramidates, and phosphorothioates ("P═S"), and phosphorodithioates ("HS-P═S"). Representative non-phosphorus containing internucleoside linking groups include but are not limited to methylenemethylimino (—CH.sub.2—N(CH.sub.3)—O—CH.sub.2—), thiodiester (—O—C(═O)—S—), thionocarbamate (—O—C(═O)(NH)—S—); siloxane (—O—SiH.sub.2—O—); and N,N′-dimethylhydrazine (—CH.sub.2—N(CH.sub.3)—N(CH.sub.3)—). Modified internucleoside linkages, compared to naturally occurring phosphate linkages, can be used to alter, typically increase, nuclease resistance of the oligonucleotide. In certain embodiments, internucleoside linkages having a chiral atom can be prepared as a racemic mixture, or as separate enantiomers. Representative chiral internucleoside linkages include but are not limited to alkylphosphonates and phosphorothioates. Methods of preparation of phosphorous-containing and non-phosphorous-containing internucleoside linkages are well known to those skilled in the art.

[0350] Neutral internucleoside linkages include, without limitation, phosphotriesters, methylphosphonates, MMI (3′-CH.sub.2—N(CH.sub.3)—O-5′), amide-3 (3′-CH.sub.2—C(═O)—N(H)—5′), amide-4 (3′-CH.sub.2—N(H)—C(═O)—5′), formacetyl (3′-O—CH.sub.2—O-5′), methoxypropyl, and thioformacetal (3′-S—CH.sub.2—O-5′). Further neutral internucleoside linkages include nonionic linkages comprising siloxane (dialkylsiloxane), carboxylate ester, carboxamide, sulfide, sulfonate ester and amides (See for example: *Carbohydrate Modifications in Antisense Research*; Y. S. Sanghvi and P. D. Cook, Eds., ACS Symposium Series 580; Chapters 3 and 4, 40-65). Further neutral internucleoside linkages include nonionic linkages comprising mixed N, O, S and CH.sub.2 component parts.

[0351] In certain embodiments, antisense compounds targeted to a DGAT2 nucleic acid comprise one or more modified internucleoside linkages. In certain embodiments, the modified internucleoside linkages are phosphorothioate linkages. In certain embodiments, each internucleoside linkage of an antisense compound is a phosphorothioate internucleoside linkage.

[0352] In certain embodiments, oligonucleotides comprise modified internucleoside linkages arranged along the oligonucleotide or region thereof in a defined pattern or modified internucleoside linkage motif. In certain embodiments, internucleoside linkages are arranged in a gapped motif. In such embodiments, the internucleoside linkages in each of two wing regions are different from the internucleoside linkages in the gap region. In certain embodiments the internucleoside linkages in the wings are phosphodiester and the internucleoside linkages in the gap are phosphorothioate. The nucleoside motif is independently selected, so such oligonucleotides having a gapped internucleoside linkage motif may or may not have a gapped nucleoside motif and if it does have a gapped nucleoside motif, the wing and gap lengths may or may not be the same.

[0353] In certain embodiments, oligonucleotides comprise a region having an alternating internucleoside linkage motif. In certain embodiments, oligonucleotides of the present invention comprise a region of uniformly modified internucleoside linkages. In certain such embodiments, the oligonucleotide comprises a region that is uniformly linked by phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide is uniformly linked by phosphorothioate. In certain embodiments, each internucleoside linkage of the oligonucleotide is selected from phosphodiester and phosphorothioate. In certain embodiments, each internucleoside linkage of the oligonucleotide is selected from phosphodiester and phosphorothioate and at least one internucleoside linkage is phosphorothioate.

[0354] In certain embodiments, the oligonucleotide comprises at least 6 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least 8 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least 10 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 6 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 8 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 10 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least block of at least one 12 consecutive phosphorothioate internucleoside linkages. In certain such embodiments, at least one such block is located at the 3′ end of the oligonucleotide. In certain such embodiments, at least one such block is located within 3 nucleosides of the 3′ end of the oligonucleotide.

[0355] In certain embodiments, oligonucleotides comprise one or more methylphosphonate linkages. In certain embodiments, oligonucleotides having a gapmer nucleoside motif comprise a linkage motif comprising all phosphorothioate linkages except for one or two methylphosphonate linkages. In certain embodiments, one methylphosphonate linkage is in the central gap of an oligonucleotide having a gapmer nucleoside motif.

[0356] In certain embodiments, it is desirable to arrange the number of phosphorothioate internucleoside linkages and phosphodiester internucleoside linkages to maintain nuclease resistance. In certain embodiments, it is desirable to arrange the number and position of phosphorothioate internucleoside linkages and the number and position of phosphodiester internucleoside linkages to maintain nuclease resistance. In certain embodiments, the number of phosphorothioate internucleoside linkages may be decreased and the number of phosphodiester internucleoside linkages may be increased. In certain embodiments, the number of phosphorothioate internucleoside linkages may be decreased and the number of phosphodiester internucleoside linkages may be increased while still maintaining nuclease resistance. In certain embodiments it is desirable to decrease the number of phosphorothioate internucleoside linkages while retaining nuclease resistance. In certain embodiments it is desirable to increase the number of phosphodiester internucleoside linkages while retaining nuclease resistance.

Modified Sugar Moieties

[0357] Antisense compounds can optionally contain one or more nucleosides wherein the sugar group has been modified. Such sugar modified nucleosides may impart enhanced nuclease stability, increased binding affinity, or some other beneficial biological property to the antisense compounds.

[0358] In certain embodiments, modified oligonucleotides comprise one or more modified nucleosides comprising a modified sugar moiety. Such modified oligonucleotides comprising one or more sugar-modified nucleosides may have desirable properties, such as enhanced nuclease stability or increased binding affinity with a target nucleic acid relative to oligonucleotides lacking such sugar-modified nucleosides. In certain embodiments, modified sugar moieties are linearly modified sugar moieties. In certain embodiments, modified sugar moieties are bicyclic or tricyclic sugar moieties. In certain embodiments, modified sugar moieties are sugar surrogates. Such sugar surrogates may comprise one or more substitutions corresponding to those of substituted sugar moieties.

[0359] In certain embodiments, modified sugar moieties are linearly modified sugar moieties comprising a furanosyl ring with one or more acyclic substituent, including but not limited to substituents at the 2′ and/or 5′ positions. Examples of 2′-substituent groups suitable for linearly modified sugar moieties include but are not limited to: 2′-F, 2′—OCH.sub.3 (“OMe” or “O-methyl”), and 2′-O(CH.sub.2).sub.2OCH.sub.3 (“MOE”). In certain embodiments, 2′-substituent groups are selected from among: halo, allyl, amino, azido, SH, CN, OCN, CF.sub.3, OCF.sub.3, O—C.sub.1-C.sub.10 alkoxy, O—C.sub.1-C.sub.10 substituted alkoxy, O—C.sub.1-C.sub.10 alkyl, O—C.sub.1-C.sub.10 substituted alkyl, S-alkyl, N(R.sub.m)-alkyl, O-alkenyl, S-alkenyl, N(R.sub.m)-alkenyl, O-alkynyl, S-alkynyl, N(R.sub.m)-alkynyl, O-alkylenyl-O-alkyl, alkynyl, alkaryl, aralkyl, O-alkaryl, O-aralkyl, O(CH.sub.2).sub.2SCH.sub.3, O(CH.sub.2).sub.2ON(R.sub.m)(R.sub.n) or OCH.sub.2C(═O)—N(R.sub.m)(R.sub.n), where each R.sub.m and R.sub.n is, independently, H, an amino protecting group, or substituted or unsubstituted C.sub.1-C.sub.10 alkyl. Certain embodiments of these 2′-substituent groups can be further substituted with one or more substituent groups independently selected from among: hydroxyl, amino, alkoxy, carboxy, benzyl, phenyl, nitro (NO.sub.2), thiol, thioalkoxy, thioalkyl, halogen, alkyl, aryl, alkenyl and alkynyl. Examples of 5′-substituent groups suitable for linearly modified sugar moieties include but are not limited to: 5′-methyl (R or S), 5′-vinyl, and 5′-methoxy. In certain embodiments, linearly modified sugars comprise more than one non-bridging sugar substituent, for example, 2′-F-5′-methyl sugar moieties (see, e.g., PCT International Application WO 2008/101157, for additional 2′,5′-bis substituted sugar moieties and nucleosides).

[0360] In certain embodiments, a 2′-substituted nucleoside or 2′-linearly modified nucleoside comprises a sugar moiety comprising a linear 2′-substituent group selected from: F, NH.sub.2, N.sub.3, OCF.sub.3, OCH.sub.3, O(CH.sub.2).sub.3NH.sub.2, CH.sub.2CH═CH.sub.2, OCH.sub.2CH—CH.sub.2, OCH.sub.2CH.sub.2OCH.sub.3, O(CH.sub.2).sub.2SCH.sub.3, O(CH.sub.2).sub.2ON(R.sub.m)(R.sub.n), O(CH.sub.2).sub.2O(CH.sub.2).sub.2N(CH.sub.3).sub.2, and N-substituted acetamide (OCH.sub.2C(═O)—N(R.sub.m)(R.sub.n)), where each R.sub.m and R.sub.n is, independently, H, an amino protecting group, or substituted or unsubstituted C.sub.1-C.sub.10 alkyl.

[0361] In certain embodiments, a 2′-substituted nucleoside or 2′-linearly modified nucleoside comprises a sugar moiety comprising a linear 2′-substituent group selected from: F, OCF.sub.3, OCH.sub.3, OCH.sub.2CH.sub.2OCH.sub.3, O(CH.sub.2).sub.2SCH.sub.3, O(CH.sub.2).sub.2ON(CH.sub.3).sub.2, O(CH.sub.2).sub.2O(CH.sub.2).sub.2N(CH.sub.3).sub.2, and OCH.sub.2C(═O)—N(H) CH.sub.3 (“NMA”).

[0362] In certain embodiments, a 2′-substituted nucleoside or 2′-linearly modified nucleoside comprises a sugar moiety comprising a linear 2′-substituent group selected from: F, OCH.sub.3, and OCH.sub.2CH.sub.2OCH.sub.3.

[0363] Nucleosides comprising modified sugar moieties, such as linearly modified sugar moieties, are referred to by the position(s) of the substitution(s) on the sugar moiety of the nucleoside. For example, nucleosides comprising 2′-substituted or 2-modified sugar moieties are referred to as 2′-substituted nucleosides or 2-modified nucleosides.

[0364] Certain modified sugar moieties comprise a bridging sugar substituent that forms a second ring resulting in a bicyclic sugar moiety. In certain such embodiments, the bicyclic sugar moiety comprises a bridge between the 4′ and the 2′ furanose ring atoms. Examples of such 4′ to 2′ bridging sugar substituents include but are not limited to: 4′-CH.sub.2-2′,4′—(CH.sub.2).sub.2—2′,4′—(CH.sub.2).sub.3—2′,4′-CH.sub.2—O-2′ (“LNA”), 4′-CH.sub.2—S-2′,4′—(CH.sub.2).sub.2—O-2′ (“ENA”), 4′-CH(CH.sub.3)—O-2′ (referred to as “constrained ethyl” or “cEt” when in the S configuration), 4′-CH.sub.2—O—CH.sub.2-2′,4′-CH.sub.2—N(R)-2′,4′-CH(CH.sub.2OCH.sub.3)—O-2′ (“constrained MOE” or “cMOE”) and analogs thereof (see, e.g., U.S. Pat. No. 7,399,845), 4′-C(CH.sub.3)(CH.sub.3)—O-2′ and analogs thereof (see, e.g., WO2009/006478), 4′-CH.sub.2—N(OCH.sub.3)-2′ and analogs thereof (see, e.g., WO2008/150729), 4′-CH.sub.2—O—N(CH.sub.3)-2′ (see, e.g., US2004/0171570), 4′-CH.sub.2—C(H)(CH.sub.3)-2′ (see, e.g.,

Chattopadhyaya, et al., *J. Org. Chem.*, 2009, 74, 118-134), 4′-CH.sub.2—C(═CH.sub.2)-2′ and analogs thereof (see, published PCT International Application WO 2008/154401), 4′-C(R.sub.aR.sub.b)—N(R)—O-2′,4′-C(R.sub.aR.sub.b)—O—N(R)-2′,4′-CH.sub.2—O—N(R)-2′, and 4′-CH.sub.2—N(R)—O-2′, wherein each R, R.sub.a, and R.sub.b is, independently, H, a protecting group, or C.sub.1-C.sub.12 alkyl (see, e.g. U.S. Pat. No. 7,427,672).
[0365] In certain embodiments, such 4′ to 2′ bridges independently comprise from 1 to 4 linked groups independently selected from:—[C(R.sub.a)(R.sub.b)].sub.n—, —[C(R.sub.a)(R.sub.b)].sub.n—O—, —C(R.sub.a)═C(R.sub.b)—, —C(R.sub.a)═N—, —C(═NR.sub.a)—, —C(═O)—, —C(═S)—, —O—, —Si(R.sub.a).sub.2—, —S(═O).sub.x—, and —N(R.sub.a)—;
[0366] wherein: [0367] x is 0, 1, or 2; [0368] n is 1, 2, 3, or 4; [0369] each R.sub.a and R.sub.b is, independently, H, a protecting group, hydroxyl, C.sub.1-C.sub.12 alkyl, substituted C.sub.1-C.sub.12 alkyl, C.sub.2-C.sub.12 alkenyl, substituted C.sub.2-C.sub.12 alkenyl, C.sub.2-C.sub.12 alkynyl, substituted C.sub.2-C.sub.12 alkynyl, C.sub.5-C.sub.20 aryl, substituted C.sub.5-C.sub.20 aryl, heterocycle radical, substituted heterocycle radical, heteroaryl, substituted heteroaryl, C.sub.5-C.sub.7 alicyclic radical, substituted C.sub.5-C.sub.7 alicyclic radical, halogen, OJ.sub.1, NJ.sub.1J.sub.2, SJ.sub.1, N.sub.3, COOJ.sub.1, acyl (C(═O)—H), substituted acyl, CN, sulfonyl (S(═O).sub.2-J.sub.1), or sulfoxyl (S(═O)-J.sub.1); and each J.sub.1 and J.sub.2 is, independently, H, C.sub.1-C.sub.12 alkyl, substituted C.sub.1-C.sub.12 alkyl, C.sub.2-C.sub.12 alkenyl, substituted C.sub.2-C.sub.12 alkenyl, C.sub.2-C.sub.12 alkynyl, substituted C.sub.2-C.sub.12 alkynyl, C.sub.5-C.sub.20 aryl, substituted C.sub.5-C.sub.20 aryl, acyl (C(═O)—H), substituted acyl, a heterocycle radical, a substituted heterocycle radical, C.sub.1-C.sub.12 aminoalkyl, substituted C.sub.1-C.sub.12 aminoalkyl, or a protecting group.
[0370] Additional bicyclic sugar moieties are known in the art, for example: Freier et al., *Nucleic Acids Research*, 1997, 25 (22), 4429-4443, Albaek et al., *J. Org. Chem.*, 2006, 71, 7731-7740, Singh et al., *Chem. Commun.*, 1998, 4, 455-456; Koshkin et al., *Tetrahedron*, 1998, 54, 3607-3630; Wahlestedt et al., *Proc. Natl. Acad. Sci. U.S.A*, 2000, 97, 5633-5638; Kumar et al., *Bioorg. Med. Chem. Lett.*, 1998, 8, 2219-2222; Singh et al., *J. Org. Chem.*, 1998, 63, 10035-10039; Srivastava et al., *J. Am. Chem. Soc.*, 20017, 129, 8362-8379; Elayadi et al., *Curr. Opinion Invens. Drugs*, 2001, 2, 558-561; Braasch et al., *Chem. Biol.*, 2001, 8, 1-7; Orum et al., *Curr. Opinion Mol. Ther.*, 2001, 3, 239-243; U.S. Pat. Nos. 7,053,207, 6,268,490, 6,770,748, 6,794,499, 7,034,133, 6,525,191, 6,670,461, and 7,399,845; WO 2004/106356, WO 1994/14226, WO 2005/021570, and WO 2007/134181; U.S. Patent Publication Nos. US2004/0171570, US2007/0287831, and US2008/0039618; U.S. patents Ser. Nos. 12/129,154, 60/989,574, 61/026,995, 61/026,998, 61/056,564, 61/086,231, 61/097,787, and 61/099,844; and PCT International Applications Nos. PCT/US2008/064591, PCT/US2008/066154, and PCT/US2008/068922.
[0371] In certain embodiments, bicyclic sugar moieties and nucleosides incorporating such bicyclic sugar moieties are further defined by isomeric configuration. For example, an LNA nucleoside (described above) may be in the α-L configuration or in the β-D configuration.
##STR00007##
[0372] α-L-methyleneoxy (4′-CH.sub.2—O-2′) or α-L-LNA bicyclic nucleosides have been incorporated into antisense oligonucleotides that showed antisense activity (Frieden et al., *Nucleic Acids Research*, 2003, 21, 6365-6372). Herein, general descriptions of bicyclic nucleosides include both isomeric configurations. When the positions of specific bicyclic nucleosides (e.g., LNA or cEt) are identified in exemplified embodiments herein, they are in the B-D configuration, unless otherwise specified.
[0373] In certain embodiments, modified sugar moieties comprise one or more non-bridging sugar substituent and one or more bridging sugar substituent (e.g., 5′-substituted and 4′-2′ bridged sugars). (see, e.g., WO 2007/134181, wherein LNA nucleosides are further substituted with, for example, a 5′-methyl or a 5′-vinyl group, and see, e.g., U.S. Pat. Nos. 7,547,684; 7,750,131; 8,030,467; 8,268,980; 7,666,854; and 8,088,746).

[0374] In certain embodiments, modified sugar moieties are sugar surrogates. In certain such embodiments, the oxygen atom of the sugar moiety is replaced, e.g., with a sulfur, carbon or nitrogen atom. In certain such embodiments, such modified sugar moieties also comprise bridging and/or non-bridging substituents as described above. For example, certain sugar surrogates comprise a 4′-sulfur atom and a substitution at the 2′-position (see, e.g., US2005/0130923) and/or the 5′ position.

[0375] In certain embodiments, sugar surrogates comprise rings having other than 5 atoms. For example, in certain embodiments, a sugar surrogate comprises a six-membered tetrahydropyran ("THP"). Such tetrahydropyrans may be further modified or substituted. Nucleosides comprising such modified tetrahydropyrans include but are not limited to hexitol nucleic acid ("HNA"), anitol nucleic acid ("ANA"), manitol nucleic acid ("MNA")(see Leumann, C J. *Bioorg*. & *Med. Chem.* 2002, 10, 841-854), fluoro HNA:

##STR00008##

[0376] ("F-HNA", see e.g., U.S. Pat. Nos. 8,088,904; 8,440,803; and 8,796,437, F-HNA can also be referred to as a F-THP or 3′-fluoro tetrahydropyran), and nucleosides comprising additional modified THP compounds having the formula:

##STR00009## [0377] wherein, independently, for each of said modified THP nucleoside: [0378] Bx is a nucleobase moiety;

[0379] T.sub.3 and T.sub.4 are each, independently, an internucleoside linking group linking the modified THP nucleoside to the remainder of an oligonucleotide or one of T.sub.3 and T.sub.4 is an internucleoside linking group linking the modified THP nucleoside to the remainder of an oligonucleotide and the other of T.sub.3 and T.sub.4 is H, a hydroxyl protecting group, a linked conjugate group, or a 5′ or 3′-terminal group; q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 are each, independently, H, C.sub.1-C.sub.6 alkyl, substituted C.sub.1-C.sub.6 alkyl, C.sub.2-C.sub.6 alkenyl, substituted C.sub.2-C.sub.6 alkenyl, C.sub.2-C.sub.6 alkynyl, or substituted C.sub.2-C.sub.6 alkynyl; and each of R.sub.1 and R.sub.2 is independently selected from among: hydrogen, halogen, substituted or unsubstituted alkoxy, NJ.sub.1J.sub.2, SJ.sub.1, N.sub.3, OC(═X) J.sub.1, OC(═X) NJ.sub.1J.sub.2, NJ.sub.3C(═X) NJ.sub.1J.sub.2, and CN, wherein X is O, S or NJ.sub.1, and each J.sub.1, J.sub.2, and J.sub.3 is, independently, H or C.sub.1-C.sub.6 alkyl.

[0380] In certain embodiments, modified THP nucleosides are provided wherein q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 are each H. In certain embodiments, at least one of q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 is other than H. In certain embodiments, at least one of q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 is methyl. In certain embodiments, modified THP nucleosides are provided wherein one of R.sub.1 and R.sub.2 is F. In certain embodiments, R.sub.1 is F and R.sub.2 is H, in certain embodiments, R.sub.1 is methoxy and R.sub.2 is H, and in certain embodiments, R.sub.1 is methoxyethoxy and R.sub.2 is H.

[0381] In certain embodiments, sugar surrogates comprise rings having more than 5 atoms and more than one heteroatom. For example, nucleosides comprising morpholino sugar moieties and their use in oligonucleotides have been reported (see, e.g., Braasch et al., Biochemistry, 2002, 41, 4503-4510 and U.S. Pat. Nos. 5,698,685; 5,166,315; 5,185,444; and 5,034,506). As used here, the term "morpholino" means a sugar surrogate having the following structure:

##STR00010##

[0382] In certain embodiments, morpholinos may be modified, for example by adding or altering various substituent groups from the above morpholino structure. Such sugar surrogates are referred to herein as "modified morpholinos."

[0383] In certain embodiments, sugar surrogates comprise acyclic moieties. Examples of nucleosides and oligonucleotides comprising such acyclic sugar surrogates include but are not limited to: peptide nucleic acid ("PNA"), acyclic butyl nucleic acid (see, e.g., Kumar et al., *Org.*

Biomol. Chem., 2013, 11, 5853-5865), and nucleosides and oligonucleotides described in WO2011/133876.

[0384] Many other bicyclic and tricyclic sugar and sugar surrogate ring systems are known in the art that can be used in modified nucleosides (see, e.g., Leumann, J. C, *Bioorganic* & *Medicinal Chemistry*, 2002, 10, 841-854).

Modified Nucleobases

[0385] Nucleobase (or base) modifications or substitutions are structurally distinguishable from, yet functionally interchangeable with, naturally occurring or synthetic unmodified nucleobases. Both natural and modified nucleobases are capable of participating in hydrogen bonding. Such nucleobase modifications can impart nuclease stability, binding affinity or some other beneficial biological property to antisense compounds.

[0386] In certain embodiments, modified oligonucleotides comprise one or more nucleoside comprising an unmodified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more nucleoside comprising a modified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more nucleoside that does not comprise a nucleobase, referred to as an abasic nucleoside.

[0387] In certain embodiments, modified nucleobases are selected from: 5-substituted pyrimidines, 6-azapyrimidines, alkyl or alkynyl substituted pyrimidines, alkyl substituted purines, and N-2, N-6 and O-6 substituted purines. In certain embodiments, modified nucleobases are selected from: 2-aminopropyladenine, 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-N-methylguanine, 6-N-methyladenine, 2-propyladenine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-propynyl (—C≡C—CH.sub.3) uracil, 5-propynylcytosine, 6-azauracil, 6-azocytosine, 6-azothymine, 5-ribosyluracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl, 8-aza and other 8-substituted purines, 5-halo, particularly 5-bromo, 5-trifluoromethyl, 5-halouracil, and 5-halocytosine, 7-methylguanine, 7-methyladenine, 2-F-adenine, 2-aminoadenine, 7-deazaguanine, 7-deazaadenine, 3-deazaguanine, 3-deazaadenine, 6-N-benzoyladenine, 2-N-isobutyrylguanine, 4-N-benzoylcytosine, 4-N-benzoyluracil, 5-methyl 4-N-benzoylcytosine, 5-methyl 4-N-benzoyluracil, universal bases, hydrophobic bases, promiscuous bases, size-expanded bases, and fluorinated bases. Further modified nucleobases include tricyclic pyrimidines, such as 1,3-diazaphenoxazine-2-one, 1,3-diazaphenothiazine-2-one and 9-(2-aminoethoxy)-1,3-diazaphenoxazine-2-one (G-clamp). Modified nucleobases may also include those in which the purine or pyrimidine base is replaced with other heterocycles, for example 7-deaza-adenine, 7-deazaguanosine, 2-aminopyridine and 2-pyridone. Further nucleobases include those disclosed in U.S. Pat. No. 3,687,808, those disclosed in *The Concise Encyclopedia Of Polymer Science And Engineering*, Kroschwitz, J. I., Ed., John Wiley & Sons, 1990, 858-859; Englisch et al., *Angewandte Chemie*, International Edition, 1991, 30, 613; Sanghvi, Y. S., Chapter 15, *Antisense Research and Applications*, Crooke, S. T. and Lebleu, B., Eds., CRC Press, 1993, 273-288; and those disclosed in Chapters 6 and 15, *Antisense Drug Technology*, Crooke S. T., Ed., CRC Press, 2008, 163-166 and 442-443.

[0388] Representative United States patents that teach the preparation of certain of the above noted modified nucleobases as well as other modified nucleobases include without limitation, US2003/0158403, U.S. Pat. Nos. 3,687,808; 4,845,205; 5,130,302; 5,134,066; 5,175,273; 5,367,066; 5,432,272; 5,434,257; 5,457,187; 5,459,255; 5,484,908; 5,502,177; 5,525,711; 5,552,540; 5,587,469; 5,594,121; 5,596,091; 5,614,617; 5,645,985; 5,681,941; 5,750,692; 5,763,588; 5,830,653 and 6,005,096.

[0389] In certain embodiments, antisense compounds targeted to a DGAT2 nucleic acid comprise one or more modified nucleobases. In certain embodiments, the modified nucleobase is 5-methylcytosine. In certain embodiments, each cytosine is a 5-methylcytosine.

Certain Motifs

[0390] Oligonucleotides can have a motif, e.g. a pattern of unmodified and/or modified sugar moieties, nucleobases, and/or internucleoside linkages. In certain embodiments, modified oligonucleotides comprise one or more modified nucleoside comprising a modified sugar. In certain embodiments, modified oligonucleotides comprise one or more modified nucleosides comprising a modified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more modified internucleoside linkage. In such embodiments, the modified, unmodified, and differently modified sugar moieties, nucleobases, and/or internucleoside linkages of a modified oligonucleotide define a pattern or motif. In certain embodiments, the patterns of sugar moieties, nucleobases, and internucleoside linkages are each independent of one another. Thus, a modified oligonucleotide may be described by its sugar motif, nucleobase motif and/or internucleoside linkage motif (as used herein, nucleobase motif describes the modifications to the nucleobases independent of the sequence of nucleobases).

1. Certain Sugar Motifs

[0391] In certain embodiments, oligonucleotides comprise one or more type of modified sugar and/or unmodified sugar moiety arranged along the oligonucleotide or region thereof in a defined pattern or sugar motif. In certain instances, such sugar motifs include but are not limited to any of the sugar modifications discussed herein.

[0392] In certain embodiments, modified oligonucleotides comprise or consist of a region having a gapmer motif, which comprises two external regions or “wings” and a central or internal region or “gap.” The three regions of a gapmer motif (the 5′-wing, the gap, and the 3′-wing) form a contiguous sequence of nucleosides wherein at least some of the sugar moieties of the nucleosides of each of the wings differ from at least some of the sugar moieties of the nucleosides of the gap. Specifically, at least the sugar moieties of the nucleosides of each wing that are closest to the gap (the 3′-most nucleoside of the 5′-wing and the 5′-most nucleoside of the 3′-wing) differ from the sugar moiety of the neighboring gap nucleosides, thus defining the boundary between the wings and the gap (i.e., the wing/gap junction). In certain embodiments, the sugar moieties within the gap are the same as one another. In certain embodiments, the gap includes one or more nucleoside having a sugar moiety that differs from the sugar moiety of one or more other nucleosides of the gap. In certain embodiments, the sugar motifs of the two wings are the same as one another (symmetric gapmer). In certain embodiments, the sugar motif of the 5′-wing differs from the sugar motif of the 3′-wing (asymmetric gapmer).

[0393] In certain embodiments, the wings of a gapmer comprise 1-5 nucleosides. In certain embodiments, the wings of a gapmer comprise 2-5 nucleosides. In certain embodiments, the wings of a gapmer comprise 3-5 nucleosides. In certain embodiments, the nucleosides of a gapmer are all modified nucleosides.

[0394] In certain embodiments, the gap of a gapmer comprises 7-12 nucleosides. In certain embodiments, the gap of a gapmer comprises 7-10 nucleosides. In certain embodiments, the gap of a gapmer comprises 8-10 nucleosides. In certain embodiments, the gap of a gapmer comprises 10 nucleosides. In certain embodiment, each nucleoside of the gap of a gapmer is an unmodified 2′-deoxy nucleoside.

[0395] In certain embodiments, the gapmer is a deoxy gapmer. In such embodiments, the nucleosides on the gap side of each wing/gap junction are unmodified 2′-deoxy nucleosides and the nucleosides on the wing sides of each wing/gap junction are modified nucleosides. In certain such embodiments, each nucleoside of the gap is an unmodified 2′-deoxy nucleoside. In certain such embodiments, each nucleoside of each wing is a modified nucleoside.

[0396] In certain embodiments, modified oligonucleotides comprise or consist of a region having a fully modified sugar motif. In such embodiments, each nucleoside of the fully modified region of the modified oligonucleotide comprises a modified sugar moiety. In certain such embodiments, each nucleoside to the entire modified oligonucleotide comprises a modified sugar moiety. In certain embodiments, modified oligonucleotides comprise or consist of a region having a fully modified sugar motif, wherein each nucleoside within the fully modified region comprises the same modified sugar moiety, referred to herein as a uniformly modified sugar motif. In certain embodiments, a fully modified oligonucleotide is a uniformly modified oligonucleotide. In certain embodiments, each nucleoside of a uniformly modified comprises the same 2′-modification.

2. Certain Nucleobase Motifs

[0397] In certain embodiments, oligonucleotides comprise modified and/or unmodified nucleobases arranged along the oligonucleotide or region thereof in a defined pattern or motif. In certain embodiments, each nucleobase is modified. In certain embodiments, none of the nucleobases are modified. In certain embodiments, each purine or each pyrimidine is modified. In certain embodiments, each adenine is modified. In certain embodiments, each guanine is modified. In certain embodiments, each thymine is modified. In certain embodiments, each uracil is modified. In certain embodiments, each cytosine is modified. In certain embodiments, some or all of the cytosine nucleobases in a modified oligonucleotide are 5-methylcytosines.

[0398] In certain embodiments, modified oligonucleotides comprise a block of modified nucleobases. In certain such embodiments, the block is at the 3′-end of the oligonucleotide. In certain embodiments the block is within 3 nucleosides of the 3′-end of the oligonucleotide. In certain embodiments, the block is at the 5′-end of the oligonucleotide. In certain embodiments the block is within 3 nucleosides of the 5′-end of the oligonucleotide.

[0399] In certain embodiments, oligonucleotides having a gapmer motif comprise a nucleoside comprising a modified nucleobase. In certain such embodiments, one nucleoside comprising a modified nucleobase is in the central gap of an oligonucleotide having a gapmer motif. In certain such embodiments, the sugar moiety of said nucleoside is a 2′-deoxyribosyl moiety. In certain embodiments, the modified nucleobase is selected from: a 2-thiopyrimidine and a 5-propynepyrimidine.

3. Certain Internucleoside Linkage Motifs

[0400] In certain embodiments, oligonucleotides comprise modified and/or unmodified internucleoside linkages arranged along the oligonucleotide or region thereof in a defined pattern or motif. In certain embodiments, essentially each internucleoside linking group is a phosphate internucleoside linkage (P═O). In certain embodiments, each internucleoside linking group of a modified oligonucleotide is a phosphorothioate (P═S). In certain embodiments, each internucleoside linking group of a modified oligonucleotide is independently selected from a phosphorothioate and phosphate internucleoside linkage. In certain embodiments, the sugar motif of a modified oligonucleotide is a gapmer and the internucleoside linkages within the gap are all modified. In certain such embodiments, some or all of the internucleoside linkages in the wings are unmodified phosphate linkages. In certain embodiments, the terminal internucleoside linkages are modified.

Certain Oligonucleotides

[0401] Oligonucleotides are characterized by their motifs and overall lengths. In certain embodiments, such parameters are each independent of one another. Thus, unless otherwise indicated, each internucleoside linkage of an oligonucleotide having a gapmer motif may be modified or unmodified and may or may not follow the gapmer modification pattern of the sugar modifications. For example, the internucleoside linkages within the wing regions of a gapmer may be the same or different from one another and may be the same or different from the internucleoside linkages of the gap region. Likewise, such gapmer oligonucleotides may comprise one or more modified nucleobase independent of the gapmer pattern of the sugar modifications. Furthermore, unless otherwise indicated, each internucleoside linkage and each nucleobase of a fully modified oligonucleotide may be modified or unmodified. One of skill in the art will appreciate that such motifs may be combined to create a variety of oligonucleotides. Herein, if a description of an oligonucleotide is silent with respect to one or more parameter, such parameter is not limited. Thus, a modified oligonucleotide described only as having a gapmer motif without further description may have any length, internucleoside linkage motif, and nucleobase motif.

Unless otherwise indicated, all modifications are independent of nucleobase sequence.

[0402] In certain embodiments, oligonucleotides have a nucleobase sequence that is complementary to a second oligonucleotide or a target nucleic acid. In certain such embodiments, a region of an oligonucleotide has a nucleobase sequence that is complementary to a second oligonucleotide or a target nucleic acid. In certain embodiments, the nucleobase sequence of a region or entire length of an oligonucleotide is at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or 100% complementary to the second oligonucleotide or target nucleic acid. In certain embodiments, antisense compounds comprise two oligomeric compounds, wherein the two oligonucleotides of the oligomeric compounds are at least 80%, at least 90%, or 100% complementary to each other. In certain embodiments, one or both oligonucleotides of a double-stranded antisense compound comprise two nucleosides that are not complementary to the other oligonucleotide.

Conjugated Groups and Terminal Groups

[0403] In certain embodiments, antisense compounds and oligomeric compounds comprise conjugate groups and/or terminal groups. In certain such embodiments, oligonucleotides are covalently attached to one or more conjugate group. In certain embodiments, conjugate groups modify one or more properties of the attached oligonucleotide, including but not limited to pharmacodynamics, pharmacokinetics, stability, binding, absorption, cellular distribution, cellular uptake, charge and clearance. In certain embodiments, conjugate groups impart a new property on the attached oligonucleotide, e.g., fluorophores or reporter groups that enable detection of the oligonucleotide. Conjugate groups and/or terminal groups may be added to oligonucleotides having any of the modifications or motifs described above. Thus, for example, an antisense compound or oligomeric compound comprising an oligonucleotide having a gapmer motif may also comprise a conjugate group.

[0404] Conjugate groups include, without limitation, intercalators, reporter molecules, polyamines, polyamides, peptides, carbohydrates, vitamin moieties, polyethylene glycols, thioethers, polyethers, cholesterols, thiocholesterols, cholic acid moieties, folate, lipids, phospholipids, biotin, phenazine, phenanthridine, anthraquinone, adamantane, acridine, fluoresceins, rhodamines, coumarins, fluorophores, and dyes. Certain conjugate groups have been described previously, for example: cholesterol moiety (Letsinger et al., Proc. Natl. Acad. Sci. USA, 1989, 86, 6553-6556), cholic acid (Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1994, 4, 1053-1060), a thioether, e.g., hexyl-S-tritylthiol (Manoharan et al., *Ann. N.Y. Acad. Sci.*, 1992, 660, 306-309; Manoharan et al., *Bioorg. Med. Chem. Let.*, 1993, 3, 2765-2770), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20, 533-538), an aliphatic chain, e.g., do-decan-diol or undecyl residues (Saison-Behmoaras et al., *EMBO J.*, 1991, 10, 1111-1118; Kabanov et al., *FEBS Lett.*, 1990, 259, 327-330; Svinarchuk et al., *Biochimie*, 1993, 75, 49-54), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethyl-ammonium 1,2-di-O-hexadecyl-rac-glycero-3-H-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36, 3651-3654; Shea et al., *Nucl. Acids Res.*, 1990, 18, 3777-3783), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides & Nucleotides*, 1995, 14, 969-973), or adamantane acetic acid (Manoharan et al., *Tetrahedron Lett.*, 1995, 36, 3651-3654), a palmityl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264, 229-237), an octadecylamine or hexylamino-carbonyl-oxycholesterol moiety (Crooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277, 923-937), a tocopherol group (Nishina et al., *Molecular Therapy Nucleic Acids*, 2015, 4, e220; doi: 10.1038/mtna.2014.72 and Nishina et al., *Molecular Therapy*, 2008, 16, 734-740), or a GalNAc cluster (e.g., WO2014/179620).

[0405] In certain embodiments, a conjugate group comprises an active drug substance, for example, aspirin, warfarin, phenylbutazone, ibuprofen, suprofen, fen-bufen, ketoprofen, (S)-(+)-pranoprofen, carprofen, dansylsarcosine, 2,3,5-triiodobenzoic acid, fingolimod, flufenamic acid, folinic acid, a benzothiadiazide, chlorothiazide, a diazepine, indo-methicin, a barbiturate, a cephalosporin, a sulfa drug, an antidiabetic, an antibacterial or an antibiotic.

[0406] Conjugate groups are attached directly or via an optional conjugate linker to a parent compound, such as an oligonucleotide. In certain embodiments, conjugate groups are directly attached to oligonucleotides. In certain embodiments, conjugate groups are indirectly attached to oligonucleotides via conjugate linkers. In certain embodiments, the conjugate linker comprises a chain structure, such as a hydrocarbyl chain, or an oligomer of repeating units such as ethylene glycol or amino acid units. In certain embodiments, conjugate groups comprise a cleavable moiety. In certain embodiments, conjugate groups are attached to oligonucleotides via a cleavable moiety. In certain embodiments, conjugate linkers comprise a cleavable moiety. In certain such embodiments, conjugate linkers are attached to oligonucleotides via a cleavable moiety. In certain embodiments, oligonucleotides comprise a cleavable moiety, wherein the cleavable moiety is a nucleoside is attached to a cleavable internucleoside linkage, such as a phosphate internucleoside linkage. In certain embodiments, a conjugate group comprises a nucleoside or oligonucleotide, wherein the nucleoside or oligonucleotide of the conjugate group is indirectly attached to a parent oligonucleotide.
[0407] In certain embodiments, a conjugate linker comprises one or more groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether, and hydroxylamino. In certain such embodiments, the conjugate linker comprises groups selected from alkyl, amino, oxo, amide and ether groups. In certain embodiments, the conjugate linker comprises groups selected from alkyl and amide groups. In certain embodiments, the conjugate linker comprises groups selected from alkyl and ether groups. In certain embodiments, the conjugate linker comprises at least one phosphorus moiety. In certain embodiments, the conjugate linker comprises at least one phosphate group. In certain embodiments, the conjugate linker includes at least one neutral linking group.
[0408] In certain embodiments, conjugate linkers, including the conjugate linkers described above, are bifunctional linking moieties, e.g., those known in the art to be useful for attaching conjugate groups to parent compounds, such as the oligonucleotides provided herein. In general, a bifunctional linking moiety comprises at least two functional groups. One of the functional groups is selected to bind to a particular site on a parent compound and the other is selected to bind to a conjugate group. Examples of functional groups used in a bifunctional linking moiety include but are not limited to electrophiles for reacting with nucleophilic groups and nucleophiles for reacting with electrophilic groups. In certain embodiments, bifunctional linking moieties comprise one or more groups selected from amino, hydroxyl, carboxylic acid, thiol, alkyl, alkenyl, and alkynyl.
[0409] Examples of conjugate linkers include but are not limited to pyrrolidine, 8-amino-3,6-dioxaoctanoic acid (ADO), succinimidyl 4-(N-maleimidomethyl)cyclohexane-1-carboxylate (SMCC) and 6-aminohexanoic acid (AHEX or AHA). Other conjugate linkers include but are not limited to substituted or unsubstituted C.sub.1-C.sub.10 alkyl, substituted or unsubstituted C.sub.2-C.sub.10 alkenyl or substituted or unsubstituted C.sub.2-C.sub.10 alkynyl, wherein a nonlimiting list of preferred substituent groups includes hydroxyl, amino, alkoxy, carboxy, benzyl, phenyl, nitro, thiol, thioalkoxy, halogen, alkyl, aryl, alkenyl and alkynyl.
[0410] In certain embodiments, a cleavable moiety is a cleavable bond. In certain embodiments, a cleavable moiety comprises a cleavable bond. In certain embodiments, a cleavable moiety is a group of atoms comprising at least one cleavable bond. In certain embodiments, a cleavable moiety comprises a group of atoms having one, two, three, four, or more than four cleavable bonds. In certain embodiments, a cleavable moiety is selectively cleaved inside a cell or subcellular compartment, such as a lysosome. In certain embodiments, a cleavable moiety is selectively cleaved by endogenous enzymes, such as nucleases.
[0411] In certain embodiments, a cleavable bond is selected from among: an amide, an ester, an ether, one or both esters of a phosphodiester, a phosphate ester, a carbamate, or a disulfide. In certain embodiments, a cleavable bond is one or both of the esters of a phosphodiester. In certain embodiments, a cleavable moiety comprises a phosphate or phosphodiester. In certain

embodiments, the cleavable moiety is a phosphate linkage between an oligonucleotide and a conjugate linker or conjugate group.

[0412] In certain embodiments, a cleavable moiety is a nucleoside. In certain such embodiments, the unmodified or modified nucleoside comprises an optionally protected heterocyclic base selected from a purine, substituted purine, pyrimidine or substituted pyrimidine. In certain embodiments, a cleavable moiety is a nucleoside selected from uracil, thymine, cytosine, 4-N-benzoylcytosine, 5-methylcytosine, 4-N-benzoyl-5-methylcytosine, adenine, 6-N-benzoyladenine, guanine and 2-N-isobutyrylguanine. In certain embodiments, a cleavable moiety is 2′-deoxy nucleoside that is attached to either the 3′ or 5′-terminal nucleoside of an oligonucleotide by a phosphate internucleoside linkage and covalently attached to the conjugate linker or conjugate group by a phosphate or phosphorothioate linkage. In certain such embodiments, the cleavable moiety is 2′-deoxyadenosine.

[0413] Conjugate groups may be attached to either or both ends of an oligonucleotide and/or at any internal position. In certain embodiments, conjugate groups are attached to the 2′-position of a nucleoside of a modified oligonucleotide. In certain embodiments, conjugate groups that are attached to either or both ends of an oligonucleotide are terminal groups. In certain such embodiments, conjugate groups or terminal groups are attached at the 3′ and/or 5′-end of oligonucleotides. In certain such embodiments, conjugate groups (or terminal groups) are attached at the 3′-end of oligonucleotides. In certain embodiments, conjugate groups are attached near the 3′-end of oligonucleotides. In certain embodiments, conjugate groups (or terminal groups) are attached at the 5′-end of oligonucleotides. In certain embodiments, conjugate groups are attached near the 5′-end of oligonucleotides.

[0414] Examples of terminal groups include but are not limited to conjugate groups, capping groups, phosphate moieties, protecting groups, modified or unmodified nucleosides, and two or more nucleosides that are independently modified or unmodified.

[0415] In certain embodiments, a conjugate group is a cell-targeting moiety. In certain embodiments, a conjugate group, optional conjugate linker, and optional cleavable moiety have the general formula:

##STR00011##

wherein n is from 1 to about 3, m is 0 when n is 1, m is 1 when n is 2 or greater, j is 1 or 0, and k is 1 or 0.

[0416] In certain embodiments, n is 1, j is 1 and k is 0. In certain embodiments, n is 1, j is 0 and k is 1. In certain embodiments, n is 1, j is 1 and k is 1. In certain embodiments, n is 2, j is 1 and k is 0. In certain embodiments, n is 2, j is 0 and k is 1. In certain embodiments, n is 2, j is 1 and k is 1. In certain embodiments, n is 3, j is 1 and k is 0. In certain embodiments, n is 3, j is 0 and k is 1. In certain embodiments, n is 3, j is 1 and k is 1.

[0417] In certain embodiments, conjugate groups comprise cell-targeting moieties that have at least one tethered ligand. In certain embodiments, cell-targeting moieties comprise two tethered ligands covalently attached to a branching group. In certain embodiments, cell-targeting moieties comprise three tethered ligands covalently attached to a branching group.

[0418] In certain embodiments, the cell-targeting moiety comprises a branching group comprising one or more groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether and hydroxylamino groups. In certain embodiments, the branching group comprises a branched aliphatic group comprising groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether and hydroxylamino groups. In certain such embodiments, the branched aliphatic group comprises groups selected from alkyl, amino, oxo, amide and ether groups. In certain such embodiments, the branched aliphatic group comprises groups selected from alkyl, amino and ether groups. In certain such embodiments, the branched aliphatic group comprises groups selected from alkyl and ether groups. In certain embodiments, the branching group comprises a mono or polycyclic ring system.

[0419] In certain embodiments, each tether of a cell-targeting moiety comprises one or more groups selected from alkyl, substituted alkyl, ether, thioether, disulfide, amino, oxo, amide, phosphodiester, and polyethylene glycol, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, ether, thioether, disulfide, amino, oxo, amide, and polyethylene glycol, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, phosphodiester, ether, amino, oxo, and amide, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, ether, amino, oxo, and amid, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, amino, and oxo, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl and oxo, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl and phosphodiester, in any combination. In certain embodiments, each tether comprises at least one phosphorus linking group or neutral linking group. In certain embodiments, each tether comprises a chain from about 6 to about 20 atoms in length. In certain embodiments, each tether comprises a chain from about 10 to about 18 atoms in length. In certain embodiments, each tether comprises about 10 atoms in chain length.

[0420] In certain embodiments, each ligand of a cell-targeting moiety has an affinity for at least one type of receptor on a target cell. In certain embodiments, each ligand has an affinity for at least one type of receptor on the surface of a mammalian liver cell. In certain embodiments, each ligand has an affinity for the hepatic asialoglycoprotein receptor (ASGP-R). In certain embodiments, each ligand is a carbohydrate. In certain embodiments, each ligand is, independently selected from galactose, N-acetyl galactoseamine (GalNAc), mannose, glucose, glucosamine and fucose. In certain embodiments, each ligand is N-acetyl galactoseamine (GalNAc). In certain embodiments, the cell-targeting moiety comprises 3 GalNAc ligands. In certain embodiments, the cell-targeting moiety comprises 2 GalNAc ligands. In certain embodiments, the cell-targeting moiety comprises 1 GalNAc ligand.

[0421] In certain embodiments, each ligand of a cell-targeting moiety is a carbohydrate, carbohydrate derivative, modified carbohydrate, polysaccharide, modified polysaccharide, or polysaccharide derivative. In certain such embodiments, the conjugate group comprises a carbohydrate cluster (see, e.g., Maier et al., “Synthesis of Antisense Oligonucleotides Conjugated to a Multivalent Carbohydrate Cluster for Cellular Targeting,” *Bioconjugate Chemistry,* 2003, 14, 18-29, or Rensen et al., “Design and Synthesis of Novel N-Acetylgalactosamine-Terminated Glycolipids for Targeting of Lipoproteins to the Hepatic Asiaglycoprotein Receptor,” *J. Med. Chem.* 2004, 47, 5798-5808, which are incorporated herein by reference in their entirety). In certain such embodiments, each ligand is an amino sugar or a thio sugar. For example, amino sugars may be selected from any number of compounds known in the art, such as sialic acid, α-D-galactosamine, β-muramic acid, 2-deoxy-2-methylamino-L-glucopyranose, 4,6-dideoxy-4-formamido-2,3-di-O-methyl-D-mannopyranose, 2-deoxy-2-sulfoamino-D-glucopyranose and N-sulfo-D-glucosamine, and N-glycoloyl-α-neuraminic acid. For example, thio sugars may be selected from 5-Thio-β-D-glucopyranose, methyl 2,3,4-tri-O-acetyl-1-thio-6-O-trityl-α-D-glucopyranoside, 4-thio-β-D-galactopyranose, and ethyl 3,4,6,7-tetra-O-acetyl-2-deoxy-1,5-dithio-α-D-gluco-heptopyranoside.

[0422] In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:

##STR00012##

[0423] In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:

##STR00013##

[0424] In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:

##STR00014##

[0425] In certain embodiments, antisense compounds and oligomeric compounds comprise a conjugate group and conjugate linker described herein as "LICA-1". LICA-1 has the formula:

##STR00015##

[0426] In certain embodiments, antisense compounds and oligomeric compounds comprising LICA-1 have the formula:

##STR00016## [0427] wherein oligo is an oligonucleotide.

[0428] Representative United States patents, United States patent application publications, international patent application publications, and other publications that teach the preparation of certain of the above noted conjugate groups, oligomeric compounds and antisense compounds comprising conjugate groups, tethers, conjugate linkers, branching groups, ligands, cleavable moieties as well as other modifications include without limitation, U.S. Pat. Nos. 5,994,517, 6,300,319, 6,660,720, 6,906,182, 7,262,177, 7,491,805, 8,106,022, 7,723,509, US 2006/0148740, US 2011/0123520, WO 2013/033230 and WO 2012/037254, Biessen et al., *J. Med. Chem.* 1995, 38, 1846-1852, Lee et al., *Bioorganic* & *Medicinal Chemistry* 2011,19, 2494-2500, Rensen et al., *J. Biol. Chem.* 2001, 276, 37577-37584, Rensen et al., *J. Med. Chem.* 2004, 47, 5798-5808, Sliedregt et al., *J. Med. Chem.* 1999, 42, 609-618, and Valentijn et al., *Tetrahedron,* 1997, 53, 759-770, each of which is incorporated by reference herein in its entirety.

[0429] In certain embodiments, antisense compounds and oligomeric compounds comprise modified oligonucleotides comprising a gapmer or fully modified motif and a conjugate group comprising at least one, two, or three GalNAc ligands. In certain embodiments antisense compounds and oligomeric compounds comprise a conjugate group found in any of the following references: Lee, Carbohydr Res, 1978, 67, 509-514; Connolly et al., *J Biol Chem,* 1982, 257, 939-945; Pavia et al., *Int J Pep Protein Res,* 1983, 22, 539-548; Lee et al., *Biochem,* 1984, 23, 4255-4261; Lee et al., *Glycoconjugate J,* 1987, 4, 317-328; Toyokuni et al., *Tetrahedron Lett,* 1990, 31, 2673-2676; Biessen et al., *J Med Chem,* 1995, 38, 1538-1546; Valentijn et al., *Tetrahedron,* 1997, 53, 759-770; Kim et al., *Tetrahedron Lett,* 1997, 38, 3487-3490; Lee et al., *Bioconjug Chem,* 1997, 8, 762-765; Kato et al., *Glycobiol,* 2001, 11, 821-829; Rensen et al., *J Biol Chem,* 2001, 276, 37577-37584; Lee et al., *Methods Enzymol,* 2003, 362, 38-43; Westerlind et al., *Glycoconj J,* 2004, 21, 227-241; Lee et al., *Bioorg Med Chem Lett,* 2006, 16 (19), 5132-5135; Maierhofer et al., *Bioorg Med Chem,* 2007, 15, 7661-7676; Khorev et al., *Bioorg Med Chem,* 2008, 16, 5216-5231; Lee et al., *Bioorg Med Chem,* 2011, 19, 2494-2500; Kornilova et al., Analyt Biochem, 2012, 425, 43-46; Pujol et al., *Angew Chemie Int Ed Engl,* 2012, 51, 7445-7448; Biessen et al., *J Med Chem,* 1995, 38, 1846-1852; Sliedregt et al., *J Med Chem,* 1999, 42, 609-618; Rensen et al., *J Med Chem,* 2004, 47, 5798-5808; Rensen et al., *Arterioscler Thromb Vasc Biol,* 2006, 26, 169-175; van Rossenberg et al., *Gene Ther,* 2004, 11, 457-464; Sato et al., *J Am Chem Soc,* 2004, 126, 14013-14022; Lee et al., *J Org Chem,* 2012, 77, 7564-7571; Biessen et al., *FASEB J,* 2000, 14, 1784-1792; Rajur et al., *Bioconjug Chem,* 1997, 8, 935-940; Duff et al., *Methods Enzymol,* 2000, 313, 297-321; Maier et al., *Bioconjug Chem,* 2003, 14, 18-29; Jayaprakash et al., *Org Lett,* 2010, 12, 5410-5413; Manoharan, *Antisense Nucleic Acid Drug Dev,* 2002, 12, 103-128; Merwin et al., *Bioconjug Chem,* 1994, 5, 612-620; Tomiya et al., *Bioorg Med Chem,* 2013, 21, 5275-5281; International applications WO1998/013381; WO2011/038356; WO1997/046098; WO2008/098788; WO2004/101619; WO2012/037254; WO2011/120053; WO2011/100131; WO2011/163121; WO2012/177947; WO2013/033230; WO2013/075035; WO2012/083185; WO2012/083046; WO2009/082607; WO2009/134487; WO2010/144740; WO2010/148013; WO1997/020563; WO2010/088537; WO2002/043771; WO2010/129709; WO2012/068187; WO2009/126933; WO2004/024757; WO2010/054406; WO2012/089352; WO2012/089602; WO2013/166121; WO2013/165816; U.S. Pat. Nos. 4,751,219; 8,552,163; 6,908,903; 7,262,177; 5,994,517; 6,300,319; 8,106,022; 7,491,805; 7,491,805; 7,582,744; 8,137,695; 6,383,812; 6,525,031;

6,660,720; 7,723,509; 8,541,548; 8,344,125; 8,313,772; 8,349,308; 8,450,467; 8,501,930; 8,158,601; 7,262,177; 6,906,182; 6,620,916; 8,435,491; 8,404,862; 7,851,615; Published U.S. patent Application Publications US2011/0097264; US2011/0097265; US2013/0004427; US2005/0164235; US2006/0148740; US2008/0281044; US2010/0240730; US2003/0119724; US2006/0183886; US2008/0206869; US2011/0269814; US2009/0286973; US2011/0207799; US2012/0136042; US2012/0165393; US2008/0281041; US2009/0203135; US2012/0035115; US2012/0095075; US2012/0101148; US2012/0128760; US2012/0157509; US2012/0230938; US2013/0109817; US2013/0121954; US2013/0178512; US2013/0236968; US2011/0123520; US2003/0077829; US2008/0108801; and US2009/0203132; each of which is incorporated by reference in its entirety.

[0430] In certain embodiments, a modified oligonucleotide targeting DGAT2 described herein further comprises a GalNAc conjugate group. In certain embodiments, the GalNAc conjugate group is 5′-Trishexylamino-(THA)-C6 GalNAc.sub.3. In certain embodiments, the 5′-Trishexylamino-(THA)-C6 GalNAc.sub.3 conjugate has the formula

##STR00017##

[0431] In certain embodiments, the modified oligonucleotide is linked to the 5′-Trishexylamino-(THA)-C6 GalNAc.sub.3 conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

Compositions and Methods for Formulating Pharmaceutical Compositions

[0432] In certain embodiments, the present invention provides pharmaceutical compositions comprising one or more antisense compound or a salt thereof. In certain such embodiments, the pharmaceutical composition comprises a suitable pharmaceutically acceptable diluent or carrier. In certain embodiments, a pharmaceutical composition comprises a sterile saline solution and one or more antisense compound. In certain embodiments, such pharmaceutical composition consists of a sterile saline solution and one or more antisense compound. In certain embodiments, the sterile saline is pharmaceutical grade saline. In certain embodiments, a pharmaceutical composition comprises one or more antisense compound and sterile water. In certain embodiments, a pharmaceutical composition consists of one antisense compound and sterile water. In certain embodiments, the sterile water is pharmaceutical grade water. In certain embodiments, a pharmaceutical composition comprises one or more antisense compound and phosphate-buffered saline (PBS). In certain embodiments, a pharmaceutical composition consists of one or more antisense compound and sterile PBS. In certain embodiments, the sterile PBS is pharmaceutical grade PBS. Compositions and methods for the formulation of pharmaceutical compositions are dependent upon a number of criteria, including, but not limited to, route of administration, extent of disease, or dose to be administered.

[0433] An antisense compound targeted to DGAT2 nucleic acid can be utilized in pharmaceutical compositions by combining the antisense compound with a suitable pharmaceutically acceptable diluent or carrier. In certain embodiments, a pharmaceutically acceptable diluent is water, such as sterile water suitable for injection. Accordingly, in one embodiment, employed in the methods described herein is a pharmaceutical composition comprising an antisense compound targeted to DGAT2 nucleic acid and a pharmaceutically acceptable diluent. In certain embodiments, the pharmaceutically acceptable diluent is water. In certain embodiments, the antisense compound is an antisense oligonucleotide provided herein.

[0434] Pharmaceutical compositions comprising antisense compounds encompass any pharmaceutically acceptable salts, esters, or salts of such esters, or any other oligonucleotide which, upon administration to an animal, including a human, is capable of providing (directly or indirectly) the biologically active metabolite or residue thereof. Accordingly, for example, the disclosure is also drawn to pharmaceutically acceptable salts of antisense compounds, prodrugs, pharmaceutically acceptable salts of such prodrugs, and other bioequivalents. Suitable pharmaceutically acceptable salts include, but are not limited to, sodium and potassium salts.

[0435] A prodrug can include the incorporation of additional nucleosides at one or both ends of an antisense compound which are cleaved by endogenous nucleases within the body, to form the active antisense compound.

[0436] In certain embodiments, the compounds or compositions further comprise a pharmaceutically acceptable carrier or diluent.

Certain Compounds

[0437] In vitro screens in human HepG2 cells were performed with about 5,000 antisense oligonucleotides that exhibited 100% complementarity to a human DGAT2 gene sequence. The new compounds were compared with previously designed compounds, which have been previously determined to be some of the most potent antisense compounds in vitro (see, e.g. published PCT application WO 2005/019418). From these in vitro screens, several antisense oligonucleotides that exhibited the greatest potency in reducing the expression of human DGAT2 mRNA were selected for in vivo tolerability assays.

[0438] The selected oligonucleotides were tested for tolerability in a CD1 mouse model, as well as a Sprague-Dawley rat model. In these models, body weights and organ weights, liver function markers (such as alanine transaminase, aspartate transaminase, and bilirubin), and kidney function markers (such as BUN and creatinine) were measured.

[0439] Final evaluation of all studies (Examples 1-15) led to the selection of eight oligonucleotides having a nucleobase sequence of SEQ ID NO: 1371 (ISIS 484085), SEQ ID NO: 1415 (ISIS 484129), SEQ ID NO: 4680 (ISIS484137), SEQ ID NO: 1844 (ISIS 495576), SEQ ID NO: 2959 (ISIS 501861), SEQ ID NO: 3292 (ISIS 502194), SEQ ID NO: 4198 (ISIS 525443), and SEQ ID NO: 4373 (ISIS 525612). The compounds are complementary to the regions 23242-23261, 26630-26649, 26778-26797, 15251-15270, 28026-28045, 35436-35455, 10820-10836, and 23246-23262. In certain embodiments, the compounds targeting the listed regions, as further described herein, comprise a modified oligonucleotide having some nucleobase portion of the sequence recited in the SEQ ID Nos, as further described herein. In certain embodiments, the compounds targeting the listed regions or having a nucleobase portion of a sequence recited in the listed SEQ ID Nos can be of various lengths, as further described herein, and can have one of various motifs, as further described herein. In certain embodiments, a compound targeting a region or having a nucleobase portion of a sequence in the listed SEQ ID Nos has the specific length and motif, as indicated by the ISIS Nos: 484085, 484129, 484137, 495576, 501861, 502194, 525443, and 525612.

[0440] These eight compounds were tested for activity, pharmacokinetic profile and tolerability in cynomolgus monkeys (Example 16). Specifically, ISIS 484137 was found to be potent and the most tolerable in this monkey study. Further evaluation of ISIS 484137 in a separate monkey study (Example 17) confirmed this finding.

[0441] The nucleotide sequence of ISIS 484137 is fully homologous to the rhesus monkey DGAT2 mRNA transcript. The cynomolgus monkey is regarded as a relevant preclinical safety model system for oligonucleotide therapeutics and the demonstrated pharmacologic activity of ISIS 484137 in this species makes it an appropriate species for safety assessment. General toxicology studies were conducted with ISIS 484137 in monkeys for 13 weeks of treatment and it was well tolerated with no overt adverse effects and no changes in routine laboratory parameters. ISIS 484137 reduced the expression of hepatic DGAT2 by about 70%.

[0442] The findings observed in mice and monkey toxicology studies following 13-weeks of ISIS 484137 treatment were, in general, non-specific class effects that are typical for 2′-MOE ASOs. There was no drug-related mortality or changes in clinical signs up to the highest doses tested (100 mg/kg in mice and 40 mg/kg in monkeys). There were no toxicologically significant findings at doses up to 12 mg/kg/wk for 13 weeks in the mouse and monkey studies, and therefore there is sufficient therapeutic margin to support the safe clinical use of ISIS 484137 at the proposed clinical doses and regimen.

[0443] The pharmacokinetic results confirm continuous and dose-dependent exposure to ISIS

484137 in the 13-week mouse and monkey studies. Pharmacokinetics observed in monkeys for antisense oligonucleotides typically well predict the observed plasma (and expected tissue) exposure levels in humans on the basis of mg/kg equivalent doses (Geary et al. 2003; 31:1419-1428; Yu et al. Drug Metab Dispos 2007; 35:460-468).

[0444] Thus, reduction of DGAT2 expression with ISIS 484137 provides a mechanism to reduce hepatic steatosis, thereby potentially attenuating subsequent inflammation and fibrosis. This mechanism of action could offer an attractive treatment option for patients who have significant hepatic steatosis associated with NAFLD and NASH.

EXAMPLES

[0445] The Examples below describe the screening process to identify the lead compounds targeted to DGAT2. Out of about 5,000 antisense oligonucleotides screened, ISIS 484085, ISIS 484129, ISIS 484137, ISIS 495576, ISIS 501861, ISIS 502194, ISIS 525443, and ISIS 525612 emerged as the top lead compounds. In particular, ISIS 484137 exhibited the best combination of properties in terms of potency and tolerability for DGAT2 out of about 5,000 antisense oligonucleotides.

Non-Limiting Disclosure and Incorporation by Reference

[0446] Although the sequence listing accompanying this filing identifies each sequence as either “RNA” or “DNA” as required, in reality, those sequences may be modified with any combination of chemical modifications. One of skill in the art will readily appreciate that such designation as “RNA” or “DNA” to describe modified oligonucleotides is, in certain instances, arbitrary. For example, an oligonucleotide comprising a nucleoside comprising a 2′-OH sugar moiety and a thymine base could be described as a DNA having a modified sugar (2′-OH for the natural 2′-H of DNA) or as an RNA having a modified base (thymine(methylated uracil) for natural uracil of RNA).

[0447] Accordingly, nucleic acid sequences provided herein, including, but not limited to those in the sequence listing, are intended to encompass nucleic acids containing any combination of natural or modified RNA and/or DNA, including, but not limited to such nucleic acids having modified nucleobases. By way of further example and without limitation, an oligonucleotide having the nucleobase sequence “ATCGATCG” encompasses any oligonucleotides having such nucleobase sequence, whether modified or unmodified, including, but not limited to, such compounds comprising RNA bases, such as those having sequence “AUCGAUCG” and those having some DNA bases and some RNA bases such as “AUCGATCG”.

[0448] While certain compounds, compositions and methods described herein have been described with specificity in accordance with certain embodiments, the following examples serve only to illustrate the compounds described herein and are not intended to limit the same. Each of the references recited in the present application is incorporated herein by reference in its entirety.

Example 1: Antisense Inhibition of Human Diacylglycerol Acyltransferase 2 in HepG2 Cells by MOE Gapmers

[0449] Antisense oligonucleotides were designed targeting a diacylglycerol acyltransferase 2 (DGAT2) nucleic acid and were tested for their effects on DGAT2 mRNA in vitro. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. The results for each experiment are presented in separate tables shown below. Cultured HepG2 cells at a density of 10,000 cells per well were transfected using Lipofectin reagent with 120 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2367 (forward sequence GGCCTCCCGGAGACTGA, designated herein as SEQ ID NO: 4; reverse sequence AAGTGATTTGCAGCTGGTTCCT, designated herein as SEQ ID NO: 5; probe sequence AGGTGAACTGAGCCAGCCTTCGGG, designated herein as SEQ ID NO: 6) was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells.

[0450] ISIS oligonucleotides from an earlier published application, WO 2005/019418, were also included in this assay. These ISIS oligonucleotides are ISIS 217312-217322, ISIS 217324, ISIS 217325, ISIS 217328, ISIS 217333, ISIS 217336-217339, ISIS 217341-217343, ISIS 217346-217348, ISIS 217353-217355, ISIS 334177, ISIS 366710, ISIS 366714, ISIS 366722, ISIS 366728, ISIS 366730, ISIS 366741, ISIS 366746, ISIS 369220, ISIS 369221, ISIS 369255, ISIS 370727, ISIS 370747, ISIS 370784.

[0451] The newly designed chimeric antisense oligonucleotides in the Tables below were designed as 5-10-5 MOE gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment comprises of ten 2′-deoxynucleosides and is flanked by wing segments on the 5′ direction and the 3′ direction comprising five nucleosides each. Each nucleoside in the 5′ wing segment and each nucleoside in the 3′ wing segment has a 2′-MOE modification. The internucleoside linkages throughout each gapmer are phosphorothioate (P═S) linkages. All cytosine residues throughout each gapmer are 5-methylcytosines. “Start site” indicates the 5′-most nucleoside to which the gapmer is targeted in the human gene sequence. “Stop site” indicates the 3′-most nucleoside to which the gapmer is targeted in the human gene sequence. Each gapmer listed in the Tables below is targeted to either the human DGAT2 mRNA, designated herein as SEQ ID NO: 1 (Ref Seq No. NM_032564.3) or the human DGAT2 genomic sequence, designated herein as SEQ ID NO: 2 (Ref Seq No. NT 033927.5 truncated from nucleotides 5669186 to 5712008). ‘n/a’ indicates that the antisense oligonucleotide does not target that particular gene sequence with 100% complementarity.

TABLE-US-00001 TABLE 1 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 381726 278 297 CGCAGGACCC 75 9899 9918 38 CGGAGTAGGC 217312 668 687 TGGATGGGAA 40 31600 31619 16 AGTAGTCTCG 411887 669 688 CTGGATGGGA 38 31601 31620 50 AAGTAGTCTC 411888 670 689 GCTGGATGGG 22 n/a n/a 51 AAAGTAGTCT 411889 671 690 AGCTGGATGG 31 n/a n/a 52 GAAAGTAGTC 411890 672 691 CAGCTGGATG 21 n/a n/a 53 GGAAAGTAGT 217313 673 692 CCAGCTGGAT 24 n/a n/a 17 GGGAAAGTAG 411891 674 693 ACCAGCTGGA 21 n/a n/a 54 TGGGAAAGTA 380109 675 694 CACCAGCTGG 31 n/a n/a 35 ATGGGAAAGT 411892 676 695 TCACCAGCTG 22 n/a n/a 55 GATGGGAAAG 411893 677 696 TTCACCAGCT 26 n/a n/a 56 GGATGGGAAA 217314 678 697 CTTCACCAGC 36 n/a n/a 18 TGGATGGGAA 369219 679 698 TCTTCACCAG 50 n/a n/a 32 CTGGATGGGA 411894 680 699 GTCTTCACCA 42 n/a n/a 57 GCTGGATGGG 411895 681 700 TGTCTTCACC 55 n/a n/a 58 AGCTGGATGG 411896 682 701 GTGTCTTCAC 49 n/a n/a 59 CAGCTGGATG 217315 683 702 TGTGTCTTCA 55 n/a n/a 19 CCAGCTGGAT 380110 684 703 GTGTGTCTTC 53 n/a n/a 36 ACCAGCTGGA 411897 685 704 TGTGTGTCTT 55 n/a n/a 60 CACCAGCTGG 411873 686 705 TTGTGTGTCT 63 37214 37233 46 TCACCAGCTG 411898 687 706 GTTGTGTGTC 54 37215 37234 61 TTCACCAGCT 217316 688 707 GGTTGTGTGT 64 37216 37235 20 CTTCACCAGC 381727 689 708 AGGTTGTGTG 59 37217 37236 39 TCTTCACCAG 411899 690 709 CAGGTTGTGT 68 37218 37237 62 GTCTTCACCA 411874 691 710 GCAGGTTGTG 65 37219 37238 47 TGTCTTCACC 411900 692 711 AGCAGGTTGT 55 37220 37239 63 GTGTCTTCAC 217317 693 712 CAGCAGGTTG 66 37221 37240 21 TGTGTCTTCA 381728 694 713 TCAGCAGGTT 56 37222 37241 40 GTGTGTCTTC 411901 695 714 GTCAGCAGGT 64 37223 37242 64 TGTGTGTCTT 411875 696 715 GGTCAGCAGG 45 37224 37243 48 TTGTGTGTCT 411902 697 716 TGGTCAGCAG 55 37225 37244 65 GTTGTGTGTC 217318 698 717 GTGGTCAGCA 44 37226 37245 22 GGTTGTGTGT 369220 699 718 GGTGGTCAGC 29 37227 37246 33 AGGTTGTGTG 411903 700 719 TGGTGGTCAG 51 37228 37247 66 CAGGTTGTGT 411904 701 720 CTGGTGGTCA 57

37229 37248 67 GCAGGTTGTG 411905 702 721 CCTGGTGGTC 64 37230 37249 68 AGCAGGTTGT 217319 703 722 TCCTGGTGGT 58 37231 37250 23 CAGCAGGTTG 381729 704 723 TTCCTGGTGG 61 37232 37251 41 TCAGCAGGTT 411906 705 724 GTTCCTGGTG 63 37233 37252 69 GTCAGCAGGT 411907 706 725 AGTTCCTGGT 58 37234 37253 70 GGTCAGCAGG 411908 707 726 TAGTTCCTGG 48 37235 37254 71 TGGTCAGCAG 217320 708 727 ATAGTTCCTG 46 37236 37255 24 GTGGTCAGCA 381730 709 728 TATAGTTCCT 56 37237 37256 42 GGTGGTCAGC 411909 710 729 ATATAGTTCC 53 37238 37257 72 TGGTGGTCAG 411876 711 730 GATATAGTTC 57 37239 37258 49 CTGGTGGTCA 411910 712 731 AGATATAGTT 56 37240 37259 73 CCTGGTGGTC 217321 713 732 AAGATATAGT 41 37241 37260 25 TCCTGGTGGT 381731 714 733 AAAGATATAG 54 37242 37261 43 TTCCTGGTGG 411911 715 734 CAAAGATATA 61 37243 37262 74 GTTCCTGGTG 411912 716 735 CCAAAGATAT 66 37244 37263 75 AGTTCCTGGT 411913 717 736 TCCAAAGATA 63 37245 37264 76 TAGTTCCTGG 217322 718 737 ATCCAAAGAT 39 37246 37265 26 ATAGTTCCTG 369221 719 738 TATCCAAAGA 13 37247 37266 34 TATAGTTCCT 411914 720 739 GTATCCAAAG 8 37248 37267 77 ATATAGTTCC 411915 721 740 GGTATCCAAA 43 37249 37268 78 GATATAGTTC 411916 722 741 TGGTATCCAA 42 37250 37269 79 AGATATAGTT 217324 752 771 AAGGCACCCA 46 37280 37299 27 GGCCCATGAT 381732 753 772 GAAGGCACCC 38 37281 37300 44 AGGCCCATGA 380114 754 773 AGAAGGCACC 38 37282 37301 37 CAGGCCCATG 411917 755 774 CAGAAGGCAC 33 37283 37302 80 CCAGGCCCAT 217325 875 894 CCTCCAGACA 29 37403 37422 28 TCAGGTACTC 217333 992 1011 TTGCCAGGCA 37 38145 38164 29 TGGAGCTCAG 381733 993 1012 CTTGCCAGGC 51 38146 38165 45 ATGGAGCTCA 411918 994 1013 TCTTGCCAGG 47 38147 38166 81 CATGGAGCTC 411919 995 1014 TTCTTGCCAG 54 38148 38167 82 GCATGGAGCT 217336 1139 1158 TGGACCCATC 52 39186 39205 30 GGCCCCAGGA 411920 1140 1159 CTGGACCCAT 41 39187 39206 83 CGGCCCCAGG 411921 1141 1160 TCTGGACCCA 20 39188 39207 84 TCGGCCCCAG 411922 1142 1161 TTCTGGACCC 27 39189 39208 85 ATCGGCCCCA 411923 1143 1162 CTTCTGGACC 30 39190 39209 86 CATCGGCCCC 217337 1144 1163 TCTTCTGGAC 37 39191 39210 31 CCATCGGCCC 411924 1145 1164 TTCTTCTGGA 44 39192 39211 87 CCCATCGGCC 411925 1146 1165 CTTCTTCTGG 27 39193 39212 88 ACCCATCGGC 411926 1147 1166 ACTTCTTCTG 28 39194 39213 89 GACCCATCGG 411927 1148 1167 AACTTCTTCT 28 39195 39214 90 GGACCCATCG

TABLE-US-00002 TABLE 2 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 217338 1149 1168 GAACTTCTTC 19 39196 39215 91 TGGACCCATC 411928 1150 1169 GGAACTTCTT 36 39197 39216 104 CTGGACCCAT 411929 1151 1170 TGGAACTTCT 43 39198 39217 105 TCTGGACCCA 411930 1152 1171 CTGGAACTTC 25 39199 39218 106 TTCTGGACCC 411931 1153 1172 TCTGGAACTT 43 39200 39219 107 CTTCTGGACC 217339 1154 1173 TTCTGGAACT 37 39201 39220 92 TCTTCTGGAC 217341 1226 1245 GGCACCAGCC 30 39273 39292 93 CCCAGGTGTC 411932 1227 1246 GGGCACCAGC 34 39274 39293 108 CCCCAGGTGT 411933 1228 1247 AGGGCACCAG 29 39275 39294 109 CCCCCAGGTG 411934 1229 1248 TAGGGCACCA 32 39276 39295 110 GCCCCCAGGT 411935 1230 1249 GTAGGGCACC 23 39277 39296 111 AGCCCCCAGG 217342 1231 1250 AGTAGGGCAC 0 39278 39297 94 CAGCCCCCAG 411936 1232 1251 GAGTAGGGCA 0 39279 39298 112 CCAGCCCCCA 411937 1233 1252 GGAGTAGGGC 17 39280 39299 113 ACCAGCCCCC 411938 1234 1253 TGGAGTAGGG 28 39281 39300 114 CACCAGCCCC 411939 1235 1254 TTGGAGTAGG 16 39282 39301 115 GCACCAGCCC 217343 1236 1255 CTTGGAGTAG 35 39283 39302 95 GGCACCAGCC 411940 1237 1256 GCTTGGAGTA 45

39284 39303 116 GGGCACCAGC 411941 1238 1257 GGCTTGGAGT 56 39285 39304 117 AGGGCACCAG 366741 1245 1264 GGTGATGGGC 36 39292 39311 102 TTGGAGTAGG 411942 1246 1265 TGGTGATGGG 28 39293 39312 118 CTTGGAGTAG 411943 1247 1266 GTGGTGATGG 23 39294 39313 119 GCTTGGAGTA 217346 1338 1357 CAGGGCCTCC 37 41309 41328 96 ATGTACATGG 369255 1339 1358 CCAGGGCCTC 38 41310 41329 103 CATGTACATG 411944 1340 1359 ACCAGGGCCT 52 41311 41330 120 CCATGTACAT 411945 1341 1360 CACCAGGGCC 53 41312 41331 121 TCCATGTACA 411946 1342 1361 TCACCAGGGC 37 41313 41332 122 CTCCATGTAC 217347 1343 1362 TTCACCAGGG 20 41314 41333 97 CCTCCATGTA 411947 1344 1363 CTTCACCAGG 46 41315 41334 123 GCCTCCATGT 411948 1345 1364 GCTTCACCAG 31 41316 41335 124 GGCCTCCATG 411949 1346 1365 AGCTTCACCA 42 41317 41336 125 GGGCCTCCAT 411950 1347 1366 GAGCTTCACC 52 41318 41337 126 AGGGCCTCCA 217348 1348 1367 AGAGCTTCAC 46 41319 41338 98 CAGGGCCTCC 411951 1349 1368 AAGAGCTTCA 44 41320 41339 127 CCAGGGCCTC 217353 1498 1517 AACCCACAGA 4 41469 41488 99 CACCCATGAC 411952 1499 1518 TAACCCACAG 14 41470 41489 128 ACACCCATGA 411953 1500 1519 ATAACCCACA 24 41471 41490 129 GACACCCATG 411954 1501 1520 AATAACCCAC 0 41472 41491 130 AGACACCCAT 411955 1502 1521 AAATAACCCA 10 41473 41492 131 CAGACACCCA 217354 1503 1522 TAAATAACCC 6 41474 41493 100 ACAGACACCC 411956 1504 1523 TTAAATAACC 0 41475 41494 132 CACAGACACC 411957 1505 1524 TTTAAATAAC 14 41476 41495 133 CCACAGACAC 411958 1506 1525 TTTTAAATAA 10 41477 41496 134 CCCACAGACA 411959 1507 1526 CTTTTAAATA 0 41478 41497 135 ACCCACAGAC 217355 1508 1527 TCTTTTAAAT 10 41479 41498 101 AACCCACAGA 411960 1509 1528 TTCTTTTAAA 8 41480 41499 136 TAACCCACAG 411961 1510 1529 TTTCTTTTAA 1 41481 41500 137 ATAACCCACA 411962 1511 1530 ATTTCTTTTA 1 41482 41501 138 AATAACCCAC 411963 1512 1531 AATTTCTTTT 0 41483 41502 139 AAATAACCCA 411964 1513 1532 TAATTTCTTT 0 41484 41503 140 TAAATAACCC 411965 1514 1533 ATAATTTCTT 0 41485 41504 141 TTAAATAACC 411966 1515 1534 TATAATTTCT 0 41486 41505 142 TTTAAATAAC 411967 1516 1535 TTATAATTTC 0 41487 41506 143 TTTTAAATAA
TABLE-US-00003 TABLE 3 Inhibition of DGAT2 mRNA by MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence Motif tion Site Site NO 413166 254 273 ATGAGGGTCT 5-10-5 7 9875 9894 156 TCATGGCTGA 413167 266 285 GAGTAGGCGG 5-10-5 11 9887 9906 157 CTATGAGGGT 413168 269 288 CCGGAGTAGG 5-10-5 37 9890 9909 158 CGGCTATGAG 413169 272 291 ACCCCGGAGT 5-10-5 46 9893 9912 159 AGGCGGCTAT 334177 275 294 AGGACCCCGG 5-10-5 52 9896 9915 145 AGTAGGCGGC 413170 304 323 GGTCAGCCTC 5-10-5 11 9925 9944 160 GGCCTGACGC 413171 307 326 TCCGGTCAGC 5-10-5 20 9928 9947 161 CTCGGCCTGA 413172 310 329 GGCTCCGGTC 5-10-5 0 9931 9950 162 AGCCTCGGCC 413173 331 350 CAGGTCCTCC 5-10-5 29 9952 9971 163 GTGAGAGCGC 413174 339 358 CGACAGCGCA 5-10-5 48 9960 9979 164 GGTCCTCCGT 413175 348 367 CCCCTCGCGC 5-10-5 25 9969 9988 165 GACAGCGCAG 413176 355 374 TCCCAGACCC 5-10-5 36 9976 9995 166 CTCGCGCGAC 413177 361 380 CCCATCTCCC 5-10-5 12 9982 10001 167 AGACCCCTCG 413178 367 386 CAGTGCCCCA 5-10-5 0 n/a n/a 168 TCTCCCAGAC 413179 370 389 ATCCAGTGCC 5-10-5 16 n/a n/a 169 CCATCTCCCA 413180 376 395 TGCTGGATCC 5-10-5 44 n/a n/a 170 AGTGCCCCAT 413181 379 398 GGATGCTGGA 5-10-5 32 n/a n/a 171 TCCAGTGCCC 413182 382 401 AGAGGATGCT 5-10-5 45 25508 25527 172 GGATCCAGTG 413183 385 404 CGGAGAGGAT 5-10-5 47 25511 25530 173 GCTGGATCCA 413184 396 415

GTCCTGGAGG 5-10-5 32 25522 25541 174 GCGGAGAGGA 413185 404 423
GAGAAGAGGT 5-10-5 10 25530 25549 175 CCTGGAGGGC 366710 425 444
GACCTATTGA 5-10-5 36 25551 25570 146 GCCAGGTGAC 370727 443 462
AGCTGCTTTT 2-16-2 45 25569 25588 152 CCACCTTGGA 366714 445 464
GTAGCTGCTT 5-10-5 43 25571 25590 147 TTCCACCTTG 413186 448 467
CCTGTAGCTG 5-10-5 29 25574 25593 176 CTTTTCCACC 413187 451 470
TGACCTGTAG 5-10-5 35 25577 25596 177 CTGCTTTTCC 413188 455 474
GAGATGACCT 5-10-5 34 25581 25600 178 GTAGCTGCTT 413189 458 477
ACTGAGATGA 5-10-5 33 25584 25603 179 CCTGTAGCTG 413190 461 480
AGCACTGAGA 5-10-5 36 25587 25606 180 TGACCTGTAG 413191 467 486
CACTGGAGCA 5-10-5 27 25593 25612 181 CTGAGATGAC 413192 473 492
AGGACCCACT 5-10-5 40 25599 25618 182 GGAGCACTGA 413193 476 495
GACAGGACCC 5-10-5 44 25602 25621 183 ACTGGAGCAC 380089 482 501
AGGAAGGACA 5-10-5 38 25608 25627 155 GGACCCACTG 413194 494 513
ACTCCCAGTA 5-10-5 3 n/a n/a 184 CAAGGAAGGA 413195 497 516 GCCACTCCCA 5-10-5 0 n/a n/a 185 GTACAAGGAA 413196 500 519 CAGGCCACTC 5-10-5 1 n/a n/a 186 CCAGTACAAG 413197 503 522 CTGCAGGCCA 5-10-5 40 n/a n/a 187 CTCCCAGTAC 413198 506 525 GCACTGCAGG 5-10-5 37 n/a n/a 188 CCACTCCCAG 413199 510 529 GATGGCACTG 5-10-5 28 31077 31096 189 CAGGCCACTC 366722 530 549 GTGCAGAATA 5-10-5 43 31097 31116 148 TGTACATGAG 413200 554 573
AGCACAGCGA 5-10-5 35 31121 31140 190 TGAGCCAGCA 413201 570 589
CAGCCAAGTG 5-10-5 41 31137 31156 191 AAGTAGAGCA 413202 573 592
CACCAGCCAA 5-10-5 23 31140 31159 192 GTGAAGTAGA 413203 576 595
AAACACCAGC 5-10-5 9 31143 31162 193 CAAGTGAAGT 413204 579 598
GTCAAACACC 5-10-5 15 31146 31165 194 AGCCAAGTGA 413205 585 604
GTTCCAGTCA 5-10-5 31 31152 31171 195 AACACCAGCC 413206 588 607
TGTGTTCCAG 5-10-5 23 31155 31174 196 TCAAACACCA 413207 591 610
GGGTGTGTTC 5-10-5 6 31158 31177 197 CAGTCAAACA 413208 594 613
CTTGGGTGTG 5-10-5 5 31161 31180 198 TTCCAGTCAA 413209 615 634
CTGTGACCTC 5-10-5 25 31547 31566 199 CTGCCACCTT 413210 618 637
CCACTGTGAC 5-10-5 30 31550 31569 200 CTCCTGCCAC 413211 621 640
GACCCACTGT 5-10-5 25 31553 31572 201 GACCTCCTGC 413212 625 644
TTCGGACCCA 5-10-5 38 31557 31576 202 CTGTGACCTC 413213 629 648
CAGTTTCGGA 5-10-5 39 31561 31580 203 CCCACTGTGA 413214 632 651
GCCCAGTTTC 5-10-5 51 31564 31583 204 GGACCCACTG 413215 635 654
ACAGCCCAGT 5-10-5 27 31567 31586 205 TTCGGACCCA 413216 638 657
CACACAGCCC 5-10-5 24 31570 31589 206 AGTTTCGGAC 413217 642 661
GCGCCACACA 5-10-5 31 31574 31593 207 GCCCAGTTTC 413218 658 677
AGTAGTCTCG 5-10-5 39 31590 31609 208 AAAGTAGCGC 413219 661 680
GAAAGTAGTC 5-10-5 0 31593 31612 209 TCGAAAGTAG 413220 665 684
ATGGGAAAGT 5-10-5 0 31597 31616 210 AGTCTCGAAA 366728 757 776
TGCAGAAGGC 5-10-5 46 37285 37304 149 ACCCAGGCCC 413221 780 799
TTCTGTGGCC 5-10-5 25 37308 37327 211 TCTGTGCTGA 413222 783 802
CACTTCTGTG 5-10-5 38 37311 37330 212 GCCTCTGTGC 413223 786 805
GCTCACTTCT 5-10-5 41 37314 37333 213 GTGGCCTCTG 413224 789 808
CTTGCTCACT 5-10-5 48 37317 37336 214 TCTGTGGCCT 413225 792 811
CTTCTTGCTC 5-10-5 25 37320 37339 215 ACTTCTGTGG 413226 798 817
TGGGAACTTC 5-10-5 43 37326 37345 216 TTGCTCACTT 413227 801 820
GCCTGGGAAC 5-10-5 47 37329 37348 217 TTCTTGCTCA 413228 804 823
TATGCCTGGG 5-10-5 26 37332 37351 218 AACTTCTTGC 413229 808 827

GCCGTATGCC 5-10-5 29 37336 37355 219 TGGGAACTTC 413230 811 830 AAGGCCGTAT 5-10-5 25 37339 37358 220 GCCTGGGAAC 413231 814 833 GGTAAGGCCG 5-10-5 41 37342 37361 221 TATGCCTGGG 217328 938 957 GCATTGCCAC 5-10-5 41 38091 38110 144 TCCCATTCTT 366730 979 998 AGCTCAGAGA 5-10-5 39 38132 38151 150 CTCAGCCGCA 370747 982 1001 TGGAGCTCAG 2-16-2 37 38135 38154 153 AGACTCAGCC 366746 1366 1385 TGGTCTTGTG 5-10-5 41 41337 41356 151 CTTGTCGAAG 370784 2128 2147 GCTGCATCCA 2-16-2 53 42099 42118 154 TGTCATCAGC

TABLE-US-00004 TABLE 4 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in NO: NO: 1 1 hi 2 2 SEQ ISIS Start Stop bi Start Stop ID NO Site Site Sequence tion Site Site NO 413232 817 836 CCAGGTAAGG 78 37345 37364 225 CCGTATGCCT 413233 820 839 TAGCCAGGTA 56 37348 37367 226 AGGCCGTATG 413234 823 842 GTGTAGCCAG 50 37351 37370 227 GTAAGGCCGT 413235 827 846 GCCAGTGTAG 9 37355 37374 228 CCAGGTAAGG 413236 861 880 GTACTCCCTC 72 37389 37408 229 AACACAGGCA 413237 864 883 CAGGTACTCC 52 37392 37411 230 CTCAACACAG 413238 867 886 CATCAGGTAC 35 37395 37414 231 TCCCTCAACA 413239 870 889 AGACATCAGG 35 37398 37417 232 TACTCCCTCA 413240 878 897 ATACCTCCAG 50 n/a n/a 233 ACATCAGGTA 413241 881 900 CAGATACCTC 38 n/a n/a 234 CAGACATCAG 413242 887 906 ACAGGGCAGA 58 n/a n/a 235 TACCTCCAGA 413243 910 929 AATAGTCTAT 63 38063 38082 236 GGTGTCCCGG 413244 913 932 GCAAATAGTC 60 38066 38085 237 TATGGTGTCC 413245 916 935 AAAGCAAATA 46 38069 38088 238 GTCTATGGTG 413246 919 938 TTGAAAGCAA 42 38072 38091 239 ATAGTCTATG 413247 922 941 TCTTTGAAAG 40 38075 38094 240 CAAATAGTCT 413248 925 944 CATTCTTTGA 15 38078 38097 241 AAGCAAATAG 413249 928 947 TCCCATTCTT 31 38081 38100 242 TGAAAGCAAA 413250 932 951 CCACTCCCAT 34 38085 38104 243 TCTTTGAAAG 413251 935 954 TTGCCACTCC 39 38088 38107 244 CATTCTTTGA 217328 938 957 GCATTGCCAC 70 38091 38110 144 TCCCATTCTT 413252 976 995 TCAGAGACTC 43 38129 38148 245 AGCCGCACCC 413253 987 1006 AGGCATGGAG 69 38140 38159 246 CTCAGAGACT 413254 1002 1021 GACTGCATTC 60 38155 38174 247 TTGCCAGGCA 413255 1005 1024 GGTGACTGCA 25 38158 38177 248 TTCTTGCCAG 413256 1012 1031 TCCGCAGGGT 48 38165 38184 249 GACTGCATTC 369241 1019 1038 TTGCGGTTCC 70 38172 38191 222 GCAGGGTGAC 413257 1022 1041 CCCTTGCGGT 36 38175 38194 250 TCCGCAGGGT 413258 1025 1044 AAGCCCTTGC 67 38178 38197 251 GGTTCCGCAG 413259 1028 1047 ACAAAGCCCT 35 38181 38200 252 TGCGGTTCCG 413260 1034 1053 AGTTTCACAA 25 38187 38206 253 AGCCCTTGCG 413261 1037 1056 GCCAGTTTCA 34 38190 38209 254 CAAAGCCCTT 413262 1040 1059 AGGGCCAGTT 33 38193 38212 255 TCACAAAGCC 413263 1043 1062 CGCAGGGCCA 31 38196 38215 256 GTTTCACAAA 413264 1088 1107 TCATTCTCTC 18 39135 39154 257 CAAAGGAGTA 413265 1091 1110 ACTTCATTCT 45 39138 39157 258 CTCCAAAGGA 413266 1095 1114 GTACACTTCA 82 39142 39161 259 TTCTCTCCAA 413267 1101 1120 CTGCTTGTAC 58 39148 39167 260 ACTTCATTCT 413268 1105 1124 TCACCTGCTT 40 39152 39171 261 GTACACTTCA 413269 1111 1130 CGAAGATCAC 63 39158 39177 262 CTGCTTGTAC 413270 1114 1133 CCTCGAAGAT 53 39161 39180 263 CACCTGCTTG 413271 1117 1136 CCTCCTCGAA 49 39164 39183 264 GATCACCTGC 413272 1120 1139 AGCCCTCCTC 39 39167 39186 265 GAAGATCACC 413273 1123 1142 AGGAGCCCTC 31 39170 39189 266 CTCGAAGATC 413274 1126 1145 CCCAGGAGCC 54 39173 39192 267 CTCCTCGAAG 413275 1129 1148 GGCCCCAGGA 40 39176 39195 268 GCCCTCCTCG 411877 1136 1155 ACCCATCGGC 59 39183 39202 223 CCCAGGAGCC 413276 1157 1176 TATTTCTGGA 22 39204 39223 269 ACTTCTTCTG 413277 1160 1179 ATGTATTTCT 45 39207 39226 270 GGAACTTCTT 413278

1171 1190 GGGCGAAACC 24 39218 39237 271 AATGTATTTC 413279 1208 1227 TCGGAGGAGA 60 39255 39274 272 AGAGGCCTCG 413280 1211 1230 GTGTCGGAGG 52 39258 39277 273 AGAAGAGGCC 413281 1215 1234 CCAGGTGTCG 53 39262 39281 274 GAGGAGAAGA 413282 1222 1241 CCAGCCCCCA 50 39269 39288 275 GGTGTCGGAG 413283 1250 1269 ACAGTGGTGA 52 39297 39316 276 TGGGCTTGGA 413284 1273 1292 GGATGGTGAT 67 41244 41263 277 GGGCTCTCCC 411879 1330 1349 CCATGTACAT 52 41301 41320 224 GGTGTGGTAC 413285 1334 1353 GCCTCCATGT 63 41305 41324 278 ACATGGTGTG 413286 1363 1382 TCTTGTGCTT 62 41334 41353 279 GTCGAAGAGC 413287 1399 1418 CCTCCAGGAC 66 41370 41389 280 CTCAGTCTCC 413288 1403 1422 TTCACCTCCA 46 41374 41393 281 GGACCTCAGT 413289 1407 1426 TCAGTTCACC 46 41378 41397 282 TCCAGGACCT 413290 1412 1431 CTGGCTCAGT 90 41383 41402 283 TCACCTCCAG 413291 1537 1556 TGTAATGGTT 26 41508 41527 284 TAGCAAAATT 413292 1555 1574 TTAAAAAAGA 0 41526 41545 285 CCTAACATTG 413293 1559 1578 CTTCTTAAAA 26 41530 41549 286 AAGACCTAAC 413294 1563 1582 TTTCCTTCTT 9 41534 41553 287 AAAAAAGACC 413295 1567 1586 ACTTTTTCCT 0 41538 41557 288 TCTTAAAAAA 413296 1572 1591 TACTGACTTT 39 41543 41562 289 TTCCTTCTTA 413297 1616 1635 CCACCACCTA 55 41587 41606 290 GAACAGGGCA 413298 1653 1672 AGGTTAGCTG 59 41624 41643 291 AGCCACCCAG 413299 1840 1859 GTCCTGCAGT 54 41811 41830 292 TTCAGGACTA 413300 1844 1863 ACTGGTCCTG 55 41815 41834 293 CAGTTTCAGG 413301 1860 1879 TCCCCTTGGC 47 41831 41850 294 AGAGAAACTG 413302 1864 1883 CTCCTCCCCT 43 41835 41854 295 TGGCAGAGAA 413303 1868 1887 CCAACTCCTC 28 41839 41858 296 CCCTTGGCAG 413304 1872 1891 CTCTCCAACT 25 41843 41862 297 CCTCCCCTTG 413305 1875 1894 GTGCTCTCCA 34 41846 41865 298 ACTCCTCCCC

TABLE-US-00005 TABLE 5 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 413347 n/a n/a AGCCTGCCAC 37 10050 10069 339 AGGGCCCTTT 413348 n/a n/a AGGACAGGGC 44 10396 10415 340 AGACACACCT 413349 n/a n/a GATTTGTACT 34 10703 10722 341 TGAATCCAGG 413350 n/a n/a ACACGCAGAC 46 10747 10766 342 CAGGACAGCT 413351 n/a n/a TTTGGATAGT 41 10943 10962 343 CGATTTACCA 413352 n/a n/a AAGATTCATA 6 11045 11064 344 ACTATATCTA 413353 n/a n/a AAAAACATGA 31 11558 11577 345 CAGCCAGGGT 413354 n/a n/a GACTTCCCTT 5 11983 12002 346 CACAGAATCC 413355 n/a n/a CAGGCAGGTG 21 12250 12269 347 TCAGAGGGCT 413356 n/a n/a TAGCCTGGCT 42 12892 12911 348 TTGATAACCC 413357 n/a n/a CATAGGCCAG 35 13206 13225 349 GAGGAAGAGT 413358 n/a n/a TTGCTTAAAC 17 13541 13560 350 AGATAAGCAC 413359 n/a n/a AATGTCACAA 4 14428 14447 351 GTTCACAAAC 413360 n/a n/a CTGACCTCAG 31 14721 14740 352 GGTGATCAAG 413361 n/a n/a CCAAGCAGGA 19 15143 15162 353 GCTGGACAGA 413362 n/a n/a TCATTTCATA 34 15215 15234 354 GATGAGGAGA 413363 n/a n/a AGGAGTTTGT 39 15320 15339 355 GTTTCCCATT 413364 n/a n/a GGCAGGGCTT 48 15359 15378 356 AGAATGGCTA 413365 n/a n/a ACCATTTTCA 30 15572 15591 357 CAAGGAGAAG 413366 n/a n/a TTCAAAAAGT 47 15738 15757 358 AGCTACTGCA 413367 n/a n/a TCCAGGACCC 50 15813 15832 359 TGGACATGAT 413368 n/a n/a AGAGCCAGCA 20 16251 16270 360 CACAGCTATG 413369 n/a n/a AGTCTAGTTG 16 16545 16564 361 GAAAAGTAGA 413370 n/a n/a AGATGCCACT 24 17722 17741 362 TCATCAAGGC 413371 n/a n/a GAAGTCAGGC 26 17744 17763 363 CAAGTGCCAA 413372 n/a n/a TGATTCTTAC 22 18221 18240 364 CTGCAATGAG 413373 n/a n/a CCAAGTAAGC 32 18329 18348 365 CTCGGTGTCC 413374 n/a n/a TGAGTAGTCA 30 19019 19038 366 AAGGTGGCTT 413375 n/a n/a ATGCCTGAGG 45 19907 19926 367 GCAGCAGTGT 413376 n/a n/a TGACCAGGAA 15 19967 19986 368 GGCCACACCT 413377 n/a n/a GGCTTCACCG 39

20409 20428 369 TCCCACAGCA 413378 n/a n/a CAGGACTTGG 35 20883 20902 370 TACCTGATTC 413379 n/a n/a TGGCTGGGAG 10 21131 21150 371 GAGTCCAGCA 413380 n/a n/a GGGTCAAGGT 33 21660 21679 372 CACTCAGCCA 413381 n/a n/a TATTTGAAGA 0 22021 22040 373 TAAAGTCAGA 413382 n/a n/a GGGATGATAA 37 22093 22112 374 ACACTAAGGT 217328 938 957 GCATTGCCAC 59 38091 38110 144 TCCCATTCTT 413306 1992 2011 CTGGAGGCCA 23 41963 41982 299 GTCCAGGCTC 413307 1995 2014 ATCCTGGAGG 18 41966 41985 300 CCAGTCCAGG 413308 2006 2025 CCCCCATCCT 7 41977 41996 301 CATCCTGGAG 413309 2010 2029 GCCACCCCCA 0 41981 42000 302 TCCTCATCCT 413310 2014 2033 CATTGCCACC 9 41985 42004 303 CCCATCCTCA 413311 2040 2059 GGGCAGTCCT 37 42011 42030 304 TTCCCCTGCA 413312 2081 2100 TAGCTCATGG 35 42052 42071 305 TGGCGGCATC 413313 2087 2106 TCCACCTAGC 21 42058 42077 306 TCATGGTGGC 413314 2091 2110 TTACTCCACC 6 42062 42081 307 TAGCTCATGG 413315 2099 2118 AAAACCAGTT 0 42070 42089 308 ACTCCACCTA 413316 2102 2121 AGAAAAACCA 4 42073 42092 309 GTTACTCCAC 413317 2113 2132 TCAGCCACCC 0 42084 42103 310 AAGAAAAACC 413318 2120 2139 CATGTCATCA 16 42091 42110 311 GCCACCCAAG 413319 2128 2147 GCTGCATCCA 43 42099 42118 154 TGTCATCAGC 413320 2131 2150 TGTGCTGCAT 36 42102 42121 312 CCATGTCATC 413321 2136 2155 GAGTCTGTGC 48 42107 42126 313 TGCATCCATG 413322 2143 2162 CAAGGCTGAG 24 42114 42133 314 TCTGTGCTGC 413323 2146 2165 GGCCAAGGCT 17 42117 42136 315 GAGTCTGTGC 413324 2149 2168 CCAGGCCAAG 27 42120 42139 316 GCTGAGTCTG 413325 2182 2201 AAGGTAAACT 37 42153 42172 317 GAGGCCACCA 413326 2185 2204 GGGAAGGTAA 32 42156 42175 318 ACTGAGGCCA 413327 2236 2255 AGGCCCCTTC 43 42207 42226 319 TGAAGAGGGA 413328 2239 2258 GCCAGGCCCC 36 42210 42229 320 TTCTGAAGAG 413329 2242 2261 AAGGCCAGGC 32 42213 42232 321 CCCTTCTGAA 413330 2246 2265 TCAGAAGGCC 20 42217 42236 322 AGGCCCCTTC 413331 2252 2271 TGCTGCTCAG 42 42223 42242 323 AAGGCCAGGC 413332 2257 2276 TAATCTGCTG 36 42228 42247 324 CTCAGAAGGC 413333 2265 2284 TTTGGAACTA 50 42236 42255 325 ATCTGCTGCT 413334 2274 2293 GCCACCTGCT 41 42245 42264 326 TTGGAACTAA 413335 2301 2320 ACAGAAAAGT 40 42272 42291 327 GAGGCTTGGG 413336 2304 2323 GGCACAGAAA 32 42275 42294 328 AGTGAGGCTT 413337 2307 2326 GAAGGCACAG 11 42278 42297 329 AAAAGTGAGG 413338 2310 2329 CAGGAAGGCA 28 42281 42300 330 CAGAAAAGTG 413339 2313 2332 CCTCAGGAAG 40 42284 42303 331 GCACAGAAAA 413340 2317 2336 ACCCCCTCAG 25 42288 42307 332 GAAGGCACAG 413341 2323 2342 GGCCCAACCC 19 42294 42313 333 CCTCAGGAAG 413342 2373 2392 TCTCATCAAG 6 42344 42363 334 AGATAACAGA 413343 2376 2395 TGATCTCATC 17 42347 42366 335 AAGAGATAAC 413344 2399 2418 TACAAAAGTC 40 42370 42389 336 TGACATGGTG 413345 2402 2421 ATATACAAAA 21 42373 42392 337 GTCTGACATG 413346 2406 2425 AGGCATATAC 38 42377 42396 338 AAAAGTCTGA

TABLE-US-00006 TABLE 6 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ ID ID NO: NO: 2 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 413383 22161 22180 ACCCAGAGATGGTGATAAGG 53 375 413384 22521 22540 AAGTGAACCCTCACTTCCCA 35 376 413385 22840 22859 ACATATCCCCAACTGAAACA 21 377 413386 22849 22868 TCACTGATGACATATCCCCA 52 378 413387 22862 22881 GAGTATAGAAATTTCACTGA 44 379 413388 22877 22896 TGCAGCTTGGAGGAGGAGTA 36 380 413389 23153 23172 GCATTTTAACAAGATCCCCA 34 381 413390 23167 23186 CTGAATCAAAATCTGCATTT 28 382 413391 23182 23201 TCCAGACTGGATTTACTGAA 60 383 413392 23550 23569 TTATCCTTGAGGGTCTCATA 45 384 413393 23560 23579 CATCACATGCTTATCCTTGA 41 385 413394 23669 23688 TTCCCCAGAGCAGCTCTGAG 27 386 413395 23726 23745 GCGGCAGAGCCAGGACTGAA 35 387 413396 23812 23831 GTCATCCAGTATAGCCTAAT

44 388 413397 23956 23975 TAAAGAGGCTGGGCCATAGA 21 389 413398 24122 24141 TCAAACCTAGGTTCAGACTT 47 390 413399 24204 24223 AGGAAGGAAATGCTAGGCCT 64 391 413400 24642 24661 ACAGGAAGCCTGGTGACTGC 54 392 413401 24790 24809 CAGGCAGCCTGAAGGACACT 49 393 413402 25074 25093 AGGACAGAGGTTCAACATCC 53 394 413403 25234 25253 AGACCACATGAAGTATCTAA 50 395 413404 25376 25395 GCTATAGAATCAGACAGACC 46 396 413405 25488 25507 CCTACAGCAGAGGGAAGATG 13 397 413406 25493 25512 CAGTGCCTACAGCAGAGGGA 0 398 413407 25504 25523 GATGCTGGATCCAGTGCCTA 53 399 413408 25953 25972 GTGGGAAAGGCACAGGCTTT 53 400 413409 26393 26412 GATGGCAACCTAAGGAGTGA 37 401 413410 26893 26912 GTTCTTGGGCCTAAAGGTGA 34 402 413411 27564 27583 ATTAGCAGTAGCTCAGGAGA 48 403 413412 27677 27696 CTGGTCCAGGGAGAGACCAA 25 404 413413 27711 27730 AGAGGGTTCCATGGCACAAA 67 405 413414 28186 28205 AATTTCCTTACAGGGTATTG 46 406 413415 28461 28480 TCATGAGAAGCTTAGAAGGC 58 407 413416 28524 28543 CAACAGGCTCTGGTTCCATG 56 408 413417 28920 28939 CAGGCCCCACTGCTTAGAGG 48 409 413418 29446 29465 AAATGGTAGCCCCTTGTTGC 32 410 413419 29503 29522 TTTCCCATGGGAGAAGATAA 55 411 413420 30361 30380 AGCGCAGAGCCCATCAGCCT 48 412 413421 30620 30639 CAGGGTGACTTTGCCCCATT 43 413 413422 30974 30993 TGGCACTAAGCTAGGCACAC 62 414 413423 31184 31203 AGGGAGGCCTTGCACTTACC 13 415 413424 31240 31259 TGAGGCCCTTCAGCTTGTGC 49 416 413425 31370 31389 AAACCCTCGACTGAGTGTGA 37 417 413426 31531 31550 CCTTCCAGGGAATAAAATAC 3 418 413427 31534 31553 CCACCTTCCAGGGAATAAAA 22 419 413428 31537 31556 CTGCCACCTTCCAGGGAATA 27 420 413429 31620 31639 AACACTCACAGCACTTTACC 4 421 413430 31769 31788 TGGATACTCAGAAGAGCAGT 37 422 413431 31935 31954 CAGAGTTATCCTCAATTCAC 34 423 413432 32145 32164 ACACCAGGATCTCAGTCACT 42 424 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 70 425 413434 32528 32547 GAAACAGGCAGTAGGAAATC 23 426 413435 32644 32663 ATGGAGTGACAGGGCAGGAA 59 427 413436 32939 32958 AGGCCACAGTGGCAACAGAG 54 428 413437 33067 33086 GTTCTTTTGGAAGGGTGGAG 52 429 413438 33164 33183 GGAGCCCTCACAGGGCCAGG 38 430 413439 33364 33383 GGACAGGAGGGTCACACACA 45 431 413440 33671 33690 AGATGGACAGGTGATTCTAA 50 432 413441 33739 33758 TTAAGCTTTGTGACCTTGGG 67 433 413442 34101 34120 GGGAAGGATACCGCCAATGA 57 434 413443 34311 34330 CAGTGGGCCCCAGGTGGCTC 46 435 413444 34642 34661 GGTGGGAAACTTGGAAACTT 37 436 413445 35355 35374 ACAATTCCTGGATAACAAGG 41 437 413446 35362 35381 TAGAAATACAATTCCTGGAT 71 438 413447 35568 35587 TGTCCTTATCAAAATCCCTC 38 439 413448 36267 36286 GCTGAGAGAGACAATGAGTA 54 440 413449 36858 36877 GATTATTCTAAAACTCAAAT 0 441 413450 37202 37221 ACCAGCTGCAAGGATGACCT 49 442 413451 37457 37476 GAGGCTCAGGCCTTGACAAC 12 443 413452 37604 37623 TGTTATCCGAGTTGAATTCT 45 444 413453 37818 37837 GTTTTGGGAACTCATGCATT 50 445 413454 37837 37856 CAGCTAATGATACAAGGTTG 55 446 413455 37880 37899 GCTATTCATTTTTCTGAGCC 47 447 217328 38091 38110 GCATTGCCACTCCCATTCTT 67 144 413456 39033 39052 AGAGGCCCTGGACACTGGCC 32 448 413457 39040 39059 TCAGCCTAGAGGCCCTGGAC 26 449 413458 39300 39319 CCAACAGTGGTGATGGGCTT 60 450 413459 39305 39324 GCTTACCAACAGTGGTGATG 21 451

Example 2: Antisense Inhibition of Human Diacylglycerol Acyltransferase 2 in HepG2 Cells by MOE Gapmers

[0452] Antisense oligonucleotides were designed targeting a diacylglycerol acyltransferase 2 (DGAT2) nucleic acid and were tested for their effects on DGAT2 mRNA in vitro. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. The results for each experiment are presented in separate tables shown below. Cultured HepG2 cells at a density of 10,000 cells per well were transfected using Lipofectin reagent with 120 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB (forward sequence AACTGGCCCTGCGTCATG, designated herein as SEQ ID NO: 7; reverse sequence CTTGTACACTTCATTCTCTCCAAAGG, designated herein as SEQ ID NO: 8; probe sequence CTGACCTGGTTCCC, designated herein as SEQ ID NO: 9) was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells.

[0453] ISIS oligonucleotides from an earlier published application, WO 2005/019418, were also included in this assay. These ISIS oligonucleotides are ISIS 217316, ISIS 217317, ISIS 217328, ISIS 217329, ISIS 334177, ISIS 334178, ISIS 366730, ISIS 366731, and ISIS 369255. As shown in the Tables below, several newly designed antisense oligonucleotides demonstrated similar or greater potency that any of these benchmark oligonucleotides.

[0454] The newly designed chimeric antisense oligonucleotides in the Tables below were designed as 5-10-5 MOE, 3-14-3 MOE, or 2-13-5 MOE gapmers. The 5-10-5 MOE gapmers are 20 nucleosides in length, wherein the central gap segment comprises of ten 2′-deoxynucleosides and is flanked by wing segments on the 5′ direction and the 3′ direction comprising five nucleosides each. The 3-14-3 MOE gapmers are 20 nucleosides in length, wherein the central gap segment comprises of fourteen 2′-deoxynucleosides and is flanked by wing segments on the 5′ direction and the 3′ direction comprising three nucleosides each. The 2-13-5 MOE gapmers are 20 nucleosides in length, wherein the central gap segment comprises of thirteen 2′-deoxynucleosides and is flanked by wing segments on the 5′ direction and the 3′ direction comprising two and three nucleosides respectively. Each nucleoside in the 5′ wing segment and each nucleoside in the 3′ wing segment has a 2′-MOE modification. The internucleoside linkages throughout each gapmer are phosphorothioate (P═S) linkages. All cytosine residues throughout each gapmer are 5-methylcytosines. “Start site” indicates the 5′-most nucleoside to which the gapmer is targeted in the human gene sequence. “Stop site” indicates the 3′-most nucleoside to which the gapmer is targeted in the human gene sequence. Each gapmer listed in the Tables below is targeted to either the human DGAT2 mRNA, designated herein as SEQ ID NO: 1 (Ref Seq No. NM_032564.3) or the human DGAT2 genomic sequence, designated herein as SEQ ID NO: 2 (Ref Seq No. NT_033927.5 truncated from nucleotides 5669186 to 5712008). ‘n/a’ indicates that the antisense oligonucleotide does not target that particular gene sequence with 100% complementarity.

TABLE-US-00007 TABLE 8 Inhibition of DGAT2 mRNA by MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Motif Sequence tion Site Site NO 423460 n/a n/a 5-10-5 TGGACAAGTC 72 32428 32447 464 CTGCCCATCT 423520 n/a n/a 3-14-3 TGGACAAGTC 69 32428 32447 464 CTGCCCATCT 423598 n/a n/a 2-13-5 TGGACAAGTC 51 32428 32447 464 CTGCCCATCT 423461 n/a n/a 5-10-5 CTGGACAAGT 77 32429 32448 465 CCTGCCCATC 423521 n/a n/a 3-14-3 CTGGACAAGT 67 32429 32448 465 CCTGCCCATC 423599 n/a n/a 2-13-5 CTGGACAAGT 58 32429 32448 465 CCTGCCCATC 423462 n/a n/a 5-10-5 CCTGGACAAG 73 32430 32449 466 TCCTGCCCAT 423522 n/a n/a 3-14-3 CCTGGACAAG 70 32430 32449 466 TCCTGCCCAT 423600 n/a n/a 2-13-5 CCTGGACAAG 59 32430 32449 466 TCCTGCCCAT 413433 n/a n/a 5-10-5 GCCTGGACAA 86 32431 32450 425 GTCCTGCCCA 423523 n/a n/a 3-14-3 GCCTGGACAA 84 32431 32450 425 GTCCTGCCCA 423601 n/a n/a 2-13-5 GCCTGGACAA 78 32431 32450 425 GTCCTGCCCA 423463 n/a n/a 5-10-5 AGCCTGGACA 79 32432 32451 467 AGTCCTGCCC 423524 n/a n/a 3-14-3 AGCCTGGACA 82 32432 32451 467 AGTCCTGCCC 423602 n/a n/a 2-13-5 AGCCTGGACA 75 32432 32451 467 AGTCCTGCCC 423464 n/a n/a 5-10-5 CAGCCTGGAC 84 32433 32452 468 AAGTCCTGCC 423525 n/a n/a 3-14-3 CAGCCTGGAC 72 32433 32452 468 AAGTCCTGCC 423603 n/a n/a 2-13-5 CAGCCTGGAC 72 32433 32452 468 AAGTCCTGCC 423465 n/a n/a 5-

10-5 GCAGCCTGGA 70 32434 32453 469 CAAGTCCTGC 423526 n/a n/a 3-14-3 GCAGCCTGGA 83 32434 32453 469 CAAGTCCTGC 423604 n/a n/a 2-13-5 GCAGCCTGGA 77 32434 32453 469 CAAGTCCTGC 423466 n/a n/a 5-10-5 TGCAGCCTGG 67 32435 32454 470 ACAAGTCCTG 423527 n/a n/a 3-14-3 TGCAGCCTGG 77 32435 32454 470 ACAAGTCCTG 413191 467 486 5-10-5 CACTGGAGCA 48 25593 25612 181 CTGAGATGAC 423502 467 486 3-14-3 CACTGGAGCA 55 25593 25612 181 CTGAGATGAC 423580 467 486 2-13-5 CACTGGAGCA 61 25593 25612 181 CTGAGATGAC 423449 468 487 5-10-5 CCACTGGAGC 71 25594 25613 452 ACTGAGATGA 423503 468 487 3-14-3 CCACTGGAGC 62 25594 25613 452 ACTGAGATGA 423581 468 487 2-13-5 CCACTGGAGC 54 25594 25613 452 ACTGAGATGA 423450 469 488 5-10-5 CCCACTGGAG 78 25595 25614 453 CACTGAGATG 423504 469 488 3-14-3 CCCACTGGAG 75 25595 25614 453 CACTGAGATG 423582 469 488 2-13-5 CCCACTGGAG 70 25595 25614 453 CACTGAGATG 334178 470 489 5-10-5 ACCCACTGGA 72 25596 25615 454 GCACTGAGAT 423505 470 489 3-14-3 ACCCACTGGA 59 25596 25615 454 GCACTGAGAT 423583 470 489 2-13-5 ACCCACTGGA 52 25596 25615 454 GCACTGAGAT 423451 471 490 5-10-5 GACCCACTGG 69 25597 25616 455 AGCACTGAGA 423506 471 490 3-14-3 GACCCACTGG 56 25597 25616 455 AGCACTGAGA 423584 471 490 2-13-5 GACCCACTGG 70 25597 25616 455 AGCACTGAGA 423452 472 491 5-10-5 GGACCCACTG 84 25598 25617 456 GAGCACTGAG 423507 472 491 3-14-3 GGACCCACTG 82 25598 25617 456 GAGCACTGAG 423585 472 491 2-13-5 GGACCCACTG 86 25598 25617 456 GAGCACTGAG 413192 473 492 5-10-5 AGGACCCACT 70 25599 25618 182 GGAGCACTGA 423508 473 492 3-14-3 AGGACCCACT 82 25599 25618 182 GGAGCACTGA 423586 473 492 2-13-5 AGGACCCACT 75 25599 25618 182 GGAGCACTGA 423453 474 493 5-10-5 CAGGACCCAC 80 25600 25619 457 TGGAGCACTG 423509 474 493 3-14-3 CAGGACCCAC 71 25600 25619 457 TGGAGCACTG 423587 474 493 2-13-5 CAGGACCCAC 78 25600 25619 457 TGGAGCACTG 423454 475 494 5-10-5 ACAGGACCCA 79 25601 25620 458 CTGGAGCACT 423510 475 494 3-14-3 ACAGGACCCA 61 25601 25620 458 CTGGAGCACT 423588 475 494 2-13-5 ACAGGACCCA 68 25601 25620 458 CTGGAGCACT 413193 476 495 5-10-5 GACAGGACCC 69 25602 25621 183 ACTGGAGCAC 423511 476 495 3-14-3 GACAGGACCC 58 25602 25621 183 ACTGGAGCAC 423589 476 495 2-13-5 GACAGGACCC 73 25602 25621 183 ACTGGAGCAC 413212 625 644 5-10-5 TTCGGACCCA 75 31557 31576 202 CTGTGACCTC 423512 625 644 3-14-3 TTCGGACCCA 64 31557 31576 202 CTGTGACCTC 423590 625 644 2-13-5 TTCGGACCCA 71 31557 31576 202 CTGTGACCTC 423455 626 645 5-10-5 TTTCGGACCC 65 31558 31577 459 ACTGTGACCT 423513 626 645 3-14-3 TTTCGGACCC 61 31558 31577 459 ACTGTGACCT 423591 626 645 2-13-5 TTTCGGACCC 68 31558 31577 459 ACTGTGACCT 423456 627 646 5-10-5 GTTTCGGACC 65 31559 31578 460 CACTGTGACC 423514 627 646 3-14-3 GTTTCGGACC 71 31559 31578 460 CACTGTGACC 423592 627 646 2-13-5 GTTTCGGACC 70 31559 31578 460 CACTGTGACC 423457 628 647 5-10-5 AGTTTCGGAC 63 31560 31579 461 CCACTGTGAC 423515 628 647 3-14-3 AGTTTCGGAC 60 31560 31579 461 CCACTGTGAC 423593 628 647 2-13-5 AGTTTCGGAC 64 31560 31579 461 CCACTGTGAC 413213 629 648 5-10-5 CAGTTTCGGA 74 31561 31580 203 CCCACTGTGA 423516 629 648 3-14-3 CAGTTTCGGA 58 31561 31580 203 CCCACTGTGA 423594 629 648 2-13-5 CAGTTTCGGA 77 31561 31580 203 CCCACTGTGA 423458 630 649 5-10-5 CCAGTTTCGG 83 31562 31581 462 ACCCACTGTG 423517 630 649 3-14-3 CCAGTTTCGG 82 31562 31581 462 ACCCACTGTG 423595 630 649 2-13-5 CCAGTTTCGG 80 31562 31581 462 ACCCACTGTG 423459 631 650 5-10-5 CCCAGTTTCG 85 31563 31582 463 GACCCACTGT 423518 631 650 3-14-3 CCCAGTTTCG 76 31563 31582 463 GACCCACTGT 423596 631 650 2-13-5 CCCAGTTTCG 70 31563 31582 463 GACCCACTGT 413214 632 651 5-10-5 GCCCAGTTTC 89 31564 31583 204 GGACCCACTG 423519 632 651 3-14-3 GCCCAGTTTC 81 31564 31583 204 GGACCCACTG 423597 632 651 2-13-5 GCCCAGTTTC 82 31564 31583

204 GGACCCACTG 217328 938 957 5-10-5 GCATTGCCAC 82 38091 38110 144 TCCCATTCTT

TABLE-US-00008 TABLE 9 Inhibition of DGAT2 mRNA by MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ ID ID SEQ SEQ NO: NO: ID ID 1 1 NO: 2 NO: 2 SEQ ISIS Start Stop % Start Stop ID NO Site Site Motif Sequence inhibition Site Site NO 334177 275 294 5-10-5 AGGACCCCGGAGTAGGCGGC 76 9896 9915 145 423528 275 294 3-14-3 AGGACCCCGGAGTAGGCGGC 76 9896 9915 145 423606 275 294 2-13-5 AGGACCCCGGAGTAGGCGGC 81 9896 9915 145 423467 276 295 5-10-5 CAGGACCCCGGAGTAGGCGG 75 9897 9916 471 423529 276 295 3-14-3 CAGGACCCCGGAGTAGGCGG 53 9897 9916 471 423607 276 295 2-13-5 CAGGACCCCGGAGTAGGCGG 76 9897 9916 471 423468 277 296 5-10-5 GCAGGACCCCGGAGTAGGCG 58 9898 9917 472 423530 277 296 3-14-3 GCAGGACCCCGGAGTAGGCG 70 9898 9917 472 423608 277 296 2-13-5 GCAGGACCCCGGAGTAGGCG 68 9898 9917 472 381726 278 297 5-10-5 CGCAGGACCCCGGAGTAGGC 70 9899 9918 38 423531 278 297 3-14-3 CGCAGGACCCCGGAGTAGGC 65 9899 9918 38 423609 278 297 2-13-5 CGCAGGACCCCGGAGTAGGC 71 9899 9918 38 423469 279 298 5-10-5 GCGCAGGACCCCGGAGTAGG 79 9900 9919 473 423532 279 298 3-14-3 GCGCAGGACCCCGGAGTAGG 71 9900 9919 473 423610 279 298 2-13-5 GCGCAGGACCCCGGAGTAGG 72 9900 9919 473 423470 280 299 5-10-5 CGCGCAGGACCCCGGAGTAG 74 9901 9920 474 423533 280 299 3-14-3 CGCGCAGGACCCCGGAGTAG 69 9901 9920 474 423611 280 299 2-13-5 CGCGCAGGACCCCGGAGTAG 67 9901 9920 474 423471 281 300 5-10-5 CCGCGCAGGACCCCGGAGTA 75 9902 9921 475 423534 281 300 3-14-3 CCGCGCAGGACCCCGGAGTA 71 9902 9921 475 411873 686 705 5-10-5 TTGTGTGTCTTCACCAGCTG 73 37214 37233 46 423542 686 705 3-14-3 TTGTGTGTCTTCACCAGCTG 60 37214 37233 46 423620 686 705 2-13-5 TTGTGTGTCTTCACCAGCTG 71 37214 37233 46 411898 687 706 5-10-5 GTTGTGTGTCTTCACCAGCT 68 37215 37234 61 423543 687 706 3-14-3 GTTGTGTGTCTTCACCAGCT 56 37215 37234 61 423621 687 706 2-13-5 GTTGTGTGTCTTCACCAGCT 65 37215 37234 61 217316 688 707 5-10-5 GGTTGTGTGTCTTCACCAGC 70 37216 37235 20 423544 688 707 3-14-3 GGTTGTGTGTCTTCACCAGC 70 37216 37235 20 423622 688 707 2-13-5 GGTTGTGTGTCTTCACCAGC 68 37216 37235 20 381727 689 708 5-10-5 AGGTTGTGTGTCTTCACCAG 74 37217 37236 39 423545 689 708 3-14-3 AGGTTGTGTGTCTTCACCAG 61 37217 37236 39 423623 689 708 2-13-5 AGGTTGTGTGTCTTCACCAG 61 37217 37236 39 411899 690 709 5-10-5 CAGGTTGTGTGTCTTCACCA 76 37218 37237 62 423546 690 709 3-14-3 CAGGTTGTGTGTCTTCACCA 68 37218 37237 62 423624 690 709 2-13-5 CAGGTTGTGTGTCTTCACCA 62 37218 37237 62 411874 691 710 5-10-5 GCAGGTTGTGTGTCTTCACC 73 37219 37238 47 423547 691 710 3-14-3 GCAGGTTGTGTGTCTTCACC 69 37219 37238 47 423625 691 710 2-13-5 GCAGGTTGTGTGTCTTCACC 56 37219 37238 47 411900 692 711 5-10-5 AGCAGGTTGTGTGTCTTCAC 77 37220 37239 63 423548 692 711 3-14-3 AGCAGGTTGTGTGTCTTCAC 63 37220 37239 63 423626 692 711 2-13-5 AGCAGGTTGTGTGTCTTCAC 44 37220 37239 63 217317 693 712 5-10-5 CAGCAGGTTGTGTGTCTTCA 75 37221 37240 21 423549 693 712 3-14-3 CAGCAGGTTGTGTGTCTTCA 47 37221 37240 21 423627 693 712 2-13-5 CAGCAGGTTGTGTGTCTTCA 57 37221 37240 21 381728 694 713 5-10-5 TCAGCAGGTTGTGTGTCTTC 56 37222 37241 40 423550 694 713 3-14-3

TCAGCAGGTTGTGTGTCTTC 54 37222 37241 40 423628 694 713 2-13-5
TCAGCAGGTTGTGTGTCTTC 57 37222 37241 40 411901 695 714 5-10-5
GTCAGCAGGTTGTGTGTCTT 59 37223 37242 64 423551 695 714 3-14-3
GTCAGCAGGTTGTGTGTCTT 61 37223 37242 64 423629 695 714 2-13-5
GTCAGCAGGTTGTGTGTCTT 58 37223 37242 64 411875 696 715 5-10-5
GGTCAGCAGGTTGTGTGTCT 62 37224 37243 48 423552 696 715 3-14-3
GGTCAGCAGGTTGTGTGTCT 57 37224 37243 48 423630 696 715 2-13-5
GGTCAGCAGGTTGTGTGTCT 45 37224 37243 48 411902 697 716 5-10-5
TGGTCAGCAGGTTGTGTGTC 43 37225 37244 65 423553 697 716 3-14-3
TGGTCAGCAGGTTGTGTGTC 57 37225 37244 65 423631 697 716 2-13-5
TGGTCAGCAGGTTGTGTGTC 57 37225 37244 65 413231 814 833 5-10-5
GGTAAGGCCGTATGCCTGGG 71 37342 37361 221 423535 814 833 3-14-3
GGTAAGGCCGTATGCCTGGG 51 37342 37361 221 423613 814 833 2-13-5
GGTAAGGCCGTATGCCTGGG 53 37342 37361 221 423472 815 834 5-10-5
AGGTAAGGCCGTATGCCTGG 72 37343 37362 476 423536 815 834 3-14-3
AGGTAAGGCCGTATGCCTGG 67 37343 37362 476 423614 815 834 2-13-5
AGGTAAGGCCGTATGCCTGG 63 37343 37362 476 423473 816 835 5-10-5
CAGGTAAGGCCGTATGCCTG 63 37344 37363 477 423537 816 835 3-14-3
CAGGTAAGGCCGTATGCCTG 72 37344 37363 477 423615 816 835 2-13-5
CAGGTAAGGCCGTATGCCTG 59 37344 37363 477 413232 817 836 5-10-5
CCAGGTAAGGCCGTATGCCT 76 37345 37364 225 423538 817 836 3-14-3
CCAGGTAAGGCCGTATGCCT 71 37345 37364 225 423616 817 836 2-13-5
CCAGGTAAGGCCGTATGCCT 64 37345 37364 225 423474 818 837 5-10-5
GCCAGGTAAGGCCGTATGCC 68 37346 37365 478 423539 818 837 3-14-3
GCCAGGTAAGGCCGTATGCC 61 37346 37365 478 423617 818 837 2-13-5
GCCAGGTAAGGCCGTATGCC 66 37346 37365 478 423475 819 838 5-10-5
AGCCAGGTAAGGCCGTATGC 70 37347 37366 479 423540 819 838 3-14-3
AGCCAGGTAAGGCCGTATGC 51 37347 37366 479 423618 819 838 2-13-5
AGCCAGGTAAGGCCGTATGC 67 37347 37366 479 413233 820 839 5-10-5
TAGCCAGGTAAGGCCGTATG 60 37348 37367 226 423541 820 839 3-14-3
TAGCCAGGTAAGGCCGTATG 49 37348 37367 226 423619 820 839 2-13-5
TAGCCAGGTAAGGCCGTATG 51 37348 37367 226 217328 938 957 5-10-5
GCATTGCCACTCCCATTCTT 82 38091 38110 144
TABLE-US-00009 TABLE 10 Inhibition of DGAT2 mRNA by MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Motif Sequence tion Site Site NO 413251 935 954 5-10-5 TTGCCACTCC 57 38088 38107 244 CATTCTTTGA 423476 935 954 3-14-3 TTGCCACTCC 59 38088 38107 244 CATTCTTTGA 423554 935 954 2-13-5 TTGCCACTCC 57 38088 38107 244 CATTCTTTGA 423435 936 955 5-10-5 ATTGCCACTC 35 38089 38108 480 CCATTCTTTG 423477 936 955 3-14-3 ATTGCCACTC 40 38089 38108 480 CCATTCTTTG 423555 936 955 2-13-5 ATTGCCACTC 53 38089 38108 480 CCATTCTTTG 423436 937 956 5-10-5 CATTGCCACT 44 38090 38109 481 CCCATTCTTT 423478 937 956 3-14-3 CATTGCCACT 46 38090 38109 481 CCCATTCTTT 423556 937 956 2-13-5 CATTGCCACT 40 38090 38109 481 CCCATTCTTT 217328 938 957 5-10-5 GCATTGCCAC 85 38091 38110 144 TCCCATTCTT 423479 938 957 3-14-3 GCATTGCCAC 85 38091 38110 144 TCCCATTCTT 423557 938 957 2-13-5 GCATTGCCAC 61 38091 38110 144 TCCCATTCTT 380133 939 958 5-10-5 AGCATTGCCA 74 38092 38111 482 CTCCCATTCT 423480 939 958 3-14-3 AGCATTGCCA 67 38092 38111 482 CTCCCATTCT 423558 939 958 2-13-5 AGCATTGCCA 50 38092 38111 482 CTCCCATTCT 423437 940 959 5-10-5 TAGCATTGCC 71 38093 38112 483 ACTCCCATTC 423481 940 959 3-14-3

TAGCATTGCC 68 38093 38112 483 ACTCCCATTC 423559 940 959 2-13-5 TAGCATTGCC 60 38093 38112 483 ACTCCCATTC 423438 941 960 5-10-5 ATAGCATTGC 57 38094 38113 484 CACTCCCATT 423482 941 960 3-14-3 ATAGCATTGC 71 38094 38113 484 CACTCCCATT 423560 941 960 2-13-5 ATAGCATTGC 57 38094 38113 484 CACTCCCATT 423439 942 961 5-10-5 GATAGCATTG 66 38095 38114 485 CCACTCCCAT 423483 942 961 3-14-3 GATAGCATTG 75 38095 38114 485 CCACTCCCAT 423561 942 961 2-13-5 GATAGCATTG 62 38095 38114 485 CCACTCCCAT 217329 943 962 5-10-5 TGATAGCATT 64 38096 38115 486 GCCACTCCCA 423484 943 962 3-14-3 TGATAGCATT 66 38096 38115 486 GCCACTCCCA 423562 943 962 2-13-5 TGATAGCATT 69 38096 38115 486 GCCACTCCCA 413252 976 995 5-10-5 TCAGAGACTC 55 38129 38148 245 AGCCGCACCC 423485 976 995 3-14-3 TCAGAGACTC 41 38129 38148 245 AGCCGCACCC 423563 976 995 2-13-5 TCAGAGACTC 56 38129 38148 245 AGCCGCACCC 423440 977 996 5-10-5 CTCAGAGACT 80 38130 38149 487 CAGCCGCACC 423486 977 996 3-14-3 CTCAGAGACT 70 38130 38149 487 CAGCCGCACC 423564 977 996 2-13-5 CTCAGAGACT 43 38130 38149 487 CAGCCGCACC 423441 978 997 5-10-5 GCTCAGAGAC 86 38131 38150 488 TCAGCCGCAC 423487 978 997 3-14-3 GCTCAGAGAC 69 38131 38150 488 TCAGCCGCAC 423565 978 997 2-13-5 GCTCAGAGAC 82 38131 38150 488 TCAGCCGCAC 366730 979 998 5-10-5 AGCTCAGAGA 85 38132 38151 150 CTCAGCCGCA 423488 979 998 3-14-3 AGCTCAGAGA 77 38132 38151 150 CTCAGCCGCA 423566 979 998 2-13-5 AGCTCAGAGA 86 38132 38151 150 CTCAGCCGCA 423442 980 999 5-10-5 GAGCTCAGAG 81 38133 38152 489 ACTCAGCCGC 423489 980 999 3-14-3 GAGCTCAGAG 89 38133 38152 489 ACTCAGCCGC 423567 980 999 2-13-5 GAGCTCAGAG 89 38133 38152 489 ACTCAGCCGC 423443 981 1000 5-10-5 GGAGCTCAGA 73 38134 38153 490 GACTCAGCCG 423490 981 1000 3-14-3 GGAGCTCAGA 81 38134 38153 490 GACTCAGCCG 423568 981 1000 2-13-5 GGAGCTCAGA 65 38134 38153 490 GACTCAGCCG 423444 982 1001 5-10-5 TGGAGCTCAG 77 38135 38154 153 AGACTCAGCC 423491 982 1001 3-14-3 TGGAGCTCAG 76 38135 38154 153 AGACTCAGCC 423569 982 1001 2-13-5 TGGAGCTCAG 82 38135 38154 153 AGACTCAGCC 423445 983 1002 5-10-5 ATGGAGCTCA 78 38136 38155 491 GAGACTCAGC 423492 983 1002 3-14-3 ATGGAGCTCA 71 38136 38155 491 GAGACTCAGC 423570 983 1002 2-13-5 ATGGAGCTCA 73 38136 38155 491 GAGACTCAGC 366731 984 1003 5-10-5 CATGGAGCTC 65 38137 38156 492 AGAGACTCAG 423493 984 1003 3-14-3 CATGGAGCTC 62 38137 38156 492 AGAGACTCAG 423571 984 1003 2-13-5 CATGGAGCTC 71 38137 38156 492 AGAGACTCAG 423446 985 1004 5-10-5 GCATGGAGCT 69 38138 38157 493 CAGAGACTCA 423494 985 1004 3-14-3 GCATGGAGCT 51 38138 38157 493 CAGAGACTCA 423572 985 1004 2-13-5 GCATGGAGCT 77 38138 38157 493 CAGAGACTCA 423447 986 1005 5-10-5 GGCATGGAGC 77 38139 38158 494 TCAGAGACTC 423495 986 1005 3-14-3 GGCATGGAGC 74 38139 38158 494 TCAGAGACTC 423573 986 1005 2-13-5 GGCATGGAGC 77 38139 38158 494 TCAGAGACTC 413253 987 1006 5-10-5 AGGCATGGAG 83 38140 38159 246 CTCAGAGACT 423496 987 1006 3-14-3 AGGCATGGAG 66 38140 38159 246 CTCAGAGACT 423574 987 1006 2-13-5 AGGCATGGAG 78 38140 38159 246 CTCAGAGACT 423448 988 1007 5-10-5 CAGGCATGGA 57 38141 38160 495 GCTCAGAGAC 423497 988 1007 3-14-3 CAGGCATGGA 76 38141 38160 495 GCTCAGAGAC 423575 988 1007 2-13-5 CAGGCATGGA 70 38141 38160 495 GCTCAGAGAC 369255 1339 1358 5-10-5 CCAGGGCCTC 83 41310 41329 103 CATGTACATG 423498 1339 1358 3-14-3 CCAGGGCCTC 86 41310 41329 103 CATGTACATG 423576 1339 1358 2-13-5 CCAGGGCCTC 69 41310 41329 103 CATGTACATG 411944 1340 1359 5-10-5 ACCAGGGCCT 85 41311 41330 120 CCATGTACAT 423499 1340 1359 3-14-3 ACCAGGGCCT 84 41311 41330 120 CCATGTACAT 423577 1340 1359 2-13-5 ACCAGGGCCT 50 41311 41330 120 CCATGTACAT 411945 1341

1360 5-10-5 CACCAGGGCC 80 41312 41331 121 TCCATGTACA 423500 1341 1360 3-14-3 CACCAGGGCC 41 41312 41331 121 TCCATGTACA 423578 1341 1360 2-13-5 CACCAGGGCC 69 41312 41331 121 TCCATGTACA 411946 1342 1361 5-10-5 TCACCAGGGC 65 41313 41332 122 CTCCATGTAC 423501 1342 1361 3-14-3 TCACCAGGGC 68 41313 41332 122 CTCCATGTAC 423579 1342 1361 2-13-5 TCACCAGGGC 3 41313 41332 122 CTCCATGTAC

Example 3: Antisense Inhibition of Human DGAT2 in HepG2 Cells by MOE Gapmers

[0455] Antisense oligonucleotides were designed targeting a DGAT2 nucleic acid and were tested for their effects on DGAT2 mRNA in vitro. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. Previously disclosed oligonucleotides, ISIS 217317, 217328, ISIS 366722, ISIS 366710, ISIS 366714, ISIS 366728, ISIS 366746, and ISIS 369255 were included in this study as benchmark oligonucleotides. The results for each experiment are presented in separate tables shown below. Cultured HepG2 cells at a density of 20,000 cells per well were transfected using electroporation with 4,500 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988 MGB was used to measure mRNA levels. The potency of some oligonucleotides was measured with human primer probe set RTS2367 (forward sequence GGCCTCCCGGAGACTGA, designated herein as SEQ ID NO: 4; reverse sequence AAGTGATTTGCAGCTGGTTCCT, designated herein as SEQ ID NO: 5; probe sequence AGGTGAACTGAGCCAGCCTTCGGG, designated herein as SEQ ID NO: 6). DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. The results show several antisense oligonucleotides demonstrate greater potency than the benchmark oligonucleotides.

[0456] The newly designed chimeric antisense oligonucleotides in the Tables below were designed as 5-10-5 MOE gapmers. The 5-10-5 MOE gapmers are 20 nucleosides in length, wherein the central gap segment comprises of ten 2′-deoxynucleosides and is flanked by wing segments on the 5′ direction and the 3′ direction comprising five nucleosides each. Each nucleoside in the 5′ wing segment and each nucleoside in the 3′ wing segment has a 2′-MOE modification. The internucleoside linkages throughout each gapmer are phosphorothioate (P═S) linkages. All cytosine residues throughout each gapmer are 5-methylcytosines. “Start site” indicates the 5′-most nucleoside to which the gapmer is targeted in the human gene sequence. “Stop site” indicates the 3′-most nucleoside to which the gapmer is targeted in the human gene sequence. Each gapmer listed in the Tables below is targeted to SEQ ID NO: 1, SEQ ID NO: 2, or both. ‘n/a’ indicates that the antisense oligonucleotide does not target that particular gene sequence with 100% complementarity. In case the sequence alignment for a target gene sequence in a particular table is not shown, it is understood that none of the oligonucleotides presented in that table align with 100% complementarity with that target gene sequence.

TABLE-US-00010 TABLE 7 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ SEQ ID ID NO: NO: 2 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 334177 9896 9915 AGGACCCCGGAGTAGGCGGC 76 145 366710 25551 25570 GACCTATTGAGCCAGGTGAC 60 146 366722 31097 31116 GTGCAGAATATGTACATGAG 50 148 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 80 425 523014 35670 35689 TTCTACTCAAAAATATACAT 23 4084 495875 36249 36268 TAAGGGTATTCTGCTCATAA 73 2148 495876 36250 36269 GTAAGGGTATTCTGCTCATA 83 2149 495877 36251 36270 AGTAAGGGTATTCTGCTCAT 82 2150 495878 36252 36271 GAGTAAGGGTATTCTGCTCA 89 2151 523015 36254 36273 ATGAGTAAGGGTATTCTGCT 68 4085 523016 36255 36274 AATGAGTAAGGGTATTCTGC 65 4086 523017 36256 36275 CAATGAGTAAGGGTATTCTG 39 4087 523018 36257 36276 ACAATGAGTAAGGGTATTCT 25 4088 495879 36259 36278

AGACAATGAGTAAGGGTATT 64 2152 495880 36260 36279 GAGACAATGAGTAAGGGTAT 62 2153 495881 36261 36280 AGAGACAATGAGTAAGGGTA 64 2154 495882 36262 36281 GAGAGACAATGAGTAAGGGT 66 2155 523019 36513 36532 CAGCCCTGTGCCAGCCAGCC 37 4089 523020 36514 36533 ACAGCCCTGTGCCAGCCAGC 17 4090 523021 36515 36534 GACAGCCCTGTGCCAGCCAG 52 4091 523022 36516 36535 CGACAGCCCTGTGCCAGCCA 62 4092 495886 36518 36537 AGCGACAGCCCTGTGCCAGC 57 2159 495887 36519 36538 AAGCGACAGCCCTGTGCCAG 46 2160 495888 36520 36539 CAAGCGACAGCCCTGTGCCA 28 2161 495889 36521 36540 ACAAGCGACAGCCCTGTGCC 46 2162 523023 36579 36598 ACTTTCCCACAGTCAGCTGG 59 4093 523024 36580 36599 AACTTTCCCACAGTCAGCTG 51 4094 523025 36581 36600 AAACTTTCCCACAGTCAGCT 68 4095 523026 36582 36601 GAAACTTTCCCACAGTCAGC 73 4096 523027 36584 36603 TGGAAACTTTCCCACAGTCA 68 4097 523028 36585 36604 ATGGAAACTTTCCCACAGTC 52 4098 523029 36586 36605 AATGGAAACTTTCCCACAGT 23 4099 523030 36587 36606 AAATGGAAACTTTCCCACAG 36 4100 523031 36842 36861 AAATCCAACCCTTGTCACTC 28 4101 523032 36843 36862 CAAATCCAACCCTTGTCACT 22 4102 523033 36844 36863 TCAAATCCAACCCTTGTCAC 21 4103 523034 36845 36864 CTCAAATCCAACCCTTGTCA 38 4104 523035 37120 37139 TGGATGTGTCATTTCCCCTG 69 4105 217328 38091 38110 GCATTGCCACTCCCATTCTT 73 144 423444 38135 38154 TGGAGCTCAGAGACTCAGCC 56 153 522979 39671 39690 AGTGCAAGAGTTACCTCCTC 67 4106 522980 39672 39691 CAGTGCAAGAGTTACCTCCT 67 4107 522981 39673 39692 GCAGTGCAAGAGTTACCTCC 60 4108 522982 39674 39693 AGCAGTGCAAGAGTTACCTC 63 4109 522983 39681 39700 ATCAGTTAGCAGTGCAAGAG 46 4110 522984 39682 39701 TATCAGTTAGCAGTGCAAGA 43 4111 522985 39683 39702 CTATCAGTTAGCAGTGCAAG 52 4112 522986 39684 39703 CCTATCAGTTAGCAGTGCAA 64 4113 522987 39686 39705 TTCCTATCAGTTAGCAGTGC 61 4114 522988 39687 39706 ATTCCTATCAGTTAGCAGTG 47 4115 522989 39688 39707 AATTCCTATCAGTTAGCAGT 16 4116 522990 39689 39708 TAATTCCTATCAGTTAGCAG 11 4117 522991 39814 39833 TCTATGCTGCAGTCATATTA 28 4118 522992 39815 39834 GTCTATGCTGCAGTCATATT 20 4119 522993 39816 39835 GGTCTATGCTGCAGTCATAT 45 4120 522994 39817 39836 AGGTCTATGCTGCAGTCATA 40 4121 522995 39819 39838 ACAGGTCTATGCTGCAGTCA 56 4122 522996 39820 39839 GACAGGTCTATGCTGCAGTC 70 4123 522997 39821 39840 TGACAGGTCTATGCTGCAGT 65 4124 522998 39822 39841 CTGACAGGTCTATGCTGCAG 54 4125 522999 39829 39848 CACTCTTCTGACAGGTCTAT 54 4126 523000 39830 39849 CCACTCTTCTGACAGGTCTA 70 4127 523001 39831 39850 TCCACTCTTCTGACAGGTCT 69 4128 523002 39832 39851 TTCCACTCTTCTGACAGGTC 84 4129 523003 39834 39853 TTTTCCACTCTTCTGACAGG 35 4130 523004 39835 39854 GTTTTCCACTCTTCTGACAG 25 4131 523005 39836 39855 TGTTTTCCACTCTTCTGACA 24 4132 523006 40329 40348 ATCTCTTCCATCTCCACTGC 30 4133 523007 40330 40349 CATCTCTTCCATCTCCACTG 32 4134 523008 40331 40350 ACATCTCTTCCATCTCCACT 39 4135 523009 40332 40351 CACATCTCTTCCATCTCCAC 40 4136 523010 40334 40353 TGCACATCTCTTCCATCTCC 50 4137 523011 40335 40354 ATGCACATCTCTTCCATCTC 35 4138 523012 40336 40355 CATGCACATCTCTTCCATCT 43 4139 523013 40337 40356 TCATGCACATCTCTTCCATC 28 4140 369255 41310 41329 CCAGGGCCTCCATGTACATG 33 103 411947 41315 41334 CTTCACCAGGGCCTCCATGT 22 123 366746 41337 41356 TGGTCTTGTGCTTGTCGAAG 46 151

TABLE-US-00011 TABLE 11 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NOs: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 413433 n/a n/a GCCTGGACAA 82 32431 32450 425 GTCCTGCCCA 472154 105 124 GACACCTGGA 51 9726 9745 573 GCTGCGGCCG 472155 106 125 GGACACCTGG

79 9727 9746 574 AGCTGCGGCC 472156 107 126 AGGACACCTG 79 9728 9747 575 GAGCTGCGGC 472157 108 127 TAGGACACCT 56 9729 9748 576 GGAGCTGCGG 472158 110 129 GCTAGGACAC 76 9731 9750 577 CTGGAGCTGC 472159 111 130 GGCTAGGACA 63 9732 9751 578 CCTGGAGCTG 472160 161 180 CAGGGCTTCG 46 9782 9801 579 CGCAGAGCAC 472161 162 181 CCAGGGCTTC 42 9783 9802 580 GCGCAGAGCA 472162 164 183 GGCCAGGGCT 6 9785 9804 581 TCGCGCAGAG 472163 253 272 TGAGGGTCTT 40 9874 9893 582 CATGGCTGAA 472164 255 274 TATGAGGGTC 69 9876 9895 583 TTCATGGCTG 472165 256 275 CTATGAGGGT 62 9877 9896 584 CTTCATGGCT 472166 257 276 GCTATGAGGG 53 9878 9897 585 TCTTCATGGC 472167 259 278 CGGCTATGAG 59 9880 9899 586 GGTCTTCATG 472168 260 279 GCGGCTATGA 54 9881 9900 587 GGGTCTTCAT 472169 261 280 GGCGGCTATG 57 9882 9901 588 AGGGTCTTCA 472170 262 281 AGGCGGCTAT 69 9883 9902 589 GAGGGTCTTC 472171 264 283 GTAGGCGGCT 32 9885 9904 590 ATGAGGGTCT 472172 265 284 AGTAGGCGGC 23 9886 9905 591 TATGAGGGTC 472173 267 286 GGAGTAGGCG 15 9888 9907 592 GCTATGAGGG 472174 268 287 CGGAGTAGGC 48 9889 9908 593 GGCTATGAGG 472175 270 289 CCCGGAGTAG 80 9891 9910 594 GCGGCTATGA 472176 271 290 CCCCGGAGTA 75 9892 9911 595 GGCGGCTATG 472177 273 292 GACCCCGGAG 69 9894 9913 596 TAGGCGGCTA 472178 274 293 GGACCCCGGA 76 9895 9914 597 GTAGGCGGCT 334177 275 294 AGGACCCCGG 74 9896 9915 145 AGTAGGCGGC 472179 305 324 CGGTCAGCCT 30 9926 9945 598 CGGCCTGACG 472180 306 325 CCGGTCAGCC 5 9927 9946 599 TCGGCCTGAC 472181 308 327 CTCCGGTCAG 43 9929 9948 600 CCTCGGCCTG 472182 309 328 GCTCCGGTCA 61 9930 9949 601 GCCTCGGCCT 472183 311 330 TGGCTCCGGT 48 9932 9951 602 CAGCCTCGGC 472184 312 331 CTGGCTCCGG 28 9933 9952 603 TCAGCCTCGG 472185 313 332 GCTGGCTCCG 35 9934 9953 604 GTCAGCCTCG 472186 314 333 CGCTGGCTCC 44 9935 9954 605 GGTCAGCCTC 472187 315 334 GCGCTGGCTC 43 9936 9955 606 CGGTCAGCCT 472188 332 351 GCAGGTCCTC 63 9953 9972 607 CGTGAGAGCG 472189 333 352 CGCAGGTCCT 68 9954 9973 608 CCGTGAGAGC 472190 334 353 GCGCAGGTCC 40 9955 9974 609 TCCGTGAGAG 472191 335 354 AGCGCAGGTC 39 9956 9975 610 CTCCGTGAGA 472192 336 355 CAGCGCAGGT 54 9957 9976 611 CCTCCGTGAG 472193 337 356 ACAGCGCAGG 36 9958 9977 612 TCCTCCGTGA 472194 338 357 GACAGCGCAG 63 9959 9978 613 GTCCTCCGTG 472195 340 359 GCGACAGCGC 62 9961 9980 614 AGGTCCTCCG 472196 341 360 CGCGACAGCG 65 9962 9981 615 CAGGTCCTCC 472197 342 361 GCGCGACAGC 57 9963 9982 616 GCAGGTCCTC 472198 343 362 CGCGCGACAG 58 9964 9983 617 CGCAGGTCCT 472199 344 363 TCGCGCGACA 53 9965 9984 618 GCGCAGGTCC 472200 345 364 CTCGCGCGAC 52 9966 9985 619 AGCGCAGGTC 472201 346 365 CCTCGCGCGA 26 9967 9986 620 CAGCGCAGGT 472202 347 366 CCCTCGCGCG 32 9968 9987 621 ACAGCGCAGG 472203 349 368 ACCCCTCGCG 67 9970 9989 622 CGACAGCGCA 472204 350 369 GACCCCTCGC 72 9971 9990 623 GCGACAGCGC 472205 351 370 AGACCCCTCG 74 9972 9991 624 CGCGACAGCG 472206 352 371 CAGACCCCTC 67 9973 9992 625 GCGCGACAGC 472207 353 372 CCAGACCCCT 53 9974 9993 626 CGCGCGACAG 472208 354 373 CCCAGACCCC 70 9975 9994 627 TCGCGCGACA 472209 356 375 CTCCCAGACC 50 9977 9996 628 CCTCGCGCGA 472210 357 376 TCTCCCAGAC 31 9978 9997 629 CCCTCGCGCG 472211 358 377 ATCTCCCAGA 54 9979 9998 630 CCCCTCGCGC 472212 359 378 CATCTCCCAG 51 9980 9999 631 ACCCCTCGCG 472213 360 379 CCATCTCCCA 42 9981 10000 632 GACCCCTCGC 472214 362 381 CCCCATCTCC 48 9983 10002 633 CAGACCCCTC 472215 363 382 GCCCCATCTC 32 n/a n/a 634 CCAGACCCCT 472216 364 383 TGCCCATCT 46 n/a n/a 635 CCCAGACCCC 472217 365

384 GTGCCCCATC 43 n/a n/a 636 TCCCAGACCC 472218 366 385 AGTGCCCCAT 19 n/a n/a 637 CTCCCAGACC 472219 368 387 CCAGTGCCCC 9 n/a n/a 638 ATCTCCCAGA 472220 369 388 TCCAGTGCCC 0 n/a n/a 639 CATCTCCCAG 472221 371 390 GATCCAGTGC 0 n/a n/a 640 CCCATCTCCC 472222 372 391 GGATCCAGTG 2 n/a n/a 641 CCCCATCTCC 472223 374 393 CTGGATCCAG 31 n/a n/a 642 TGCCCCATCT 472224 375 394 GCTGGATCCA 39 n/a n/a 643 GTGCCCCATC 472225 377 396 ATGCTGGATC 36 n/a n/a 644 CAGTGCCCCA 472226 378 397 GATGCTGGAT 18 n/a n/a 645 CCAGTGCCCC 472227 380 399 AGGATGCTGG 41 25506 25525 646 ATCCAGTGCC 472228 381 400 GAGGATGCTG 17 25507 25526 647 GATCCAGTGC 472229 383 402 GAGAGGATGC 6 25509 25528 648 TGGATCCAGT

TABLE-US-00012 TABLE 12 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NOs: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 413433 n/a n/a GCCTGGACAA 75 32431 32450 425 GTCCTGCCCA 472373 782 801 ACTTCTGTGG 31 37310 37329 649 CCTCTGTGCT 472374 784 803 TCACTTCTGT 42 37312 37331 650 GGCCTCTGTG 472375 785 804 CTCACTTCTG 53 37313 37332 651 TGGCCTCTGT 472376 787 806 TGCTCACTTC 66 37315 37334 652 TGTGGCCTCT 472377 788 807 TTGCTCACTT 78 37316 37335 653 CTGTGGCCTC 472378 790 809 TCTTGCTCAC 57 37318 37337 654 TTCTGTGGCC 472379 791 810 TTCTTGCTCA 43 37319 37338 655 CTTCTGTGGC 472380 793 812 ACTTCTTGCT 41 37321 37340 656 CACTTCTGTG 472381 794 813 AACTTCTTGC 14 37322 37341 657 TCACTTCTGT 472382 796 815 GGAACTTCTT 39 37324 37343 658 GCTCACTTCT 472383 797 816 GGGAACTTCT 47 37325 37344 659 TGCTCACTTC 472384 799 818 CTGGGAACTT 65 37327 37346 660 CTTGCTCACT 472385 800 819 CCTGGGAACT 0 37328 37347 661 TCTTGCTCAC 472386 802 821 TGCCTGGGAA 54 37330 37349 662 CTTCTTGCTC 472387 803 822 ATGCCTGGGA 24 37331 37350 663 ACTTCTTGCT 472388 805 824 GTATGCCTGG 26 37333 37352 664 GAACTTCTTG 472389 806 825 CGTATGCCTG 46 37334 37353 665 GGAACTTCTT 472390 807 826 CCGTATGCCT 41 37335 37354 666 GGGAACTTCT 472391 809 828 GGCCGTATGC 47 37337 37356 667 CTGGGAACTT 472392 810 829 AGGCCGTATG 45 37338 37357 668 CCTGGGAACT 472393 812 831 TAAGGCCGTA 38 37340 37359 669 TGCCTGGGAA 472394 813 832 GTAAGGCCGT 53 37341 37360 670 ATGCCTGGGA 472395 821 840 GTAGCCAGGT 53 37349 37368 671 AAGGCCGTAT 472396 822 841 TGTAGCCAGG 41 37350 37369 672 TAAGGCCGTA 472397 824 843 AGTGTAGCCA 36 37352 37371 673 GGTAAGGCCG 472398 825 844 CAGTGTAGCC 63 37353 37372 674 AGGTAAGGCC 472399 826 845 CCAGTGTAGC 45 37354 37373 675 CAGGTAAGGC 472400 862 881 GGTACTCCCT 42 37390 37409 676 CAACACAGGC 472401 863 882 AGGTACTCCC 11 37391 37410 677 TCAACACAGG 472402 865 884 TCAGGTACTC 20 37393 37412 678 CCTCAACACA 472403 866 885 ATCAGGTACT 13 37394 37413 679 CCCTCAACAC 472404 868 887 ACATCAGGTA 11 37396 37415 680 CTCCCTCAAC 472405 869 888 GACATCAGGT 27 37397 37416 681 ACTCCCTCAA 472406 871 890 CAGACATCAG 39 37399 37418 682 GTACTCCCTC 472407 872 891 CCAGACATCA 46 37400 37419 683 GGTACTCCCT 472408 873 892 TCCAGACATC 32 37401 37420 684 AGGTACTCCC 472409 874 893 CTCCAGACAT 29 37402 37421 685 CAGGTACTCC 472410 879 898 GATACCTCCA 33 n/a n/a 686 GACATCAGGT 472411 880 899 AGATACCTCC 17 n/a n/a 687 AGACATCAGG 472412 882 901 GCAGATACCT 9 n/a n/a 688 CCAGACATCA 472413 883 902 GGCAGATACC 44 n/a n/a 689 TCCAGACATC 472414 885 904 AGGGCAGATA 0 n/a n/a 690 CCTCCAGACA 472415 886 905 CAGGGCAGAT 16 n/a n/a 691 ACCTCCAGAC 472416 888 907 GACAGGGCAG 1 n/a n/a 692 ATACCTCCAG 472417 889 908 TGACAGGGCA 31 n/a n/a 693 GATACCTCCA 472418 911 930 AAATAGTCTA 25 38064 38083 694 TGGTGTCCCG

472419 912 931 CAAATAGTCT 32 38065 38084 695 ATGGTGTCCC 472420 914 933 AGCAAATAGT 54 38067 38086 696 CTATGGTGTC 472421 915 934 AAGCAAATAG 22 38068 38087 697 TCTATGGTGT 472422 917 936 GAAAGCAAAT 24 38070 38089 698 AGTCTATGGT 472423 918 937 TGAAAGCAAA 19 38071 38090 699 TAGTCTATGG 472424 920 939 TTTGAAAGCA 17 38073 38092 700 AATAGTCTAT 472425 921 940 CTTTGAAAGC 0 38074 38093 701 AAATAGTCTA 472426 923 942 TTCTTTGAAA 24 38076 38095 702 GCAAATAGTC 472427 924 943 ATTCTTTGAA 9 38077 38096 703 AGCAAATAGT 472428 926 945 CCATTCTTTG 31 38079 38098 704 AAAGCAAATA 472429 927 946 CCCATTCTTT 48 38080 38099 705 GAAAGCAAAT 380132 929 948 CTCCCATTCT 75 38082 38101 706 TTGAAAGCAA 472430 930 949 ACTCCCATTC 35 38083 38102 707 TTTGAAAGCA 472431 931 950 CACTCCCATT 35 38084 38103 708 CTTTGAAAGC 472432 933 952 GCCACTCCCA 69 38086 38105 709 TTCTTTGAAA 472433 934 953 TGCCACTCCC 73 38087 38106 710 ATTCTTTGAA 217328 938 957 GCATTGCCAC 76 38091 38110 144 TCCCATTCTT 472434 944 963 ATGATAGCAT 65 38097 38116 711 TGCCACTCCC 472435 945 964 GATGATAGCA 50 38098 38117 712 TTGCCACTCC 472436 946 965 TGATGATAGC 56 38099 38118 713 ATTGCCACTC 380134 947 966 ATGATGATAG 34 38100 38119 714 CATTGCCACT 472437 949 968 CGATGATGAT 56 38102 38121 715 AGCATTGCCA 472438 950 969 ACGATGATGA 51 38103 38122 716 TAGCATTGCC 472439 951 970 CACGATGATG 56 38104 38123 717 ATAGCATTGC 472440 952 971 CCACGATGAT 63 38105 38124 718 GATAGCATTG 472441 974 993 AGAGACTCAG 63 38127 38146 719 CCGCACCCCC 472442 975 994 CAGAGACTCA 66 38128 38147 720 GCCGCACCCC 366730 979 998 AGCTCAGAGA 67 38132 38151 150 CTCAGCCGCA 423442 980 999 GAGCTCAGAG 52 38133 38152 489 ACTCAGCCGC 423444 982 1001 TGGAGCTCAG 63 38135 38154 153 AGACTCAGCC 472443 989 1008 CCAGGCATGG 47 38142 38161 721 AGCTCAGAGA

TABLE-US-00013 TABLE 13 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 413433 n/a n/a GCCTGGACAA 94 32431 32450 425 GTCCTGCCCA 472230 384 403 GGAGAGGATG 38 25510 25529 722 CTGGATCCAG 472231 386 405 GCGGAGAGGA 41 25512 25531 723 TGCTGGATCC 472232 387 406 GGCGGAGAGG 28 25513 25532 724 ATGCTGGATC 472233 389 408 AGGGCGGAGA 0 25515 25534 725 GGATGCTGGA 472234 390 409 GAGGGCGGAG 66 25516 25535 726 AGGATGCTGG 472235 391 410 GGAGGGCGGA 28 25517 25536 727 GAGGATGCTG 472236 392 411 TGGAGGGCGG 35 25518 25537 728 AGAGGATGCT 472237 394 413 CCTGGAGGGC 9 25520 25539 729 GGAGAGGATG 472238 395 414 TCCTGGAGGG 0 25521 25540 730 CGGAGAGGAT 472239 397 416 GGTCCTGGAG 0 25523 25542 731 GGCGGAGAGG 472240 398 417 AGGTCCTGGA 32 25524 25543 732 GGGCGGAGAG 472241 399 418 GAGGTCCTGG 33 25525 25544 733 AGGGCGGAGA 472242 400 419 AGAGGTCCTG 0 25526 25545 734 GAGGGCGGAG 472243 401 420 AAGAGGTCCT 35 25527 25546 735 GGAGGGCGGA 472244 402 421 GAAGAGGTCC 16 25528 25547 736 TGGAGGGCGG 472245 403 422 AGAAGAGGTC 53 25529 25548 737 CTGGAGGGCG 366710 425 444 GACCTATTGA 62 25551 25570 146 GCCAGGTGAC 472246 426 445 GGACCTATTG 63 25552 25571 738 AGCCAGGTGA 472247 427 446 TGGACCTATT 58 25553 25572 739 GAGCCAGGTG 472248 428 447 TTGGACCTAT 71 25554 25573 740 TGAGCCAGGT 472249 429 448 CTTGGACCTA 63 25555 25574 741 TTGAGCCAGG 472250 431 450 ACCTTGGACC 80 25557 25576 742 TATTGAGCCA 472251 432 451 CACCTTGGAC 70 25558 25577 743 CTATTGAGCC 472252 433 452 CCACCTTGGA 74 25559 25578 744 CCTATTGAGC 472253 434 453 TCCACCTTGG 68 25560 25579 745 ACCTATTGAG 472254 436 455 TTTCCACCTT 41 25562 25581 746 GGACCTATTG 472255 437 456 TTTTCCACCT 43 25563 25582 747 TGGACCTATT 472256

438 457 CTTTTCCACC 48 25564 25583 748 TTGGACCTAT 472257 439 458 GCTTTTCCAC 75 25565 25584 749 CTTGGACCTA 472258 441 460 CTGCTTTTCC 83 25567 25586 750 ACCTTGGACC 472259 442 461 GCTGCTTTTC 83 25568 25587 751 CACCTTGGAC 472260 443 462 AGCTGCTTTT 84 25569 25588 152 CCACCTTGGA 472261 444 463 TAGCTGCTTT 76 25570 25589 752 TCCACCTTGG 366714 445 464 GTAGCTGCTT 70 25571 25590 147 TTCCACCTTG 472262 446 465 TGTAGCTGCT 39 25572 25591 753 TTTCCACCTT 472263 447 466 CTGTAGCTGC 72 25573 25592 754 TTTTCCACCT 380086 449 468 ACCTGTAGCT 70 25575 25594 755 GCTTTTCCAC 472264 450 469 GACCTGTAGC 77 25576 25595 756 TGCTTTTCCA 472265 452 471 ATGACCTGTA 57 25578 25597 757 GCTGCTTTTC 472266 453 472 GATGACCTGT 74 25579 25598 758 AGCTGCTTTT 472267 454 473 AGATGACCTG 67 25580 25599 759 TAGCTGCTTT 472268 456 475 TGAGATGACC 77 25582 25601 760 TGTAGCTGCT 472269 457 476 CTGAGATGAC 78 25583 25602 761 CTGTAGCTGC 472270 459 478 CACTGAGATG 80 25585 25604 762 ACCTGTAGCT 472271 460 479 GCACTGAGAT 77 25586 25605 763 GACCTGTAGC 472272 462 481 GAGCACTGAG 71 25588 25607 764 ATGACCTGTA 472273 463 482 GGAGCACTGA 75 25589 25608 765 GATGACCTGT 472274 465 484 CTGGAGCACT 72 25591 25610 766 GAGATGACCT 472275 466 485 ACTGGAGCAC 62 25592 25611 767 TGAGATGACC 423453 474 493 CAGGACCCAC 92 25600 25619 457 TGGAGCACTG 472276 477 496 GGACAGGACC 72 25603 25622 768 CACTGGAGCA 472277 478 497 AGGACAGGAC 74 25604 25623 769 CCACTGGAGC 472278 480 499 GAAGGACAGG 53 25606 25625 770 ACCCACTGGA 472279 481 500 GGAAGGACAG 48 25607 25626 771 GACCCACTGG 472280 483 502 AAGGAAGGAC 54 25609 25628 772 AGGACCCACT 472281 484 503 CAAGGAAGGA 40 25610 25629 773 CAGGACCCAC 472282 485 504 ACAAGGAAGG 65 25611 25630 774 ACAGGACCCA 472283 487 506 GTACAAGGAA 55 25613 25632 775 GGACAGGACC 472284 488 507 AGTACAAGGA 35 25614 25633 776 AGGACAGGAC 472285 489 508 CAGTACAAGG 40 25615 25634 777 AAGGACAGGA 472286 490 509 CCAGTACAAG 32 25616 25635 778 GAAGGACAGG 472287 492 511 TCCCAGTACA 58 n/a n/a 779 AGGAAGGACA 472288 493 512 CTCCCAGTAC 33 n/a n/a 780 AAGGAAGGAC 472289 495 514 CACTCCCAGT 47 n/a n/a 781 ACAAGGAAGG 472290 496 515 CCACTCCCAG 4 n/a n/a 782 TACAAGGAAG 472291 498 517 GGCCACTCCC 12 n/a n/a 783 AGTACAAGGA 472292 499 518 AGGCCACTCC 2 n/a n/a 784 CAGTACAAGG 472293 501 520 GCAGGCCACT 0 n/a n/a 785 CCCAGTACAA 472294 502 521 TGCAGGCCAC 0 n/a n/a 786 TCCCAGTACA 472295 504 523 ACTGCAGGCC 43 n/a n/a 787 ACTCCCAGTA 472296 505 524 CACTGCAGGC 14 n/a n/a 788 CACTCCCAGT 472297 507 526 GGCACTGCAG 73 n/a n/a 789 GCCACTCCCA 472298 508 527 TGGCACTGCA 25 n/a n/a 790 GGCCACTCCC 472299 509 528 ATGGCACTGC 46 31076 31095 791 AGGCCACTCC 472300 511 530 GGATGGCACT 43 31078 31097 792 GCAGGCCACT 472301 512 531 AGGATGGCAC 61 31079 31098 793 TGCAGGCCAC 472302 513 532 GAGGATGGCA 69 31080 31099 794 CTGCAGGCCA

TABLE-US-00014 TABLE 14 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 413433 n/a n/a GCCTGGACAA 90 32431 32450 425 GTCCTGCCCA 472303 514 533 TGAGGATGGC 45 31081 31100 795 ACTGCAGGCC 472304 516 535 CATGAGGATG 24 31083 31102 796 GCACTGCAGG 472305 517 536 ACATGAGGAT 19 31084 31103 797 GGCACTGCAG 472306 518 537 TACATGAGGA 59 31085 31104 798 TGGCACTGCA 472307 521 540 ATGTACATGA 32 31088 31107 799 GGATGGCACT 472308 522 541 TATGTACATG 22 31089 31108 800 AGGATGGCAC 472309 523 542 ATATGTACAT 39 31090 31109 801 GAGGATGGCA 472310 524 543 AATATGTACA 34 31091 31110 802 TGAGGATGGC 472311 526 545 AGAATATGTA 1 31093 31112 803 CATGAGGATG 472312 527 546 CAGAATATGT 48 31094 31113 804 ACATGAGGAT 472313 528 547 GCAGAATATG 44 31095 31114 805

TACATGAGGA 472314 529 548 TGCAGAATAT 43 31096 31115 806 GTACATGAGG 366722 530 549 GTGCAGAATA 65 31097 31116 148 TGTACATGAG 472315 552 571 CACAGCGATG 65 31119 31138 807 AGCCAGCAAT 472316 553 572 GCACAGCGAT 84 31120 31139 808 GAGCCAGCAA 472317 555 574 GAGCACAGCG 87 31122 31141 809 ATGAGCCAGC 472318 556 575 AGAGCACAGC 83 31123 31142 810 GATGAGCCAG 472319 558 577 GTAGAGCACA 73 31125 31144 811 GCGATGAGCC 472320 559 578 AGTAGAGCAC 68 31126 31145 812 AGCGATGAGC 472321 560 579 AAGTAGAGCA 48 31127 31146 813 CAGCGATGAG 472322 561 580 GAAGTAGAGC 22 31128 31147 814 ACAGCGATGA 472323 563 582 GTGAAGTAGA 64 31130 31149 815 GCACAGCGAT 472324 564 583 AGTGAAGTAG 51 31131 31150 816 AGCACAGCGA 472325 565 584 AAGTGAAGTA 39 31132 31151 817 GAGCACAGCG 472326 566 585 CAAGTGAAGT 26 31133 31152 818 AGAGCACAGC 472327 568 587 GCCAAGTGAA 64 31135 31154 819 GTAGAGCACA 472328 569 588 AGCCAAGTGA 65 31136 31155 820 AGTAGAGCAC 472329 571 590 CCAGCCAAGT 51 31138 31157 821 GAAGTAGAGC 472330 572 591 ACCAGCCAAG 47 31139 31158 822 TGAAGTAGAG 472331 574 593 ACACCAGCCA 35 31141 31160 823 AGTGAAGTAG 472332 575 594 AACACCAGCC 34 31142 31161 824 AAGTGAAGTA 472333 577 596 CAAACACCAG 33 31144 31163 825 CCAAGTGAAG 472334 578 597 TCAAACACCA 34 31145 31164 826 GCCAAGTGAA 472335 580 599 AGTCAAACAC 39 31147 31166 827 CAGCCAAGTG 472336 581 600 CAGTCAAACA 37 31148 31167 828 CCAGCCAAGT 472337 583 602 TCCAGTCAAA 53 31150 31169 829 CACCAGCCAA 472338 584 603 TTCCAGTCAA 52 31151 31170 830 ACACCAGCCA 472339 586 605 TGTTCCAGTC 38 31153 31172 831 AAACACCAGC 472340 587 606 GTGTTCCAGT 43 31154 31173 832 CAAACACCAG 472341 589 608 GTGTGTTCCA 26 31156 31175 833 GTCAAACACC 472342 590 609 GGTGTGTTCC 41 31157 31176 834 AGTCAAACAC 472343 592 611 TGGGTGTGTT 7 31159 31178 835 CCAGTCAAAC 472344 593 612 TTGGGTGTGT 15 31160 31179 836 TCCAGTCAAA 472345 595 614 TCTTGGGTGT 54 31162 31181 837 GTTCCAGTCA 472346 596 615 TTCTTGGGTG 47 31163 31182 838 TGTTCCAGTC 472347 616 635 ACTGTGACCT 66 31548 31567 839 CCTGCCACCT 472348 617 636 CACTGTGACC 55 31549 31568 840 TCCTGCCACC 472349 619 638 CCCACTGTGA 79 31551 31570 841 CCTCCTGCCA 472350 620 639 ACCCACTGTG 77 31552 31571 842 ACCTCCTGCC 472351 622 641 GGACCCACTG 80 31554 31573 843 TGACCTCCTG 472352 623 642 CGGACCCACT 63 31555 31574 844 GTGACCTCCT 472353 624 643 TCGGACCCAC 82 31556 31575 845 TGTGACCTCC 423459 631 650 CCCAGTTTCG 49 31563 31582 463 GACCCACTGT 472354 633 652 AGCCCAGTTT 79 31565 31584 846 CGGACCCACT 472355 634 653 CAGCCCAGTT 48 31566 31585 847 TCGGACCCAC 472356 636 655 CACAGCCCAG 72 31568 31587 848 TTTCGGACCC 472357 637 656 ACACAGCCCA 63 31569 31588 849 GTTTCGGACC 472358 639 658 CCACACAGCC 75 31571 31590 850 CAGTTTCGGA 472359 640 659 GCCACACAGC 74 31572 31591 851 CCAGTTTCGG 472360 641 660 CGCCACACAG 77 31573 31592 852 CCCAGTTTCG 472361 659 678 AAGTAGTCTC 7 31591 31610 853 GAAAGTAGCG 472362 660 679 AAAGTAGTCT 0 31592 31611 854 CGAAAGTAGC 472363 662 681 GGAAAGTAGT 15 31594 31613 855 CTCGAAAGTA 472364 663 682 GGGAAAGTAG 0 31595 31614 856 TCTCGAAAGT 472365 664 683 TGGGAAAGTA 57 31596 31615 857 GTCTCGAAAG 472366 666 685 GATGGGAAAG 17 31598 31617 858 TAGTCTCGAA 472367 667 686 GGATGGGAAA 21 31599 31618 859 GTAGTCTCGA 411897 685 704 TGTGTGTCTT 41 n/a n/a 60 CACCAGCTGG 217317 693 712 CAGCAGGTTG 57 37221 37240 21 TGTGTCTTCA 411901 695 714 GTCAGCAGGT 45 37223 37242 64 TGTGTGTCTT 411902 697 716 TGGTCAGCAG 28 37225 37244 65 GTTGTGTGTC 472368 724 743 GGTGGTATCC 50 37252 37271 860 AAAGATATAG 472369 725 744 GGGTGGTATC 27 37253 37272 861 CAAAGATATA 366728 757 776 TGCAGAAGGC 63 37285 37304 149 ACCCAGGCCC 472370 758 777 TTGCAGAAGG 31 37286 37305 862 CACCCAGGCC 472371 759 778 GTTGCAGAAG 46 37287 37306 863 GCACCCAGGC 472372 781 800 CTTCTGTGGC 33

37309 37328 864 CTCTGTGCTG
TABLE-US-00015 TABLE 15 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 413433 n/a n/a GCCTGGACAA 89 32431 32450 425 GTCCTGCCCA 472521 1210 1229 TGTCGGAGGA 32 39257 39276 865 GAAGAGGCCT 472522 1212 1231 GGTGTCGGAG 0 39259 39278 866 GAGAAGAGGC 472523 1213 1232 AGGTGTCGGA 0 39260 39279 867 GGAGAAGAGG 472524 1214 1233 CAGGTGTCGG 9 39261 39280 868 AGGAGAAGAG 472525 1216 1235 CCCAGGTGTC 3 39263 39282 869 GGAGGAGAAG 472526 1217 1236 CCCCAGGTGT 36 39264 39283 870 CGGAGGAGAA 472527 1219 1238 GCCCCCAGGT 10 39266 39285 871 GTCGGAGGAG 472528 1220 1239 AGCCCCCAGG 13 39267 39286 872 TGTCGGAGGA 472529 1221 1240 CAGCCCCCAG 32 39268 39287 873 GTGTCGGAGG 472530 1223 1242 ACCAGCCCCC 40 39270 39289 874 AGGTGTCGGA 472531 1251 1270 AACAGTGGTG 17 39298 39317 875 ATGGGCTTGG 472532 1252 1271 CAACAGTGGT 28 39299 39318 876 GATGGGCTTG 472533 1323 1342 CATGGTGTGG 42 41294 41313 877 TACAGGTCGA 472534 1324 1343 ACATGGTGTG 63 41295 41314 878 GTACAGGTCG 411878 1325 1344 TACATGGTGT 63 41296 41315 879 GGTACAGGTC 472535 1326 1345 GTACATGGTG 70 41297 41316 880 TGGTACAGGT 472536 1328 1347 ATGTACATGG 56 41299 41318 881 TGTGGTACAG 472537 1329 1348 CATGTACATG 50 41300 41319 882 GTGTGGTACA 472538 1331 1350 TCCATGTACA 53 41302 41321 883 TGGTGTGGTA 472539 1332 1351 CTCCATGTAC 68 41303 41322 884 ATGGTGTGGT 472540 1333 1352 CCTCCATGTA 63 41304 41323 885 CATGGTGTGG 472541 1335 1354 GGCCTCCATG 48 41306 41325 886 TACATGGTGT 472542 1336 1355 GGGCCTCCAT 51 41307 41326 887 GTACATGGTG 472543 1337 1356 AGGGCCTCCA 56 41308 41327 888 TGTACATGGT 369255 1339 1358 CCAGGGCCTC 35 41310 41329 103 CATGTACATG 411944 1340 1359 ACCAGGGCCT 47 41311 41330 120 CCATGTACAT 411947 1344 1363 CTTCACCAGG 22 41315 41334 123 GCCTCCATGT 411948 1345 1364 GCTTCACCAG 43 41316 41335 124 GGCCTCCATG 411950 1347 1366 GAGCTTCACC 47 41318 41337 126 AGGGCCTCCA 411951 1349 1368 AAGAGCTTCA 51 41320 41339 127 CCAGGGCCTC 472544 1352 1371 TCGAAGAGCT 60 41323 41342 889 TCACCAGGGC 472545 1353 1372 GTCGAAGAGC 73 41324 41343 890 TTCACCAGGG 472546 1355 1374 TTGTCGAAGA 45 41326 41345 891 GCTTCACCAG 472547 1356 1375 CTTGTCGAAG 65 41327 41346 892 AGCTTCACCA 472548 1357 1376 GCTTGTCGAA 86 41328 41347 893 GAGCTTCACC 472549 1358 1377 TGCTTGTCGA 68 41329 41348 894 AGAGCTTCAC 472550 1360 1379 TGTGCTTGTC 70 41331 41350 895 GAAGAGCTTC 472551 1361 1380 TTGTGCTTGT 40 41332 41351 896 CGAAGAGCTT 472552 1362 1381 CTTGTGCTTG 77 41333 41352 897 TCGAAGAGCT 366746 1366 1385 TGGTCTTGTG 37 41337 41356 151 CTTGTCGAAG 472553 1368 1387 CTTGGTCTTG 58 41339 41358 898 TGCTTGTCGA 472554 1369 1388 ACTTGGTCTT 60 41340 41359 899 GTGCTTGTCG 472555 1370 1389 AACTTGGTCT 61 41341 41360 900 TGTGCTTGTC 472556 1372 1391 CGAACTTGGT 55 41343 41362 901 CTTGTGCTTG 472557 1373 1392 CCGAACTTGG 63 41344 41363 902 TCTTGTGCTT 472558 1374 1393 GCCGAACTTG 59 41345 41364 903 GTCTTGTGCT 472559 1375 1394 GGCCGAACTT 60 41346 41365 904 GGTCTTGTGC 472560 1377 1396 GAGGCCGAAC 53 41348 41367 905 TTGGTCTTGT 472561 1378 1397 GGAGGCCGAA 35 41349 41368 906 CTTGGTCTTG 472562 1400 1419 ACCTCCAGGA 66 41371 41390 907 CCTCAGTCTC 472563 1401 1420 CACCTCCAGG 19 41372 41391 908 ACCTCAGTCT 472564 1402 1421 TCACCTCCAG 40 41373 41392 909 GACCTCAGTC 472565 1404 1423 GTTCACCTCC 60 41375 41394 910 AGGACCTCAG 472566 1405 1424 AGTTCACCTC 23 41376 41395 911 CAGGACCTCA 472567 1406 1425 CAGTTCACCT 13 41377 41396 912 CCAGGACCTC 472568 1408 1427 CTCAGTTCAC 67 41379 41398 913 CTCCAGGACC 472569 1409 1428 GCTCAGTTCA 71

41380 41399 914 CCTCCAGGAC 472570 1410 1429 GGCTCAGTTC 84 41381 41400 915 ACCTCCAGGA 472571 1411 1430 TGGCTCAGTT 74 41382 41401 916 CACCTCCAGG 411955 1502 1521 AAATAACCCA 81 41473 41492 131 CAGACACCCA 411958 1506 1525 TTTTAAATAA 68 41477 41496 134 CCCACAGACA 472572 1538 1557 TTGTAATGGT 7 41509 41528 917 TTAGCAAAAT 472573 1539 1558 ATTGTAATGG 47 41510 41529 918 TTTAGCAAAA 472574 1540 1559 CATTGTAATG 18 41511 41530 919 GTTTAGCAAA 472575 1541 1560 ACATTGTAAT 40 41512 41531 920 GGTTTAGCAA 472576 1543 1562 TAACATTGTA 58 41514 41533 921 ATGGTTTAGC 472577 1544 1563 CTAACATTGT 49 41515 41534 922 AATGGTTTAG 472578 1545 1564 CCTAACATTG 67 41516 41535 923 TAATGGTTTA 472579 1546 1565 ACCTAACATT 27 41517 41536 924 GTAATGGTTT 472580 1548 1567 AGACCTAACA 54 41519 41538 925 TTGTAATGGT 472581 1549 1568 AAGACCTAAC 46 41520 41539 926 ATTGTAATGG 472582 1550 1569 AAAGACCTAA 34 41521 41540 927 CATTGTAATG 472583 1551 1570 AAAAGACCTA 33 41522 41541 928 ACATTGTAAT 472584 1553 1572 AAAAAAGACC 53 41524 41543 929 TAACATTGTA 472585 1554 1573 TAAAAAAGAC 58 41525 41544 930 CTAACATTGT 472586 1556 1575 CTTAAAAAAG 25 41527 41546 931 ACCTAACATT 472587 1557 1576 TCTTAAAAAA 23 41528 41547 932 GACCTAACAT

Example 4: Antisense Inhibition of Human DGAT2 in HepG2 Cells by MOE Gapmers

[0457] Antisense oligonucleotides were designed targeting a DGAT2 nucleic acid and were tested for their effects on DGAT2 mRNA in vitro. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. ISIS 413433, which consistently demonstrated higher potency than any of the previously disclosed oligonucleotides in the studies above was included in this study as a benchmark oligonucleotide. Antisense oligonucleotides that demonstrated about the same or greater potency than ISIS 413433 were therefore considered for further experimentation.

[0458] The results for each experiment are presented in separate tables shown below. Cultured HepG2 cells at a density of 20,000 cells per well were transfected using electroporation with 4,500 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988 MGB was used to measure mRNA levels. The potency of some oligonucleotides was measured with human primer probe set RTS2367 (forward sequence GGCCTCCCGGAGACTGA, designated herein as SEQ ID NO: 4; reverse sequence AAGTGATTTGCAGCTGGTTCCT, designated herein as SEQ ID NO: 5; probe sequence AGGTGAACTGAGCCAGCCTTCGGG, designated herein as SEQ ID NO: 6). DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. The results show several antisense oligonucleotides demonstrate greater potency than the benchmark oligonucleotides.

[0459] The newly designed chimeric antisense oligonucleotides in the Tables below were designed as 5-10-5 MOE gapmers or 3-10-4 MOE gapmers. The 5-10-5 MOE gapmers are 20 nucleosides in length, wherein the central gap segment comprises of ten 2′-deoxynucleosides and is flanked by wing segments on the 5′ direction and the 3′ direction comprising five nucleosides each. The 3-10-4 MOE gapmers are 17 nucleosides in length, wherein the central gap segment comprises of ten 2′-deoxynucleosides and is flanked by wing segments on the 5′ direction and the 3′ direction comprising three and four nucleosides respectively. Each nucleoside in the 5′ wing segment and each nucleoside in the 3′ wing segment has a 2′-MOE modification. The internucleoside linkages throughout each gapmer are phosphorothioate (P═S) linkages. All cytosine residues throughout each gapmer are 5-methylcytosines. “Start site” indicates the 5′-most nucleoside to which the gapmer is targeted in the human gene sequence. “Stop site” indicates the 3′-most nucleoside to which the gapmer is targeted human gene sequence. Each gapmer listed in the Tables below is targeted to SEQ ID NO: 1, SEQ ID NO: 2, or both. 'n/a' indicates that the antisense oligonucleotide does not target that particular gene sequence with 100% complementarity. In case the sequence alignment for a target gene in a particular table is not shown, it is understood that none of the oligonucleotides presented in that table align with 100% complementarity with that target gene. TABLE-US-00016 TABLE 16 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ SEQ SEQ SEQ ID ID ID ID % NO: NO: NO: NO: NO: in- 1 1 2 2 hi- SEQ ISIS Start Stop Start Stop bi- ID NO Site Site Sequence Site Site tion NO 483799 221 240 ACGGCCCCGC 9842 9861 8 569 GGGAAGCCGC 483800 226 245 CAGTCACGGC 9847 9866 46 570 CCCGCGGGAA 483801 231 250 CCGCCCAGTC 9852 9871 0 571 ACGGCCCCGC 483802 236 255 GAAGCCCGCC 9857 9876 50 572 CAGTCACGGC 413433 n/a n/a GCCTGGACAA 32431 32450 85 425 GTCCTGCCCA 483803 n/a n/a ACCCCATCTC 9984 10003 75 496 CCAGACCCCT 483804 n/a n/a CACTCACCCC 9989 10008 46 497 ATCTCCCAGA 483805 n/a n/a CAGGTCCATA 10014 10033 51 498 ACCCCTGCGC 483806 n/a n/a TCTCGCAGGT 10019 10038 6 499 CCATAACCCC 483807 n/a n/a AATCTTCTCG 10024 10043 50 500 CAGGTCCATA 483808 n/a n/a CAGAAAATCT 10029 10048 49 501 TCTCGCAGGT 483809 n/a n/a CTTTCCAGAA 10034 10053 37 502 AATCTTCTCG 483810 n/a n/a GGGCCCTTTC 10039 10058 40 503 CAGAAAATCT 483811 n/a n/a CCACAGGGCC 10044 10063 85 504 CTTTCCAGAA 483812 n/a n/a GCCTGCCACA 10049 10068 76 505 GGGCCCTTTC 483813 n/a n/a CACCAGCCTG 10054 10073 80 506 CCACAGGGCC 483814 n/a n/a CACGGATGAG 10160 10179 73 507 GGAAACAAGC 483815 n/a n/a AATATCCCTA 10205 10224 35 508 ATAACTAAGA 483816 n/a n/a TCTCGAATAT 10210 10229 84 509 CCCTAATAAC 483817 n/a n/a GGAGTTCTCG 10215 10234 96 510 AATATCCCTA 483818 n/a n/a AAACCCATAC 10327 10346 76 511 CATCCTGCCC 483819 n/a n/a TGGTGGCTGT 10351 10370 49 512 CTCAGGAGAC 483820 n/a n/a CCGTCTGGTG 10356 10375 36 513 GCTGTCTCAG 483821 n/a n/a GGCAGACACA 10389 10408 64 514 CCTGTTCCAG 483822 n/a n/a GACAGGGCAG 10394 10413 64 515 ACACACCTGT 483823 n/a n/a CAGAGGACAG 10399 10418 69 516 GGCAGACACA 483824 n/a n/a CCAGAGCCTG 10430 10449 45 517 GGCGAGAGGA 483825 n/a n/a CAGGGTCCTC 10490 10509 91 518 TCCGCTGCCT 483826 n/a n/a CTGAGGACCT 10522 10541 83 519 CAGTTCTACC 483827 n/a n/a ATTCACTGAG 10527 10546 67 520 GACCTCAGTT 483828 n/a n/a GCGCGATTCA 10532 10551 88 521 CTGAGGACCT 483829 n/a n/a ACTCTGCGCG 10537 10556 73 522 ATTCACTGAG 483830 n/a n/a CTCCCACTTC 10645 10664 83 523 AGTTTCTCCA 483831 n/a n/a GCTTCCTCCC 10650 10669 86 524 ACTTCAGTTT 483832 n/a n/a GGCATGCTTC 10655 10674 89 525 CTCCCACTTC 483833 n/a n/a ACTTAGGCAT 10660 10679 89 526 GCTTCCTCCC 483834 n/a n/a GGAAAACTTA 10665 10684 92 527 GGCATGCTTC 483835 n/a n/a GCTAAGGAAA 10670 10689 85 528 ACTTAGGCAT 483836 n/a n/a CATCAGCTAA 10675 10694 53 529 GGAAAACTTA 483837 n/a n/a CAAGGCATCA 10680 10699 27 530 GCTAAGGAAA 483838 n/a n/a ATTTGTACTT 10702 10721 77 531 GAATCCAGGG 483839 n/a n/a GTGATGACTT 10727 10746 87 532 CCCAGGGTCT 483840 n/a n/a CAGCTGTGAT 10732 10751 84 533 GACTTCCCAG 483841 n/a n/a CAGGACAGCT 10737 10756 80 534 GTGATGACTT 483842 n/a n/a CAGACCAGGA 10742 10761 72 535 CAGCTGTGAT 483843 n/a n/a CACACGCAGA 10748 10767 54 536 CCAGGACAGC 483844 n/a n/a AGACACACAC 10753 10772 44 537 GCAGACCAGG 483845 n/a n/a ATGCAAGACA 10758 10777 45 538 CACACGCAGA 483846 n/a n/a GCTTCATGCA 10763 10782 78 539 AGACACACAC 483847 n/a n/a ATGTCAGAGA 10814 10833 67 540 GGCTCAGCAA 483848 n/a n/a AATCCATGTC 10819 10838 93 541 AGAGAGGCTC 483849 n/a n/a AGAAAAATCC 10824 10843 37 542 ATGTCAGAGA 483850 n/a n/a ACTAAAGAAA 10829 10848 49 543 AATCCATGTC 483851 n/a n/a TAGTTACTAA 10834 10853 21 544 AGAAAAATCC 483852 n/a n/a GGATAGTCGA 10940 10959 88 545 TTTACCAGAA 483853 n/a n/a TCTTTGGATA 10945 10964 66 546 GTCGATTTAC 483854 n/a n/a ACTAAAATGC 10968 10987 2 547 TAATAGGAAG 483855 n/a n/a GATATACTAA 10973 10992 9 548 AATGCTAATA 483856 n/a n/a CAAGAGATAT 10978 10997 32 549 ACTAAAATGC 483857 n/a n/a ATGATCAAGA 10983 11002 36 550 GATATACTAA 483858 n/a n/a AGAAAATGAT 10988 11007 6 551 CAAGAGATAT 483859 n/a n/a TATAAAGAAA 10993 11012 6 552 ATGATCAAGA 483860 n/a n/a ATGTGTATAA 10998 11017 0 553 AGAAAATGAT 483861 n/a n/a ATGCAATGTG 11003 11022 36 554 TATAAAGAAA 483862 n/a n/a TATCTATGCA 11008 11027 75 555 ATGTGTATAA 483863 n/a n/a ATTCATAACT 11042 11061 52 556 ATATCTACAG 483864 n/a n/a ACAAGATTCA 11047 11066 31 557 TAACTATATC 483865 n/a n/a AGAAAACAAG 11052 11071 33 558 ATTCATAACT 483866 n/a n/a CCTCGATGTT 11074 11093 90 559 ACATTAAGGG 483867 n/a n/a GGAAACCTCG 11079 11098 59 560 ATGTTACATT 483868 n/a n/a ATTGCAACCA 11123 11142 60 561 CTAGGACATT 483869 n/a n/a GGGTTATTGC 11128 11147 91 562 AACCACTAGG 483870 n/a n/a CACAGTTAAA 11158 11177 86 563 GTGTGGTACA 483871 n/a n/a TAGGTCACAG 11163 11182 60 564 TTAAAGTGTG 483872 n/a n/a ATGCCTAGGT 11168 11187 74 565 CACAGTTAAA 483873 n/a n/a TGCCAATGCC 11173 11192 90 566 TAGGTCACAG 483874 n/a n/a AGCAATGCCA 11178 11197 91 567 ATGCCTAGGT 483875 n/a n/a CACAGCGATA 11199 11218 87 568 ATCACACAAG

TABLE-US-00017 TABLE 17 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 413433 n/a n/a GCCTGGACAA 89 32431 32450 425 GTCCTGCCCA 472665 1870 1889 CTCCAACTCC 56 41841 41860 933 TCCCCTTGGC 472666 1871 1890 TCTCCAACTC 37 41842 41861 934 CTCCCCTTGG 472667 1873 1892 GCTCTCCAAC 63 41844 41863 935 TCCTCCCCTT 472668 1874 1893 TGCTCTCCAA 65 41845 41864 936 CTCCTCCCCT 472669 1917 1936 CATTCCAGAT 63 41888 41907 937 GCCTACTACT 472670 1918 1937 GCATTCCAGA 68 41889 41908 938 TGCCTACTAC 472671 1919 1938 AGCATTCCAG 81 41890 41909 939 ATGCCTACTA 472672 1920 1939 GAGCATTCCA 77 41891 41910 940 GATGCCTACT 472673 1993 2012 CCTGGAGGCC 33 41964 41983 941 AGTCCAGGCT 472674 1994 2013 TCCTGGAGGC 22 41965 41984 942 CAGTCCAGGC 472675 1996 2015 CATCCTGGAG 14 41967 41986 943 GCCAGTCCAG 472676 1997 2016 TCATCCTGGA 47 41968 41987 944 GGCCAGTCCA 472677 1999 2018 CCTCATCCTG 57 41970 41989 945 GAGGCCAGTC 472678 2000 2019 TCCTCATCCT 37 41971 41990 946 GGAGGCCAGT 472679 2001 2020 ATCCTCATCC 50 41972 41991 947 TGGAGGCCAG 472680 2002 2021 CATCCTCATC 56 41973 41992 948 CTGGAGGCCA 472681 2004 2023 CCCATCCTCA 29 41975 41994 949 TCCTGGAGGC 472682 2005 2024 CCCCATCCTC 21 41976 41995 950 ATCCTGGAGG 472683 2007 2026 ACCCCCATCC 24 41978 41997 951 TCATCCTGGA 472684 2008 2027 CACCCCCATC 8 41979 41998 952 CTCATCCTGG 472685 2009 2028 CCACCCCCAT 39 41980 41999 953 CCTCATCCTG 472686 2011 2030 TGCCACCCCC 46 41982 42001 954 ATCCTCATCC 472687 2012 2031 TTGCCACCCC 48 41983 42002 955 CATCCTCATC 472688 2013 2032 ATTGCCACCC 57 41984 42003 956 CCATCCTCAT 472689 2015 2034 TCATTGCCAC 43 41986 42005 957 CCCCATCCTC 472690 2016 2035 GTCATTGCCA 29 41987 42006 958 CCCCCATCCT 472691 2017 2036 TGTCATTGCC 51 41988 42007 959 ACCCCCATCC 472692 2019 2038 GGTGTCATTG 74 41990 42009 960 CCACCCCCAT 472693 2079 2098 GCTCATGGTG 77 42050 42069 961 GCGGCATCCT 472694 2080 2099 AGCTCATGGT 66 42051 42070 962 GGCGGCATCC 472695 2082 2101 CTAGCTCATG 65 42053 42072 963 GTGGCGGCAT 472696 2083 2102 CCTAGCTCAT 68 42054 42073 964 GGTGGCGGCA 472697 2085 2104 CACCTAGCTC 49 42056 42075 965 ATGGTGGCGG 472698 2086 2105 CCACCTAGCT 44 42057 42076 966 CATGGTGGCG 472699 2088 2107 CTCCACCTAG 18 42059 42078 967 CTCATGGTGG 472700 2089 2108 ACTCCACCTA 61 42060 42079 968 GCTCATGGTG 472701 2090 2109 TACTCCACCT 46 42061 42080 969 AGCTCATGGT 472702 2092 2111 GTTACTCCAC 51 42063 42082 970

CTAGCTCATG 472703 2093 2112 AGTTACTCCA 58 42064 42083 971 CCTAGCTCAT 472704 2094 2113 CAGTTACTCC 66 42065 42084 972 ACCTAGCTCA 472705 2095 2114 CCAGTTACTC 80 42066 42085 973 CACCTAGCTC 472706 2097 2116 AACCAGTTAC 71 42068 42087 974 TCCACCTAGC 472707 2098 2117 AAACCAGTTA 68 42069 42088 975 CTCCACCTAG 472708 2100 2119 AAAAACCAGT 67 42071 42090 976 TACTCCACCT 472709 2101 2120 GAAAAACCAG 68 42072 42091 977 TTACTCCACC 472710 2103 2122 AAGAAAAACC 53 42074 42093 978 AGTTACTCCA 472711 2104 2123 CAAGAAAAAC 61 42075 42094 979 CAGTTACTCC 472712 2106 2125 CCCAAGAAAA 67 42077 42096 980 ACCAGTTACT 472713 2107 2126 ACCCAAGAAA 71 42078 42097 981 AACCAGTTAC 472714 2108 2127 CACCCAAGAA 56 42079 42098 982 AAACCAGTTA 472715 2109 2128 CCACCCAAGA 61 42080 42099 983 AAAACCAGTT 472716 2111 2130 AGCCACCCAA 62 42082 42101 984 GAAAAACCAG 472717 2112 2131 CAGCCACCCA 69 42083 42102 985 AGAAAAACCA 472718 2114 2133 ATCAGCCACC 58 42085 42104 986 CAAGAAAAAC 472719 2115 2134 CATCAGCCAC 46 42086 42105 987 CCAAGAAAAA 472720 2116 2135 TCATCAGCCA 64 42087 42106 988 CCCAAGAAAA 472721 2118 2137 TGTCATCAGC 48 42089 42108 989 CACCCAAGAA 472722 2119 2138 ATGTCATCAG 50 42090 42109 990 CCACCCAAGA 472723 2121 2140 CCATGTCATC 79 42092 42111 991 AGCCACCCAA 472724 2122 2141 TCCATGTCAT 79 42093 42112 992 CAGCCACCCA 472725 2123 2142 ATCCATGTCA 83 42094 42113 993 TCAGCCACCC 472726 2124 2143 CATCCATGTC 80 42095 42114 994 ATCAGCCACC 472727 2126 2145 TGCATCCATG 81 42097 42116 995 TCATCAGCCA 472728 2127 2146 CTGCATCCAT 78 42098 42117 996 GTCATCAGCC 413319 2128 2147 GCTGCATCCA 74 42099 42118 154 TGTCATCAGC 472729 2129 2148 TGCTGCATCC 80 42100 42119 997 ATGTCATCAG 472730 2130 2149 GTGCTGCATC 68 42101 42120 998 CATGTCATCA 472731 2132 2151 CTGTGCTGCA 64 42103 42122 999 TCCATGTCAT 472732 2133 2152 TCTGTGCTGC 80 42104 42123 1000 ATCCATGTCA 472733 2134 2153 GTCTGTGCTG 83 42105 42124 1001 CATCCATGTC 472734 2135 2154 AGTCTGTGCT 68 42106 42125 1002 GCATCCATGT 472735 2137 2156 TGAGTCTGTG 71 42108 42127 1003 CTGCATCCAT 472736 2138 2157 CTGAGTCTGT 78 42109 42128 1004 GCTGCATCCA 472737 2139 2158 GCTGAGTCTG 80 42110 42129 1005 TGCTGCATCC 472738 2140 2159 GGCTGAGTCT 84 42111 42130 1006 GTGCTGCATC 472739 2141 2160 AGGCTGAGTC 76 42112 42131 1007 TGTGCTGCAT 472740 2142 2161 AAGGCTGAGT 81 42113 42132 1008 CTGTGCTGCA

TABLE-US-00018 TABLE 18 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO: NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 413433 n/a n/a GCCTGGACAA 89 32431 32450 425 GTCCTGCCCA 411901 695 714 GTCAGCAGGT 64 37223 37242 64 TGTGTGTCTT 472741 2144 2163 CCAAGGCTGA 60 42115 42134 1009 GTCTGTGCTG 472742 2145 2164 GCCAAGGCTG 63 42116 42135 1010 AGTCTGTGCT 472743 2147 2166 AGGCCAAGGC 55 42118 42137 1011 TGAGTCTGTG 472744 2148 2167 CAGGCCAAGG 39 42119 42138 1012 CTGAGTCTGT 472745 2150 2169 TCCAGGCCAA 56 42121 42140 1013 GGCTGAGTCT 472746 2151 2170 CTCCAGGCCA 62 42122 42141 1014 AGGCTGAGTC 472747 2152 2171 GCTCCAGGCC 77 42123 42142 1015 AAGGCTGAGT 472748 2153 2172 TGCTCCAGGC 43 42124 42143 1016 CAAGGCTGAG 472749 2155 2174 TGTGCTCCAG 63 42126 42145 1017 GCCAAGGCTG 472750 2156 2175 ATGTGCTCCA 69 42127 42146 1018 GGCCAAGGCT 472751 2181 2200 AGGTAAACTG 35 42152 42171 1019 AGGCCACCAG 472752 2183 2202 GAAGGTAAAC 72 42154 42173 1020 TGAGGCCACC 472753 2184 2203 GGAAGGTAAA 45 42155 42174 1021 CTGAGGCCAC 472754 2214 2233 TCTTCCTCAC 49 42185 42204 1022 ATCCAGAATC 472755 2215 2234 CTCTTCCTCA 62 42186 42205 1023 CATCCAGAAT 472756 2237 2256 CAGGCCCCTT 73 42208 42227 1024 CTGAAGAGGG 472757 2238 2257 CCAGGCCCCT 68 42209 42228 1025

TCTGAAGAGG 472758 2240 2259 GGCCAGGCCC 41 42211 42230 1026 CTTCTGAAGA 472759 2241 2260 AGGCCAGGCC 51 42212 42231 1027 CCTTCTGAAG 472760 2243 2262 GAAGGCCAGG 32 42214 42233 1028 CCCCTTCTGA 472761 2244 2263 AGAAGGCCAG 28 42215 42234 1029 GCCCCTTCTG 472762 2245 2264 CAGAAGGCCA 24 42216 42235 1030 GGCCCCTTCT 472763 2247 2266 CTCAGAAGGC 54 42218 42237 1031 CAGGCCCCTT 472764 2248 2267 GCTCAGAAGG 72 42219 42238 1032 CCAGGCCCCT 472765 2250 2269 CTGCTCAGAA 60 42221 42240 1033 GGCCAGGCCC 472766 2251 2270 GCTGCTCAGA 81 42222 42241 1034 AGGCCAGGCC 472767 2253 2272 CTGCTGCTCA 51 42224 42243 1035 GAAGGCCAGG 472768 2254 2273 TCTGCTGCTC 79 42225 42244 1036 AGAAGGCCAG 472769 2255 2274 ATCTGCTGCT 63 42226 42245 1037 CAGAAGGCCA 472770 2256 2275 AATCTGCTGC 69 42227 42246 1038 TCAGAAGGCC 472771 2258 2277 CTAATCTGCT 57 42229 42248 1039 GCTCAGAAGG 472772 2259 2278 ACTAATCTGC 63 42230 42249 1040 TGCTCAGAAG 472773 2260 2279 AACTAATCTG 52 42231 42250 1041 CTGCTCAGAA 472774 2261 2280 GAACTAATCT 45 42232 42251 1042 GCTGCTCAGA 472775 2262 2281 GGAACTAATC 70 42233 42252 1043 TGCTGCTCAG 472776 2263 2282 TGGAACTAAT 73 42234 42253 1044 CTGCTGCTCA 472777 2264 2283 TTGGAACTAA 70 42235 42254 1045 TCTGCTGCTC 472778 2266 2285 CTTTGGAACT 69 42237 42256 1046 AATCTGCTGC 472779 2267 2286 GCTTTGGAAC 70 42238 42257 1047 TAATCTGCTG 472780 2268 2287 TGCTTTGGAA 76 42239 42258 1048 CTAATCTGCT 472781 2269 2288 CTGCTTTGGA 71 42240 42259 1049 ACTAATCTGC 472782 2270 2289 CCTGCTTTGG 55 42241 42260 1050 AACTAATCTG 472783 2272 2291 CACCTGCTTT 65 42243 42262 1051 GGAACTAATC 472784 2273 2292 CCACCTGCTT 78 42244 42263 1052 TGGAACTAAT 472785 2275 2294 GGCCACCTGC 66 42246 42265 1053 TTTGGAACTA 472786 2276 2295 GGGCCACCTG 67 42247 42266 1054 CTTTGGAACT 472787 2296 2315 AAAGTGAGGC 58 42267 42286 1055 TTGGGTTCGG 472788 2297 2316 AAAAGTGAGG 54 42268 42287 1056 CTTGGGTTCG 472789 2299 2318 AGAAAAGTGA 65 42270 42289 1057 GGCTTGGGTT 472790 2300 2319 CAGAAAAGTG 61 42271 42290 1058 AGGCTTGGGT 472791 2302 2321 CACAGAAAAG 70 42273 42292 1059 TGAGGCTTGG 472792 2303 2322 GCACAGAAAA 81 42274 42293 1060 GTGAGGCTTG 472793 2305 2324 AGGCACAGAA 82 42276 42295 1061 AAGTGAGGCT 472794 2306 2325 AAGGCACAGA 58 42277 42296 1062 AAAGTGAGGC 472795 2308 2327 GGAAGGCACA 29 42279 42298 1063 GAAAAGTGAG 472796 2309 2328 AGGAAGGCAC 24 42280 42299 1064 AGAAAAGTGA 472797 2311 2330 TCAGGAAGGC 67 42282 42301 1065 ACAGAAAAGT 472798 2312 2331 CTCAGGAAGG 32 42283 42302 1066 CACAGAAAAG 472799 2314 2333 CCCTCAGGAA 48 42285 42304 1067 GGCACAGAAA 472800 2315 2334 CCCCTCAGGA 55 42286 42305 1068 AGGCACAGAA 472801 2316 2335 CCCCCTCAGG 57 42287 42306 1069 AAGGCACAGA 472802 2318 2337 AACCCCCTCA 57 42289 42308 1070 GGAAGGCACA 472803 2319 2338 CAACCCCCTC 63 42290 42309 1071 AGGAAGGCAC 472804 2320 2339 CCAACCCCCT 66 42291 42310 1072 CAGGAAGGCA 472805 2321 2340 CCCAACCCCC 67 42292 42311 1073 TCAGGAAGGC 472806 2322 2341 GCCCAACCCC 1 42293 42312 1074 CTCAGGAAGG 472807 2372 2391 CTCATCAAGA 47 42343 42362 1075 GATAACAGAA 472808 2374 2393 ATCTCATCAA 56 42345 42364 1076 GAGATAACAG 472809 2375 2394 GATCTCATCA 48 42346 42365 1077 AGAGATAACA 472810 2377 2396 ATGATCTCAT 69 42348 42367 1078 CAAGAGATAA 472811 2398 2417 ACAAAAGTCT 82 42369 42388 1079 GACATGGTGC 472812 2400 2419 ATACAAAAGT 74 42371 42390 1080 CTGACATGGT 472813 2401 2420 TATACAAAAG 52 42372 42391 1081 TCTGACATGG 472814 2403 2422 CATATACAAA 45 42374 42393 1082 AGTCTGACAT 472815 2404 2423 GCATATACAA 78 42375 42394 1083 AAGTCTGACA 472816 2405 2424 GGCATATACA 81 42376 42395 1084 AAAGTCTGAC

TABLE-US-00019 TABLE 19 Inhibition of DGAT2 mRNA by 3-10-4 MOE gapmers targeting SEQ ID NOs: 1 and 2 SEQ SEQ SEQ SEQ ID ID % ID ID NO:

NO: in- NO: NO: 1 1 hi- 2 2 SEQ ISIS Start Stop bi- Start Stop ID NO Site Site Sequence tion Site Site NO 496047 1000 1016 CATTCTTGC 42 38153 38169 1102 CAGGCATG 496048 1003 1019 CTGCATTCT 61 38156 38172 1103 TGCCAGGC 496049 1004 1020 ACTGCATTC 43 38157 38173 1104 TTGCCAGG 496050 1005 1021 GACTGCATT 31 38158 38174 1105 CTTGCCAG 496051 1015 1031 TCCGCAGGG 59 38168 38184 1106 TGACTGCA 496052 1016 1032 TTCCGCAGG 60 38169 38185 1107 GTGACTGC 496053 1017 1033 GTTCCGCAG 64 38170 38186 1108 GGTGACTG 496054 1096 1112 ACACTTCAT 39 39143 39159 1109 TCTCTCCA 496055 1097 1113 TACACTTCA 39 39144 39160 1110 TTCTCTCC 496056 1098 1114 GTACACTTC 44 39145 39161 1111 ATTCTCTC 496057 1099 1115 TGTACACTT 36 39146 39162 1112 CATTCTCT 496058 1100 1116 TTGTACACT 28 39147 39163 1113 TCATTCTC 496059 1101 1117 CTTGTACAC 35 39148 39164 1114 TTCATTCT 496060 1102 1118 GCTTGTACA 48 39149 39165 1115 CTTCATTC 496061 1103 1119 TGCTTGTAC 40 39150 39166 1116 ACTTCATT 496062 1104 1120 CTGCTTGTA 63 39151 39167 1117 CACTTCAT 496063 1105 1121 CCTGCTTGT 62 39152 39168 1118 ACACTTCA 496064 1106 1122 ACCTGCTTG 47 39153 39169 1119 TACACTTC 496065 1107 1123 CACCTGCTT 39 39154 39170 1120 GTACACTT 496066 1108 1124 TCACCTGCT 43 39155 39171 1121 TGTACACT 496067 1109 1125 ATCACCTGC 23 39156 39172 1122 TTGTACAC 496068 1119 1135 CTCCTCGAA 9 39166 39182 1123 GATCACCT 496069 1120 1136 CCTCCTCGA 26 39167 39183 1124 AGATCACC 496070 1121 1137 CCCTCCTCG 25 39168 39184 1125 AAGATCAC 496071 1357 1373 TGTCGAAGA 36 41328 41344 1126 GCTTCACC 496072 1358 1374 TTGTCGAAG 7 41329 41345 1127 AGCTTCAC 496073 1359 1375 CTTGTCGAA 22 41330 41346 1128 GAGCTTCA 496074 1362 1378 GTGCTTGTC 44 41333 41349 1129 GAAGAGCT 496075 1363 1379 TGTGCTTGT 33 41334 41350 1130 CGAAGAGC 496076 1364 1380 TTGTGCTTG 10 41335 41351 1131 TCGAAGAG 496077 1410 1426 TCAGTTCAC 11 41381 41397 1132 CTCCAGGA 496078 1411 1427 CTCAGTTCA 11 41382 41398 1133 CCTCCAGG 496079 1412 1428 GCTCAGTTC 25 41383 41399 1134 ACCTCCAG 496080 1413 1429 GGCTCAGTT 56 41384 41400 1135 CACCTCCA 496081 1502 1518 TAACCCACA 61 41473 41489 1136 GACACCCA 496082 1503 1519 ATAACCCAC 55 41474 41490 1137 AGACACCC 496083 1504 1520 AATAACCCA 59 41475 41491 1138 CAGACACC 496084 1627 1643 AGATTTAGC 17 41598 41614 1139 CACCACCT 496085 1628 1644 CAGATTTAG 58 41599 41615 1140 CCACCACC 496086 1629 1645 CCAGATTTA 60 41600 41616 1141 GCCACCAC 496087 1630 1646 CCCAGATTT 62 41601 41617 1142 AGCCACCA 496088 1631 1647 GCCCAGATT 28 41602 41618 1143 TAGCCACC 496089 1853 1869 AGAGAAACT 0 41824 41840 1144 GGTCCTGC 496090 1854 1870 CAGAGAAAC 0 41825 41841 1145 TGGTCCTG 496091 1855 1871 GCAGAGAAA 42 41826 41842 1146 CTGGTCCT 496092 2123 2139 CATGTCATC 52 42094 42110 1147 AGCCACCC 496093 2124 2140 CCATGTCAT 62 42095 42111 1148 CAGCCACC 496094 2125 2141 TCCATGTCA 60 42096 42112 1149 TCAGCCAC 496095 2134 2150 TGTGCTGCA 53 42105 42121 1150 TCCATGTC 496096 2135 2151 CTGTGCTGC 21 42106 42122 1151 ATCCATGT 496097 2136 2152 TCTGTGCTG 33 42107 42123 1152 CATCCATG 496098 2140 2156 TGAGTCTGT 60 42111 42127 1153 GCTGCATC 496099 2141 2157 CTGAGTCTG 58 42112 42128 1154 TGCTGCAT 496100 2142 2158 GCTGAGTCT 72 42113 42129 1155 GTGCTGCA 496101 2305 2321 CACAGAAAA 53 42276 42292 1156 GTGAGGCT 496102 2306 2322 GCACAGAAA 49 42277 42293 1157 AGTGAGGC 496103 2307 2323 GGCACAGAA 28 42278 42294 1158 AAGTGAGG 496104 2398 2414 AAAGTCTGA 65 42369 42385 1159 CATGGTGC 496105 2399 2415 AAAAGTCTG 54 42370 42386 1160 ACATGGTG 496106 2400 2416 CAAAAGTCT 39 42371 42387 1161 GACATGGT 496036 474 490 GACCCACTG 49 25600 25616 1091 GAGCACTG 496037 475 491 GGACCCACT 76 25601 25617 1092 GGAGCACT 496038 476 492 AGGACCCAC 72 25602 25618 1093 TGGAGCAC 496039 553 569 CAGCGATGA 67 31120 31136 1094 GCCAGCAA 496040 554 570 ACAGCGATG 67 31121 31137 1095 AGCCAGCA 496041

555 571 CACAGCGAT 78 31122 31138 1096 GAGCCAGC 496042 556 572 GCACAGCGA 61 31123 31139 1097 TGAGCCAG 496043 557 573 AGCACAGCG 70 31124 31140 1098 ATGAGCCA 496044 558 574 GAGCACAGC 48 31125 31141 1099 GATGAGCC 496045 998 1014 TTCTTGCCA 39 38151 38167 1100 GGCATGGA 496046 999 1015 ATTCTTGCC 31 38152 38168 1101 AGGCATGG 413433 n/a n/a GCCTGGACA 68 32431 32450 425 AGTCCTGC CCA 496107 n/a n/a CAGGGCCCT 34 10044 10060 1085 TTCCAGAA 496108 n/a n/a ACAGGGCCC 39 10045 10061 1086 TTTCCAGA 496109 n/a n/a CACAGGGCC 39 10046 10062 1087 CTTTCCAG 496110 n/a n/a CGAATATCC 14 10210 10226 1088 CTAATAAC 496111 n/a n/a TCGAATATC 9 10211 10227 1089 CCTAATAA 496112 n/a n/a CTCGAATAT 17 10212 10228 1090 CCCTAATA

TABLE-US-00020 TABLE 20 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 483876 14332 14351 GTGAGATCCCCAGAGAGGAA 26 1162 483877 14337 14356 AAGGAGTGAGATCCCCAGAG 34 1163 483878 14360 14379 TTCTGACCCTATTTTCCAGA 52 1164 483879 14385 14404 CCCTCCATCCTTGAGAATCT 76 1165 483880 14427 14446 ATGTCACAAGTTCACAAACA 49 1166 483881 14432 14451 CATGAATGTCACAAGTTCAC 38 1167 483882 14437 14456 ACTAACATGAATGTCACAAG 35 1168 483883 14442 14461 CAAGGACTAACATGAATGTC 0 1169 483884 14447 14466 AATGACAAGGACTAACATGA 24 1170 483885 14452 14471 GGACAAATGACAAGGACTAA 12 1171 483886 14457 14476 ACACAGGACAAATGACAAGG 62 1172 483887 14462 14481 GTGTAACACAGGACAAATGA 82 1173 483888 14467 14486 TGAATGTGTAACACAGGACA 75 1174 483889 14503 14522 CAGGGTCTGCCACTCTCTAC 88 1175 483890 14508 14527 TAGAGCAGGGTCTGCCACTC 85 1176 483891 14532 14551 TCAGTGAAGCCTTCCTGGAT 31 1177 483892 14537 14556 CTTCCTCAGTGAAGCCTTCC 72 1178 483893 14542 14561 AGTCCCTTCCTCAGTGAAGC 59 1179 483894 14547 14566 CCCCAAGTCCCTTCCTCAGT 51 1180 483895 14648 14667 CACAGTCATCTTGTGTACAT 93 1181 483896 14653 14672 CTCTGCACAGTCATCTTGTG 59 1182 483897 14658 14677 GATCACTCTGCACAGTCATC 83 1183 483898 14663 14682 GCTCAGATCACTCTGCACAG 94 1184 483899 14668 14687 TCATTGCTCAGATCACTCTG 71 1185 483900 14673 14692 TCCTGTCATTGCTCAGATCA 81 1186 483901 14678 14697 GTCTTTCCTGTCATTGCTCA 86 1187 483902 14715 14734 TCAGGGTGATCAAGCTCCCC 70 1188 483903 14720 14739 TGACCTCAGGGTGATCAAGC 75 1189 483904 14725 14744 TTCCCTGACCTCAGGGTGAT 49 1190 483905 14730 14749 AAGACTTCCCTGACCTCAGG 46 1191 483906 14735 14754 CCAGGAAGACTTCCCTGACC 65 1192 483907 14740 14759 CTCTTCCAGGAAGACTTCCC 68 1193 483908 14745 14764 GTCACCTCTTCCAGGAAGAC 94 1194 483909 14750 14769 TGAATGTCACCTCTTCCAGG 84 1195 483910 14755 14774 CGGACTGAATGTCACCTCTT 95 1196 483911 14760 14779 AGATCCGGACTGAATGTCAC 79 1197 483912 14765 14784 TTTCCAGATCCGGACTGAAT 79 1198 483913 14770 14789 TCATCTTTCCAGATCCGGAC 95 1199 483914 14775 14794 TCTATTCATCTTTCCAGATC 55 1200 483915 14802 14821 TGAATGTTCTTGCCTGTCTT 48 1201 483916 14807 14826 GTACCTGAATGTTCTTGCCT 85 1202 483917 14812 14831 TTCCTGTACCTGAATGTTCT 71 1203 483918 14852 14871 TGCCTCTCCCATCCAAATCT 75 1204 483919 14882 14901 GTTGGTCACTCAGAGGCCCT 86 1205 483920 14953 14972 ACAGGCCTGGATCCAGCATC 77 1206 483921 14986 15005 TCTTCCTCCTGCTGCAGACA 86 1207 483922 14991 15010 TCAAGTCTTCCTCCTGCTGC 68 1208 483923 14996 15015 AGCTCTCAAGTCTTCCTCCT 87 1209 483924 15001 15020 CCATGAGCTCTCAAGTCTTC 82 1210 483925 15006 15025 CTTTCCCATGAGCTCTCAAG 68 1211 483926 15011 15030 GGCTCCTTTCCCATGAGCTC 77 1212 483927 15016 15035 CACCAGGCTCCTTTCCCATG 84 1213 483928 15021 15040 AACTGCACCAGGCTCCTTTC 70 1214 483929 15026 15045 AAACTAACTGCACCAGGCTC 65 1215 483930 15031 15050 CCAATAAACTAACTGCACCA

36 1216 483931 15036 15055 GGAGGCCAATAAACTAACTG 37 1217 483932 15041 15060 GTGCTGGAGGCCAATAAACT 60 1218 483933 15046 15065 TCAAAGTGCTGGAGGCCAAT 66 1219 483934 15073 15092 AGCCAGCAGCTCCATACCAG 70 1220 483935 15078 15097 AAATCAGCCAGCAGCTCCAT 56 1221 483936 15083 15102 CCCTCAAATCAGCCAGCAGC 62 1222 483937 15088 15107 TGAGGCCCTCAAATCAGCCA 52 1223 483938 15093 15112 GCCCATGAGGCCCTCAAATC 44 1224 483939 15098 15117 GCCCTGCCCATGAGGCCCTC 44 1225 483940 15103 15122 CCTGGGCCCTGCCCATGAGG 28 1226 483941 15144 15163 GCCAAGCAGGAGCTGGACAG 35 1227 483942 15178 15197 TAGTGGCTTGCTGAGGCTGC 67 1228 483943 15183 15202 AAGGGTAGTGGCTTGCTGAG 31 1229 483944 15188 15207 TAAGGAAGGGTAGTGGCTTG 5 1230 483945 15199 15218 GAGACTGAGGGTAAGGAAGG 31 1231 483946 15204 15223 ATGAGGAGACTGAGGGTAAG 41 1232 483947 15209 15228 CATAGATGAGGAGACTGAGG 42 1233 483948 15214 15233 CATTTCATAGATGAGGAGAC 71 1234 483949 15219 15238 TTGCTCATTTCATAGATGAG 78 1235 483950 15224 15243 CACTTTTGCTCATTTCATAG 74 1236 483951 15247 15266 ATAATCTGCACAGGTTCTTA 47 1237 483952 15252 15271 GCACCATAATCTGCACAGGT 92 1238 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 87 425

TABLE-US-00021 TABLE 21 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 483953 17938 17957 GGGTCTAGGTGAGTGAGAAA 42 1239 483954 17960 17979 ATCTGGACCTAATGTCTGCC 71 1240 483955 17965 17984 GGCCCATCTGGACCTAATGT 41 1241 483956 17970 17989 ACCTGGGCCCATCTGGACCT 61 1242 483957 18042 18061 CTCTTGGGCTCCTTGAGGCA 33 1243 483958 18047 18066 ACAGCCTCTTGGGCTCCTTG 33 1244 483959 18052 18071 TGTGAACAGCCTCTTGGGCT 49 1245 483960 18057 18076 AGAGGTGTGAACAGCCTCTT 0 1246 483961 18078 18097 GCCAGAAAGATGCCAGCTAA 67 1247 483962 18083 18102 AGAGAGCCAGAAAGATGCCA 57 1248 483963 18113 18132 AAAAGGGCCAGAATGTCTGG 24 1249 483964 18136 18155 TAAGGAATCTAAATAACTTA 21 1250 483965 18141 18160 TGTCATAAGGAATCTAAATA 16 1251 483966 18146 18165 AGGATTGTCATAAGGAATCT 54 1252 483967 18151 18170 AATCCAGGATTGTCATAAGG 59 1253 483968 18156 18175 GCTTTAATCCAGGATTGTCA 80 1254 483969 18161 18180 CCTTAGCTTTAATCCAGGAT 86 1255 483970 18166 18185 GTCCTCCTTAGCTTTAATCC 80 1256 483971 18171 18190 TCAGTGTCCTCCTTAGCTTT 65 1257 483972 18176 18195 GGGACTCAGTGTCCTCCTTA 81 1258 483973 18181 18200 TCCCTGGGACTCAGTGTCCT 77 1259 483974 18220 18239 GATTCTTACCTGCAATGAGA 36 1260 483975 18225 18244 GCTCTGATTCTTACCTGCAA 70 1261 483976 18230 18249 CCCTGGCTCTGATTCTTACC 60 1262 483977 18235 18254 TCAAACCCTGGCTCTGATTC 70 1263 483978 18240 18259 AGGATTCAAACCCTGGCTCT 72 1264 483979 18286 18305 ATGCCTGCCCTGCCTAGGAA 75 1265 483980 18291 18310 AAATAATGCCTGCCCTGCCT 37 1266 483981 18296 18315 GGGTAAAATAATGCCTGCCC 65 1267 483982 18301 18320 TTGATGGGTAAAATAATGCC 9 1268 483983 18306 18325 CCTGTTTGATGGGTAAAATA 60 1269 483984 18311 18330 CCTCTCCTGTTTGATGGGTA 89 1270 483985 18316 18335 GGTGTCCTCTCCTGTTTGAT 78 1271 483986 18321 18340 GCCTCGGTGTCCTCTCCTGT 84 1272 483987 18326 18345 AGTAAGCCTCGGTGTCCTCT 87 1273 483988 18331 18350 TACCAAGTAAGCCTCGGTGT 87 1274 483989 18336 18355 TTAATTACCAAGTAAGCCTC 74 1275 483990 18369 18388 TTCATAAAAAGTAAAGATCT 51 1276 483991 18374 18393 AGAGCTTCATAAAAAGTAAA 0 1277 483992 18379 18398 AGCCAAGAGCTTCATAAAAA 87 1278 483993 18417 18436 TCACCAGATTGTTGTGGGAA 84 1279 483994 18422 18441 CTACCTCACCAGATTGTTGT 75 1280 483995 18427 18446 AATACCTACCTCACCAGATT 63 1281 483996 18432 18451 GGGCTAATACCTACCTCACC 81 1282 483997 18450 18469 TTCCTCATCTATACAGTGGG

85 1283 483998 18455 18474 TCAGCTTCCTCATCTATACA 52 1284 483999 18460 18479 AAGCTTCAGCTTCCTCATCT 35 1285 484000 18465 18484 TATACAAGCTTCAGCTTCCT 44 1286 484001 18470 18489 CTTCCTATACAAGCTTCAGC 77 1287 484002 18475 18494 AGTCACTTCCTATACAAGCT 64 1288 484003 18513 18532 AGAACCTGGCCTCTAACTCG 42 1289 484004 18581 18600 CATACTGGCATCTGGCAGGG 84 1290 484005 18586 18605 ACCTCCATACTGGCATCTGG 66 1291 484006 18591 18610 ACCTCACCTCCATACTGGCA 79 1292 484007 18747 18766 ATGAAGAAAAATAATTTGGG 0 1293 484008 18821 18840 TTAACTTCTGCTGAGGGAGA 63 1294 484009 18826 18845 GATGCTTAACTTCTGCTGAG 68 1295 484010 18831 18850 GTTAGGATGCTTAACTTCTG 78 1296 484011 18836 18855 GTTAAGTTAGGATGCTTAAC 65 1297 484012 18869 18888 ATAGGCCTGGATGCCCAAGT 78 1298 20779 20798 484013 18880 18899 CAGAGGCAGCTATAGGCCTG 69 1299 20790 20809 484014 18885 18904 TGCCTCAGAGGCAGCTATAG 30 1300 484015 18890 18909 TGTCCTGCCTCAGAGGCAGC 65 1301 484016 18895 18914 GATGATGTCCTGCCTCAGAG 61 1302 484017 18900 18919 CAGCAGATGATGTCCTGCCT 85 1303 484018 18982 19001 GGGACAGAGCTAAGACCCAA 58 1304 484019 19013 19032 GTCAAAGGTGGCTTCCTGGG 74 1305 484020 19018 19037 GAGTAGTCAAAGGTGGCTTC 78 1306 484021 19023 19042 GGAATGAGTAGTCAAAGGTG 27 1307 484022 19028 19047 AAGTTGGAATGAGTAGTCAA 14 1308 484023 19041 19060 GCTGAGAATAAAGAAGTTGG 55 1309 484024 19046 19065 TAGAAGCTGAGAATAAAGAA 36 1310 484025 19051 19070 TGAAATAGAAGCTGAGAATA 20 1311 484026 19112 19131 GGAGTTCTAATGAGGCAGAC 50 1312 484027 19117 19136 TGCAGGGAGTTCTAATGAGG 53 1313 484028 19122 19141 AGCCTTGCAGGGAGTTCTAA 62 1314 484029 19127 19146 AACAAAGCCTTGCAGGGAGT 30 1315 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 76 425

TABLE-US-00022 TABLE 22 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 484030 22439 22458 ATGGCAAGTGCTTCCGTGGG 79 1316 484031 22444 22463 ACCAAATGGCAAGTGCTTCC 82 1317 484032 22449 22468 CAAATACCAAATGGCAAGTG 34 1318 484033 22474 22493 CTGCAAGGTGGTGAGCTCTA 28 1319 484034 22513 22532 CCTCACTTCCCATCTGCCTG 24 1320 484035 22518 22537 TGAACCCTCACTTCCCATCT 14 1321 484036 22523 22542 CCAAGTGAACCCTCACTTCC 12 1322 484037 22528 22547 GTCTACCAAGTGAACCCTCA 20 1323 484038 22533 22552 ATGCAGTCTACCAAGTGAAC 39 1324 484039 22538 22557 GGGAAATGCAGTCTACCAAG 55 1325 484040 22543 22562 CAAGTGGGAAATGCAGTCTA 53 1326 484041 22548 22567 GCCAACAAGTGGGAAATGCA 85 1327 484042 22573 22592 CCAACTACCGCCCACAGCTC 43 1328 484043 22578 22597 CAGCCCCAACTACCGCCCAC 29 1329 484044 22583 22602 TTATCCAGCCCCAACTACCG 10 1330 484045 22588 22607 CTGCCTTATCCAGCCCCAAC 43 1331 484046 22593 22612 TCACCCTGCCTTATCCAGCC 51 1332 484047 22713 22732 TCTTGTTCAACCCTCACCCC 45 1333 484048 22766 22785 CTTATTAGGTGTCTTAAAAT 43 1334 484049 22771 22790 CTAAGCTTATTAGGTGTCTT 85 1335 484050 22776 22795 CTCTGCTAAGCTTATTAGGT 69 1336 484051 22820 22839 ACTGAGAAGACCTAAAAATT 0 1337 484052 22825 22844 AAACAACTGAGAAGACCTAA 45 1338 484053 22830 22849 AACTGAAACAACTGAGAAGA 32 1339 484054 22835 22854 TCCCCAACTGAAACAACTGA 65 1340 484055 22841 22860 GACATATCCCCAACTGAAAC 67 1341 484056 22846 22865 CTGATGACATATCCCCAACT 71 1342 484057 22851 22870 TTTCACTGATGACATATCCC 69 1343 484058 22856 22875 AGAAATTTCACTGATGACAT 56 1344 484059 22861 22880 AGTATAGAAATTTCACTGAT 18 1345 484060 22866 22885 GGAGGAGTATAGAAATTTCA 53 1346 484061 22871 22890 TTGGAGGAGGAGTATAGAAA 48 1347 484062 22876 22895 GCAGCTTGGAGGAGGAGTAT 32 1348 484063 22881 22900 CCCATGCAGCTTGGAGGAGG 68 1349 484064 22886 22905 CAGCCCCCATGCAGCTTGGA

84 1350 484065 22891 22910 AGGGCCAGCCCCATGCAGC 67 1351 484066 22916 22935 TTATCTAGCTCCCTACCTTG 57 1352 484067 22968 22987 TGAAGCCAGTGGTTTGAGGG 30 1353 484068 22992 23011 TGCTTCCCGTTGCAGACAGG 34 1354 484069 23095 23114 GTCCAAACACTCAGGTAGGG 72 1355 484070 23100 23119 CTTCAGTCCAAACACTCAGG 85 1356 484071 23128 23147 TTAAGTCTGAGAAACACCAC 77 1357 484072 23133 23152 GGTGATTAAGTCTGAGAAAC 64 1358 484073 23138 23157 CCCCAGGTGATTAAGTCTGA 73 1359 484074 23143 23162 AAGATCCCCAGGTGATTAAG 26 1360 484075 23148 23167 TTAACAAGATCCCCAGGTGA 22 1361 484076 23154 23173 TGCATTTTAACAAGATCCCC 0 1362 484077 23159 23178 AAATCTGCATTTTAACAAGA 20 1363 484078 23164 23183 AATCAAAATCTGCATTTTAA 40 1364 484079 23169 23188 TACTGAATCAAAATCTGCAT 53 1365 484080 23174 23193 GGATTTACTGAATCAAAATC 31 1366 484081 23179 23198 AGACTGGATTTACTGAATCA 65 1367 484082 23184 23203 GCTCCAGACTGGATTTACTG 72 1368 484083 23232 23251 AGCAGCATCAGCATCACCTG 79 1369 484084 23237 23256 GGAACAGCAGCATCAGCATC 39 1370 484085 23242 23261 GTCTGGGAACAGCAGCATCA 90 1371 484086 23271 23290 TACTCTAGCACTTTCCTACT 53 1372 484087 23276 23295 AAATGTACTCTAGCACTTTC 41 1373 484088 23281 23300 AGACAAAATGTACTCTAGCA 58 1374 484089 23305 23324 GGCTGGATGCCCTGGCTAGA 79 1375 484090 23310 23329 AGGCAGGCTGGATGCCCTGG 55 1376 484091 23315 23334 ATCTGAGGCAGGCTGGATGC 59 1377 484092 23417 23436 ATCTTCCAGCAGCTGGCTCT 28 1378 484093 23422 23441 AAGTCATCTTCCAGCAGCTG 41 1379 484094 23427 23446 TGGACAAGTCATCTTCCAGC 85 1380 484095 23470 23489 AAGTCCAAGCACAGGCTTGA 44 1381 484096 23475 23494 AAGTGAAGTCCAAGCACAGG 39 1382 484097 23480 23499 GAGGGAAGTGAAGTCCAAGC 63 1383 484098 23547 23566 TCCTTGAGGGTCTCATAACC 66 1384 484099 23552 23571 GCTTATCCTTGAGGGTCTCA 84 1385 484100 23557 23576 CACATGCTTATCCTTGAGGG 82 1386 484101 23562 23581 TTCATCACATGCTTATCCTT 75 1387 484102 23567 23586 ATGAGTTCATCACATGCTTA 78 1388 484103 23597 23616 TCAACCTGCCCAGCAGTGGG 62 1389 484104 23641 23660 CTCTGTACACAGGACACAGT 70 1390 484105 23646 23665 AAAGGCTCTGTACACAGGAC 79 1391 484106 23651 23670 AGTGCAAAGGCTCTGTACAC 66 1392 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 86 425

TABLE-US-00023 TABLE 23 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % Unmodified Modified NO Start Site Stop Site Sequence inhibition SEQ ID NO SEQ ID NO 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 84 425 N/A 484107 26390 26409 GGCAACCTAAGGAGTGAGGG 71 1393 N/A 484108 26395 26414 TGGATGGCAACCTAAGGAGT 40 1394 N/A 484109 26400 26419 TGGCCTGGATGGCAACCTAA 57 1395 N/A 484110 26479 26498 GCTATGCTGAGAGCACAGGG 79 1396 N/A 484111 26484 26503 TACCTGCTATGCTGAGAGCA 71 1397 N/A 484112 26489 26508 GAAGCTACCTGCTATGCTGA 46 1398 N/A 484113 26494 26513 CTGAGGAAGCTACCTGCTAT 36 1399 N/A 484114 26516 26535 CAGGTTCATCTGCCTTGACG 70 1400 N/A 484115 26521 26540 TGGAGCAGGTTCATCTGCCT 83 1401 N/A 484116 26526 26545 TGCTCTGGAGCAGGTTCATC 81 1402 N/A 484117 26531 26550 TGTGATGCTCTGGAGCAGGT 80 1403 N/A 484118 26536 26555 CACTCTGTGATGCTCTGGAG 74 1404 N/A 484119 26541 26560 GAATGCACTCTGTGATGCTC 68 1405 N/A 484120 26566 26585 CAGAGGGACTGGCTCACAGG 21 1406 N/A 484121 26590 26609 TACAGTCCCGAAGAGTGGGT 38 1407 N/A 484122 26595 26614 GCCTATACAGTCCCGAAGAG 51 1408 N/A 484123 26600 26619

TACCAGCCTATACAGTCCCG 60 1409 N/A 484124 26605 26624
TCCCCTACCAGCCTATACAG 59 1410 N/A 484125 26610 26629
GATGATCCCCTACCAGCCTA 68 1411 N/A 484126 26615 26634
GTCCTGATGATCCCCTACCA 80 1412 N/A 484127 26620 26639
GGTAAGTCCTGATGATCCCC 86 1413 N/A 484128 26625 26644
GACATGGTAAGTCCTGATGA 54 1414 N/A 484129 26630 26649
GCACTGACATGGTAAGTCCT 87 1415 N/A 484130 26635 26654
GCTCAGCACTGACATGGTAA 85 1416 N/A 484131 26640 26659
CAGCTGCTCAGCACTGACAT 80 1417 N/A 484132 26645 26664
AAGGACAGCTGCTCAGCACT 60 1418 N/A 484133 26711 26730
GTGAAGTTCTATCCCTTGGC 89 1419 N/A 484134 26716 26735
CACCTGTGAAGTTCTATCCC 69 1420 N/A 484135 26721 26740
TTTCTCACCTGTGAAGTTCT 79 1421 N/A 484136 26755 26774
AGCAGGTAGTTGATGAACCT 70 1422 N/A 484137 26778 26797
TGCCATTTAATGAGCTTCAC 86 1423 4680 484138 26783 26802
AACTCTGCCATTTAATGAGC 55 1424 N/A 484139 26788 26807
ATCCCAACTCTGCCATTTAA 81 1425 N/A 484140 26811 26830
GAATGCACTGAGTTTCTGCT 89 1426 N/A 484141 26849 26868
CACTAATTCTGGGCTTCCAG 87 1427 N/A 484142 26854 26873
CAGTTCACTAATTCTGGGCT 79 1428 N/A 484143 26859 26878
AGCCCCAGTTCACTAATTCT 52 1429 N/A 484144 26864 26883
GTTTCAGCCCCAGTTCACTA 54 1430 N/A 484145 26869 26888
TGGCTGTTTCAGCCCCAGTT 70 1431 N/A 484146 26874 26893
AAGACTGGCTGTTTCAGCCC 79 1432 N/A 484147 26879 26898
AGGTGAAGACTGGCTGTTTC 76 1433 N/A 484148 26884 26903
CCTAAAGGTGAAGACTGGCT 91 1434 N/A 484149 26889 26908
TTGGGCCTAAAGGTGAAGAC 64 1435 N/A 484150 26894 26913
CGTTCTTGGGCCTAAAGGTG 57 1436 N/A 484151 26899 26918
AAGCCCGTTCTTGGGCCTAA 23 1437 N/A 484152 26926 26945
TTAAGGGTTCATGGATCCCC 66 1438 N/A 484153 26931 26950
TAATTTTAAGGGTTCATGGA 29 1439 N/A 484154 27005 27024
CTGTAAGACTTTGAGGACAG 60 1440 N/A 484155 27010 27029
TGTCCCTGTAAGACTTTGAG 43 1441 N/A 484156 27046 27065
GTTCTTCCAACCAGTGTTTG 84 1442 N/A 484157 27051 27070
GCTCAGTTCTTCCAACCAGT 94 1443 N/A 484158 27056 27075
AGTTTGCTCAGTTCTTCCAA 85 1444 N/A 484159 27061 27080
TGTTTAGTTTGCTCAGTTCT 68 1445 N/A 484160 27101 27120
AAATGCAAGCCATGTTAGGG 54 1446 N/A 484161 27106 27125
TGGCCAAATGCAAGCCATGT 72 1447 N/A 484162 27111 27130
GTAAGTGGCCAAATGCAAGC 67 1448 N/A 484163 27116 27135
TTCTAGTAAGTGGCCAAATG 57 1449 N/A 484164 27174 27193
CAGGCCTCCACTTCCTTTTT 66 1450 N/A 484165 27211 27230
TCTTCTCATGTGACCCCAAG 69 1451 N/A 484166 27216 27235
TACTTTCTTCTCATGTGACC 59 1452 N/A 484167 27221 27240
CCCAGTACTTTCTTCTCATG 85 1453 N/A 484168 27226 27245
CTGGTCCCAGTACTTTCTTC 63 1454 N/A 484169 27231 27250
TTGTCCTGGTCCCAGTACTT 86 1455 N/A 484170 27236 27255
GGTTCTTGTCCTGGTCCCAG 90 1456 N/A 484171 27241 27260
CCCTGGGTTCTTGTCCTGGT 82 1457 N/A 484172 27246 27265
GGAGTCCCTGGGTTCTTGTC 74 1458 N/A 484173 27251 27270

AGGCTGGAGTCCCTGGGTTC 53 1459 N/A 484174 27256 27275
CTGGGAGGCTGGAGTCCCTG 55 1460 N/A 484175 27352 27371
ACCAGCCAGGCCCAGACCCA 34 1461 N/A 484176 27357 27376
TCCCTACCAGCCAGGCCCAG 17 1462 N/A 484177 27362 27381
TCAGCTCCCTACCAGCCAGG 47 1463 N/A 484178 27367 27386
GCTGCTCAGCTCCCTACCAG 74 1464 N/A 484179 27372 27391
TACAAGCTGCTCAGCTCCCT 67 1465 N/A 484180 27377 27396
TGTTCTACAAGCTGCTCAGC 73 1466 N/A 484181 27382 27401
GCTGGTGTTCTACAAGCTGC 92 1467 N/A 484182 27387 27406
CGTGAGCTGGTGTTCTACAA 80 1468 N/A 484183 27437 27456
TCTGATCAATTATTAACCTA 21 1469 N/A
TABLE-US-00024 TABLE 24 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 484184 29480 29499 TGGATGTAGACTCTCCATGA 61 1470 484185 29485 29504 AAGGCTGGATGTAGACTCTC 72 1471 484186 29490 29509 AAGATAAGGCTGGATGTAGA 28 1472 484187 29495 29514 GGGAGAAGATAAGGCTGGAT 50 1473 484188 29500 29519 CCCATGGGAGAAGATAAGGC 62 1474 484189 29505 29524 GGTTTCCCATGGGAGAAGAT 50 1475 484190 29538 29557 CATGCTCTCTTCTCACCATG 71 1476 484191 29543 29562 GATGTCATGCTCTCTTCTCA 66 1477 484192 29548 29567 CTCTGGATGTCATGCTCTCT 82 1478 484193 29570 29589 CCAGGTGCTGTAGGCTGCCT 84 1479 484194 29575 29594 TGGTCCCAGGTGCTGTAGGC 75 1480 484195 29580 29599 CCTGGTGGTCCCAGGTGCTG 62 1481 484196 29607 29626 AGGCCAACCCTTGCTGTGTG 63 1482 484197 29612 29631 AAGGGAGGCCAACCCTTGCT 60 1483 484198 29635 29654 TAGGACTTTTTCCACTGCCC 72 1484 484199 29640 29659 CCTTCTAGGACTTTTTCCAC 41 1485 484200 29663 29682 GTTTGGTGGGAGAAGCATGG 25 1486 484201 29668 29687 CTCATGTTTGGTGGGAGAAG 45 1487 484202 29673 29692 AGGTACTCATGTTTGGTGGG 77 1488 484203 29678 29697 GCAGCAGGTACTCATGTTTG 86 1489 484204 29683 29702 CAAGGGCAGCAGGTACTCAT 81 1490 484205 29762 29781 ACATATCATCTCTTGCCTGT 59 1491 484206 29767 29786 TATCTACATATCATCTCTTG 31 1492 484207 29772 29791 CATACTATCTACATATCATC 10 1493 484208 29777 29796 AATATCATACTATCTACATA 21 1494 484209 29782 29801 TCCCCAATATCATACTATCT 84 1495 484210 29804 29823 ACCTCAGCTCTTCAAGAAGT 61 1496 484211 29809 29828 CTCAGACCTCAGCTCTTCAA 76 1497 484212 29814 29833 CCTATCTCAGACCTCAGCTC 58 1498 484213 29819 29838 TAAGGCCTATCTCAGACCTC 66 1499 484214 29824 29843 ACCTTTAAGGCCTATCTCAG 4 1500 484215 29829 29848 ACCCAACCTTTAAGGCCTAT 86 1501 484216 29834 29853 TTTTTACCCAACCTTTAAGG 53 1502 484217 29839 29858 TCCATTTTTTACCCAACCTT 87 1503 484218 29844 29863 CTCTTTCCATTTTTTACCCA 80 1504 484219 29849 29868 CTTCTCTCTTTCCATTTTTT 75 1505 484220 29854 29873 CAGGGCTTCTCTCTTTCCAT 83 1506 484221 29859 29878 CTCAGCAGGGCTTCTCTCTT 66 1507 484222 29864 29883 CTGCCCTCAGCAGGGCTTCT 0 1508 484223 29869 29888 ACTAGCTGCCCTCAGCAGGG 28 1509 484224 29902 29921 TTGTGCCATGCCTGCTTTAT 39 1510 484225 30358 30377 GCAGAGCCCATCAGCCTCCC 44 1511 484226 30363 30382 AAAGCGCAGAGCCCATCAGC 48 1512 484227 30407 30426 CTGGCTCCCTTCCAACATAA 64 1513 484228 30412 30431 GAGGGCTGGCTCCCTTCCAA 41 1514 484229 30417 30436 AGGAGGAGGGCTGGCTCCCT 12 1515 484230 30422 30441
TGCCCAGGAGGAGGGCTGGC 4 1516 484231 30483 30502
TAGGCTTTAGAAATTCACCA 91 1517 484232 30514 30533 CAGGTCTGACTCTACAGTCC 78 1518 484233 30519 30538 AAACACAGGTCTGACTCTAC 55 1519 484234 30524 30543 GATTCAAACACAGGTCTGAC 62 1520 484235 30529 30548 GCCAGGATTCAAACACAGGT 82 1521 484236 30534 30553 GCAGAGCCAGGATTCAAACA 55 1522 484237 30539 30558

CAGTGGCAGAGCCAGGATTC 66 1523 484238 30544 30563
CAGGACAGTGGCAGAGCCAG 27 1524 484239 30549 30568
CCCAGCAGGACAGTGGCAGA 39 1525 484240 30554 30573
GGTCACCCAGCAGGACAGTG 65 1526 484241 30559 30578
CCCAAGGTCACCCAGCAGGA 76 1527 484242 30564 30583
ACTTGCCCAAGGTCACCCAG 68 1528 484243 30569 30588 AGATAACTTGCCCAAGGTCA 62 1529 484244 30610 30629 TTGCCCCATTTTAGAGATAA 26 1530 484245 30615 30634 TGACTTTGCCCCATTTTAGA 0 1531 484246 30621 30640 GCAGGGTGACTTTGCCCCAT 35 1532 484247 30644 30663 TGAGCCACTGTCTCAAGTGT 42 1533 484248 30669 30688 TGCCTCTGCATCTCAAGACT 75 1534 484249 30674 30693 CCCAGTGCCTCTGCATCTCA 83 1535 484250 30712 30731 CCAGCCCAGACACTGTGGAT 60 1536 484251 30717 30736 AGCACCCAGCCCAGACACTG 59 1537 484252 30749 30768 TACTGTCCCCTCCTTGTGTG 16 1538 484253 30773 30792 TTCAGCTTCTTGTGAAGCTG 0 1539 484254 30778 30797 TAGGCTTCAGCTTCTTGTGA 25 1540 484255 30783 30802 GGAGATAGGCTTCAGCTTCT 58 1541 484256 30788 30807 CCAAAGGAGATAGGCTTCAG 49 1542 484257 30887 30906 AGCAGATTCCTGGGACAAGA 62 1543 484258 30922 30941 GAGAAATAGCTACAGGAATG 36 1544 484259 30927 30946 ACCCTGAGAAATAGCTACAG 22 1545 484260 30932 30951
CACAAACCCTGAGAAATAGC 0 1546 413433 32431 32450
GCCTGGACAAGTCCTGCCCA 88 425
TABLE-US-00025 TABLE 25 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 484261 32253 32272
AAGACTGGGCTTAGGCTAAG 50 1547 484262 32303 32322
TCAGGAAAAGGAGAAGGAAC 36 1548 484263 32336 32355 AGCTTCAAATTTTCTGCAGT 72 1549 484264 32399 32418 AGTTTCTTTACCTCCAAAAT 48 1550 413433 32431 32450
GCCTGGACAAGTCCTGCCCA 87 425 423463 32432 32451
AGCCTGGACAAGTCCTGCCC 90 467 423464 32433 32452
CAGCCTGGACAAGTCCTGCC 86 468 423465 32434 32453
GCAGCCTGGACAAGTCCTGC 74 469 484265 32436 32455
ATGCAGCCTGGACAAGTCCT 74 1551 484266 32441 32460 AGACTATGCAGCCTGGACAA 63 1552 484267 32446 32465 ATACTAGACTATGCAGCCTG 82 1553 484268 32451 32470 CCATCATACTAGACTATGCA 87 1554 484269 32456 32475 TGTTGCCATCATACTAGACT 83 1555 484270 32461 32480 TGCAATGTTGCCATCATACT 89 1556 484271 32466 32485 GTGGTTGCAATGTTGCCATC 90 1557 484272 32471 32490 GGATGGTGGTTGCAATGTTG 55 1558 484273 32476 32495 AGCCTGGATGGTGGTTGCAA 86 1559 484274 32481 32500 CAATAAGCCTGGATGGTGGT 79 1560 484275 32486 32505 GAATTCAATAAGCCTGGATG 66 1561 484276 32510 32529 TCAGTGGAAAAGAACCTGGG 71 1562 484277 32515 32534
GGAAATCAGTGGAAAAGAAC 55 1563 484278 32520 32539
CAGTAGGAAATCAGTGGAAA 54 1564 484279 32525 32544 ACAGGCAGTAGGAAATCAGT 51 1565 484280 32530 32549 GAGAAACAGGCAGTAGGAAA 56 1566 484281 32586 32605
GACCCAGGAAGCTGGAATCA 66 1567 484282 32591 32610
AGGGAGACCCAGGAAGCTGG 60 1568 484283 32619 32638
ACTGAGATCTCCAGCAGCAA 90 1569 484284 32645 32664
AATGGAGTGACAGGGCAGGA 82 1570 484285 32670 32689
AGGGAGCAATGGTGGCAGGT 71 1571 484286 32675 32694
TGGACAGGGAGCAATGGTGG 46 1572 484287 32680 32699
TGCACTGGACAGGGAGCAAT 36 1573 484288 32685 32704
AGCCCTGCACTGGACAGGGA 41 1574 484289 32706 32725 TTGTGTCCCCATGCCCAGCA 58 1575 484290 32711 32730 CTGACTTGTGTCCCCATGCC 78 1576 484291 32716 32735 CAGGGCTGACTTGTGTCCCC 52 1577 484292 32752 32771 ACAAAGTCTATCAGGATGCA

83 1578 484293 32757 32776 AAGTGACAAAGTCTATCAGG 83 1579 484294 32781 32800 GTCTGCCCATGGCCCCATGG 31 1580 484295 32786 32805 AGAAAGTCTGCCCATGGCCC 45 1581 484296 32791 32810 GCTTGAGAAAGTCTGCCCAT 35 1582 484297 32796 32815 AGCAGGCTTGAGAAAGTCTG 18 1583 484298 32801 32820 GGCTCAGCAGGCTTGAGAAA 60 1584 484299 32806 32825 GATGAGGCTCAGCAGGCTTG 68 1585 484300 32811 32830 TTGCAGATGAGGCTCAGCAG 48 1586 484301 32816 32835 CCATTTTGCAGATGAGGCTC 80 1587 484302 32821 32840 CAGCTCCATTTTGCAGATGA 77 1588 484303 32878 32897 AGAACAGTCTATCATCACTC 34 1589 484304 32899 32918 GAGGTGAGAGAGAAAGCAGA 11 1590 484305 32904 32923 CCCCTGAGGTGAGAGAGAAA 46 1591 484306 32909 32928 CCTGGCCCCTGAGGTGAGAG 38 1592 484307 32914 32933 TGGAGCCTGGCCCCTGAGGT 12 1593 484308 32919 32938 AACACTGGAGCCTGGCCCCT 18 1594 484309 32924 32943 CAGAGAACACTGGAGCCTGG 35 1595 484310 32929 32948 GGCAACAGAGAACACTGGAG 62 1596 484311 32934 32953 ACAGTGGCAACAGAGAACAC 0 1597 484312 32940 32959 CAGGCCACAGTGGCAACAGA 3 1598 484313 32945 32964 AGGACCAGGCCACAGTGGCA 31 1599 484314 32950 32969 TCCAGAGGACCAGGCCACAG 22 1600 484315 32973 32992 GGGCTACTGGCCTCCTGGAG 47 1601 484316 33038 33057 CAAAGGACACCTGCCTGTCC 46 1602 484317 33043 33062 GATAGCAAAGGACACCTGCC 48 1603 484318 33065 33084 TCTTTTGGAAGGGTGGAGAT 16 1604 484319 33070 33089 TTGGTTCTTTTGGAAGGGTG 62 1605 484320 33096 33115 GATGTGAGAAGGACACAGGG 79 1606 484321 33101 33120 ACAGAGATGTGAGAAGGACA 77 1607 484322 33106 33125 TTGGAACAGAGATGTGAGAA 71 1608 484323 33111 33130 ACTTCTTGGAACAGAGATGT 69 1609 484324 33161 33180 GCCCTCACAGGGCCAGGTTG 77 1610 484325 33170 33189 CCCCAGGGAGCCCTCACAGG 68 1611 484326 33175 33194 TAGTGCCCCAGGGAGCCCTC 51 1612 484327 33180 33199 TGTCCTAGTGCCCCAGGGAG 86 1613 484328 33217 33236 TCAACACCCCAGCCTGCCTC 56 1614 484329 33222 33241 GAGGCTCAACACCCCAGCCT 47 1615 484330 33360 33379 AGGAGGGTCACACACACCCC 56 1616 484331 33388 33407 GCAAGAGGCCAGGAAAAGGG 53 1617 484332 33393 33412 TACTGGCAAGAGGCCAGGAA 53 1618 484333 33398 33417 ATGATTACTGGCAAGAGGCC 45 1619 484334 33403 33422 ATTACATGATTACTGGCAAG 30 1620

TABLE-US-00026 TABLE 26 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 94 425 484335 35661 35680 AAAATATACATGCTCTTTTT 52 1621 484336 35666 35685 ACTCAAAAATATACATGCTC 88 1622 484337 35671 35690 TTTCTACTCAAAAATATACA 28 1623 484338 35676 35695 TCCTTTTTCTACTCAAAAAT 64 1624 484339 36215 36234 TTCTGATTAGAAAAATAATA 15 1625 484340 36220 36239 TTTTATTCTGATTAGAAAAA 33 1626 484341 36243 36262 TATTCTGCTCATAAAAACAT 44 1627 484342 36248 36267 AAGGGTATTCTGCTCATAAA 78 1628 484343 36253 36272 TGAGTAAGGGTATTCTGCTC 89 1629 484344 36258 36277 GACAATGAGTAAGGGTATTC 88 1630 484345 36263 36282 AGAGAGACAATGAGTAAGGG 46 1631 484346 36268 36287 GGCTGAGAGAGACAATGAGT 72 1632 484347 36318 36337 AGCCTGTGACACCTACCCTG 68 1633 484348 36354 36373 ATGCAGTGTCCCTAAACCCT 82 1634 484349 36359 36378 GGGTGATGCAGTGTCCCTAA 73 1635 484350 36409 36428 TGGCTTCCTCAGGGCCCACC 90 1636 484351 36454 36473 CCTGACCTGTGGTACTAAGG 68 1637 484352 36512 36531 AGCCCTGTGCCAGCCAGCCT 74 1638 484353 36517 36536 GCGACAGCCCTGTGCCAGCC 92 1639 484354 36522 36541 GACAAGCGACAGCCCTGTGC 60 1640 484355 36573 36592 CCACAGTCAGCTGGGCTCAG 72 1641 484356 36578 36597 CTTTCCCACAGTCAGCTGGG

70 1642 484357 36583 36602 GGAAACTTTCCCACAGTCAG 89 1643 484358 36588 36607 CAAATGGAAACTTTCCCACA 56 1644 484359 36617 36636 GGCCTGGAAAAGGGACACTG 71 1645 484360 36622 36641 CCCCTGGCCTGGAAAAGGGA 48 1646 484361 36627 36646 CTACTCCCCTGGCCTGGAAA 21 1647 484362 36632 36651 ACCTCCTACTCCCCTGGCCT 67 1648 484363 36637 36656 AGCCCACCTCCTACTCCCCT 68 1649 484364 36642 36661 CAGGCAGCCCACCTCCTACT 66 1650 484365 36647 36666 TGAGACAGGCAGCCCACCTC 66 1651 484366 36652 36671 CAGAATGAGACAGGCAGCCC 56 1652 484367 36676 36695 CTCTGTGGGCTCCTCCACAG 45 1653 484368 36681 36700 CTGTGCTCTGTGGGCTCCTC 81 1654 484369 36686 36705 TGGCCCTGTGCTCTGTGGGC 42 1655 484370 36691 36710 TTACTTGGCCCTGTGCTCTG 75 1656 484371 36733 36752 ACTGTCAGTCTCTCCAAACT 73 1657 484372 36738 36757 CCCCCACTGTCAGTCTCTCC 78 1658 484373 36743 36762 CTCTGCCCCCACTGTCAGTC 72 1659 484374 36748 36767 GCAAGCTCTGCCCCCACTGT 71 1660 484375 36753 36772 TGGCTGCAAGCTCTGCCCCC 70 1661 484376 36758 36777 GGCCTTGGCTGCAAGCTCTG 76 1662 484377 36841 36860 AATCCAACCCTTGTCACTCT 88 1663 484378 36846 36865 ACTCAAATCCAACCCTTGTC 81 1664 484379 36855 36874 TATTCTAAAACTCAAATCCA 61 1665 484380 36860 36879 GTGATTATTCTAAAACTCAA 65 1666 484381 36865 36884 CCAGAGTGATTATTCTAAAA 73 1667 484382 36896 36915 CCTCCCTCTGGGCCCATCCT 74 1668 484383 36923 36942 TTTGCACTCTGCCAGCCTCC 78 1669 484384 36928 36947 TCTTCTTTGCACTCTGCCAG 69 1670 484385 36969 36988 TGCCCTTGTTCCTCTCAGTC 68 1671 484386 36974 36993 ATGTTTGCCCTTGTTCCTCT 62 1672 484387 37022 37041 GTTGGGCAGTCACCATTTGC 63 1673 484388 37027 37046 CTGGTGTTGGGCAGTCACCA 30 1674 484389 37032 37051 CAGTGCTGGTGTTGGGCAGT 30 1675 484390 37054 37073 CAGACTCTCATCCCCAGGGC 72 1676 484391 37119 37138 GGATGTGTCATTTCCCCTGG 86 1677 484392 37174 37193 AGGAAACTGGCTGGGAGAGG 20 1678 484393 37179 37198 GTCAGAGGAAACTGGCTGGG 48 1679 484394 37184 37203 CTTGGGTCAGAGGAAACTGG 37 1680 484395 37189 37208 ATGACCTTGGGTCAGAGGAA 38 1681 484396 37194 37213 CAAGGATGACCTTGGGTCAG 36 1682 484397 37199 37218 AGCTGCAAGGATGACCTTGG 48 1683 484398 37204 37223 TCACCAGCTGCAAGGATGAC 24 1684 484399 37440 37459 AACAGTGCCCAGCAGGAGCA 14 1685 484400 37445 37464 TTGACAACAGTGCCCAGCAG 19 1686 484401 37450 37469 AGGCCTTGACAACAGTGCCC 36 1687 484402 37455 37474 GGCTCAGGCCTTGACAACAG 57 1688 484403 37460 37479 GGAGAGGCTCAGGCCTTGAC 53 1689 484404 37485 37504 GGCTCCCTGTGTCACCCTGC 61 1690 484405 37490 37509 GTGTTGGCTCCCTGTGTCAC 39 1691 484406 37495 37514 ATGGTGTGTTGGCTCCCTGT 36 1692 484407 37500 37519 AAGGAATGGTGTGTTGGCTC 48 1693 484408 37505 37524 GCACCAAGGAATGGTGTGTT 30 1694 484409 37510 37529 GCCCAGCACCAAGGAATGGT 48 1695 484410 37515 37534 TGCAGGCCCAGCACCAAGGA 55 1696 484411 37520 37539 CCCAATGCAGGCCCAGCACC 67 1697

TABLE-US-00027 TABLE 27 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 495425 10040 10059 AGGGCCCTTTCCAGAAAATC 84 1698 495426 10041 10060 CAGGGCCCTTTCCAGAAAAT 29 1699 495427 10042 10061 ACAGGGCCCTTTCCAGAAAA 14 1700 495428 10043 10062 CACAGGGCCCTTTCCAGAAA 64 1701 495429 10045 10064 GCCACAGGGCCCTTTCCAGA 86 1702 495430 10046 10065 TGCCACAGGGCCCTTTCCAG 77 1703 495431 10047 10066 CTGCCACAGGGCCCTTTCCA 50 1704 495432 10048 10067 CCTGCCACAGGGCCCTTTCC 44 1705 495433 10206 10225 GAATATCCCTAATAACTAAG 18 1706 495434 10207 10226

CGAATATCCCTAATAACTAA 8 1707 495435 10208 10227 TCGAATATCCCTAATAACTA 18 1708 495436 10209 10228 CTCGAATATCCCTAATAACT 66 1709 495437 10211 10230 TTCTCGAATATCCCTAATAA 31 1710 495438 10212 10231 GTTCTCGAATATCCCTAATA 69 1711 495439 10213 10232 AGTTCTCGAATATCCCTAAT 36 1712 495440 10214 10233 GAGTTCTCGAATATCCCTAA 84 1713 495441 10216 10235 AGGAGTTCTCGAATATCCCT 79 1714 495442 10217 10236 GAGGAGTTCTCGAATATCCC 83 1715 495443 10218 10237 GGAGGAGTTCTCGAATATCC 78 1716 495444 10219 10238 GGGAGGAGTTCTCGAATATC 64 1717 495445 10491 10510 GCAGGGTCCTCTCCGCTGCC 70 1718 495446 10492 10511 TGCAGGGTCCTCTCCGCTGC 80 1719 495447 10528 10547 GATTCACTGAGGACCTCAGT 71 1720 495448 10529 10548 CGATTCACTGAGGACCTCAG 78 1721 495449 10530 10549 GCGATTCACTGAGGACCTCA 94 1722 495450 10531 10550 CGCGATTCACTGAGGACCTC 90 1723 495451 10533 10552 TGCGCGATTCACTGAGGACC 82 1724 495452 10534 10553 CTGCGCGATTCACTGAGGAC 77 1725 495453 10535 10554 TCTGCGCGATTCACTGAGGA 81 1726 495454 10536 10555 CTCTGCGCGATTCACTGAGG 85 1727 495455 10646 10665 CCTCCCACTTCAGTTTCTCC 64 1728 495456 10647 10666 TCCTCCCACTTCAGTTTCTC 71 1729 495457 10648 10667 TTCCTCCCACTTCAGTTTCT 63 1730 495458 10649 10668 CTTCCTCCCACTTCAGTTTC 70 1731 495459 10651 10670 TGCTTCCTCCCACTTCAGTT 63 1732 495460 10652 10671 ATGCTTCCTCCCACTTCAGT 55 1733 495461 10653 10672 CATGCTTCCTCCCACTTCAG 70 1734 495462 10654 10673 GCATGCTTCCTCCCACTTCA 78 1735 495463 10656 10675 AGGCATGCTTCCTCCCACTT 81 1736 495464 10657 10676 TAGGCATGCTTCCTCCCACT 85 1737 495465 10658 10677 TTAGGCATGCTTCCTCCCAC 78 1738 495466 10659 10678 CTTAGGCATGCTTCCTCCCA 83 1739 495467 10661 10680 AACTTAGGCATGCTTCCTCC 82 1740 495468 10662 10681 AAACTTAGGCATGCTTCCTC 76 1741 495469 10663 10682 AAAACTTAGGCATGCTTCCT 83 1742 495470 10664 10683 GAAAACTTAGGCATGCTTCC 87 1743 495471 10666 10685 AGGAAAACTTAGGCATGCTT 83 1744 495472 10667 10686 AAGGAAAACTTAGGCATGCT 85 1745 495473 10668 10687 TAAGGAAAACTTAGGCATGC 76 1746 495474 10669 10688 CTAAGGAAAACTTAGGCATG 59 1747 495475 10671 10690 AGCTAAGGAAAACTTAGGCA 65 1748 495476 10672 10691 CAGCTAAGGAAAACTTAGGC 70 1749 495477 10673 10692 TCAGCTAAGGAAAACTTAGG 40 1750 495478 10674 10693 ATCAGCTAAGGAAAACTTAG 25 1751 495479 10728 10747 TGTGATGACTTCCCAGGGTC 75 1752 495480 10729 10748 CTGTGATGACTTCCCAGGGT 72 1753 495481 10730 10749 GCTGTGATGACTTCCCAGGG 87 1754 495482 10731 10750 AGCTGTGATGACTTCCCAGG 80 1755 495483 10733 10752 ACAGCTGTGATGACTTCCCA 79 1756 495484 10734 10753 GACAGCTGTGATGACTTCCC 86 1757 495485 10735 10754 GGACAGCTGTGATGACTTCC 80 1758 495486 10736 10755 AGGACAGCTGTGATGACTTC 62 1759 495487 10815 10834 CATGTCAGAGAGGCTCAGCA 72 1760 495488 10816 10835 CCATGTCAGAGAGGCTCAGC 87 1761 495489 10817 10836 TCCATGTCAGAGAGGCTCAG 86 1762 495490 10818 10837 ATCCATGTCAGAGAGGCTCA 88 1763 495491 10820 10839 AAATCCATGTCAGAGAGGCT 85 1764 495492 10821 10840 AAAATCCATGTCAGAGAGGC 86 1765 495493 10822 10841 AAAAATCCATGTCAGAGAGG 57 1766 495494 10823 10842 GAAAAATCCATGTCAGAGAG 52 1767 495495 10941 10960 TGGATAGTCGATTTACCAGA 92 1768 495496 10942 10961 TTGGATAGTCGATTTACCAG 89 1769 495497 10944 10963 CTTTGGATAGTCGATTTACC 89 1770 495498 11075 11094 ACCTCGATGTTACATTAAGG 90 1771 495499 11076 11095 AACCTCGATGTTACATTAAG 72 1772 495500 11077 11096 AAACCTCGATGTTACATTAA 64 1773 495501 11078 11097 GAAACCTCGATGTTACATTA 71 1774 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 74 425

TABLE-US-00028 TABLE 28 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 495502 11124 11143 TATTGCAACCACTAGGACAT 78 1775 495503 11125 11144 TTATTGCAACCACTAGGACA

72 1776 495504 11126 11145 GTTATTGCAACCACTAGGAC 79 1777 495505 11127 11146 GGTTATTGCAACCACTAGGA 91 1778 495506 11129 11148 TGGGTTATTGCAACCACTAG 91 1779 495507 11130 11149 GTGGGTTATTGCAACCACTA 86 1780 495508 11131 11150 GGTGGGTTATTGCAACCACT 85 1781 495509 11159 11178 TCACAGTTAAAGTGTGGTAC 82 1782 495510 11160 11179 GTCACAGTTAAAGTGTGGTA 84 1783 495511 11161 11180 GGTCACAGTTAAAGTGTGGT 78 1784 495512 11162 11181 AGGTCACAGTTAAAGTGTGG 7 1785 495513 11169 11188 AATGCCTAGGTCACAGTTAA 46 1786 495514 11170 11189 CAATGCCTAGGTCACAGTTA 80 1787 495515 11171 11190 CCAATGCCTAGGTCACAGTT 88 1788 495516 11172 11191 GCCAATGCCTAGGTCACAGT 94 1789 495517 11174 11193 ATGCCAATGCCTAGGTCACA 92 1790 495518 11175 11194 AATGCCAATGCCTAGGTCAC 93 1791 495519 11176 11195 CAATGCCAATGCCTAGGTCA 92 1792 495520 11177 11196 GCAATGCCAATGCCTAGGTC 92 1793 495521 11200 11219 CCACAGCGATAATCACACAA 84 1794 495522 11201 11220 ACCACAGCGATAATCACACA 92 1795 495523 11202 11221 AACCACAGCGATAATCACAC 94 1796 495524 11203 11222 TAACCACAGCGATAATCACA 91 1797 495525 14504 14523 GCAGGGTCTGCCACTCTCTA 86 1798 495526 14505 14524 AGCAGGGTCTGCCACTCTCT 92 1799 495527 14506 14525 GAGCAGGGTCTGCCACTCTC 91 1800 495528 14507 14526 AGAGCAGGGTCTGCCACTCT 82 1801 495529 14649 14668 GCACAGTCATCTTGTGTACA 86 1802 495530 14650 14669 TGCACAGTCATCTTGTGTAC 81 1803 495531 14651 14670 CTGCACAGTCATCTTGTGTA 80 1804 495532 14652 14671 TCTGCACAGTCATCTTGTGT 64 1805 495533 14659 14678 AGATCACTCTGCACAGTCAT 81 1806 495534 14660 14679 CAGATCACTCTGCACAGTCA 87 1807 495535 14661 14680 TCAGATCACTCTGCACAGTC 90 1808 495536 14662 14681 CTCAGATCACTCTGCACAGT 86 1809 495537 14664 14683 TGCTCAGATCACTCTGCACA 87 1810 495538 14665 14684 TTGCTCAGATCACTCTGCAC 78 1811 495539 14666 14685 ATTGCTCAGATCACTCTGCA 84 1812 495540 14667 14686 CATTGCTCAGATCACTCTGC 88 1813 495541 14741 14760 CCTCTTCCAGGAAGACTTCC 80 1814 495542 14742 14761 ACCTCTTCCAGGAAGACTTC 72 1815 495543 14743 14762 CACCTCTTCCAGGAAGACTT 88 1816 495544 14744 14763 TCACCTCTTCCAGGAAGACT 86 1817 495545 14746 14765 TGTCACCTCTTCCAGGAAGA 82 1818 495546 14747 14766 ATGTCACCTCTTCCAGGAAG 42 1819 495547 14748 14767 AATGTCACCTCTTCCAGGAA 50 1820 495548 14749 14768 GAATGTCACCTCTTCCAGGA 74 1821 495549 14751 14770 CTGAATGTCACCTCTTCCAG 75 1822 495550 14752 14771 ACTGAATGTCACCTCTTCCA 81 1823 495551 14753 14772 GACTGAATGTCACCTCTTCC 81 1824 495552 14754 14773 GGACTGAATGTCACCTCTTC 88 1825 495553 14756 14775 CCGGACTGAATGTCACCTCT 90 1826 495554 14757 14776 TCCGGACTGAATGTCACCTC 91 1827 495555 14758 14777 ATCCGGACTGAATGTCACCT 91 1828 495556 14759 14778 GATCCGGACTGAATGTCACC 85 1829 495557 14766 14785 CTTTCCAGATCCGGACTGAA 67 1830 495558 14767 14786 TCTTTCCAGATCCGGACTGA 77 1831 495559 14768 14787 ATCTTTCCAGATCCGGACTG 69 1832 495560 14769 14788 CATCTTTCCAGATCCGGACT 88 1833 495561 14771 14790 TTCATCTTTCCAGATCCGGA 89 1834 495562 14772 14791 ATTCATCTTTCCAGATCCGG 94 1835 495563 14773 14792 TATTCATCTTTCCAGATCCG 92 1836 495564 14774 14793 CTATTCATCTTTCCAGATCC 84 1837 495565 14992 15011 CTCAAGTCTTCCTCCTGCTG 79 1838 495566 14993 15012 TCTCAAGTCTTCCTCCTGCT 83 1839 495567 14994 15013 CTCTCAAGTCTTCCTCCTGC 63 1840 495568 14995 15014 GCTCTCAAGTCTTCCTCCTG 86 1841 495569 14997 15016 GAGCTCTCAAGTCTTCCTCC 84 1842 495570 14998 15017 TGAGCTCTCAAGTCTTCCTC 90 1843 495571 14999 15018 ATGAGCTCTCAAGTCTTCCT 91 1844 495572 15000 15019 CATGAGCTCTCAAGTCTTCC 85 1845 495573 15248 15267 CATAATCTGCACAGGTTCTT 84 1846 495574 15249 15268 CCATAATCTGCACAGGTTCT 87 1847 495575 15250 15269 ACCATAATCTGCACAGGTTC 91 1848 495576 15251 15270 CACCATAATCTGCACAGGTT 94 1849 495577 15253 15272 TGCACCATAATCTGCACAGG 90 1850 495578 15254 15273 CTGCACCATAATCTGCACAG

86 1851 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 92 425 TABLE-US-00029 TABLE 29 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 495579 15255 15274 TCTGCACCATAATCTGCACA 78 1852 495580 15256 15275 ATCTGCACCATAATCTGCAC 73 1853 495581 18152 18171 TAATCCAGGATTGTCATAAG 53 1854 495582 18153 18172 TTAATCCAGGATTGTCATAA 54 1855 495583 18154 18173 TTTAATCCAGGATTGTCATA 40 1856 495584 18155 18174 CTTTAATCCAGGATTGTCAT 70 1857 495585 18157 18176 AGCTTTAATCCAGGATTGTC 81 1858 495586 18158 18177 TAGCTTTAATCCAGGATTGT 37 1859 495587 18159 18178 TTAGCTTTAATCCAGGATTG 60 1860 495588 18160 18179 CTTAGCTTTAATCCAGGATT 75 1861 495589 18162 18181 TCCTTAGCTTTAATCCAGGA 78 1862 495590 18163 18182 CTCCTTAGCTTTAATCCAGG 77 1863 495591 18164 18183 CCTCCTTAGCTTTAATCCAG 82 1864 495592 18165 18184 TCCTCCTTAGCTTTAATCCA 73 1865 495593 18167 18186 TGTCCTCCTTAGCTTTAATC 69 1866 495594 18168 18187 GTGTCCTCCTTAGCTTTAAT 61 1867 495595 18169 18188 AGTGTCCTCCTTAGCTTTAA 51 1868 495596 18170 18189 CAGTGTCCTCCTTAGCTTTA 61 1869 495597 18172 18191 CTCAGTGTCCTCCTTAGCTT 76 1870 495598 18173 18192 ACTCAGTGTCCTCCTTAGCT 81 1871 495599 18174 18193 GACTCAGTGTCCTCCTTAGC 83 1872 495600 18175 18194 GGACTCAGTGTCCTCCTTAG 80 1873 495601 18177 18196 TGGGACTCAGTGTCCTCCTT 64 1874 495602 18178 18197 CTGGGACTCAGTGTCCTCCT 72 1875 495603 18179 18198 CCTGGGACTCAGTGTCCTCC 77 1876 495604 18180 18199 CCCTGGGACTCAGTGTCCTC 88 1877 495605 18307 18326 TCCTGTTTGATGGGTAAAAT 77 1878 495606 18308 18327 CTCCTGTTTGATGGGTAAAA 83 1879 495607 18309 18328 TCTCCTGTTTGATGGGTAAA 78 1880 495608 18310 18329 CTCTCCTGTTTGATGGGTAA 81 1881 495609 18312 18331 TCCTCTCCTGTTTGATGGGT 83 1882 495610 18313 18332 GTCCTCTCCTGTTTGATGGG 77 1883 495611 18314 18333 TGTCCTCTCCTGTTTGATGG 69 1884 495612 18315 18334 GTGTCCTCTCCTGTTTGATG 76 1885 495613 18317 18336 CGGTGTCCTCTCCTGTTTGA 77 1886 495614 18318 18337 TCGGTGTCCTCTCCTGTTTG 72 1887 495615 18319 18338 CTCGGTGTCCTCTCCTGTTT 69 1888 495616 18320 18339 CCTCGGTGTCCTCTCCTGTT 77 1889 495617 18322 18341 AGCCTCGGTGTCCTCTCCTG 72 1890 495618 18323 18342 AAGCCTCGGTGTCCTCTCCT 84 1891 495619 18324 18343 TAAGCCTCGGTGTCCTCTCC 88 1892 495620 18325 18344 GTAAGCCTCGGTGTCCTCTC 91 1893 495621 18327 18346 AAGTAAGCCTCGGTGTCCTC 86 1894 495622 18328 18347 CAAGTAAGCCTCGGTGTCCT 81 1895 495623 18330 18349 ACCAAGTAAGCCTCGGTGTC 79 1896 495624 18418 18437 CTCACCAGATTGTTGTGGGA 79 1897 495625 18419 18438 CCTCACCAGATTGTTGTGGG 77 1898 495626 18420 18439 ACCTCACCAGATTGTTGTGG 79 1899 495627 18421 18440 TACCTCACCAGATTGTTGTG 47 1900 495628 18428 18447 TAATACCTACCTCACCAGAT 56 1901 495629 18429 18448 CTAATACCTACCTCACCAGA 49 1902 495630 18430 18449 GCTAATACCTACCTCACCAG 60 1903 495631 18431 18450 GGCTAATACCTACCTCACCA 70 1904 495632 18451 18470 CTTCCTCATCTATACAGTGG 72 1905 495633 18452 18471 GCTTCCTCATCTATACAGTG 72 1906 495634 18453 18472 AGCTTCCTCATCTATACAGT 63 1907 495635 18454 18473 CAGCTTCCTCATCTATACAG 66 1908 495636 18582 18601 CCATACTGGCATCTGGCAGG 78 1909 495637 18583 18602 TCCATACTGGCATCTGGCAG 73 1910 495638 18584 18603 CTCCATACTGGCATCTGGCA 78 1911 495639 18585 18604 CCTCCATACTGGCATCTGGC 82 1912 495640 18587 18606 CACCTCCATACTGGCATCTG 51 1913 495641 18588 18607 TCACCTCCATACTGGCATCT 71 1914 495642 18589 18608 CTCACCTCCATACTGGCATC 72 1915 495643 18590 18609 CCTCACCTCCATACTGGCAT 70 1916 495644 18592 18611 CACCTCACCTCCATACTGGC 68 1917 495645 18593 18612 CCACCTCACCTCCATACTGG 54 1918 495646 18594 18613 CCCACCTCACCTCCATACTG 45 1919 495647 18595 18614 ACCCACCTCACCTCCATACT 37 1920 495648 18827 18846

GGATGCTTAACTTCTGCTGA 67 1921 495649 18828 18847 AGGATGCTTAACTTCTGCTG 83 1922 495650 18829 18848 TAGGATGCTTAACTTCTGCT 84 1923 495651 18830 18849 TTAGGATGCTTAACTTCTGC 68 1924 495652 18832 18851 AGTTAGGATGCTTAACTTCT 50 1925 495653 18833 18852 AAGTTAGGATGCTTAACTTC 49 1926 495654 18834 18853 TAAGTTAGGATGCTTAACTT 49 1927 495655 18835 18854 TTAAGTTAGGATGCTTAACT 42 1928 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 82 425

TABLE-US-00030 TABLE 30 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 423461 32429 32448 CTGGACAAGTCCTGCCCATC 73 465 423462 32430 32449 CCTGGACAAGTCCTGCCCAT 70 466 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 87 425 495656 18870 18889 TATAGGCCTGGATGCCCAAG 70 1929 20780 20799 495657 18871 18890 CTATAGGCCTGGATGCCCAA 72 1930 20781 20800 495658 18896 18915 AGATGATGTCCTGCCTCAGA 30 1931 495659 18897 18916 CAGATGATGTCCTGCCTCAG 37 1932 495660 18898 18917 GCAGATGATGTCCTGCCTCA 65 1933 495661 18899 18918 AGCAGATGATGTCCTGCCTC 72 1934 495662 19014 19033 AGTCAAAGGTGGCTTCCTGG 67 1935 495663 19015 19034 TAGTCAAAGGTGGCTTCCTG 57 1936 495664 19016 19035 GTAGTCAAAGGTGGCTTCCT 75 1937 495665 19017 19036 AGTAGTCAAAGGTGGCTTCC 66 1938 495666 19020 19039 ATGAGTAGTCAAAGGTGGCT 73 1939 495667 19021 19040 AATGAGTAGTCAAAGGTGGC 64 1940 495668 19022 19041 GAATGAGTAGTCAAAGGTGG 31 1941 495669 20775 20794 GCCTGGATGCCCAAGTTAGA 72 1942 495670 20776 20795 GGCCTGGATGCCCAAGTTAG 73 1943 495671 20777 20796 AGGCCTGGATGCCCAAGTTA 77 1944 495672 20778 20797 TAGGCCTGGATGCCCAAGTT 57 1945 495673 22544 22563 ACAAGTGGGAAATGCAGTCT 48 1946 495674 22545 22564 AACAAGTGGGAAATGCAGTC 42 1947 495675 22546 22565 CAACAAGTGGGAAATGCAGT 35 1948 495676 22547 22566 CCAACAAGTGGGAAATGCAG 62 1949 495677 22549 22568 TGCCAACAAGTGGGAAATGC 70 1950 495678 22550 22569 CTGCCAACAAGTGGGAAATG 52 1951 495679 22551 22570 TCTGCCAACAAGTGGGAAAT 39 1952 495680 22552 22571 CTCTGCCAACAAGTGGGAAA 34 1953 495681 22767 22786 GCTTATTAGGTGTCTTAAAA 51 1954 495682 22768 22787 AGCTTATTAGGTGTCTTAAA 56 1955 495683 22769 22788 AAGCTTATTAGGTGTCTTAA 59 1956 495684 22770 22789 TAAGCTTATTAGGTGTCTTA 63 1957 495685 22772 22791 GCTAAGCTTATTAGGTGTCT 86 1958 495686 22773 22792 TGCTAAGCTTATTAGGTGTC 80 1959 495687 22774 22793 CTGCTAAGCTTATTAGGTGT 68 1960 495688 22775 22794 TCTGCTAAGCTTATTAGGTG 64 1961 495689 22882 22901 CCCCATGCAGCTTGGAGGAG 60 1962 495690 22883 22902 CCCCCATGCAGCTTGGAGGA 53 1963 495691 22884 22903 GCCCCCATGCAGCTTGGAGG 43 1964 495692 22885 22904 AGCCCCCATGCAGCTTGGAG 61 1965 495693 22887 22906 CCAGCCCCCATGCAGCTTGG 66 1966 495694 22888 22907 GCCAGCCCCCATGCAGCTTG 78 1967 495695 22889 22908 GGCCAGCCCCCATGCAGCTT 71 1968 495696 22890 22909 GGGCCAGCCCCCATGCAGCT 63 1969 495697 23096 23115 AGTCCAAACACTCAGGTAGG 82 1970 495698 23097 23116 CAGTCCAAACACTCAGGTAG 68 1971 495699 23098 23117 TCAGTCCAAACACTCAGGTA 74 1972 495700 23099 23118 TTCAGTCCAAACACTCAGGT 72 1973 495701 23238 23257 GGGAACAGCAGCATCAGCAT 77 1974 495702 23239 23258 TGGGAACAGCAGCATCAGCA 83 1975 495703 23240 23259 CTGGGAACAGCAGCATCAGC 68 1976 495704 23241 23260 TCTGGGAACAGCAGCATCAG 65 1977 495705 23243 23262 GGTCTGGGAACAGCAGCATC 84 1978 495706 23244 23263 TGGTCTGGGAACAGCAGCAT 81 1979 495707 23245 23264 GTGGTCTGGGAACAGCAGCA 84 1980 495708 23423 23442 CAAGTCATCTTCCAGCAGCT 57 1981 495709 23424 23443 ACAAGTCATCTTCCAGCAGC

36 1982 495710 23425 23444 GACAAGTCATCTTCCAGCAG 38 1983 495711 23426 23445 GGACAAGTCATCTTCCAGCA 77 1984 495712 23428 23447 CTGGACAAGTCATCTTCCAG 53 1985 495713 23429 23448 GCTGGACAAGTCATCTTCCA 66 1986 495714 23430 23449 GGCTGGACAAGTCATCTTCC 76 1987 495715 23431 23450 GGGCTGGACAAGTCATCTTC 45 1988 495716 23548 23567 ATCCTTGAGGGTCTCATAAC 58 1989 495717 23549 23568 TATCCTTGAGGGTCTCATAA 61 1990 495718 23551 23570 CTTATCCTTGAGGGTCTCAT 83 1991 495719 23553 23572 TGCTTATCCTTGAGGGTCTC 79 1992 495720 23554 23573 ATGCTTATCCTTGAGGGTCT 73 1993 495721 23555 23574 CATGCTTATCCTTGAGGGTC 64 1994 495722 23556 23575 ACATGCTTATCCTTGAGGGT 60 1995 495723 26396 26415 CTGGATGGCAACCTAAGGAG 61 1996 495724 26397 26416 CCTGGATGGCAACCTAAGGA 74 1997 495725 26398 26417 GCCTGGATGGCAACCTAAGG 61 1998 495726 26399 26418 GGCCTGGATGGCAACCTAAG 60 1999 495727 26401 26420 CTGGCCTGGATGGCAACCTA 69 2000 495728 26480 26499 TGCTATGCTGAGAGCACAGG 69 2001 495729 26481 26500 CTGCTATGCTGAGAGCACAG 64 2002 495730 26482 26501 CCTGCTATGCTGAGAGCACA 81 2003

TABLE-US-00031 TABLE 31 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 495731 26483 26502 ACCTGCTATGCTGAGAGCAC 75 2004 495732 26485 26504 CTACCTGCTATGCTGAGAGC 69 2005 495733 26486 26505 GCTACCTGCTATGCTGAGAG 54 2006 495734 26487 26506 AGCTACCTGCTATGCTGAGA 67 2007 495735 26488 26507 AAGCTACCTGCTATGCTGAG 44 2008 495736 26517 26536 GCAGGTTCATCTGCCTTGAC 88 2009 495737 26518 26537 AGCAGGTTCATCTGCCTTGA 75 2010 495738 26519 26538 GAGCAGGTTCATCTGCCTTG 84 2011 495739 26520 26539 GGAGCAGGTTCATCTGCCTT 80 2012 495740 26532 26551 CTGTGATGCTCTGGAGCAGG 56 2013 495741 26533 26552 TCTGTGATGCTCTGGAGCAG 49 2014 495742 26534 26553 CTCTGTGATGCTCTGGAGCA 66 2015 495743 26535 26554 ACTCTGTGATGCTCTGGAGC 74 2016 495744 26537 26556 GCACTCTGTGATGCTCTGGA 85 2017 495745 26538 26557 TGCACTCTGTGATGCTCTGG 80 2018 495746 26539 26558 ATGCACTCTGTGATGCTCTG 70 2019 495747 26540 26559 AATGCACTCTGTGATGCTCT 60 2020 495748 26626 26645 TGACATGGTAAGTCCTGATG 79 2021 495749 26627 26646 CTGACATGGTAAGTCCTGAT 86 2022 495750 26628 26647 ACTGACATGGTAAGTCCTGA 79 2023 495751 26629 26648 CACTGACATGGTAAGTCCTG 84 2024 495752 26631 26650 AGCACTGACATGGTAAGTCC 90 2025 495753 26632 26651 CAGCACTGACATGGTAAGTC 86 2026 495754 26633 26652 TCAGCACTGACATGGTAAGT 73 2027 495755 26634 26653 CTCAGCACTGACATGGTAAG 63 2028 495756 26779 26798 CTGCCATTTAATGAGCTTCA 86 2029 495757 26780 26799 TCTGCCATTTAATGAGCTTC 77 2030 495758 26781 26800 CTCTGCCATTTAATGAGCTT 50 2031 495759 26782 26801 ACTCTGCCATTTAATGAGCT 43 2032 495760 26784 26803 CAACTCTGCCATTTAATGAG 34 2033 495761 26785 26804 CCAACTCTGCCATTTAATGA 39 2034 495762 26786 26805 CCCAACTCTGCCATTTAATG 74 2035 495763 26787 26806 TCCCAACTCTGCCATTTAAT 76 2036 495764 26789 26808 AATCCCAACTCTGCCATTTA 62 2037 495765 26790 26809 AAATCCCAACTCTGCCATTT 61 2038 495766 26927 26946 TTTAAGGGTTCATGGATCCC 70 2039 495767 26928 26947 TTTTAAGGGTTCATGGATCC 48 2040 495768 26929 26948 ATTTTAAGGGTTCATGGATC 23 2041 495769 26930 26949 AATTTTAAGGGTTCATGGAT 9 2042 495770 27247 27266 TGGAGTCCCTGGGTTCTTGT 51 2043 495771 27248 27267 CTGGAGTCCCTGGGTTCTTG 52 2044 495772 27249 27268 GCTGGAGTCCCTGGGTTCTT 72 2045 495773 27250 27269 GGCTGGAGTCCCTGGGTTCT 64 2046 495774 27252 27271 GAGGCTGGAGTCCCTGGGTT 64 2047 495775 27253 27272 GGAGGCTGGAGTCCCTGGGT 45 2048 495776 27255 27274 TGGGAGGCTGGAGTCCCTGG 24 2049 495777 27353 27372 TACCAGCCAGGCCCAGACCC 8 2050 495778 27354 27373

CTACCAGCCAGGCCCAGACC 7 2051 495779 27355 27374
CCTACCAGCCAGGCCCAGAC 6 2052 495780 27356 27375
CCCTACCAGCCAGGCCCAGA 9 2053 495781 27378 27397
GTGTTCTACAAGCTGCTCAG 64 2054 495782 27379 27398 GGTGTTCTACAAGCTGCTCA 77 2055 495783 27380 27399 TGGTGTTCTACAAGCTGCTC 68 2056 495784 27381 27400 CTGGTGTTCTACAAGCTGCT 74 2057 495785 27383 27402 AGCTGGTGTTCTACAAGCTG 80 2058 495786 27384 27403 GAGCTGGTGTTCTACAAGCT 61 2059 495787 27385 27404 TGAGCTGGTGTTCTACAAGC 58 2060 495788 27386 27405 GTGAGCTGGTGTTCTACAAG 62 2061 495789 27438 27457 GTCTGATCAATTATTAACCT 71 2062 495790 27439 27458 GGTCTGATCAATTATTAACC 57 2063 495791 27440 27459 GGGTCTGATCAATTATTAAC 60 2064 495792 27441 27460 TGGGTCTGATCAATTATTAA 40 2065 495793 29674 29693 CAGGTACTCATGTTTGGTGG 63 2066 495794 29675 29694 GCAGGTACTCATGTTTGGTG 75 2067 495795 29676 29695 AGCAGGTACTCATGTTTGGT 73 2068 495796 29677 29696 CAGCAGGTACTCATGTTTGG 66 2069 495797 29679 29698 GGCAGCAGGTACTCATGTTT 60 2070 495798 29680 29699 GGGCAGCAGGTACTCATGTT 58 2071 495799 29681 29700 AGGGCAGCAGGTACTCATGT 63 2072 495800 29682 29701 AAGGGCAGCAGGTACTCATG 75 2073 495801 29778 29797 CAATATCATACTATCTACAT 11 2074 495802 29779 29798 CCAATATCATACTATCTACA 13 2075 495803 29780 29799 CCCAATATCATACTATCTAC 51 2076 495804 29781 29800 CCCCAATATCATACTATCTA 71 2077 495805 29783 29802 TTCCCCAATATCATACTATC 50 2078 495806 29825 29844 AACCTTTAAGGCCTATCTCA 32 2079 495807 29826 29845 CAACCTTTAAGGCCTATCTC 25 2080 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 84 425

TABLE-US-00032 TABLE 32 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 495808 29827 29846 CCAACCTTTAAGGCCTATCT 82 2081 495809 29828 29847 CCCAACCTTTAAGGCCTATC 90 2082 495810 29830 29849 TACCCAACCTTTAAGGCCTA 82 2083 495811 29831 29850 TTACCCAACCTTTAAGGCCT 75 2084 495812 29832 29851 TTTACCCAACCTTTAAGGCC 73 2085 495813 29833 29852 TTTTACCCAACCTTTAAGGC 70 2086 495814 29835 29854 TTTTTTACCCAACCTTTAAG 32 2087 495815 29836 29855 ATTTTTTACCCAACCTTTAA 27 2088 495816 29837 29856 CATTTTTTACCCAACCTTTA 54 2089 495817 29838 29857 CCATTTTTTACCCAACCTTT 81 2090 495818 29840 29859 TTCCATTTTTTACCCAACCT 89 2091 495819 29841 29860 TTTCCATTTTTTACCCAACC 88 2092 495820 29842 29861 CTTTCCATTTTTTACCCAAC 85 2093 495821 29843 29862 TCTTTCCATTTTTTACCCAA 74 2094 495822 30484 30503 TTAGGCTTTAGAAATTCACC 85 2095 495823 30485 30504 ATTAGGCTTTAGAAATTCAC 69 2096 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 89 425 495824 32437 32456 TATGCAGCCTGGACAAGTCC 65 2097 495825 32447 32466 CATACTAGACTATGCAGCCT 91 2098 495826 32448 32467 TCATACTAGACTATGCAGCC 88 2099 495827 32449 32468 ATCATACTAGACTATGCAGC 86 2100 495828 32450 32469 CATCATACTAGACTATGCAG 82 2101 495829 32452 32471 GCCATCATACTAGACTATGC 92 2102 495830 32453 32472 TGCCATCATACTAGACTATG 81 2103 495831 32454 32473 TTGCCATCATACTAGACTAT 82 2104 495832 32455 32474 GTTGCCATCATACTAGACTA 88 2105 495833 32457 32476 ATGTTGCCATCATACTAGAC 78 2106 495834 32458 32477 AATGTTGCCATCATACTAGA 68 2107 495835 32459 32478 CAATGTTGCCATCATACTAG 85 2108 495836 32460 32479 GCAATGTTGCCATCATACTA 91 2109 495837 32462 32481 TTGCAATGTTGCCATCATAC 95 2110 495838 32463 32482 GTTGCAATGTTGCCATCATA 86 2111 495839 32464 32483 GGTTGCAATGTTGCCATCAT 94 2112 495840 32465 32484 TGGTTGCAATGTTGCCATCA 94 2113 495841 32467 32486 GGTGGTTGCAATGTTGCCAT 91 2114 495842 32468 32487 TGGTGGTTGCAATGTTGCCA 92 2115 495843 32469 32488 ATGGTGGTTGCAATGTTGCC 84 2116 495844 32470 32489 GATGGTGGTTGCAATGTTGC

82 2117 495845 32472 32491 TGGATGGTGGTTGCAATGTT 51 2118 495846 32473 32492 CTGGATGGTGGTTGCAATGT 48 2119 495847 32474 32493 CCTGGATGGTGGTTGCAATG 64 2120 495848 32475 32494 GCCTGGATGGTGGTTGCAAT 76 2121 495849 32477 32496 AAGCCTGGATGGTGGTTGCA 89 2122 495850 32478 32497 TAAGCCTGGATGGTGGTTGC 71 2123 495851 32479 32498 ATAAGCCTGGATGGTGGTTG 69 2124 495852 32480 32499 AATAAGCCTGGATGGTGGTT 85 2125 495853 32620 32639 AACTGAGATCTCCAGCAGCA 93 2126 495854 32621 32640 AAACTGAGATCTCCAGCAGC 82 2127 495855 32622 32641 TAAACTGAGATCTCCAGCAG 79 2128 495856 32623 32642 TTAAACTGAGATCTCCAGCA 81 2129 495857 32753 32772 GACAAAGTCTATCAGGATGC 90 2130 495858 32754 32773 TGACAAAGTCTATCAGGATG 67 2131 495859 32755 32774 GTGACAAAGTCTATCAGGAT 73 2132 495860 32756 32775 AGTGACAAAGTCTATCAGGA 72 2133 495861 32758 32777 AAAGTGACAAAGTCTATCAG 58 2134 495862 32759 32778 GAAAGTGACAAAGTCTATCA 67 2135 495863 32760 32779 AGAAAGTGACAAAGTCTATC 63 2136 495864 33176 33195 CTAGTGCCCCAGGGAGCCCT 63 2137 495865 33177 33196 CCTAGTGCCCCAGGGAGCCC 67 2138 495866 33178 33197 TCCTAGTGCCCCAGGGAGCC 60 2139 495867 33179 33198 GTCCTAGTGCCCCAGGGAGC 83 2140 495868 33181 33200 TTGTCCTAGTGCCCCAGGGA 87 2141 495869 33182 33201 TTTGTCCTAGTGCCCCAGGG 82 2142 495870 33183 33202 TTTTGTCCTAGTGCCCCAGG 66 2143 495871 36244 36263 GTATTCTGCTCATAAAAACA 42 2144 495872 36245 36264 GGTATTCTGCTCATAAAAAC 56 2145 495873 36246 36265 GGGTATTCTGCTCATAAAAA 72 2146 495874 36247 36266 AGGGTATTCTGCTCATAAAA 82 2147 495875 36249 36268 TAAGGGTATTCTGCTCATAA 89 2148 495876 36250 36269 GTAAGGGTATTCTGCTCATA 90 2149 495877 36251 36270 AGTAAGGGTATTCTGCTCAT 85 2150 495878 36252 36271 GAGTAAGGGTATTCTGCTCA 94 2151 495879 36259 36278 AGACAATGAGTAAGGGTATT 67 2152 495880 36260 36279 GAGACAATGAGTAAGGGTAT 47 2153 495881 36261 36280 AGAGACAATGAGTAAGGGTA 60 2154 495882 36262 36281 GAGAGACAATGAGTAAGGGT 77 2155 495883 36264 36283 GAGAGAGACAATGAGTAAGG 64 2156 495884 36265 36284 TGAGAGAGACAATGAGTAAG 31 2157

TABLE-US-00033 TABLE 33 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 87 425 495885 36266 36285 CTGAGAGAGACAATGAGTAA 51 2158 495886 36518 36537 AGCGACAGCCCTGTGCCAGC 67 2159 495887 36519 36538 AAGCGACAGCCCTGTGCCAG 43 2160 495888 36520 36539 CAAGCGACAGCCCTGTGCCA 15 2161 495889 36521 36540 ACAAGCGACAGCCCTGTGCC 25 2162 495890 36523 36542 GGACAAGCGACAGCCCTGTG 44 2163 495891 36524 36543 AGGACAAGCGACAGCCCTGT 42 2164 495892 36648 36667 ATGAGACAGGCAGCCCACCT 53 2165 495893 36649 36668 AATGAGACAGGCAGCCCACC 56 2166 495894 36650 36669 GAATGAGACAGGCAGCCCAC 46 2167 495895 36651 36670 AGAATGAGACAGGCAGCCCA 56 2168 495896 36653 36672 ACAGAATGAGACAGGCAGCC 42 2169 495897 36654 36673 CACAGAATGAGACAGGCAGC 47 2170 495898 36655 36674 TCACAGAATGAGACAGGCAG 24 2171 495899 36677 36696 GCTCTGTGGGCTCCTCCACA 72 2172 495900 36678 36697 TGCTCTGTGGGCTCCTCCAC 70 2173 495901 36679 36698 GTGCTCTGTGGGCTCCTCCA 76 2174 495902 36680 36699 TGTGCTCTGTGGGCTCCTCC 81 2175 495903 36682 36701 CCTGTGCTCTGTGGGCTCCT 65 2176 495904 36683 36702 CCCTGTGCTCTGTGGGCTCC 70 2177 495905 36684 36703 GCCCTGTGCTCTGTGGGCTC

31 2178 495906 36685 36704 GGCCCTGTGCTCTGTGGGCT 32 2179 495907 36687 36706 TTGGCCCTGTGCTCTGTGGG 46 2180 495908 36688 36707 CTTGGCCCTGTGCTCTGTGG 41 2181 495909 36689 36708 ACTTGGCCCTGTGCTCTGTG 66 2182 495910 36690 36709 TACTTGGCCCTGTGCTCTGT 52 2183 495911 36734 36753 CACTGTCAGTCTCTCCAAAC 75 2184 495912 36735 36754 CCACTGTCAGTCTCTCCAAA 75 2185 495913 36736 36755 CCCACTGTCAGTCTCTCCAA 60 2186 495914 36737 36756 CCCCACTGTCAGTCTCTCCA 63 2187 495915 36739 36758 GCCCCCACTGTCAGTCTCTC 48 2188 495916 36740 36759 TGCCCCCACTGTCAGTCTCT 46 2189 495917 36741 36760 CTGCCCCCACTGTCAGTCTC 55 2190 495918 36742 36761 TCTGCCCCCACTGTCAGTCT 68 2191 495919 36754 36773 TTGGCTGCAAGCTCTGCCCC 59 2192 495920 36755 36774 CTTGGCTGCAAGCTCTGCCC 63 2193 495921 36756 36775 CCTTGGCTGCAAGCTCTGCC 65 2194 495922 36757 36776 GCCTTGGCTGCAAGCTCTGC 65 2195 495923 36759 36778 GGGCCTTGGCTGCAAGCTCT 65 2196 495924 36856 36875 TTATTCTAAAACTCAAATCC 8 2197 495925 36857 36876 ATTATTCTAAAACTCAAATC 11 2198 495926 36859 36878 TGATTATTCTAAAACTCAAA 2 2199 495927 36861 36880 AGTGATTATTCTAAAACTCA 55 2200 495928 36862 36881 GAGTGATTATTCTAAAACTC 36 2201 495929 36863 36882 AGAGTGATTATTCTAAAACT 0 2202 495930 36864 36883 CAGAGTGATTATTCTAAAAC 28 2203 495931 37023 37042 TGTTGGGCAGTCACCATTTG 1 2204 495932 37024 37043 GTGTTGGGCAGTCACCATTT 17 2205 495933 37025 37044 GGTGTTGGGCAGTCACCATT 20 2206 495934 37026 37045 TGGTGTTGGGCAGTCACCAT 2 2207 495935 37185 37204 CCTTGGGTCAGAGGAAACTG 5 2208 495936 37186 37205 ACCTTGGGTCAGAGGAAACT 12 2209 495937 37187 37206 GACCTTGGGTCAGAGGAAAC 15 2210 495938 37188 37207 TGACCTTGGGTCAGAGGAAA 2 2211 495939 37190 37209 GATGACCTTGGGTCAGAGGA 18 2212 495940 37191 37210 GGATGACCTTGGGTCAGAGG 16 2213 495941 37192 37211 AGGATGACCTTGGGTCAGAG 13 2214 495942 37193 37212 AAGGATGACCTTGGGTCAGA 8 2215 495943 37195 37214 GCAAGGATGACCTTGGGTCA 26 2216 495944 37196 37215 TGCAAGGATGACCTTGGGTC 24 2217 495945 37197 37216 CTGCAAGGATGACCTTGGGT 32 2218 495946 37198 37217 GCTGCAAGGATGACCTTGGG 16 2219 495947 37200 37219 CAGCTGCAAGGATGACCTTG 7 2220 495948 37201 37220 CCAGCTGCAAGGATGACCTT 0 2221 495949 37203 37222 CACCAGCTGCAAGGATGACC 27 2222 495950 37491 37510 TGTGTTGGCTCCCTGTGTCA 38 2223 495951 37492 37511 GTGTGTTGGCTCCCTGTGTC 11 2224 495952 37493 37512 GGTGTGTTGGCTCCCTGTGT 40 2225 495953 37494 37513 TGGTGTGTTGGCTCCCTGTG 29 2226 495954 37496 37515 AATGGTGTGTTGGCTCCCTG 49 2227 495955 37497 37516 GAATGGTGTGTTGGCTCCCT 76 2228 495956 37498 37517 GGAATGGTGTGTTGGCTCCC 44 2229 495957 37499 37518 AGGAATGGTGTGTTGGCTCC 72 2230 495958 37501 37520 CAAGGAATGGTGTGTTGGCT 74 2231 495959 37502 37521 CCAAGGAATGGTGTGTTGGC 8 2232 495960 37503 37522 ACCAAGGAATGGTGTGTTGG 21 2233 495961 37504 37523 CACCAAGGAATGGTGTGTTG 18 2234

TABLE-US-00034 TABLE 34 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 423460 32428 32447 TGGACAAGTCCTGCCCATCT 65 464 423461 32429 32448 CTGGACAAGTCCTGCCCATC 70 465 423462 32430 32449 CCTGGACAAGTCCTGCCCAT 54 466 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 80 425 495962 37506 37525 AGCACCAAGGAATGGTGTGT 6 2235 495963 37507 37526 CAGCACCAAGGAATGGTGTG 19 2236 495964 37508 37527 CCAGCACCAAGGAATGGTGT 13 2237 495965 37509 37528 CCCAGCACCAAGGAATGGTG 26 2238 495966 37511 37530

GGCCCAGCACCAAGGAATGG 14 2239 495967 37512 37531
AGGCCCAGCACCAAGGAATG 4 2240 495968 37513 37532
CAGGCCCAGCACCAAGGAAT 8 2241 495969 37514 37533
GCAGGCCCAGCACCAAGGAA 33 2242 495970 37516 37535
ATGCAGGCCCAGCACCAAGG 33 2243 495971 37517 37536
AATGCAGGCCCAGCACCAAG 14 2244 495972 37518 37537
CAATGCAGGCCCAGCACCAA 16 2245 495973 37519 37538 CCAATGCAGGCCCAGCACCA 45 2246 495974 40192 40211 CTGAACTTCCCCTCCCAAAC 16 2247 495975 40197 40216 CAAATCTGAACTTCCCCTCC 0 2248 495976 40202 40221 AAATGCAAATCTGAACTTCC 30 2249 495977 40223 40242 GCAGCTCCAAGGATCATTTT 44 2250 495978 40228 40247 ATCCAGCAGCTCCAAGGATC 45 2251 495979 40233 40252 CTTCCATCCAGCAGCTCCAA 51 2252 495980 40238 40257 CCCATCTTCCATCCAGCAGC 49 2253 495981 40243 40262 TCTAACCCATCTTCCATCCA 28 2254 495982 40248 40267 ATTTTTCTAACCCATCTTCC 21 2255 495983 40253 40272 CTTCCATTTTTCTAACCCAT 64 2256 495984 40293 40312 GCCACTGGCTACCAAAACAG 37 2257 495985 40298 40317 TCCAAGCCACTGGCTACCAA 41 2258 495986 40303 40322 CTTGGTCCAAGCCACTGGCT 43 2259 495987 40308 40327 ACTCCCTTGGTCCAAGCCAC 44 2260 495988 40313 40332 CTGCTACTCCCTTGGTCCAA 40 2261 495989 40318 40337 CTCCACTGCTACTCCCTTGG 11 2262 495990 40323 40342 TCCATCTCCACTGCTACTCC 39 2263 495991 40328 40347 TCTCTTCCATCTCCACTGCT 36 2264 495992 40333 40352 GCACATCTCTTCCATCTCCA 82 2265 495993 40338 40357 ATCATGCACATCTCTTCCAT 39 2266 495994 40359 40378 TATTTCTGAAATTTTTCCCA 17 2267 495995 40364 40383 AATGCTATTTCTGAAATTTT 10 2268 495996 40386 40405 CAATCCATTCCTATGTCCTG 24 2269 495997 40391 40410 ATACCCAATCCATTCCTATG 13 2270 495998 40396 40415 TCTCCATACCCAATCCATTC 45 2271 495999 40401 40420 CTGCATCTCCATACCCAATC 59 2272 496000 40406 40425 TCCTGCTGCATCTCCATACC 38 2273 496001 40411 40430 TCTTATCCTGCTGCATCTCC 15 2274 496002 40416 40435 TATTTTCTTATCCTGCTGCA 46 2275 496003 40421 40440 TGCTTTATTTTCTTATCCTG 47 2276 496004 40444 40463 ACCAGCATTTATGATCTGTG 67 2277 496005 40807 40826 GACATGCTTTAAAAAGTGTA 25 2278 496006 40812 40831 AATGTGACATGCTTTAAAAA 23 2279 496007 40899 40918 AGCCATTCCTGCCTCTCTGA 12 2280 496008 40904 40923 GGGCAAGCCATTCCTGCCTC 5 2281 496009 40909 40928 GCTCTGGGCAAGCCATTCCT 60 2282 496010 40914 40933 GCTCTGCTCTGGGCAAGCCA 66 2283 496011 40919 40938 CTTTTGCTCTGCTCTGGGCA 50 2284 496012 40924 40943 CTTTGCTTTTGCTCTGCTCT 67 2285 496013 40929 40948 AACATCTTTGCTTTTGCTCT 59 2286 496014 40934 40953 AAGTAAACATCTTTGCTTTT 15 2287 496015 40939 40958 GGATCAAGTAAACATCTTTG 42 2288 496016 40967 40986 TCTGCTAGGAGGGTCTATGA 20 2289 496017 40972 40991 TGCATTCTGCTAGGAGGGTC 39 2290 496018 40977 40996 CCCACTGCATTCTGCTAGGA 27 2291 496019 40982 41001 TTGAACCCACTGCATTCTGC 6 2292 496020 40987 41006 ACTGGTTGAACCCACTGCAT 0 2293 496021 40992 41011
TCAAGACTGGTTGAACCCAC 8 2294 496022 40997 41016
TGGGATCAAGACTGGTTGAA 0 2295 496023 41002 41021
GCAGATGGGATCAAGACTGG 0 2296 496024 41007 41026
AAGCTGCAGATGGGATCAAG 0 2297 496025 41012 41031
GTGCTAAGCTGCAGATGGGA 39 2298 496026 41043 41062 GCATGTGAAGGGACCCACCC 14 2299 496027 41064 41083 TGAAAAGACTGAGGCCCAGG 1 2300 496028 41069 41088 ACAGATGAAAAGACTGAGGC 21 2301 496029 41074 41093 CTATTACAGATGAAAAGACT 0 2302 496030 41079 41098 GTCCCCTATTACAGATGAAA 34 2303 496031 41084 41103 TGGTTGTCCCCTATTACAGA 39 2304 496032 41089 41108 ATCTCTGGTTGTCCCCTATT 21 2305 496033 41094 41113 GCTGCATCTCTGGTTGTCCC 65 2306 496034 41099 41118 TATGTGCTGCATCTCTGGTT 51 2307 496035 42398 42417 TCTCACTTTCATTTATTTTC 70

2308

TABLE-US-00035 TABLE 35 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 501381 24794 24813 GCTTCAGGCAGCCTGAAGGA 46 2309 501382 24799 24818 AGTGAGCTTCAGGCAGCCTG 75 2310 501383 24823 24842 GCTTCCCCAAAGGGCCCAGT 70 2311 501384 24828 24847 CCTTTGCTTCCCCAAAGGGC 71 2312 501385 24878 24897 CCTCATCTCAGTGCCCAAAG 90 2313 501386 25018 25037 AGCATCCCCAATCCCTGCTC 65 2314 501387 25023 25042 CTTTGAGCATCCCCAATCCC 91 2315 501388 25028 25047 GTGTACTTTGAGCATCCCCA 82 2316 501389 25033 25052 TCCAAGTGTACTTTGAGCAT 85 2317 501390 25069 25088 AGAGGTTCAACATCCAATTT 81 2318 501391 25075 25094 AAGGACAGAGGTTCAACATC 88 2319 501392 25080 25099 AGGCCAAGGACAGAGGTTCA 82 2320 501393 25085 25104 CTGTGAGGCCAAGGACAGAG 88 2321 501394 25090 25109 TCTGTCTGTGAGGCCAAGGA 77 2322 501395 25095 25114 TGCTATCTGTCTGTGAGGCC 77 2323 501396 25183 25202 GTCTCTCATAATTGCCAGTT 82 2324 501397 25188 25207 GGGAAGTCTCTCATAATTGC 67 2325 501398 25193 25212 GCCTTGGGAAGTCTCTCATA 85 2326 501399 25198 25217 GCTAGGCCTTGGGAAGTCTC 76 2327 501400 25226 25245 TGAAGTATCTAAAGAGGCTA 71 2328 501401 25231 25250 CCACATGAAGTATCTAAAGA 74 2329 501402 25236 25255 GGAGACCACATGAAGTATCT 75 2330 501403 25241 25260 ATTTTGGAGACCACATGAAG 68 2331 501404 25246 25265 GGGTCATTTTGGAGACCACA 91 2332 501405 25267 25286 ATGGGAATGGTATCGCACTC 85 2333 501406 25272 25291 ACACAATGGGAATGGTATCG 80 2334 501407 25297 25316 CCCCTGTGCCCTGGTTTCTA 49 2335 501408 25302 25321 ATGCTCCCCTGTGCCCTGGT 44 2336 501409 25307 25326 TCTGCATGCTCCCCTGTGCC 44 2337 501410 25312 25331 GGCTGTCTGCATGCTCCCCT 76 2338 501411 25375 25394 CTATAGAATCAGACAGACCT 83 2339 501412 25380 25399 ATCAGCTATAGAATCAGACA 90 2340 501413 25424 25443 CAAACCTCACCTCCCACAGG 33 2341 501414 25429 25448 CAGTACAAACCTCACCTCCC 70 2342 501415 25468 25487 TCAGTCCACTGTCACCCTTC 33 2343 501416 25473 25492 AGATGTCAGTCCACTGTCAC 13 2344 501417 25478 25497 AGGGAAGATGTCAGTCCACT 0 2345 501418 25483 25502 AGCAGAGGGAAGATGTCAGT 0 2346 501419 25489 25508 GCCTACAGCAGAGGGAAGAT 29 2347 501420 25494 25513 CCAGTGCCTACAGCAGAGGG 38 2348 501421 25499 25518 TGGATCCAGTGCCTACAGCA 41 2349 501422 25505 25524 GGATGCTGGATCCAGTGCCT 74 2350 501423 25618 25637 ACCCAGTACAAGGAAGGACA 78 2351 501424 25623 25642 AGCTTACCCAGTACAAGGAA 31 2352 501425 25628 25647 GGCCCAGCTTACCCAGTACA 48 2353 501426 25717 25736 CTCTTATTCCCTCACCCCTG 77 2354 501427 25782 25801 CACTTGATCTGGGACCCAAA 96 2355 501428 25787 25806 TAGAGCACTTGATCTGGGAC 80 2356 501429 25841 25860 ACTATCACACCACCTCCCAC 85 2357 501430 25846 25865 TGGGAACTATCACACCACCT 95 2358 501431 25851 25870 GTAAATGGGAACTATCACAC 64 2359 501432 25856 25875 CATCTGTAAATGGGAACTAT 49 2360 501433 25861 25880 TTTCCCATCTGTAAATGGGA 0 2361 501434 25866 25885 TGAGGTTTCCCATCTGTAAA 61 2362 501435 25900 25919 GACCTTGGGCAAGTTACCTA 88 2363 501436 25905 25924 TGTGTGACCTTGGGCAAGTT 82 2364 501437 25910 25929 AAATCTGTGTGACCTTGGGC 67 2365 501438 25915 25934 GATTCAAATCTGTGTGACCT 89 2366 501439 25920 25939 ACAGGGATTCAAATCTGTGT 73 2367 501440 25950 25969 GGAAAGGCACAGGCTTTGGG 88 2368 501441 25981 26000 AATGTCTGTTGGTGGGCAGG 77 2369 501442 26005 26024 GGCAAAGTAACATACCTGCT 94 2370 501443 26010 26029 CTTAAGGCAAAGTAACATAC 91 2371 501444 26069 26088 TGTAAGGTCACAGAGCTTGT 54 2372 501445 26102 26121 GGAAACTAACACTCAGAGAG 70 2373 501446 26107 26126

AATGAGGAAACTAACACTCA 62 2374 501447 26165 26184 TTTCTTATCTTCAATCCTCA 95 2375 501448 26170 26189 TCTCATTTCTTATCTTCAAT 92 2376 501449 26175 26194 GGTGCTCTCATTTCTTATCT 81 2377 501450 26200 26219 CCCATCATGACCAGGCCCTG 91 2378 501451 26300 26319 AGGCCTCGGGAAGCACCTGG 89 2379 501452 26305 26324 GGAGCAGGCCTCGGGAAGCA 90 2380 501453 26331 26350 ATGTGGGCTGTGCTGAGAGA 64 2381 501454 26353 26372 AGAAAAAGCAACCAAAGCAC 89 2382 501455 26358 26377 CCTGCAGAAAAAGCAACCAA 81 2383 501456 26363 26382 CCAGACCTGCAGAAAAAGCA 76 2384 501457 26385 26404 CCTAAGGAGTGAGGGTATCT 79 2385 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 48 425

TABLE-US-00036 TABLE 36 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 501303 23656 23675 CTCTGAGTGCAAAGGCTCTG 40 2386 501304 23661 23680 AGCAGCTCTGAGTGCAAAGG 48 2387 501305 23666 23685 CCCAGAGCAGCTCTGAGTGC 45 2388 501306 23671 23690 CTTTCCCCAGAGCAGCTCTG 25 2389 501307 23694 23713 TGACCCTGCTTAGAGGACCT 29 2390 501308 23699 23718 ATTCCTGACCCTGCTTAGAG 5 2391 501309 23704 23723 CCAGCATTCCTGACCCTGCT 14 2392 501310 23709 23728 GAAACCCAGCATTCCTGACC 18 2393 501311 23714 23733 GGACTGAAACCCAGCATTCC 11 2394 501312 23719 23738 AGCCAGGACTGAAACCCAGC 34 2395 501313 23724 23743 GGCAGAGCCAGGACTGAAAC 47 2396 501314 23729 23748 TAGGCGGCAGAGCCAGGACT 43 2397 501315 23734 23753 GGCAGTAGGCGGCAGAGCCA 28 2398 501316 23770 23789 ACCAGAGACAGGCACTGACT 59 2399 501317 23792 23811 TGGACAGCCAGGGAGACTGG 36 2400 501318 23797 23816 CTAATTGGACAGCCAGGGAG 20 2401 501319 23802 23821 ATAGCCTAATTGGACAGCCA 63 2402 501320 23807 23826 CCAGTATAGCCTAATTGGAC 40 2403 501321 23813 23832 GGTCATCCAGTATAGCCTAA 53 2404 501322 23855 23874 GACTTGCCCAGCAACCCATC 65 2405 501323 23860 23879 CTCCAGACTTGCCCAGCAAC 28 2406 501324 23865 23884 CCCACCTCCAGACTTGCCCA 26 2407 501325 23870 23889 ACTTTCCCACCTCCAGACTT 18 2408 501326 23875 23894 ATAGGACTTTCCCACCTCCA 61 2409 501327 23880 23899 CCTTCATAGGACTTTCCCAC 56 2410 501328 23885 23904 TCCTACCTTCATAGGACTTT 40 2411 501329 23890 23909 AAAAATCCTACCTTCATAGG 0 2412 501330 23921 23940 TGCCTCATAATTGCTCCTTC 49 2413 501331 23926 23945 AGGTCTGCCTCATAATTGCT 8 2414 501332 23931 23950 TCCAGAGGTCTGCCTCATAA 64 2415 501333 23957 23976 ATAAAGAGGCTGGGCCATAG 8 2416 501334 23981 24000 CAGGATGTGAACTCACAAAA 59 2417 501335 24024 24043 GTAGCATCTAATAACACCAC 61 2418 501336 24029 24048 GTTCAGTAGCATCTAATAAC 59 2419 501337 24034 24053 TGGGTGTTCAGTAGCATCTA 50 2420 501338 24089 24108 TATGTGACCCCAGGCAAGCT 32 2421 501339 24094 24113 CTCACTATGTGACCCCAGGC 59 2422 501340 24099 24118 TTCTACTCACTATGTGACCC 56 2423 501341 24104 24123 TTTGGTTCTACTCACTATGT 57 2424 501342 24109 24128 CAGACTTTGGTTCTACTCAC 63 2425 501343 24114 24133 AGGTTCAGACTTTGGTTCTA 58 2426 501344 24119 24138 AACCTAGGTTCAGACTTTGG 59 2427 501345 24124 24143 AGTCAAACCTAGGTTCAGAC 68 2428 501346 24129 24148 AGAAGAGTCAAACCTAGGTT 56 2429 501347 24134 24153 TTAGCAGAAGAGTCAAACCT 38 2430 501348 24139 24158 TTAGTTTAGCAGAAGAGTCA 36 2431 501349 24164 24183 GTGTGATGCATCAGAGGGAA 40 2432 501350 24169 24188 CCCTGGTGTGATGCATCAGA 50 2433 501351 24174 24193 CCTTTCCCTGGTGTGATGCA 53 2434 501352 24197 24216 AAATGCTAGGCCTCAAGATG 17 2435 501353 24202 24221 GAAGGAAATGCTAGGCCTCA

64 2436 501354 24207 24226 GGAAGGAAGGAAATGCTAGG 31 2437 501355 24212 24231 TAGGAGGAAGGAAGGAAATG 11 2438 501356 24217 24236 ACTTTTAGGAGGAAGGAAGG 0 2439 501357 24222 24241 CTTTGACTTTTAGGAGGAAG 60 2440 501358 24227 24246 AACTGCTTTGACTTTTAGGA 39 2441 501359 24232 24251 TTAACAACTGCTTTGACTTT 30 2442 501360 24237 24256 GAAAGTTAACAACTGCTTTG 37 2443 501361 24447 24466 TACCATCTGCTCTGTGAAAG 21 2444 501362 24452 24471 CTTCATACCATCTGCTCTGT 29 2445 501363 24483 24502 CAAGCACCTACTGTGTGCCC 13 2446 501364 24488 24507 TTCACCAAGCACCTACTGTG 21 2447 501365 24493 24512 ACATCTTCACCAAGCACCTA 0 2448 501366 24516 24535 CCACTACATGCCCTTTGCTC 51 2449 501367 24557 24576 CTGAAGAAGTGGGCCACCTC 52 2450 501368 24582 24601 ACCCATACCCCATTCCTTCC 22 2451 501369 24587 24606 TCCTCACCCATACCCCATTC 9 2452 501370 24636 24655 AGCCTGGTGACTGCAGGAGA 61 2453 501371 24641 24660 CAGGAAGCCTGGTGACTGCA 45 2454 501372 24666 24685 TCCAGCAGCTGATGGGCAGT 43 2455 501373 24671 24690 TGGACTCCAGCAGCTGATGG 19 2456 501374 24695 24714 CTTGGTGCCCCTAGGATGAC 33 2457 501375 24700 24719 ATTGGCTTGGTGCCCCTAGG 8 2458 501376 24705 24724 ACTTAATTGGCTTGGTGCCC 36 2459 501377 24774 24793 CACTGAGGGCTGTAATTATG 21 2460 501378 24779 24798 AAGGACACTGAGGGCTGTAA 24 2461 501379 24784 24803 GCCTGAAGGACACTGAGGGC 2 2462 501380 24789 24808 AGGCAGCCTGAAGGACACTG 14 2463 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 81 425

TABLE-US-00037 TABLE 37 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 501226 21041 21060 TGTTCCTGACTCCTGGGTGG 26 2464 501227 21046 21065 CTGACTGTTCCTGACTCCTG 61 2465 501228 21091 21110 TGCTGCAATCAGATGCCACC 65 2466 501229 21096 21115 GCCGATGCTGCAATCAGATG 68 2467 501230 21101 21120 GGGATGCCGATGCTGCAATC 69 2468 501231 21119 21138 GTCCAGCAGAAGCTGGTGGG 19 2469 501232 21124 21143 GAGGAGTCCAGCAGAAGCTG 15 2470 501233 21129 21148 GCTGGGAGGAGTCCAGCAGA 12 2471 501234 21134 21153 CTGTGGCTGGGAGGAGTCCA 14 2472 501235 21139 21158 CCCAGCTGTGGCTGGGAGGA 0 2473 501236 21144 21163 TCTGCCCCAGCTGTGGCTGG 0 2474 501237 21149 21168 CTTCCTCTGCCCCAGCTGTG 41 2475 501238 21154 21173 CAGTACTTCCTCTGCCCCAG 40 2476 501239 21159 21178 GCTGTCAGTACTTCCTCTGC 50 2477 501240 21164 21183 ACCTGGCTGTCAGTACTTCC 70 2478 501241 21169 21188 TTGCCACCTGGCTGTCAGTA 68 2479 501242 21174 21193 AGTCCTTGCCACCTGGCTGT 50 2480 501243 21179 21198 CTGCCAGTCCTTGCCACCTG 53 2481 501244 21184 21203 AAACACTGCCAGTCCTTGCC 63 2482 501245 21213 21232 GGAAGTGAGGGTTCAGTTGG 30 2483 501246 21369 21388 CTCTCTGGGCTTCAGTTTCC 45 2484 501247 21374 21393 CTCACCTCTCTGGGCTTCAG 61 2485 501248 21379 21398 GTCTCCTCACCTCTCTGGGC 61 2486 501249 21384 21403 GCTAAGTCTCCTCACCTCTC 71 2487 501250 21490 21509 ATCTGTTTTCCTTCAGAGGG 55 2488 501251 21565 21584 TTCCTTCTCTCTGCTCCCCA 55 2489 501252 21570 21589 CCAAGTTCCTTCTCTCTGCT 15 2490 501253 21575 21594 TGTCCCCAAGTTCCTTCTCT 32 2491 501254 21659 21678 GGTCAAGGTCACTCAGCCAG 62 2492 501255 21681 21700 CCCTATTTTGATGGTGATAC 57 2493 501256 21708 21727 TGGACTCCTGTGAGTTTGGC 67 2494 501257 21713 21732 GCACTTGGACTCCTGTGAGT 25 2495 501258 21718 21737 CACAGGCACTTGGACTCCTG 27 2496 501259 21723 21742 CCTTGCACAGGCACTTGGAC 18 2497 501260 21728 21747 CATGCCCTTGCACAGGCACT 14 2498 501261 21733 21752 CCCTCCATGCCCTTGCACAG 29 2499 501262 21755 21774 AGGCCCATAAAGCTTTGGGA 49 2500 501263 21760 21779 ACAAGAGGCCCATAAAGCTT 26 2501 501264 21765 21784

GATGCACAAGAGGCCCATAA 65 2502 501265 21770 21789 GGTATGATGCACAAGAGGCC 58 2503 501266 21821 21840 CTGGCCACTCTTCAGGCCCC 32 2504 501267 21826 21845 GTGCTCTGGCCACTCTTCAG 0 2505 501268 21873 21892 GATCCATATCTGGATCCCCA 2 2506 501269 21878 21897 GGTCTGATCCATATCTGGAT 20 2507 501270 21933 21952 GCTGATTACAAGCTGATTCT 77 2508 501271 21938 21957 TAACTGCTGATTACAAGCTG 39 2509 501272 21996 22015 ACAATGGAGAAACAAAAGAG 6 2510 501273 22019 22038 TTTGAAGATAAAGTCAGACC 35 2511 501274 22075 22094 GTCTATGCAGGCCTGGAATT 54 2512 501275 22080 22099 CTAAGGTCTATGCAGGCCTG 53 2513 501276 22085 22104 AAACACTAAGGTCTATGCAG 32 2514 501277 22090 22109 ATGATAAACACTAAGGTCTA 43 2515 501278 22095 22114 CTGGGATGATAAACACTAAG 53 2516 501279 22100 22119 CTTCTCTGGGATGATAAACA 0 2517 501280 22105 22124 CTTTCCTTCTCTGGGATGAT 0 2518 501281 22126 22145 AAGCCTTTGGAAAGAGAAGG 31 2519 501282 22131 22150 CCAGGAAGCCTTTGGAAAGA 31 2520 501283 22136 22155 GTCTTCCAGGAAGCCTTTGG 51 2521 501284 22141 22160 CAGCAGTCTTCCAGGAAGCC 54 2522 501285 22146 22165 TAAGGCAGCAGTCTTCCAGG 59 2523 501286 22151 22170 GGTGATAAGGCAGCAGTCTT 56 2524 501287 22156 22175 GAGATGGTGATAAGGCAGCA 78 2525 501288 22162 22181 AACCCAGAGATGGTGATAAG 23 2526 501289 22167 22186 TTCAAAACCCAGAGATGGTG 64 2527 501290 22172 22191 CATCCTTCAAAACCCAGAGA 62 2528 501291 22194 22213 CTGATAATGAGATAAAGCAA 3 2529 501292 22231 22250 CAGAGGCCCTTCCCTGCTTT 42 2530 501293 22266 22285 TGGGAGCCCCCAGTCCACCT 48 2531 501294 22271 22290 GCCAGTGGGAGCCCCCAGTC 0 2532 501295 22311 22330 TGGCCTGGGAGCTGGCCTGG 13 2533 501296 22349 22368 CTCTCAGGCAATCCCAACCT 27 2534 501297 22354 22373 AGGCCCTCTCAGGCAATCCC 62 2535 501298 22359 22378 GGCCCAGGCCCTCTCAGGCA 34 2536 501299 22364 22383 TCTCAGGCCCAGGCCCTCTC 27 2537 501300 22403 22422 CTCCTGGAGAGGGAAGCTGG 31 2538 501301 22408 22427 GGAGGCTCCTGGAGAGGGAA 0 2539 501302 22413 22432 AGTTGGGAGGCTCCTGGAGA 0 2540 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 80 425

TABLE-US-00038 TABLE 38 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 484013 18880 18899 CAGAGGCAGCTATAGGCCTG 43 1299 20790 20809 495657 18871 18890 CTATAGGCCTGGATGCCCAA 72 1930 20781 20800 501152 19132 19151 GCTGGAACAAAGCCTTGCAG 52 2541 501153 19261 19280 AAGAAGAGGTCGCAGAGTGA 17 2542 501154 19360 19379 GCTACTGGCATTGCTCCTCC 75 2543 501155 19392 19411 ACATGTGGCATTTGTTAGGT 66 2544 501156 19421 19440 GGAACATTAAATTTTAAAAC 20 2545 501157 19775 19794 TAGGATGAGGGTGATAATCC 37 2546 501158 19804 19823 CCTAACTCTAGCTTTGGTTT 63 2547 501159 19809 19828 AGTCACCTAACTCTAGCTTT 20 2548 501160 19854 19873 GCCCTGGGCCAGCTCTGCCC 15 2549 501161 19867 19886 TCTGAGAAAGAGGGCCCTGG 26 2550 501162 19872 19891 CTAAATCTGAGAAAGAGGGC 0 2551 501163 19903 19922 CTGAGGGCAGCAGTGTCTGA 35 2552 501164 19908 19927 CATGCCTGAGGGCAGCAGTG 54 2553 501165 19913 19932 TCTCACATGCCTGAGGGCAG 74 2554 501166 19963 19982 CAGGAAGGCCACACCTCGGG 23 2555 501167 19968 19987 GTGACCAGGAAGGCCACACC 12 2556 501168 19973 19992 TCAAAGTGACCAGGAAGGCC 33 2557 501169 19978 19997 TGATATCAAAGTGACCAGGA 22 2558 501170 19983 20002 ATATCTGATATCAAAGTGAC 3 2559 501171 19988 20007 GCCCAATATCTGATATCAAA 74 2560 501172 20044 20063 GACCCAGGGCCCAGCCCAGG 18 2561 501173 20049 20068 TCTCAGACCCAGGGCCCAGC 55 2562 501174 20093 20112 TGAGACTGGATTCCACCCCC 24 2563 501175 20227 20246 TACAATGCTACGCTGTTGAG

36 2564 501176 20232 20251 ATCTCTACAATGCTACGCTG 73 2565 501177 20237 20256 TCATCATCTCTACAATGCTA 65 2566 501178 20242 20261 GTCCCTCATCATCTCTACAA 22 2567 501179 20247 20266 TCCCAGTCCCTCATCATCTC 49 2568 501180 20252 20271 CAGGCTCCCAGTCCCTCATC 16 2569 501181 20257 20276 GTATTCAGGCTCCCAGTCCC 7 2570 501182 20296 20315 TGCTTGAGGTCAGGGTGTGG 65 2571 501183 20301 20320 TCACTTGCTTGAGGTCAGGG 83 2572 501184 20306 20325 CAAAATCACTTGCTTGAGGT 72 2573 501185 20311 20330 AGAGGCAAAATCACTTGCTT 59 2574 501186 20397 20416 CCACAGCACTCTCTGGGAAG 30 2575 501187 20402 20421 CCGTCCCACAGCACTCTCTG 27 2576 501188 20407 20426 CTTCACCGTCCCACAGCACT 39 2577 501189 20442 20461 GGAAAGCAATCCCCTTCTCC 41 2578 501190 20447 20466 CACCCGGAAAGCAATCCCCT 18 2579 501191 20452 20471 TTTACCACCCGGAAAGCAAT 12 2580 501192 20457 20476 GCTCTTTTACCACCCGGAAA 55 2581 501193 20462 20481 GAGCAGCTCTTTTACCACCC 66 2582 501194 20541 20560 GCAGCACTTGCTTTACACAC 60 2583 501195 20546 20565 CCCAGGCAGCACTTGCTTTA 33 2584 501196 20551 20570 CTGCTCCCAGGCAGCACTTG 0 2585 501197 20639 20658 ACAACCCTGCTACCTACAGA 23 2586 501198 20644 20663 GGAACACAACCCTGCTACCT 37 2587 501199 20668 20687 GTACAAGTTTCTCCACCTAT 80 2588 501200 20673 20692 CCTAGGTACAAGTTTCTCCA 79 2589 501201 20714 20733 GGACTGACCCCAGAGATGGG 22 2590 501202 20719 20738 ACTGAGGACTGACCCCAGAG 0 2591 501203 20724 20743 CCATCACTGAGGACTGACCC 33 2592 501204 20729 20748 CAGCCCCATCACTGAGGACT 18 2593 501205 20771 20790 GGATGCCCAAGTTAGACTGG 24 2594 495670 20776 20795 GGCCTGGATGCCCAAGTTAG 69 1943 501206 20802 20821 CCAGAGCAGCCTCAGAGGCA 55 2595 501207 20807 20826 CAAATCCAGAGCAGCCTCAG 13 2596 501208 20812 20831 ATAAGCAAATCCAGAGCAGC 18 2597 501209 20817 20836 AATCCATAAGCAAATCCAGA 60 2598 501210 20822 20841 GAGCAAATCCATAAGCAAAT 70 2599 501211 20827 20846 TAGCTGAGCAAATCCATAAG 58 2600 501212 20832 20851 CTGCATAGCTGAGCAAATCC 79 2601 501213 20837 20856 GTTTACTGCATAGCTGAGCA 84 2602 501214 20842 20861 TAGAGGTTTACTGCATAGCT 75 2603 501215 20868 20887 GATTCATCTGTGGTAAGAGG 55 2604 501216 20873 20892 TACCTGATTCATCTGTGGTA 44 2605 501217 20878 20897 CTTGGTACCTGATTCATCTG 64 2606 501218 20884 20903 CCAGGACTTGGTACCTGATT 54 2607 501219 20889 20908 GGGTGCCAGGACTTGGTACC 26 2608 501220 20913 20932 GCCATCCCACTGCCAGTGAC 23 2609 501221 20937 20956 ACAGCACCAATGTCACTTTA 55 2610 501222 20950 20969 TCTGCACACTCCCACAGCAC 52 2611 501223 20955 20974 CTCTCTCTGCACACTCCCAC 59 2612 501224 20960 20979 CTGCACTCTCTCTGCACACT 76 2613 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 77 425

TABLE-US-00039 TABLE 39 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 501075 16484 16503 GTCCTGGCCTCTACCCTGAA 57 2614 501076 16510 16529 GAAGGACTGGGATGCTAGGT 65 2615 501077 16515 16534 GGCATGAAGGACTGGGATGC 49 2616 501078 16520 16539 AGGGAGGCATGAAGGACTGG 22 2617 501079 16525 16544 GAAACAGGGAGGCATGAAGG 21 2618 501080 16530 16549 GTAGAGAAACAGGGAGGCAT 34 2619 501081 16535 16554 GAAAAGTAGAGAAACAGGGA 0 2620 501082 16540 16559 AGTTGGAAAAGTAGAGAAAC 11 2621 501083 16546 16565 GAGTCTAGTTGGAAAAGTAG 1 2622 501084 16551 16570 ACTGTGAGTCTAGTTGGAAA 33 2623 501085 16556 16575 CAGAGACTGTGAGTCTAGTT 51 2624 501086 16561 16580 CAGCACAGAGACTGTGAGTC 59 2625 501087 16566 16585 GACTGCAGCACAGAGACTGT 29 2626 501088 16571 16590 AGTCAGACTGCAGCACAGAG 25 2627 501089 16576 16595 AGTTTAGTCAGACTGCAGCA

35 2628 501090 16673 16692 GATGCCTTCCAGGAGGAAGG 19 2629 501091 16678 16697 TTAGGGATGCCTTCCAGGAG 31 2630 501092 16683 16702 TGTTCTTAGGGATGCCTTCC 44 2631 501093 16808 16827 AGGCTGAGTTCTGCTCCTGG 72 2632 501094 16813 16832 GCTAGAGGCTGAGTTCTGCT 83 2633 501095 16818 16837 CATCTGCTAGAGGCTGAGTT 67 2634 501096 16823 16842 TTGGGCATCTGCTAGAGGCT 65 2635 501097 16828 16847 GCTTCTTGGGCATCTGCTAG 75 2636 501098 16833 16852 CCTCTGCTTCTTGGGCATCT 84 2637 501099 16896 16915 TTTCTAAGTACCCTTTACAC 34 2638 501100 16901 16920 AGTGCTTTCTAAGTACCCTT 86 2639 501101 16941 16960 GGAGAGAGAAGGGTGGAACC 26 2640 501102 16949 16968 AGAGGAAGGGAGAGAGAAGG 13 2641 501103 17007 17026 GGGAAAGGTCTTCTCCAGCT 89 2642 501104 17012 17031 CAGGAGGGAAAGGTCTTCTC 60 2643 501105 17017 17036 GGAATCAGGAGGGAAAGGTC 25 2644 501106 17038 17057 GGGTCAAGCAGAGCTTTCTG 66 2645 501107 17109 17128 ATGAATAAACAGGGCCTAGA 37 2646 501108 17114 17133 CTGTGATGAATAAACAGGGC 67 2647 501109 17138 17157 TGAATGACTGAATGCTTGAG 39 2648 501110 17143 17162 TTTGCTGAATGACTGAATGC 53 2649 501111 17207 17226 TCTCATTCATCTGGGCCCTG 72 2650 501112 17244 17263 TGGGAAAAGGTATGTGTGTG 16 2651 501113 17249 17268 CATTATGGGAAAAGGTATGT 8 2652 501114 17254 17273 TTTCTCATTATGGGAAAAGG 38 2653 501115 17259 17278 AGCCCTTTCTCATTATGGGA 47 2654 501116 17264 17283 TACTCAGCCCTTTCTCATTA 43 2655 501117 17269 17288 CCCTGTACTCAGCCCTTTCT 65 2656 501118 17274 17293 CTCACCCCTGTACTCAGCCC 73 2657 501119 17279 17298 CCCATCTCACCCCTGTACTC 66 2658 501120 17284 17303 CCTGTCCCATCTCACCCCTG 13 2659 501121 17289 17308 CCCTGCCTGTCCCATCTCAC 36 2660 501122 17328 17347 AGTCTTCCTTCTCCTCCACC 75 2661 501123 17333 17352 TGGAAAGTCTTCCTTCTCCT 72 2662 501124 17338 17357 TTCTCTGGAAAGTCTTCCTT 66 2663 501125 17420 17439 TTGGTCAGTCTTTTAGCACA 64 2664 501126 17425 17444 CTAGTTTGGTCAGTCTTTTA 68 2665 501127 17450 17469 AGGTAGAGTTCTATTGCTTC 81 2666 501128 17530 17549 GTCTCCACTTGGTGCCTCAC 72 2667 501129 17535 17554 AAGCTGTCTCCACTTGGTGC 61 2668 501130 17540 17559 ACAGGAAGCTGTCTCCACTT 69 2669 501131 17545 17564 CTCACACAGGAAGCTGTCTC 44 2670 501132 17614 17633 AGCTGCCATCTCTGGAGGGT 15 2671 501133 17619 17638 GGCAAAGCTGCCATCTCTGG 52 2672 501134 17624 17643 GTCATGGCAAAGCTGCCATC 7 2673 501135 17664 17683 CCTTTCAGCGGGAGTCCACA 44 2674 501136 17687 17706 TTCCAGGCTCCTCCAGGCAG 36 2675 501137 17692 17711 CTCTCTTCCAGGCTCCTCCA 44 2676 501138 17720 17739 ATGCCACTTCATCAAGGCTG 33 2677 501139 17725 17744 AAGAGATGCCACTTCATCAA 36 2678 501140 17730 17749 TGCCAAAGAGATGCCACTTC 63 2679 501141 17735 17754 CCAAGTGCCAAAGAGATGCC 39 2680 501142 17740 17759 TCAGGCCAAGTGCCAAAGAG 30 2681 501143 17745 17764 GGAAGTCAGGCCAAGTGCCA 38 2682 501144 17750 17769 GTCTAGGAAGTCAGGCCAAG 46 2683 501145 17755 17774 GGGAGGTCTAGGAAGTCAGG 19 2684 501146 17760 17779 CCCCAGGGAGGTCTAGGAAG 32 2685 501147 17765 17784 TCCAGCCCCAGGGAGGTCTA 30 2686 501148 17770 17789 GCTCTTCCAGCCCCAGGGAG 36 2687 501149 17804 17823 AGAGTGAGGGTCAGTACATA 15 2688 501150 17809 17828 GTAGCAGAGTGAGGGTCAGT 28 2689 501151 17933 17952 TAGGTGAGTGAGAAAGTCAC 16 2690 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 78 425

TABLE-US-00040 TABLE 40 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 500994 15257 15276 CATCTGCACCATAATCTGCA 72 2691 500995 15292 15311 TTTACCCAAAAGGTTCACAG 64 2692 500996 15297 15316 ATTGATTTACCCAAAAGGTT 32 2693 500997 15318 15337 GAGTTTGTGTTTCCCATTTC 69 2694 500998 15323 15342 AAAAGGAGTTTGTGTTTCCC

81 2695 500999 15328 15347 ATAAGAAAAGGAGTTTGTGT 30 2696 501000 15333 15352 CTCTAATAAGAAAAGGAGTT 33 2697 501001 15338 15357 TTCTGCTCTAATAAGAAAAG 12 2698 501002 15343 15362 GCTAATTCTGCTCTAATAAG 28 2699 501003 15348 15367 GAATGGCTAATTCTGCTCTA 48 2700 501004 15353 15372 GCTTAGAATGGCTAATTCTG 69 2701 501005 15358 15377 GCAGGGCTTAGAATGGCTAA 59 2702 501006 15363 15382 CAGAGGCAGGGCTTAGAATG 56 2703 501007 15368 15387 GGAAGCAGAGGCAGGGCTTA 68 2704 501008 15373 15392 ACATGGGAAGCAGAGGCAGG 66 2705 501009 15378 15397 AGGTCACATGGGAAGCAGAG 57 2706 501010 15431 15450 TCCTCACTGTGCAGATGAGA 30 2707 501011 15436 15455 AGTCCTCCTCACTGTGCAGA 48 2708 501012 15441 15460 CAACCAGTCCTCCTCACTGT 33 2709 501013 15446 15465 CATCTCAACCAGTCCTCCTC 52 2710 501014 15504 15523 TTAGGGACAAGAATGGAATC 23 2711 501015 15509 15528 AGCAGTTAGGGACAAGAATG 10 2712 501016 15514 15533 CCCACAGCAGTTAGGGACAA 63 2713 501017 15568 15587 TTTTCACAAGGAGAAGCTGA 47 2714 501018 15573 15592 CACCATTTTCACAAGGAGAA 76 2715 501019 15578 15597 GAGTGCACCATTTTCACAAG 71 2716 501020 15646 15665 ATCACAATATATGGGCAAGC 82 2717 501021 15651 15670 TCCCCATCACAATATATGGG 39 2718 501022 15671 15690 CAAGGAATCTCCAAAATACG 37 2719 501023 15703 15722 ATGGCAACACTACTAGCTTG 27 2720 501024 15708 15727 CTGCCATGGCAACACTACTA 56 2721 501025 15729 15748 TAGCTACTGCAGTGGAAGGG 71 2722 501026 15734 15753 AAAAGTAGCTACTGCAGTGG 69 2723 501027 15739 15758 ATTCAAAAAGTAGCTACTGC 59 2724 501028 15744 15763 AGCACATTCAAAAAGTAGCT 59 2725 501029 15804 15823 CTGGACATGATTGCTGAGGA 63 2726 501030 15809 15828 GGACCCTGGACATGATTGCT 74 2727 501031 15814 15833 ATCCAGGACCCTGGACATGA 69 2728 501032 15839 15858 TTCCTGATACCGAACATGGA 22 2729 501033 15876 15895 GGTAAGGCCAAGGAAGTGAG 78 2730 501034 15881 15900 CAAGAGGTAAGGCCAAGGAA 80 2731 501035 15933 15952 ACCTGGGTTTTTGTGTATTC 82 2732 501036 15938 15957 CATACACCTGGGTTTTTGTG 57 2733 501037 15943 15962 TACTCCATACACCTGGGTTT 69 2734 501038 15991 16010 TTATCCACAGAAGATCTCCC 75 2735 501039 16046 16065 CTGAGTGCTGCTAACTTGGG 70 2736 501040 16051 16070 GAAATCTGAGTGCTGCTAAC 78 2737 501041 16082 16101 CACTAAATTAGGAAGCTGGG 79 2738 501043 16105 16124 CCCTCTCTAGGTTTCCCCAT 63 2739 501045 16110 16129 TCCTCCCCTCTCTAGGTTTC 34 2740 501047 16115 16134 CCTCTTCCTCCCCTCTCTAG 24 2741 501049 16120 16139 CTAAGCCTCTTCCTCCCCTC 33 2742 501050 16171 16190 GCCTGCTGCAGGGAGCCAGG 4 2743 501051 16208 16227 GGCCATCAGGACCCTGCAGG 30 2744 501052 16213 16232 AAGTGGGCCATCAGGACCCT 5 2745 501053 16227 16246 GTGTGCCAGGTGGGAAGTGG 0 2746 501054 16232 16251 GCTAGGTGTGCCAGGTGGGA 38 2747 501055 16237 16256 GCTATGCTAGGTGTGCCAGG 62 2748 501056 16242 16261 ACACAGCTATGCTAGGTGTG 22 2749 501057 16247 16266 CCAGCACACAGCTATGCTAG 25 2750 501058 16252 16271 GAGAGCCAGCACACAGCTAT 44 2751 501059 16257 16276 TACTGGAGAGCCAGCACACA 57 2752 501060 16262 16281 CAAACTACTGGAGAGCCAGC 59 2753 501061 16288 16307 TGGGACATCTGGCCCAAAGG 67 2754 501062 16293 16312 CCCACTGGGACATCTGGCCC 77 2755 501063 16305 16324 CTTAAAGCAGGGCCCACTGG 49 2756 501064 16310 16329 GTATCCTTAAAGCAGGGCCC 80 2757 501065 16364 16383 TTCTTTTCTGAAAAGATTAA 16 2758 501066 16369 16388 GTGAGTTCTTTTCTGAAAAG 48 2759 501067 16428 16447 ACTGGTGCCTAAGAGCCCTA 76 2760 501068 16433 16452 CCTCCACTGGTGCCTAAGAG 66 2761 501069 16438 16457 CACTCCCTCCACTGGTGCCT 71 2762 501070 16459 16478 AGAAGACAGCCAGTCAGAGC 67 2763 501071 16464 16483 GGCAGAGAAGACAGCCAGTC 53 2764 501072 16469 16488 CTGAAGGCAGAGAAGACAGC 11 2765 501073 16474 16493

CTACCCTGAAGGCAGAGAAG 0 2766 501074 16479 16498
GGCCTCTACCCTGAAGGCAG 1 2767 413433 32431 32450
GCCTGGACAAGTCCTGCCCA 79 425
TABLE-US-00041 TABLE 41 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 500917 12407 12426 TGTCAGAAAGGACTCCTCTG 37 2768 500918 12412 12431 ACAGCTGTCAGAAAGGACTC 47 2769 500919 12417 12436 CCTCAACAGCTGTCAGAAAG 22 2770 500920 12439 12458 ACCCACCCCAGCCTGTTGCA 56 2771 500921 12474 12493 CAAGCAGATCCACAGTGATG 16 2772 500922 12479 12498 AGTTTCAAGCAGATCCACAG 37 2773 500923 12500 12519 CCCCCAACAATCTAGCCACT 41 2774 500924 12505 12524 AGTTACCCCCAACAATCTAG 16 2775 500925 12510 12529 TTCCCAGTTACCCCCAACAA 16 2776 500926 12515 12534 CTCAGTTCCCAGTTACCCCC 44 2777 500927 12520 12539 CAACCCTCAGTTCCCAGTTA 26 2778 500928 12549 12568 TCTGACTTAGGACTAGGTTA 70 2779 500929 12554 12573 CTCATTCTGACTTAGGACTA 64 2780 500930 12559 12578 ATGTTCTCATTCTGACTTAG 44 2781 500931 12564 12583 GAGTAATGTTCTCATTCTGA 44 2782 500932 12569 12588 CATGAGAGTAATGTTCTCAT 44 2783 500933 12618 12637 GGAAGTGATGTAGGTGAGGG 23 2784 500934 12671 12690 AGCCTAGATGAGCCCTTGGT 65 2785 500935 12676 12695 TTCTCAGCCTAGATGAGCCC 69 2786 500936 12681 12700 CTCCTTTCTCAGCCTAGATG 52 2787 500937 12686 12705 TGGCCCTCCTTTCTCAGCCT 54 2788 500938 12691 12710 ACTCTTGGCCCTCCTTTCTC 26 2789 500939 12696 12715 ACATTACTCTTGGCCCTCCT 59 2790 500940 12779 12798 TTGCTGTTGAATTTGAGGGC 53 2791 500941 12784 12803 AGTACTTGCTGTTGAATTTG 45 2792 500942 12806 12825 CAGCACTAATTTTTACAACT 78 2793 500943 12834 12853 ATCTGTAACATGAGGGTTGG 36 2794 500944 12839 12858 ATCCCATCTGTAACATGAGG 69 2795 500945 12844 12863 CAGTGATCCCATCTGTAACA 67 2796 500946 12890 12909 GCCTGGCTTTGATAACCCTG 63 2797 500947 12895 12914 TTCTAGCCTGGCTTTGATAA 20 2798 500948 12917 12936 CAGACTGGGAGATGGATCTG 31 2799 500949 12922 12941 GGCCACAGACTGGGAGATGG 47 2800 500950 12927 12946 AGTCAGGCCACAGACTGGGA 72 2801 500951 12932 12951 TAAGGAGTCAGGCCACAGAC 47 2802 500952 12937 12956 TGGCTTAAGGAGTCAGGCCA 43 2803 500953 12942 12961 TCTCTTGGCTTAAGGAGTCA 71 2804 500954 12977 12996 GCCCTGCACTCAGCCCTTCA 19 2805 500955 12982 13001 ACAGAGCCCTGCACTCAGCC 54 2806 500956 13026 13045 GACAACCTTCACTCATTCAG 35 2807 500957 13098 13117 ATCCACTAGGGCAGGCTTGG 69 2808 500958 13202 13221 GGCCAGGAGGAAGAGTTTGG 39 2809 500959 13207 13226 TCATAGGCCAGGAGGAAGAG 20 2810 500960 13212 13231 TTACTTCATAGGCCAGGAGG 63 2811 500961 13239 13258 TGGCAGTGGAGGCTGTTCAC 63 2812 500962 13306 13325 GCAGGAACAGGAAGACCACC 0 2813 500963 13311 13330 ATGCTGCAGGAACAGGAAGA 51 2814 500964 13316 13335 GAGTAATGCTGCAGGAACAG 35 2815 500965 13321 13340 ATTGGGAGTAATGCTGCAGG 55 2816 500966 13441 13460 TGGCTCTCTAGATAGGGTGG 75 2817 500967 13485 13504 TAAAAGAAGATGTGGTGATT 0 2818 500968 13490 13509 AAAAATAAAAGAAGATGTGG 0 2819 500969 13495 13514 GCTATAAAAATAAAAGAAGA 0 2820 500970 13500 13519 TAAAGGCTATAAAAATAAAA 0 2821 500971 13542 13561 ATTGCTTAAACAGATAAGCA 6 2822 500972 13547 13566 AGACAATTGCTTAAACAGAT 32 2823 500973 13916 13935 GTTCCTGTTTCCAAGGGACT 59 2824 500974 13921 13940 ACAAGGTTCCTGTTTCCAAG 69 2825 500975 13926 13945 GACAGACAAGGTTCCTGTTT 72 2826 500976 13931 13950 AGAAAGACAGACAAGGTTCC 51 2827 500977 13936 13955 CACTGAGAAAGACAGACAAG 0 2828 500978 13941 13960

CACAGCACTGAGAAAGACAG 36 2829 500979 13962 13981
GCCAGGCACTGTGCTAGATG 63 2830 500980 13967 13986 CCAGTGCCAGGCACTGTGCT 37 2831 500981 13972 13991 TATTACCAGTGCCAGGCACT 13 2832 500982 13977 13996 GTACCTATTACCAGTGCCAG 64 2833 500983 13982 14001 ACTAAGTACCTATTACCAGT 7 2834 500984 14054 14073 AGCTAAAAGCAGGGCTTCCT 0 2835 500985 14177 14196 GCTCCTTGCCAGACTGTGGG 60 2836 500986 14182 14201 AGCCAGCTCCTTGCCAGACT 75 2837 500987 14187 14206 CTTGGAGCCAGCTCCTTGCC 54 2838 500988 14192 14211 TCAGCCTTGGAGCCAGCTCC 35 2839 500989 14197 14216 GCCACTCAGCCTTGGAGCCA 76 2840 500990 14239 14258 GGAAGCAGTAAGGGTGACCA 73 2841 500991 14246 14265 AGCCCTGGGAAGCAGTAAGG 11 2842 500992 14251 14270
TAATAAGCCCTGGGAAGCAG 0 2843 500993 14327 14346
ATCCCCAGAGAGGAAGTGAA 0 2844 413433 32431 32450
GCCTGGACAAGTCCTGCCCA 80 425

TABLE-US-00042 TABLE 42 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 500840 11204 11223
ATAACCACAGCGATAATCAC 45 2845 500841 11209 11228 GGCAGATAACCACAGCGATA 79 2846 500842 11228 11247 CAGCACCCACACCAAGAGGG 38 2847 500843 11318 11337 TCAGAGGCTACTGGCTGGAT 69 2848 500844 11323 11342 GGCTGTCAGAGGCTACTGGC 75 2849 500845 11346 11365 AAAAGGACTCAAGTGAGAGA 24 2850 500846 11351 11370 AACAGAAAAGGACTCAAGTG 4 2851 500847 11356 11375
CAGGGAACAGAAAAGGACTC 30 2852 500848 11361 11380
GGACACAGGGAACAGAAAAG 2 2853 500849 11366 11385
TCAAAGGACACAGGGAACAG 24 2854 500850 11371 11390
GGACATCAAAGGACACAGGG 3 2855 500851 11376 11395
CCTAAGGACATCAAAGGACA 3 2856 500852 11413 11432
GCATGAACAAGGGCAAGTCC 74 2857 500853 11457 11476 AGAACATCCCTATCCCACTG 37 2858 500854 11462 11481 CACAGAGAACATCCCTATCC 59 2859 500855 11509 11528 CCTGTGGATGGCAGAGCCGA 32 2860 500856 11514 11533 CCCAGCCTGTGGATGGCAGA 21 2861 500857 11519 11538 CAGCTCCCAGCCTGTGGATG 0 2862 500858 11553 11572 CATGACAGCCAGGGTCACTG 57 2863 500859 11559 11578
CAAAAACATGACAGCCAGGG 80 2864 500860 11564 11583
TAAGTCAAAAACATGACAGC 24 2865 500861 11569 11588 AACTCTAAGTCAAAAACATG 0 2866 500862 11574 11593 GAACAAACTCTAAGTCAAAA 5 2867 500863 11579 11598 CTAAGGAACAAACTCTAAGT 19 2868 500864 11584 11603 TTCTCCTAAGGAACAAACTC 31 2869 500865 11589 11608 ACAAGTTCTCCTAAGGAACA 28 2870 500866 11594 11613 AGAGTACAAGTTCTCCTAAG 66 2871 500867 11599 11618 TCGCTAGAGTACAAGTTCTC 81 2872 500868 11635 11654 TCCTGACATGATATTAAGTG 50 2873 500869 11640 11659 AATGTTCCTGACATGATATT 5 2874 500870 11707 11726 ATGATGTAAATCCTTGCACC 68 2875 500871 11712 11731 ATTCCATGATGTAAATCCTT 63 2876 500872 11717 11736 CCTACATTCCATGATGTAAA 45 2877 500873 11722 11741 CCCTTCCTACATTCCATGAT 56 2878 500874 11727 11746 ACCAGCCCTTCCTACATTCC 54 2879 500875 11732 11751 TTCATACCAGCCCTTCCTAC 22 2880 500876 11755 11774 AAGCTGAACTGACTGGTTTG 41 2881 500877 11760 11779 CCAGAAAGCTGAACTGACTG 0 2882 500878 11765 11784 AGATCCCAGAAAGCTGAACT 29 2883 500879 11770 11789 AAAGTAGATCCCAGAAAGCT 15 2884 500880 11775 11794 TCACCAAAGTAGATCCCAGA 58 2885 500881 11780 11799 ATCTTTCACCAAAGTAGATC 33 2886 500882 11785 11804 ACCCAATCTTTCACCAAAGT 18 2887 500883 11790 11809 ACTCCACCCAATCTTTCACC 49 2888 500884 11795 11814 CCCCTACTCCACCCAATCTT 14 2889 500885 11800 11819 GCCCTCCCCTACTCCACCCA 32 2890 500886 11805 11824 TCAGTGCCCTCCCCTACTCC 26 2891 500887 11827 11846

TGCCCAGATAACAAAATGTG 29 2892 500888 11832 11851 GGAGATGCCCAGATAACAAA 56 2893 500889 11857 11876 CAAGGATCTAGAAGGCAGGT 59 2894 500890 11862 11881 AGGACCAAGGATCTAGAAGG 28 2895 500891 11867 11886 CTTCAAGGACCAAGGATCTA 30 2896 500892 11872 11891 AGTATCTTCAAGGACCAAGG 72 2897 500893 11893 11912 AGGCAAACTAGGCCACTGGG 20 2898 500894 11898 11917 CACAGAGGCAAACTAGGCCA 4 2899 500895 11920 11939 CTCACAACAGTGGGACCTTA 69 2900 500896 11925 11944 ACCAGCTCACAACAGTGGGA 43 2901 500897 11930 11949 TGTTCACCAGCTCACAACAG 8 2902 500898 11964 11983 CATGGTCTCTACTTGAATAC 14 2903 500899 11969 11988 GAATCCATGGTCTCTACTTG 18 2904 500900 11974 11993 TCACAGAATCCATGGTCTCT 42 2905 500901 11979 11998 TCCCTTCACAGAATCCATGG 36 2906 500902 11984 12003 GGACTTCCCTTCACAGAATC 23 2907 500903 11989 12008 TCACAGGACTTCCCTTCACA 47 2908 500904 12070 12089 ACCTACCCATACTCCCACAG 31 2909 500905 12075 12094 CACCCACCTACCCATACTCC 15 2910 500906 12080 12099 GCATGCACCCACCTACCCAT 41 2911 500907 12085 12104 CCCCAGCATGCACCCACCTA 14 2912 500908 12090 12109 GCTTCCCCCAGCATGCACCC 36 2913 500909 12138 12157 GGCACAGGACAGCCCTCCAA 54 2914 500910 12169 12188 TGTCCTGATGAACTCTGCAA 54 2915 500911 12242 12261 TGTCAGAGGGCTCTTCTGAA 0 2916 500912 12247 12266 GCAGGTGTCAGAGGGCTCTT 60 2917 500913 12285 12304 TCTCATGCTAGGTCCTGTGC 80 2918 500914 12392 12411 CTCTGCTACATCAAAACTTG 25 2919 500915 12397 12416 GACTCCTCTGCTACATCAAA 38 2920 500916 12402 12421 GAAAGGACTCCTCTGCTACA 23 2921 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 75 425

TABLE-US-00043 TABLE 43 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 ISIS SEQ ID NO: 2 SEQ ID NO: 2 % SEQ NO Start Site Stop Site Sequence inhibition ID NO 501824 27442 27461 TTGGGTCTGATCAATTATTA 14 2922 501825 27447 27466 TGTGGTTGGGTCTGATCAAT 50 2923 501826 27458 27477 GGTTTCTGGGCTGTGGTTGG 20 2924 501827 27469 27488 GATGCTGGGCCGGTTTCTGG 40 2925 501828 27551 27570 CAGGAGATCTGCCTGTCCAA 48 2926 501829 27556 27575 TAGCTCAGGAGATCTGCCTG 32 2927 501830 27561 27580 AGCAGTAGCTCAGGAGATCT 50 2928 501831 27566 27585 TAATTAGCAGTAGCTCAGGA 65 2929 501832 27613 27632 AGGTTTGAGAGGCAGAGGGA 6 2930 501833 27618 27637 CCAGCAGGTTTGAGAGGCAG 29 2931 501834 27623 27642 TTCTGCCAGCAGGTTTGAGA 40 2932 501835 27628 27647 TGAGCTTCTGCCAGCAGGTT 64 2933 501836 27633 27652 CCAGGTGAGCTTCTGCCAGC 43 2934 501837 27638 27657 GCTTGCCAGGTGAGCTTCTG 61 2935 501838 27643 27662 TCTTTGCTTGCCAGGTGAGC 67 2936 501839 27673 27692 TCCAGGGAGAGACCAACAGG 7 2937 501840 27678 27697 TCTGGTCCAGGGAGAGACCA 0 2938 501841 27683 27702 AAATCTCTGGTCCAGGGAGA 26 2939 501842 27688 27707 GTGAAAAATCTCTGGTCCAG 12 2940 501843 27693 27712 AAGTGGTGAAAAATCTCTGG 5 2941 501844 27698 27717 GCACAAAGTGGTGAAAAATC 41 2942 501845 27703 27722 CCATGGCACAAAGTGGTGAA 41 2943 501846 27708 27727 GGGTTCCATGGCACAAAGTG 1 2944 501847 27761 27780 ATTAAATACTCTAAAACACT 0 2945 501848 27766 27785 CTTGTATTAAATACTCTAAA 5 2946 501849 27771 27790 TGCACCTTGTATTAAATACT 77 2947 501850 27776 27795 TCCAATGCACCTTGTATTAA 79 2948 501851 27781 27800 TGAAATCCAATGCACCTTGT 69 2949 501852 27786 27805 TCCTTTGAAATCCAATGCAC 76 2950 501853 27791 27810 TAGTTTCCTTTGAAATCCAA 64 2951 501854 27849 27868 TGTTACTACATGTCTAATCA 32 2952 501855 27854 27873 GCACCTGTTACTACATGTCT 87 2953 501856 27859 27878 AAAAGGCACCTGTTACTACA 57 2954 501857 27864 27883 TAATAAAAAGGCACCTGTTA 32 2955 501858 27992 28011 AAATAACAAAGGTATTTTCA 1 2956 501859 27997 28016 CAAATAAATAACAAAGGTAT 14 2957 501860 28002 28021

TGACACAAATAAATAACAAA 18 2958 501861 28026 28045 TCACAGAATTATCAGCAGTA 92 2959 501862 28031 28050 ATAAATCACAGAATTATCAG 0 2960 501863 28054 28073 CATTTCCTTCACTTACCAAT 41 2961 501864 28078 28097 CACTTATCTCTTATGGAAAT 47 2962 501865 28083 28102 ATTTTCACTTATCTCTTATG 35 2963 501866 28088 28107 TCTAAATTTTCACTTATCTC 48 2964 501867 28093 28112 TGACATCTAAATTTTCACTT 63 2965 501868 28098 28117 ACAAATGACATCTAAATTTT 0 2966 501869 28103 28122 GGGAAACAAATGACATCTAA 58 2967 501870 28127 28146 ACACAATATCCATGGACTTG 59 2968 501871 28132 28151 CTGACACACAATATCCATGG 75 2969 501872 28185 28204 ATTTCCTTACAGGGTATTGG 53 2970 501873 28210 28229 GGCCAAGCCCCATTCCTAAG 7 2971 501874 28215 28234 TTGCTGGCCAAGCCCCATTC 0 2972 501875 28220 28239 GCCACTTGCTGGCCAAGCCC 22 2973 501876 28269 28288 GGTGAAGAGCATGGACTCTG 41 2974 501877 28293 28312 GCAGGGAAGGCAGCAGTGTG 39 2975 501878 28298 28317 CAATGGCAGGGAAGGCAGCA 55 2976 501879 28358 28377 ACTTGTGATACAAAATTCTG 41 2977 501880 28363 28382 AAGACACTTGTGATACAAAA 66 2978 501881 28454 28473 AAGCTTAGAAGGCCCTGCTT 44 2979 501882 28459 28478 ATGAGAAGCTTAGAAGGCCC 41 2980 501883 28464 28483 AGCTCATGAGAAGCTTAGAA 74 2981 501884 28469 28488 TGCTGAGCTCATGAGAAGCT 82 2982 501885 28474 28493 ACTGTTGCTGAGCTCATGAG 37 2983 501886 28479 28498 AAACCACTGTTGCTGAGCTC 78 2984 501887 28484 28503 AGTAAAAACCACTGTTGCTG 60 2985 501888 28489 28508 GCTGCAGTAAAAACCACTGT 66 2986 501889 28515 28534 CTGGTTCCATGCTCTTAGGT 63 2987 501890 28520 28539 AGGCTCTGGTTCCATGCTCT 71 2988 501891 28525 28544 ACAACAGGCTCTGGTTCCAT 38 2989 501892 28530 28549 TCTGAACAACAGGCTCTGGT 19 2990 501893 28535 28554 TGTCCTCTGAACAACAGGCT 52 2991 501894 28540 28559 ATCCTTGTCCTCTGAACAAC 36 2992 501895 28545 28564 GCCTAATCCTTGTCCTCTGA 54 2993 501896 28550 28569 TCAGAGCCTAATCCTTGTCC 53 2994 501897 28555 28574 CTTTCTCAGAGCCTAATCCT 39 2995 501898 28560 28579 CCTTCCTTTCTCAGAGCCTA 50 2996 501899 28565 28584 AATGACCTTCCTTTCTCAGA 19 2997 501900 28570 28589 CACCAAATGACCTTCCTTTC 66 2998 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 79 425

TABLE-US-00044 TABLE 44 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID NO: 2 NO: 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 501901 28575 28594 AAATCCACCAAATGACCTTC 42 2999 501902 28580 28599 GAACTAAATCCACCAAATGA 13 3000 501903 28585 28604 AGGATGAACTAAATCCACCA 65 3001 501904 28590 28609 GCAAAAGGATGAACTAAATC 4 3002 501905 28595 28614 GAAGAGCAAAAGGATGAACT 0 3003 501906 28600 28619 CACAGGAAGAGCAAAAGGAT 0 3004 501907 28605 28624 CCAAACACAGGAAGAGCAAA 0 3005 501908 28610 28629 AGAAACCAAACACAGGAAGA 33 3006 501909 28615 28634 GCCCCAGAAACCAAACACAG 20 3007 501910 28620 28639 CTCCAGCCCCAGAAACCAAA 45 3008 501911 28625 28644 AATCTCTCCAGCCCCAGAAA 36 3009 501912 28787 28806 CCTGGCCTTTCTCAGCTGGG 25 3010 501913 28792 28811 TCTAGCCTGGCCTTTCTCAG 0 3011 501914 28797 28816 TGAATTCTAGCCTGGCCTTT 44 3012 501915 28802 28821 GAGACTGAATTCTAGCCTGG 48 3013 501916 28807 28826 ATCCAGAGACTGAATTCTAG 71 3014 501917 28829 28848 AGAAGAAGAGGCCTGATGGG 29 3015 501918 28834 28853 GATGGAGAAGAAGAGGCCTG 59 3016 501919 28839 28858 GCCTGGATGGAGAAGAAGAG 0 3017 501920 28844 28863 AGGCAGCCTGGATGGAGAAG 5 3018 501921 28849 28868 TGCTGAGGCAGCCTGGATGG 26 3019 501922 28854 28873 TCTGCTGCTGAGGCAGCCTG 12 3020 501923 28859 28878 CTTACTCTGCTGCTGAGGCA 47 3021 501924 28864 28883 TTGTCCTTACTCTGCTGCTG 55 3022 501925 28901 28920 GGCTGGTCTCTCTGGGAAGG 25 3023 501926 28906 28925 TAGAGGGCTGGTCTCTCTGG 40 3024 501927 28911 28930 CTGCTTAGAGGGCTGGTCTC 40 3025 501928 28916 28935

CCCCACTGCTTAGAGGGCTG 34 3026 501929 28921 28940 CCAGGCCCCACTGCTTAGAG 26 3027 501930 28926 28945 GAGCTCCAGGCCCCACTGCT 7 3028 501931 28986 29005 GGAAACCTAGCTTCCAGAAA 53 3029 501932 29010 29029 TTTAGGCTATGGTCTGAGCA 78 3030 501933 29015 29034 TGAGGTTTAGGCTATGGTCT 68 3031 501934 29044 29063 GATGCTCCAGGTGGGCCAGA 68 3032 501935 29049 29068 AGGTGGATGCTCCAGGTGGG 0 3033 501936 29054 29073 CCTCTAGGTGGATGCTCCAG 18 3034 501937 29059 29078 GGCATCCTCTAGGTGGATGC 22 3035 501938 29064 29083 CTAGTGGCATCCTCTAGGTG 50 3036 501939 29069 29088 CTCCTCTAGTGGCATCCTCT 33 3037 501940 29074 29093 CCAGGCTCCTCTAGTGGCAT 55 3038 501941 29079 29098 GGCATCCAGGCTCCTCTAGT 53 3039 501942 29103 29122 GACTCTAGCCCCCCAGACTC 29 3040 501943 29139 29158 TCTGCCTGATTCCCTTTCTT 30 3041 501944 29144 29163 AGCAGTCTGCCTGATTCCCT 81 3042 501945 29149 29168 TGTTCAGCAGTCTGCCTGAT 50 3043 501946 29154 29173 CTTACTGTTCAGCAGTCTGC 70 3044 501947 29159 29178 TCATACTTACTGTTCAGCAG 65 3045 501948 29164 29183 CAAAGTCATACTTACTGTTC 29 3046 501949 29169 29188 GCCTACAAAGTCATACTTAC 17 3047 501950 29193 29212 GGTGAATAGCTATGTCTAAA 80 3048 501951 29198 29217 AGCTTGGTGAATAGCTATGT 76 3049 501952 29222 29241 AGCAAACTGTGAAAAGCTTA 59 3050 501953 29227 29246 TTAAAAGCAAACTGTGAAAA 2 3051 501954 29232 29251 GCCTGTTAAAAGCAAACTGT 48 3052 501955 29237 29256 CAAGAGCCTGTTAAAAGCAA 12 3053 501956 29242 29261 GCCTACAAGAGCCTGTTAAA 30 3054 501957 29247 29266 GTGCAGCCTACAAGAGCCTG 69 3055 501958 29252 29271 AGCATGTGCAGCCTACAAGA 57 3056 501959 29257 29276 AGGGAAGCATGTGCAGCCTA 76 3057 501960 29262 29281 TTTCTAGGGAAGCATGTGCA 76 3058 501961 29267 29286 ACAAGTTTCTAGGGAAGCAT 40 3059 501962 29272 29291 GGAAGACAAGTTTCTAGGGA 52 3060 501963 29277 29296 AGAAGGGAAGACAAGTTTCT 30 3061 501964 29282 29301 ATCGCAGAAGGGAAGACAAG 28 3062 501965 29327 29346 AGTGACGAGATGTCCAATTT 59 3063 501966 29361 29380 ACAACTCTCTTGTTGGGAGG 71 3064 501967 29366 29385 AGGGTACAACTCTCTTGTTG 59 3065 501968 29371 29390 AAAACAGGGTACAACTCTCT 73 3066 501969 29376 29395 AGCTAAAAACAGGGTACAAC 13 3067 501970 29383 29402 CCAGGGTAGCTAAAAACAGG 3 3068 501971 29388 29407 TCTCCCCAGGGTAGCTAAAA 0 3069 501972 29393 29412 CAGCCTCTCCCCAGGGTAGC 33 3070 501973 29417 29436 TAGCCCTGTTCTAGACTCCT 36 3071 501974 29440 29459 TAGCCCCTTGTTGCCCCCCA 43 3072 501975 29445 29464 AATGGTAGCCCCTTGTTGCC 11 3073 501976 29450 29469 AGGGAAATGGTAGCCCCTTG 63 3074 501977 29475 29494 GTAGACTCTCCATGAGCCTA 60 3075 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 77 425

TABLE-US-00045 TABLE 45 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID NO: 2 NO: 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 501978 30937 30956 GGGCCCACAAACCCTGAGAA 20 3076 501979 30962 30981 AGGCACACCCAGGGCCATGG 68 3077 501980 30967 30986 AAGCTAGGCACACCCAGGGC 48 3078 501981 30972 30991 GCACTAAGCTAGGCACACCC 77 3079 501982 30977 30996 CTGTGGCACTAAGCTAGGCA 78 3080 501983 30982 31001 GTTTACTGTGGCACTAAGCT 80 3081 501984 30987 31006 TGAGTGTTTACTGTGGCACT 56 3082 501985 31026 31045 ACTGGGCTTCATCTCCCCTC 0 3083 501986 31031 31050 GTCCTACTGGGCTTCATCTC 28 3084 501987 31074 31093 GGCACTGCAGGCCACTCCTG 41 3085 501988 31185 31204 AAGGGAGGCCTTGCACTTAC 26 3086 501989 31236 31255 GCCCTTCAGCTTGTGCAGGG 36 3087 501990 31241 31260 ATGAGGCCCTTCAGCTTGTG 57 3088 501991 31246 31265 TCAGGATGAGGCCCTTCAGC 67 3089 501992 31251 31270 AGCACTCAGGATGAGGCCCT 50 3090 501993 31274 31293 ACAAAGTGGGTGTTAAAAGA 22 3091 501994 31279 31298 TTTTCACAAAGTGGGTGTTA 40 3092 501995 31315 31334 GCTTTTAACCTCCCCCCAAA 6 3093 501996 31366 31385

CCTCGACTGAGTGTGAACTC 66 3094 501997 31371 31390 CAAACCCTCGACTGAGTGTG 46 3095 501998 31376 31395 TTATGCAAACCCTCGACTGA 24 3096 501999 31490 31509 ACATTTGGGCAAGGCAGACA 14 3097 502000 31525 31544 AGGGAATAAAATACAGAGTT 0 3098 502001 31530 31549 CTTCCAGGGAATAAAATACA 0 3099 502002 31535 31554 GCCACCTTCCAGGGAATAAA 0 3100 502003 31540 31559 CTCCTGCCACCTTCCAGGGA 0 3101 502004 31545 31564 GTGACCTCCTGCCACCTTCC 54 3102 502005 31606 31625 TTTACCTGGATGGGAAAGTA 0 3103 502006 31612 31631 CAGCACTTTACCTGGATGGG 2 3104 502007 31617 31636 ACTCACAGCACTTTACCTGG 0 3105 502008 31622 31641 ACAACACTCACAGCACTTTA 1 3106 502009 31688 31707 ACTTGCTTCCCTGTGGGTGG 0 3107 502010 31693 31712 TCTAAACTTGCTTCCCTGTG 16 3108 502011 31698 31717 CTTGGTCTAAACTTGCTTCC 60 3109 502012 31703 31722 ACCAACTTGGTCTAAACTTG 57 3110 502013 31768 31787 GGATACTCAGAAGAGCAGTG 79 3111 502014 31792 31811 CCTCAGGGCTGGCCCAAAGA 61 3112 502015 31797 31816 CAGGACCTCAGGGCTGGCCC 74 3113 502016 31802 31821 CCTGTCAGGACCTCAGGGCT 40 3114 502017 31807 31826 ATTTCCCTGTCAGGACCTCA 46 3115 502018 31828 31847 TGAAAGCCAAACTGAGCCAC 55 3116 502019 31866 31885 AGCTTGCAGACCAGGCAGGG 60 3117 502020 31871 31890 AGCCCAGCTTGCAGACCAGG 55 3118 502021 31876 31895 TCACCAGCCCAGCTTGCAGA 30 3119 502022 31881 31900 GTGCCTCACCAGCCCAGCTT 49 3120 502023 31904 31923 ACATGCATCAGGGCCAGATG 35 3121 502024 31932 31951 AGTTATCCTCAATTCACCAG 66 3122 502025 31937 31956 GCCAGAGTTATCCTCAATTC 76 3123 502026 31942 31961 ATCCTGCCAGAGTTATCCTC 71 3124 502027 31947 31966 TCAGGATCCTGCCAGAGTTA 63 3125 502028 31952 31971 AACCTTCAGGATCCTGCCAG 60 3126 502029 31957 31976 GGGAAAACCTTCAGGATCCT 51 3127 502030 31975 31994 ACAGGTCTTTCCCCTGTGGG 31 3128 502031 31980 31999 GCCAGACAGGTCTTTCCCCT 65 3129 502032 32052 32071 CTTCAGCACCCCTACCTTCT 45 3130 502033 32057 32076 CCACTCTTCAGCACCCCTAC 53 3131 502034 32062 32081 TGCCTCCACTCTTCAGCACC 60 3132 502035 32110 32129 ATATACACCACCCTGCACCC 30 3133 502036 32115 32134 AGCAAATATACACCACCCTG 67 3134 502037 32120 32139 ACAACAGCAAATATACACCA 73 3135 502038 32125 32144 GACTCACAACAGCAAATATA 68 3136 502039 32130 32149 TCACTGACTCACAACAGCAA 60 3137 502040 32135 32154 CTCAGTCACTGACTCACAAC 83 3138 502041 32140 32159 AGGATCTCAGTCACTGACTC 68 3139 502042 32146 32165 CACACCAGGATCTCAGTCAC 61 3140 502043 32167 32186 CCAGCCACTGCCCCCAGGCA 38 3141 502044 32172 32191 GTTACCCAGCCACTGCCCCC 33 3142 502045 32177 32196 GCAGGGTTACCCAGCCACTG 76 3143 502046 32182 32201 AGGATGCAGGGTTACCCAGC 80 3144 502047 32187 32206 AGTGAAGGATGCAGGGTTAC 13 3145 502048 32192 32211 AATGCAGTGAAGGATGCAGG 52 3146 502049 32223 32242 AGGAGCTGGCCCTGCCACCC 48 3147 502050 32228 32247 GCAGAAGGAGCTGGCCCTGC 64 3148 502051 32233 32252 GATGAGCAGAAGGAGCTGGC 56 3149 502052 32238 32257 CTAAGGATGAGCAGAAGGAG 42 3150 502053 32243 32262 TTAGGCTAAGGATGAGCAGA 60 3151 502054 32248 32267 TGGGCTTAGGCTAAGGATGA 53 3152 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 79 425

TABLE-US-00046 TABLE 46 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID NO: 2 NO: 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 77 425 502055 33408 33427 TCTTCATTACATGATTACTG 66 3153 502056 33413 33432 GCAGATCTTCATTACATGAT 81 3154 502057 33449 33468 ACCAGGAAGGGAGTAGGTGG 0 3155 502058 33454 33473 GTCCCACCAGGAAGGGAGTA 0 3156 502059 33459 33478 CCAGAGTCCCACCAGGAAGG 0 3157 502060 33464 33483 AAAGACCAGAGTCCCACCAG 15 3158 502061 33489 33508

GGGCCAGATGAGCTGTTCTG 26 3159 502062 33513 33532 GTGCCAACAGAAGGGATACA 68 3160 502063 33518 33537 CACCTGTGCCAACAGAAGGG 36 3161 502064 33523 33542 AACCCCACCTGTGCCAACAG 53 3162 502065 33557 33576 GGCTGGTGGTCTCGGTGGCT 61 3163 502066 33562 33581 CAGGTGGCTGGTGGTCTCGG 38 3164 502067 33567 33586 TCTTCCAGGTGGCTGGTGGT 32 3165 502068 33572 33591 GCTCCTCTTCCAGGTGGCTG 61 3166 502069 33577 33596 TGTCTGCTCCTCTTCCAGGT 63 3167 502070 33582 33601 GGCACTGTCTGCTCCTCTTC 64 3168 502071 33636 33655 TTCCTAGTCTCCTCTCCAAG 24 3169 502072 33658 33677 ATTCTAAGCTTAGAGAGAGT 28 3170 502073 33663 33682 AGGTGATTCTAAGCTTAGAG 53 3171 502074 33668 33687 TGGACAGGTGATTCTAAGCT 66 3172 502075 33673 33692 GCAGATGGACAGGTGATTCT 72 3173 502076 33678 33697 ATGAGGCAGATGGACAGGTG 66 3174 502077 33683 33702 GTGAAATGAGGCAGATGGAC 52 3175 502078 33688 33707 TATCAGTGAAATGAGGCAGA 16 3176 502079 33693 33712 AAGCCTATCAGTGAAATGAG 8 3177 502080 33698 33717 TCAGTAAGCCTATCAGTGAA 0 3178 502081 33703 33722 GTGCCTCAGTAAGCCTATCA 56 3179 502082 33708 33727 TCTCTGTGCCTCAGTAAGCC 32 3180 502083 33729 33748 TGACCTTGGGATAGTCCCTC 84 3181 502084 33734 33753 CTTTGTGACCTTGGGATAGT 61 3182 502085 33740 33759 CTTAAGCTTTGTGACCTTGG 69 3183 502086 33745 33764 TACTACTTAAGCTTTGTGAC 59 3184 502087 33750 33769 CCTGCTACTACTTAAGCTTT 30 3185 502088 33755 33774 CTAGTCCTGCTACTACTTAA 43 3186 502089 33760 33779 CTACACTAGTCCTGCTACTA 16 3187 502090 33765 33784 GGTTCCTACACTAGTCCTGC 59 3188 502091 33802 33821 CAGGAGTAAGACCATGGGCC 59 3189 502092 33807 33826 TAACACAGGAGTAAGACCAT 57 3190 502093 33812 33831 GAAAGTAACACAGGAGTAAG 45 3191 502094 33817 33836 ATGGTGAAAGTAACACAGGA 54 3192 502095 33846 33865 TGCTTAAGTTTACACAGCAC 66 3193 502096 33851 33870 AGGCTTGCTTAAGTTTACAC 62 3194 502097 33856 33875 AGCAAAGGCTTGCTTAAGTT 65 3195 502098 33861 33880 AGAAGAGCAAAGGCTTGCTT 79 3196 502099 33866 33885 CCCACAGAAGAGCAAAGGCT 80 3197 502100 33871 33890 TCAGACCCACAGAAGAGCAA 69 3198 502101 33893 33912 AGCAGTGCATAGAGGAAAAA 16 3199 502102 33898 33917 ACCACAGCAGTGCATAGAGG 60 3200 502103 33903 33922 GTCCCACCACAGCAGTGCAT 49 3201 502104 33908 33927 GGCTTGTCCCACCACAGCAG 42 3202 502105 33913 33932 AGATAGGCTTGTCCCACCAC 54 3203 502106 33918 33937 TGCTCAGATAGGCTTGTCCC 71 3204 502107 33958 33977 CAAGTCACTCTTTATACCCT 53 3205 502108 33963 33982 CCTATCAAGTCACTCTTTAT 18 3206 502109 33968 33987 ACATTCCTATCAAGTCACTC 58 3207 502110 34017 34036 CCCAACACGATGCCCAGGCC 70 3208 502111 34053 34072 TCCTCTTGCTGCCTGTTTCC 34 3209 502112 34058 34077 TTGTGTCCTCTTGCTGCCTG 59 3210 502113 34063 34082 TGCTCTTGTGTCCTCTTGCT 68 3211 502114 34098 34117 AAGGATACCGCCAATGAAGG 65 3212 502115 34155 34174 TCATGTCACACTCCCCTCCC 58 3213 502116 34179 34198 CATCTGACCAGCTCTGATCT 52 3214 502117 34184 34203 GTAGGCATCTGACCAGCTCT 68 3215 502118 34189 34208 AGAATGTAGGCATCTGACCA 57 3216 502119 34211 34230 TGCCCTTGCTGTAGGACCCA 84 3217 502120 34216 34235 GTCAATGCCCTTGCTGTAGG 70 3218 502121 34221 34240 TGCAAGTCAATGCCCTTGCT 44 3219 502122 34226 34245 CACAGTGCAAGTCAATGCCC 62 3220 502123 34231 34250 TGGGACACAGTGCAAGTCAA 55 3221 502124 34267 34286 AGGCTCTGATGGGACATTTG 65 3222 502125 34301 34320 CAGGTGGCTCTTTAAATCAT 59 3223 502126 34306 34325 GGCCCCAGGTGGCTCTTTAA 32 3224 502127 34312 34331 CCAGTGGGCCCCAGGTGGCT 63 3225 502128 34317 34336 GTCACCCAGTGGGCCCCAGG 46 3226 502129 34340 34359 CCTGGGCTGCTGGATGAAGA 0 3227 502130 34345 34364 TTTTCCCTGGGCTGCTGGAT 28 3228 502131 34350 34369 TGCACTTTTCCCTGGGCTGC 70 3229

TABLE-US-00047 TABLE 47 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID NO: 2 NO: 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 78 425 502132 34391 34410 TCACAGCAAATTATCCTGCA 49 3230 502133 34396 34415 GGAGGTCACAGCAAATTATC 41 3231 502134 34401 34420 CCTGTGGAGGTCACAGCAAA 52 3232 502135 34460 34479 TAAGCCATTCCTTGGATGAC 72 3233 502136 34465 34484 AGGTCTAAGCCATTCCTTGG 58 3234 502137 34470 34489 CTGAAAGGTCTAAGCCATTC 59 3235 502138 34553 34572 TTTGTCCCCTTCCTATGGCT 49 3236 502139 34605 34624 CAGGCCTGGGATAGTTACCC 8 3237 502140 34610 34629 ATGGCCAGGCCTGGGATAGT 33 3238 502141 34615 34634 AGCTGATGGCCAGGCCTGGG 7 3239 502142 34620 34639 TCCTGAGCTGATGGCCAGGC 40 3240 502143 34625 34644 CTTGCTCCTGAGCTGATGGC 20 3241 502144 34630 34649 GGAAACTTGCTCCTGAGCTG 62 3242 502145 34635 34654 AACTTGGAAACTTGCTCCTG 52 3243 502146 34640 34659 TGGGAAACTTGGAAACTTGC 66 3244 502147 34676 34695 GGCATTGAGGGAAGAGCTGG 42 3245 502148 34681 34700 AGGCAGGCATTGAGGGAAGA 53 3246 502149 34686 34705 AAGGCAGGCAGGCATTGAGG 62 3247 502150 34691 34710 TGAAAAAGGCAGGCAGGCAT 63 3248 502151 34696 34715 TTTGATGAAAAAGGCAGGCA 49 3249 502152 34701 34720 CTAGTTTTGATGAAAAAGGC 53 3250 502153 34706 34725 TTGTGCTAGTTTTGATGAAA 41 3251 502154 34739 34758 TAGCTCTCAGTGAAGCTGGA 83 3252 502155 34826 34845 GAGCAGATCTCAAAATCTCG 70 3253 502156 34868 34887 ACAGCAGCAGTCTAATCCAT 73 3254 502157 34873 34892 GGGAAACAGCAGCAGTCTAA 49 3255 502158 34878 34897 TAAATGGGAAACAGCAGCAG 62 3256 502159 34883 34902 CCAAATAAATGGGAAACAGC 15 3257 502160 34888 34907 ACTCCCCAAATAAATGGGAA 13 3258 502161 34893 34912 CAGCTACTCCCCAAATAAAT 0 3259 502162 34898 34917 ACTCTCAGCTACTCCCCAAA 56 3260 502163 34903 34922 AACCAACTCTCAGCTACTCC 76 3261 502164 34931 34950 CAAACAGATTAAAGTTGCTC 80 3262 502165 35006 35025 AATGTGAGCAAGACTCCCTC 58 3263 502166 35167 35186 TTGGGAGCCTCCTGGCAGAG 0 3264 502167 35192 35211 GCCCAGGTTTTCTGTGACTC 65 3265 502168 35197 35216 CAAGAGCCCAGGTTTTCTGT 53 3266 502169 35220 35239 GTCACTGGCCACCAGCAGAA 44 3267 502170 35225 35244 GCAGAGTCACTGGCCACCAG 59 3268 502171 35230 35249 TGGAAGCAGAGTCACTGGCC 48 3269 502172 35263 35282 TGCAAAGTGCCCCTTCCCTG 67 3270 502173 35268 35287 AGTGCTGCAAAGTGCCCCTT 61 3271 502174 35273 35292 ACCTGAGTGCTGCAAAGTGC 61 3272 502175 35278 35297 CTCCCACCTGAGTGCTGCAA 83 3273 502176 35283 35302 TGACACTCCCACCTGAGTGC 60 3274 502177 35288 35307 ATCAATGACACTCCCACCTG 59 3275 502178 35318 35337 TTTTGGCTGCCCTGCCTCAA 36 3276 502179 35323 35342 GGTCTTTTTGGCTGCCCTGC 82 3277 502180 35345 35364 GATAACAAGGAATGAACACG 8 3278 502181 35350 35369 TCCTGGATAACAAGGAATGA 36 3279 502182 35356 35375 TACAATTCCTGGATAACAAG 0 3280 502183 35361 35380 AGAAATACAATTCCTGGATA 9 3281 502184 35366 35385 CTTCTAGAAATACAATTCCT 37 3282 502185 35371 35390 ACAAACTTCTAGAAATACAA 0 3283 502186 35376 35395 GTGAAACAAACTTCTAGAAA 62 3284 502187 35401 35420 TTTGTCCACATATCTGATTG 40 3285 502188 35406 35425 TTATCTTTGTCCACATATCT 44 3286 502189 35411 35430 ATACCTTATCTTTGTCCACA 81 3287 502190 35416 35435 AATAAATACCTTATCTTTGT 0 3288 502191 35421 35440 GCTTCAATAAATACCTTATC 88 3289 502192 35426 35445 GTAATGCTTCAATAAATACC 67 3290 502193 35431 35450 TAGAAGTAATGCTTCAATAA 0 3291 502194 35436 35455 CCTCTTAGAAGTAATGCTTC 75 3292 502195 35441 35460 ATTTCCCTCTTAGAAGTAAT 30 3293 502196 35446 35465 CCAAAATTTCCCTCTTAGAA 39 3294 502197 35519 35538 TATTCTACAGCACAGCTCTC 24 3295 502198 35560 35579 TCAAAATCCCTCAGTTCCAT 25 3296 502199 35565 35584 CCTTATCAAAATCCCTCAGT

53 3297 502200 35570 35589 TCTGTCCTTATCAAAATCCC 23 3298 502201 35575 35594 AGCTATCTGTCCTTATCAAA 8 3299 502202 35607 35626 ATCATTTTTATATCATTTTC 0 3300 502203 35612 35631 TTGATATCATTTTTATATCA 0 3301 502204 35617 35636 ATGGGTTGATATCATTTTTA 61 3302 502205 35622 35641 GTTATATGGGTTGATATCAT 48 3303 502206 35627 35646 TTGAGGTTATATGGGTTGAT 68 3304 502207 35632 35651 CAAAATTGAGGTTATATGGG 19 3305 502208 35656 35675 ATACATGCTCTTTTTCAAAA 45 3306

TABLE-US-00048 TABLE 48 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID % inhi- NO: 2 NO: 2 bition % inhi- SEQ ISIS Start Stop (RTS2988_ bition ID NO Site Site Sequence MGB) (RTS2367) NO 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 72 77 425 502209 37525 37544 CATCTCCCAATGCAGGCCCA 37 34 3307 502210 37555 37574 GCCCTGACCCACCATGTCTG 0 27 3308 502211 37560 37579 CCTCAGCCCTGACCCACCAT 19 51 3309 502212 37565 37584 CTCCTCCTCAGCCCTGACCC 0 8 3310 502213 37570 37589 CAGCTCTCCTCCTCAGCCCT 20 33 3311 502214 37575 37594 ATGGACAGCTCTCCTCCTCA 31 46 3312 502215 37580 37599 ACCATATGGACAGCTCTCCT 57 68 3313 502216 37695 37714 GTGCCATGGACAACAATCAG 0 0 3314 502217 37700 37719 TTACAGTGCCATGGACAACA 0 23 3315 502218 37705 37724 GACAATTACAGTGCCATGGA 15 7 3316 502219 37740 37759 AATGCATCAGTCTGGTGGGA 53 59 3317 502220 37745 37764 ATCCCAATGCATCAGTCTGG 66 73 3318 502221 37804 37823 TGCATTTGGCAAACATTCCC 36 50 3319 502222 37809 37828 ACTCATGCATTTGGCAAACA 42 0 3320 502223 37814 37833 TGGGAACTCATGCATTTGGC 62 65 3321 502224 37819 37838 TGTTTTGGGAACTCATGCAT 3 25 3322 502225 37824 37843 AAGGTTGTTTTGGGAACTCA 44 42 3323 502226 37829 37848 GATACAAGGTTGTTTTGGGA 29 44 3324 502227 37834 37853 CTAATGATACAAGGTTGTTT 29 41 3325 502228 37839 37858 TGCAGCTAATGATACAAGGT 68 67 3326 502229 37844 37863 TAAAATGCAGCTAATGATAC 21 18 3327 502230 37849 37868 ATCTGTAAAATGCAGCTAAT 13 38 3328 502231 37854 37873 TCCTCATCTGTAAAATGCAG 48 43 3329 502232 37859 37878 CAGTTTCCTCATCTGTAAAA 8 19 3330 502233 37864 37883 AGCCTCAGTTTCCTCATCTG 42 44 3331 502234 37869 37888 TTCTGAGCCTCAGTTTCCTC 34 54 3332 502235 37874 37893 CATTTTTCTGAGCCTCAGTT 46 45 3333 502236 37879 37898 CTATTCATTTTTCTGAGCCT 56 65 3334 502237 37884 37903 GTAAGCTATTCATTTTTCTG 50 53 3335 502238 37889 37908 TCTGTGTAAGCTATTCATTT 44 55 3336 502239 37894 37913 CTGGTTCTGTGTAAGCTATT 55 55 3337 502240 37899 37918 GTTGTCTGGTTCTGTGTAAG 31 38 3338 502241 37904 37923 TTGGAGTTGTCTGGTTCTGT 52 52 3339 502242 37989 38008 TGGATCCACTCCCCATCCCC 0 30 3340 502243 37994 38013 GCCCATGGATCCACTCCCCA 0 0 3341 502244 37999 38018 GGGTTGCCCATGGATCCACT 0 10 3342 502245 38004 38023 AGTCAGGGTTGCCCATGGAT 9 13 3343 502246 38270 38289 CCAAGGAGTCTGCTGCCCTG 15 16 3344 502247 38292 38311 GTCTGGGTCCTGTCTTCAGG 17 15 3345 502248 38326 38345 CAGAATGACTGACTCCCTTC 10 19 3346 502249 38356 38375 GCAGTCTGGGCTCCAACATC 0 5 3347 502250 38361 38380 ACTGTGCAGTCTGGGCTCCA 0 7 3348 502251 38366 38385 CCCACACTGTGCAGTCTGGG 5 10 3349 502252 38371 38390 CTTGGCCCACACTGTGCAGT 4 8 3350 502253 38376 38395 GCAAACTTGGCCCACACTGT 2 0 3351 502254 38381 38400 CATGGGCAAACTTGGCCCAC 21 22 3352 502255 38386 38405 ACACACATGGGCAAACTTGG 0 0 3353 502256 38391 38410

CCCAGACACACATGGGCAAA 0 0 3354 502257 38396 38415
GCCCACCCAGACACACATGG 0 5 3355 502258 38401 38420
TCTGTGCCCACCCAGACACA 0 13 3356 502259 38406 38425
TTGCATCTGTGCCCACCCAG 5 22 3357 502260 38411 38430 ACAGGTTGCATCTGTGCCCA 33 45 3358 502261 38478 38497 GGCCCCAGGTGGTCCAGTCC 0 4 3359 502262 38483 38502
ATTCTGGCCCCAGGTGGTCC 0 10 3360 502263 38505 38524
TGGTCTCCCAGCAAAATGAT 0 0 3361 502264 38510 38529 CCTCCTGGTCTCCCAGCAAA 0 0 3362 502265 38515 38534 CCTGACCTCCTGGTCTCCCA 0 7 3363 502266 38520 38539
TCCTTCCTGACCTCCTGGTC 3 16 3364 502267 38525 38544 CACCCTCCTTCCTGACCTCC 0 3 3365 502268 38595 38614 TTCCAAAGCCTGAGGTAAGG 18 36 3366 502269 38600 38619
TCTTCTTCCAAAGCCTGAGG 14 40 3367 502270 38605 38624
CAGCCTCTTCTTCCAAAGCC 10 9 3368 502271 38630 38649
CTGGCCCAGGGCTGAGCCTG 0 0 3369 502272 38676 38695
TCCCTGAAGGCAGGGACTGG 0 10 3370 502273 38681 38700
GGTGCTCCCTGAAGGCAGGG 24 28 3371 502274 38686 38705
CCCTTGGTGCTCCCTGAAGG 0 15 3372 502275 38726 38745
CACACTGTGTTGTCACTGTC 41 44 3373 502276 38731 38750
CTCAGCACACTGTGTTGTCA 38 48 3374 502277 38736 38755
ACAGTCTCAGCACACTGTGT 6 21 3375 502278 38759 38778
AGCCCTGGACTCACCATTGT 0 4 3376 502279 38764 38783 CTCTCAGCCCTGGACTCACC 6 8 3377 502280 38809 38828 GCCTCTGCCAGGAAGTTCTC 8 6 3378 502281 38814 38833
GGCCTGCCTCTGCCAGGAAG 0 51 3379 502282 38819 38838
TGCAGGGCCTGCCTCTGCCA 9 2 3380 502283 38843 38862 GAATGCCTGTTTTCCACCTC 34 53 3381 502284 38848 38867 CTCTGGAATGCCTGTTTTCC 19 27 3382 502285 38874 38893 AGGCAAAGGGAAGGCTGAGC 4 23 3383
TABLE-US-00049 TABLE 49 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID % inhi- NO: 2 NO: 2 bition % inhi- SEQ ISIS Start Stop (RTS2988_ bition ID NO Site Site Sequence MGB) (RTS2367) NO 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 74 73 425 502286 38879 38898
CCCCCAGGCAAAGGGAAGGC 0 16 3384 502287 38928 38947
GCCTTCTGCCCCACTGCTAG 0 6 3385 502288 38933 38952 GATGGGCCTTCTGCCCCACT 0 0 3386 502289 38938 38957 GTTCTGATGGGCCTTCTGCC 0 15 3387 502290 38943 38962
ACCAGGTTCTGATGGGCCTT 0 42 3388 502291 38948 38967 TCTCTACCAGGTTCTGATGG 0 2 3389 502292 39030 39049 GGCCCTGGACACTGGCCAAG 0 12 3390 502293 39035 39054
CTAGAGGCCCTGGACACTGG 0 16 3391 502294 39041 39060
GTCAGCCTAGAGGCCCTGGA 18 40 3392 502295 39082 39101
CCAGGGTCTGTCATACCCTA 0 0 3393 502296 39087 39106 AGAGGCCAGGGTCTGTCATA 0 0 3394 502297 39097 39116 TCTGGAAGGGAGAGGCCAGG 0 0 3395 502298 39301 39320
ACCAACAGTGGTGATGGGCT 7 24 3396 502299 39306 39325
GGCTTACCAACAGTGGTGAT 0 0 3397 502300 39337 39356 GTTCAGGACAGCCCTTGGTC 25 35 3398 502301 39342 39361 CCTGTGTTCAGGACAGCCCT 26 31 3399 502302 39347 39366 GGCACCCTGTGTTCAGGACA 41 52 3400 502303 39377 39396
AATCCCGTCTCTACTGCTGA 18 39 3401 502304 39401 39420
TCAGAGCCAGGTGGCCTGCA 30 27 3402 502305 39406 39425
GGCCATCAGAGCCAGGTGGC 0 33 3403 502306 39411 39430
GGCATGGCCATCAGAGCCAG 0 12 3404 502307 39416 39435
CTAAGGGCATGGCCATCAGA 0 23 3405 502308 39421 39440
CATGGCTAAGGGCATGGCCA 0 10 3406 502309 39426 39445
GTCCTCATGGCTAAGGGCAT 26 29 3407 502310 39431 39450
TCAAAGTCCTCATGGCTAAG 0 37 3408 502311 39436 39455

ACACTTCAAAGTCCTCATGG 33 52 3409 502312 39441 39460
ACCCAACACTTCAAAGTCCT 44 57 3410 502313 39446 39465
TCAGCACCCAACACTTCAAA 35 47 3411 502314 39514 39533
ATCACAGAGCTTGGTTCATC 61 62 3412 502315 39539 39558
GACTGCAGATTTTCTTTCCT 18 40 3413 502316 39577 39596 GTCTTCCCTGATAGACTAGT 43 46 3414 502317 39614 39633 TCCTCATCACATTCCCCCAC 37 59 3415 502318 39665 39684 AGAGTTACCTCCTCCCTGGG 14 25 3416 502319 39670 39689
GTGCAAGAGTTACCTCCTCC 71 81 3417 502320 39675 39694
TAGCAGTGCAAGAGTTACCT 61 67 3418 502321 39680 39699
TCAGTTAGCAGTGCAAGAGT 38 53 3419 502322 39685 39704
TCCTATCAGTTAGCAGTGCA 73 75 3420 502323 39690 39709 ATAATTCCTATCAGTTAGCA 32 35 3421 502324 39695 39714 GCTAGATAATTCCTATCAGT 25 17 3422 502325 39700 39719 ATTTTGCTAGATAATTCCTA 0 0 3423 502326 39705 39724 CCTCTATTTTGCTAGATAAT 1 26 3424 502327 39727 39746 TTCTGATAAAAATTCTCTTC 10 13 3425 502328 39732 39751 ATTGTTTCTGATAAAAATTC 0 0 3426 502329 39737 39756 AGGCTATTGTTTCTGATAAA 34 31 3427 502330 39813 39832 CTATGCTGCAGTCATATTAA 35 26 3428 502331 39818 39837 CAGGTCTATGCTGCAGTCAT 58 45 3429 502332 39823 39842
TCTGACAGGTCTATGCTGCA 46 50 3430 502333 39828 39847
ACTCTTCTGACAGGTCTATG 47 50 3431 502334 39833 39852
TTTCCACTCTTCTGACAGGT 54 72 3432 502335 39906 39925 ATTCTTTAGAATCCTCTAAA 0 10 3433 502336 39911 39930 AGGTAATTCTTTAGAATCCT 41 62 3434 502337 39965 39984 AAGGGCCCATGTGCTCCTGG 14 33 3435 502338 39970 39989
CTGCCAAGGGCCCATGTGCT 10 30 3436 502339 39975 39994
GTCCACTGCCAAGGGCCCAT 37 51 3437 502340 39980 39999
CTCAAGTCCACTGCCAAGGG 0 9 3438 502341 39985 40004
GGCCCCTCAAGTCCACTGCC 0 14 3439 502342 39990 40009
GCTTTGGCCCCTCAAGTCCA 13 27 3440 502343 40041 40060
TAAAAGTACAAGGCAGGACA 26 25 3441 502344 40063 40082
TCCCTGTTGCTTTGTCTCTA 20 26 3442 502345 40068 40087 CTGCCTCCCTGTTGCTTTGT 24 31 3443 502346 40073 40092 TCCTGCTGCCTCCCTGTTGC 0 21 3444 502347 40078 40097 AATGTTCCTGCTGCCTCCCT 0 0 3445 502348 40083 40102 ATGGAAATGTTCCTGCTGCC 33 33 3446 502349 40088 40107 TGTGCATGGAAATGTTCCTG 39 38 3447 502350 40093 40112 ACACCTGTGCATGGAAATGT 7 24 3448 502351 40098 40117
CAGCCACACCTGTGCATGGA 56 54 3449 502352 40103 40122
CTCCCCAGCCACACCTGTGC 10 21 3450 502353 40108 40127
AGCCCCTCCCCAGCCACACC 2 0 3451 502354 40130 40149 CTTCACATTGCCCACAGGAC 39 41 3452 502355 40135 40154 AATTCCTTCACATTGCCCAC 14 19 3453 502356 40140 40159 GAGCAAATTCCTTCACATTG 0 0 3454 502357 40145 40164
GTGAAGAGCAAATTCCTTCA 0 13 3455 502358 40150 40169
TCAAGGTGAAGAGCAAATTC 0 7 3456 502359 40155 40174
CATTCTCAAGGTGAAGAGCA 32 20 3457 502360 40160 40179
CTCTCCATTCTCAAGGTGAA 0 0 3458 502361 40182 40201 CCTCCCAAACACTCTCTGGT 11 11 3459 502362 40187 40206 CTTCCCCTCCCAAACACTCT 2 15 3460
TABLE-US-00050 TABLE 50 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID NO: 2 NO: 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 507659 26616 26635
AGTCCTGATGATCCCCTACC 44 3384 507660 26617 26636 AAGTCCTGATGATCCCCTAC 54 3385 507661 26618 26637 TAAGTCCTGATGATCCCCTA 53 3386 507662 26619 26638 GTAAGTCCTGATGATCCCCT 59 3387 507663 26621 26640 TGGTAAGTCCTGATGATCCC 73 3388 507664 26622 26641 ATGGTAAGTCCTGATGATCC 49 3389 507665 26623 26642

CATGGTAAGTCCTGATGATC 68 3390 507666 26624 26643 ACATGGTAAGTCCTGATGAT 35 3391 495752 26631 26650 AGCACTGACATGGTAAGTCC 67 3392 495753 26632 26651 CAGCACTGACATGGTAAGTC 75 3393 495754 26633 26652 TCAGCACTGACATGGTAAGT 50 3394 495755 26634 26653 CTCAGCACTGACATGGTAAG 35 3395 507667 26636 26655 TGCTCAGCACTGACATGGTA 60 3396 507668 26637 26656 CTGCTCAGCACTGACATGGT 56 3397 507669 26638 26657 GCTGCTCAGCACTGACATGG 69 3398 507670 26639 26658 AGCTGCTCAGCACTGACATG 54 3399 507671 26712 26731 TGTGAAGTTCTATCCCTTGG 52 3400 507672 26713 26732 CTGTGAAGTTCTATCCCTTG 40 3401 507673 26714 26733 CCTGTGAAGTTCTATCCCTT 28 3402 507674 26715 26734 ACCTGTGAAGTTCTATCCCT 27 3403 495756 26779 26798 CTGCCATTTAATGAGCTTCA 80 3404 495757 26780 26799 TCTGCCATTTAATGAGCTTC 56 3405 495758 26781 26800 CTCTGCCATTTAATGAGCTT 22 3406 495759 26782 26801 ACTCTGCCATTTAATGAGCT 13 3407 507675 26812 26831 GGAATGCACTGAGTTTCTGC 64 3408 507676 26813 26832 GGGAATGCACTGAGTTTCTG 39 3409 507677 26850 26869 TCACTAATTCTGGGCTTCCA 66 3410 507678 26851 26870 TTCACTAATTCTGGGCTTCC 51 3411 507679 26852 26871 GTTCACTAATTCTGGGCTTC 70 3412 507680 26853 26872 AGTTCACTAATTCTGGGCTT 37 3413 507681 26880 26899 AAGGTGAAGACTGGCTGTTT 4 3414 507682 26881 26900 AAAGGTGAAGACTGGCTGTT 43 3415 507683 26882 26901 TAAAGGTGAAGACTGGCTGT 46 3416 507684 26883 26902 CTAAAGGTGAAGACTGGCTG 70 3417 507685 26885 26904 GCCTAAAGGTGAAGACTGGC 42 3418 507686 26886 26905 GGCCTAAAGGTGAAGACTGG 18 3419 507687 26887 26906 GGGCCTAAAGGTGAAGACTG 31 3420 507688 26888 26907 TGGGCCTAAAGGTGAAGACT 37 3421 507689 27047 27066 AGTTCTTCCAACCAGTGTTT 66 3422 507690 27048 27067 CAGTTCTTCCAACCAGTGTT 69 3423 507691 27049 27068 TCAGTTCTTCCAACCAGTGT 63 3424 507692 27050 27069 CTCAGTTCTTCCAACCAGTG 83 3425 507693 27052 27071 TGCTCAGTTCTTCCAACCAG 79 3426 507694 27053 27072 TTGCTCAGTTCTTCCAACCA 80 3427 507695 27054 27073 TTTGCTCAGTTCTTCCAACC 78 3428 507696 27055 27074 GTTTGCTCAGTTCTTCCAAC 85 3429 507697 27057 27076 TAGTTTGCTCAGTTCTTCCA 30 3430 507698 27058 27077 TTAGTTTGCTCAGTTCTTCC 55 3431 507699 27059 27078 TTTAGTTTGCTCAGTTCTTC 27 3432 507700 27060 27079 GTTTAGTTTGCTCAGTTCTT 48 3433 507701 27217 27236 GTACTTTCTTCTCATGTGAC 9 3434 507702 27218 27237 AGTACTTTCTTCTCATGTGA 0 3435 507703 27219 27238 CAGTACTTTCTTCTCATGTG 0 3436 507704 27220 27239 CCAGTACTTTCTTCTCATGT 53 3437 507705 27222 27241 TCCCAGTACTTTCTTCTCAT 67 3438 507706 27223 27242 GTCCCAGTACTTTCTTCTCA 57 3439 507707 27224 27243 GGTCCCAGTACTTTCTTCTC 58 3440 507708 27225 27244 TGGTCCCAGTACTTTCTTCT 49 3441 507709 27227 27246 CCTGGTCCCAGTACTTTCTT 65 3442 507710 27228 27247 TCCTGGTCCCAGTACTTTCT 77 3443 507711 27229 27248 GTCCTGGTCCCAGTACTTTC 69 3444 507712 27230 27249 TGTCCTGGTCCCAGTACTTT 64 3445 507713 27232 27251 CTTGTCCTGGTCCCAGTACT 61 3446 507714 27233 27252 TCTTGTCCTGGTCCCAGTAC 61 3447 507715 27234 27253 TTCTTGTCCTGGTCCCAGTA 67 3448 507716 27235 27254 GTTCTTGTCCTGGTCCCAGT 76 3449 507717 27237 27256 GGGTTCTTGTCCTGGTCCCA 80 3450 507718 27238 27257 TGGGTTCTTGTCCTGGTCCC 71 3451 507719 27239 27258 CTGGGTTCTTGTCCTGGTCC 62 3452 507720 27240 27259 CCTGGGTTCTTGTCCTGGTC 55 3453 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 69 425 507721 35662 35681 AAAAATATACATGCTCTTTT 0 3461 507722 35663 35682 CAAAAATATACATGCTCTTT 29 3462 507723 35664 35683 TCAAAAATATACATGCTCTT 45 3463 507724 35665 35684 CTCAAAAATATACATGCTCT 74 3464 507725 35667 35686 TACTCAAAAATATACATGCT 51 3465 507726 35668 35687 CTACTCAAAAATATACATGC 63 3466 507727 35669 35688 TCTACTCAAAAATATACATG 0 3467

TABLE-US-00051 TABLE 51 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID NO: 2 NO: 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 522363 12807 12826 CCAGCACTAATTTTTACAAC 72 3468 522364 12808 12827 TCCAGCACTAATTTTTACAA 49 3469 522365 12809 12828 ATCCAGCACTAATTTTTACA 65 3470 522366 13442 13461 GTGGCTCTCTAGATAGGGTG 80 3471 522367 14193 14212 CTCAGCCTTGGAGCCAGCTC 59 3472 522368 14194 14213 ACTCAGCCTTGGAGCCAGCT 72 3473 522369 14195 14214 CACTCAGCCTTGGAGCCAGC 63 3474 522370 14196 14215 CCACTCAGCCTTGGAGCCAG 71 3475 522371 14198 14217 TGCCACTCAGCCTTGGAGCC 67 3476 522372 14199 14218 TTGCCACTCAGCCTTGGAGC 62 3477 522373 15319 15338 GGAGTTTGTGTTTCCCATTT 83 3478 522374 15321 15340 AAGGAGTTTGTGTTTCCCAT 83 3479 522375 15322 15341 AAAGGAGTTTGTGTTTCCCA 83 3480 522376 15324 15343 GAAAAGGAGTTTGTGTTTCC 50 3481 522377 15325 15344 AGAAAAGGAGTTTGTGTTTC 17 3482 522378 15326 15345 AAGAAAAGGAGTTTGTGTTT 30 3483 522379 15327 15346 TAAGAAAAGGAGTTTGTGTT 21 3484 522380 15569 15588 ATTTTCACAAGGAGAAGCTG 40 3485 522381 15570 15589 CATTTTCACAAGGAGAAGCT 44 3486 522382 15571 15590 CCATTTTCACAAGGAGAAGC 32 3487 522383 15574 15593 GCACCATTTTCACAAGGAGA 84 3488 522384 15575 15594 TGCACCATTTTCACAAGGAG 77 3489 522385 15576 15595 GTGCACCATTTTCACAAGGA 72 3490 522386 15577 15596 AGTGCACCATTTTCACAAGG 65 3491 522387 15647 15666 CATCACAATATATGGGCAAG 44 3492 522388 15648 15667 CCATCACAATATATGGGCAA 51 3493 522389 15649 15668 CCCATCACAATATATGGGCA 60 3494 522390 15650 15669 CCCCATCACAATATATGGGC 55 3495 522391 15877 15896 AGGTAAGGCCAAGGAAGTGA 69 3496 522392 15878 15897 GAGGTAAGGCCAAGGAAGTG 50 3497 522393 15879 15898 AGAGGTAAGGCCAAGGAAGT 43 3498 522394 15880 15899 AAGAGGTAAGGCCAAGGAAG 57 3499 522395 15934 15953 CACCTGGGTTTTTGTGTATT 66 3500 522396 15935 15954 ACACCTGGGTTTTTGTGTAT 61 3501 522397 15936 15955 TACACCTGGGTTTTTGTGTA 37 3502 522398 15937 15956 ATACACCTGGGTTTTTGTGT 69 3503 522399 16047 16066 TCTGAGTGCTGCTAACTTGG 77 3504 522400 16048 16067 ATCTGAGTGCTGCTAACTTG 65 3505 522401 16049 16068 AATCTGAGTGCTGCTAACTT 62 3506 522402 16050 16069 AAATCTGAGTGCTGCTAACT 60 3507 522403 16052 16071 TGAAATCTGAGTGCTGCTAA 68 3508 522404 16053 16072 TTGAAATCTGAGTGCTGCTA 74 3509 522405 16054 16073 GTTGAAATCTGAGTGCTGCT 78 3510 522406 16083 16102 GCACTAAATTAGGAAGCTGG 77 3511 522407 16084 16103 AGCACTAAATTAGGAAGCTG 65 3512 522408 16289 16308 CTGGGACATCTGGCCCAAAG 72 3513 522409 16290 16309 ACTGGGACATCTGGCCCAAA 71 3514 522410 16291 16310 CACTGGGACATCTGGCCCAA 66 3515 522411 16292 16311 CCACTGGGACATCTGGCCCA 71 3516 522412 16294 16313 GCCCACTGGGACATCTGGCC 61 3517 522413 16295 16314 GGCCCACTGGGACATCTGGC 68 3518 522414 16306 16325 CCTTAAAGCAGGGCCCACTG 50 3519 522415 16307 16326 TCCTTAAAGCAGGGCCCACT 58 3520 522416 16308 16327 ATCCTTAAAGCAGGGCCCAC 67 3521 522417 16309 16328 TATCCTTAAAGCAGGGCCCA 63 3522 522418 16429 16448 CACTGGTGCCTAAGAGCCCT 70 3523 522419 16430 16449 CCACTGGTGCCTAAGAGCCC 63 3524 522420 16431 16450 TCCACTGGTGCCTAAGAGCC 61 3525 522421 16432 16451 CTCCACTGGTGCCTAAGAGC 73 3526 522422 16809 16828 GAGGCTGAGTTCTGCTCCTG 77 3527 522423 16810 16829 AGAGGCTGAGTTCTGCTCCT 59 3528 522424 16811 16830 TAGAGGCTGAGTTCTGCTCC 71 3529 522425 16812 16831 CTAGAGGCTGAGTTCTGCTC 60 3530 522426 16814 16833 TGCTAGAGGCTGAGTTCTGC 65 3531 522427 16815 16834 CTGCTAGAGGCTGAGTTCTG 71 3532 522428 16816 16835 TCTGCTAGAGGCTGAGTTCT 74 3533 522429 16817 16836 ATCTGCTAGAGGCTGAGTTC 67 3534 522430 16824 16843 CTTGGGCATCTGCTAGAGGC 69 3535 522431 16825 16844 TCTTGGGCATCTGCTAGAGG 66 3536 522432 16826 16845 TTCTTGGGCATCTGCTAGAG 65 3537 522433 16827 16846

CTTCTTGGGCATCTGCTAGA 62 3538 522434 16829 16848 TGCTTCTTGGGCATCTGCTA 71 3539 522435 16830 16849 CTGCTTCTTGGGCATCTGCT 81 3540 522436 16831 16850 TCTGCTTCTTGGGCATCTGC 80 3541 522437 16832 16851 CTCTGCTTCTTGGGCATCTG 82 3542 522438 16834 16853 TCCTCTGCTTCTTGGGCATC 67 3543 522439 16835 16854 CTCCTCTGCTTCTTGGGCAT 68 3544 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 82 425

TABLE-US-00052 TABLE 52 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID NO: 2 NO: 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 522440 16836 16855 TCTCCTCTGCTTCTTGGGCA 78 3545 522441 16897 16916 CTTTCTAAGTACCCTTTACA 37 3546 522442 16898 16917 GCTTTCTAAGTACCCTTTAC 71 3547 522443 16899 16918 TGCTTTCTAAGTACCCTTTA 62 3548 522444 16900 16919 GTGCTTTCTAAGTACCCTTT 85 3549 522445 16902 16921 CAGTGCTTTCTAAGTACCCT 86 3550 522446 17008 17027 AGGGAAAGGTCTTCTCCAGC 74 3551 522447 17009 17028 GAGGGAAAGGTCTTCTCCAG 67 3552 522448 17010 17029 GGAGGGAAAGGTCTTCTCCA 62 3553 522449 17011 17030 AGGAGGGAAAGGTCTTCTCC 60 3554 522450 17208 17227 GTCTCATTCATCTGGGCCCT 80 3555 522451 17209 17228 TGTCTCATTCATCTGGGCCC 65 3556 522452 17210 17229 ATGTCTCATTCATCTGGGCC 72 3557 522453 17270 17289 CCCCTGTACTCAGCCCTTTC 49 3558 522454 17271 17290 ACCCCTGTACTCAGCCCTTT 60 3559 522455 17272 17291 CACCCCTGTACTCAGCCCTT 51 3560 522456 17273 17292 TCACCCCTGTACTCAGCCCT 61 3561 522457 17275 17294 TCTCACCCCTGTACTCAGCC 70 3562 522458 17276 17295 ATCTCACCCCTGTACTCAGC 58 3563 522459 17277 17296 CATCTCACCCCTGTACTCAG 51 3564 522460 17278 17297 CCATCTCACCCCTGTACTCA 54 3565 522461 17329 17348 AAGTCTTCCTTCTCCTCCAC 65 3566 522462 17330 17349 AAAGTCTTCCTTCTCCTCCA 71 3567 522463 17331 17350 GAAAGTCTTCCTTCTCCTCC 75 3568 522464 17332 17351 GGAAAGTCTTCCTTCTCCTC 75 3569 522465 17451 17470 CAGGTAGAGTTCTATTGCTT 80 3570 522466 17452 17471 CCAGGTAGAGTTCTATTGCT 77 3571 522467 19361 19380 GGCTACTGGCATTGCTCCTC 74 3572 522468 19362 19381 TGGCTACTGGCATTGCTCCT 56 3573 522469 19363 19382 GTGGCTACTGGCATTGCTCC 68 3574 522470 19364 19383 GGTGGCTACTGGCATTGCTC 74 3575 522471 19909 19928 ACATGCCTGAGGGCAGCAGT 68 3576 522472 19910 19929 CACATGCCTGAGGGCAGCAG 69 3577 522473 19911 19930 TCACATGCCTGAGGGCAGCA 81 3578 522474 19912 19931 CTCACATGCCTGAGGGCAGC 79 3579 522475 19984 20003 AATATCTGATATCAAAGTGA 14 3580 522476 19985 20004 CAATATCTGATATCAAAGTG 14 3581 522477 19986 20005 CCAATATCTGATATCAAAGT 41 3582 522478 19987 20006 CCCAATATCTGATATCAAAG 74 3583 522479 19989 20008 TGCCCAATATCTGATATCAA 73 3584 522480 19990 20009 TTGCCCAATATCTGATATCA 65 3585 522481 20297 20316 TTGCTTGAGGTCAGGGTGTG 62 3586 522482 20298 20317 CTTGCTTGAGGTCAGGGTGT 72 3587 522483 20299 20318 ACTTGCTTGAGGTCAGGGTG 68 3588 522484 20300 20319 CACTTGCTTGAGGTCAGGGT 84 3589 522485 20302 20321 ATCACTTGCTTGAGGTCAGG 82 3590 522486 20303 20322 AATCACTTGCTTGAGGTCAG 77 3591 522487 20304 20323 AAATCACTTGCTTGAGGTCA 75 3592 522488 20305 20324 AAAATCACTTGCTTGAGGTC 71 3593 522489 20669 20688 GGTACAAGTTTCTCCACCTA 75 3594 522490 20670 20689 AGGTACAAGTTTCTCCACCT 74 3595 522491 20671 20690 TAGGTACAAGTTTCTCCACC 47 3596 522492 20672 20691 CTAGGTACAAGTTTCTCCAC 72 3597 522493 20828 20847 ATAGCTGAGCAAATCCATAA 63 3598 522494 20829 20848 CATAGCTGAGCAAATCCATA 73 3599 522495 20830 20849 GCATAGCTGAGCAAATCCAT 80 3600 522496 20831 20850 TGCATAGCTGAGCAAATCCA 72 3601 522497 20833 20852 ACTGCATAGCTGAGCAAATC 63 3602 522498 20834 20853 TACTGCATAGCTGAGCAAAT 67 3603 522499 20835 20854 TTACTGCATAGCTGAGCAAA 65 3604 522500 20836 20855 TTTACTGCATAGCTGAGCAA

65 3605 522501 20838 20857 GGTTTACTGCATAGCTGAGC 85 3606 522502 20839 20858 AGGTTTACTGCATAGCTGAG 65 3607 522503 20840 20859 GAGGTTTACTGCATAGCTGA 72 3608 522504 20841 20860 AGAGGTTTACTGCATAGCTG 75 3609 522505 20843 20862 ATAGAGGTTTACTGCATAGC 71 3610 522506 20844 20863 CATAGAGGTTTACTGCATAG 65 3611 522507 20956 20975 ACTCTCTCTGCACACTCCCA 62 3612 522508 20957 20976 CACTCTCTCTGCACACTCCC 61 3613 522509 20958 20977 GCACTCTCTCTGCACACTCC 80 3614 522510 20959 20978 TGCACTCTCTCTGCACACTC 76 3615 522511 21934 21953 TGCTGATTACAAGCTGATTC 64 3616 522512 21935 21954 CTGCTGATTACAAGCTGATT 67 3617 522513 21936 21955 ACTGCTGATTACAAGCTGAT 61 3618 522514 21937 21956 AACTGCTGATTACAAGCTGA 55 3619 522515 22152 22171 TGGTGATAAGGCAGCAGTCT 53 3620 522516 22153 22172 ATGGTGATAAGGCAGCAGTC 62 3621 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 81 425

TABLE-US-00053 TABLE 53 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID NO: 2 NO: 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 522517 22154 22173 GATGGTGATAAGGCAGCAGT 67 3622 522518 22155 22174 AGATGGTGATAAGGCAGCAG 71 3623 522519 22157 22176 AGAGATGGTGATAAGGCAGC 58 3624 522520 22158 22177 CAGAGATGGTGATAAGGCAG 38 3625 522521 22159 22178 CCAGAGATGGTGATAAGGCA 60 3626 522522 22160 22179 CCCAGAGATGGTGATAAGGC 67 3627 522523 23856 23875 AGACTTGCCCAGCAACCCAT 49 3628 522524 23857 23876 CAGACTTGCCCAGCAACCCA 58 3629 522525 23858 23877 CCAGACTTGCCCAGCAACCC 61 3630 522526 23859 23878 TCCAGACTTGCCCAGCAACC 57 3631 522527 24120 24139 AAACCTAGGTTCAGACTTTG 51 3632 522528 24121 24140 CAAACCTAGGTTCAGACTTT 48 3633 522529 24123 24142 GTCAAACCTAGGTTCAGACT 84 3634 522530 24125 24144 GAGTCAAACCTAGGTTCAGA 66 3635 522531 24126 24145 AGAGTCAAACCTAGGTTCAG 48 3636 522532 24127 24146 AAGAGTCAAACCTAGGTTCA 50 3637 522533 24128 24147 GAAGAGTCAAACCTAGGTTC 61 3638 522534 25019 25038 GAGCATCCCCAATCCCTGCT 72 3639 522535 25020 25039 TGAGCATCCCCAATCCCTGC 63 3640 522536 25021 25040 TTGAGCATCCCCAATCCCTG 56 3641 522537 25022 25041 TTTGAGCATCCCCAATCCCT 48 3642 522538 25024 25043 ACTTTGAGCATCCCCAATCC 42 3643 522539 25025 25044 TACTTTGAGCATCCCCAATC 53 3644 522540 25026 25045 GTACTTTGAGCATCCCCAAT 70 3645 522541 25027 25046 TGTACTTTGAGCATCCCCAA 68 3646 522542 25029 25048 AGTGTACTTTGAGCATCCCC 77 3647 522543 25030 25049 AAGTGTACTTTGAGCATCCC 70 3648 522544 25031 25050 CAAGTGTACTTTGAGCATCC 73 3649 522545 25032 25051 CCAAGTGTACTTTGAGCATC 76 3650 522546 25034 25053 CTCCAAGTGTACTTTGAGCA 82 3651 522547 25070 25089 CAGAGGTTCAACATCCAATT 74 3652 522548 25071 25090 ACAGAGGTTCAACATCCAAT 74 3653 522549 25072 25091 GACAGAGGTTCAACATCCAA 73 3654 522550 25073 25092 GGACAGAGGTTCAACATCCA 82 3655 522551 25076 25095 CAAGGACAGAGGTTCAACAT 53 3656 522552 25077 25096 CCAAGGACAGAGGTTCAACA 72 3657 522553 25078 25097 GCCAAGGACAGAGGTTCAAC 84 3658 522554 25079 25098 GGCCAAGGACAGAGGTTCAA 82 3659 522555 25081 25100 GAGGCCAAGGACAGAGGTTC 80 3660 522556 25082 25101 TGAGGCCAAGGACAGAGGTT 82 3661 522557 25083 25102 GTGAGGCCAAGGACAGAGGT 67 3662 522558 25084 25103 TGTGAGGCCAAGGACAGAGG 63 3663 522559 25086 25105 TCTGTGAGGCCAAGGACAGA 54 3664 522560 25087 25106 GTCTGTGAGGCCAAGGACAG 58 3665 522561 25088 25107 TGTCTGTGAGGCCAAGGACA 71 3666 522562 25089 25108 CTGTCTGTGAGGCCAAGGAC 74 3667 522563 25184 25203 AGTCTCTCATAATTGCCAGT 69 3668 522564 25185 25204 AAGTCTCTCATAATTGCCAG 69 3669 522565 25186 25205

GAAGTCTCTCATAATTGCCA 71 3670 522566 25187 25206 GGAAGTCTCTCATAATTGCC 46 3671 522567 25189 25208 TGGGAAGTCTCTCATAATTG 37 3672 522568 25190 25209 TTGGGAAGTCTCTCATAATT 22 3673 522569 25191 25210 CTTGGGAAGTCTCTCATAAT 23 3674 522570 25192 25211 CCTTGGGAAGTCTCTCATAA 35 3675 522571 25194 25213 GGCCTTGGGAAGTCTCTCAT 51 3676 522572 25195 25214 AGGCCTTGGGAAGTCTCTCA 55 3677 522573 25196 25215 TAGGCCTTGGGAAGTCTCTC 61 3678 522574 25197 25216 CTAGGCCTTGGGAAGTCTCT 43 3679 522575 25242 25261 CATTTTGGAGACCACATGAA 31 3680 522576 25243 25262 TCATTTTGGAGACCACATGA 41 3681 522577 25244 25263 GTCATTTTGGAGACCACATG 60 3682 522578 25245 25264 GGTCATTTTGGAGACCACAT 70 3683 522579 25268 25287 AATGGGAATGGTATCGCACT 84 3684 522580 25269 25288 CAATGGGAATGGTATCGCAC 87 3685 522581 25270 25289 ACAATGGGAATGGTATCGCA 85 3686 522582 25271 25290 CACAATGGGAATGGTATCGC 86 3687 522583 25377 25396 AGCTATAGAATCAGACAGAC 53 3688 522584 25378 25397 CAGCTATAGAATCAGACAGA 56 3689 522585 25379 25398 TCAGCTATAGAATCAGACAG 48 3690 522586 25381 25400 CATCAGCTATAGAATCAGAC 64 3691 522587 25783 25802 GCACTTGATCTGGGACCCAA 87 3692 522588 25784 25803 AGCACTTGATCTGGGACCCA 88 3693 522589 25785 25804 GAGCACTTGATCTGGGACCC 86 3694 522590 25786 25805 AGAGCACTTGATCTGGGACC 76 3695 522591 25788 25807 GTAGAGCACTTGATCTGGGA 74 3696 522592 25789 25808 GGTAGAGCACTTGATCTGGG 79 3697 522593 25790 25809 GGGTAGAGCACTTGATCTGG 61 3698 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 84 425

TABLE-US-00054 TABLE 54 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID NO: 2 NO: 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 522594 25842 25861 AACTATCACACCACCTCCCA 44 3699 522595 25843 25862 GAACTATCACACCACCTCCC 42 3700 522596 25844 25863 GGAACTATCACACCACCTCC 47 3701 522597 25845 25864 GGGAACTATCACACCACCTC 61 3702 522598 25847 25866 ATGGGAACTATCACACCACC 64 3703 522599 25848 25867 AATGGGAACTATCACACCAC 33 3704 522600 25849 25868 AAATGGGAACTATCACACCA 37 3705 522601 25850 25869 TAAATGGGAACTATCACACC 16 3706 522602 25901 25920 TGACCTTGGGCAAGTTACCT 74 3707 522603 25902 25921 GTGACCTTGGGCAAGTTACC 76 3708 522604 25903 25922 TGTGACCTTGGGCAAGTTAC 77 3709 522605 25904 25923 GTGTGACCTTGGGCAAGTTA 78 3710 522606 25906 25925 CTGTGTGACCTTGGGCAAGT 80 3711 522607 25907 25926 TCTGTGTGACCTTGGGCAAG 81 3712 522608 25908 25927 ATCTGTGTGACCTTGGGCAA 81 3713 522609 25909 25928 AATCTGTGTGACCTTGGGCA 87 3714 522610 25911 25930 CAAATCTGTGTGACCTTGGG 85 3715 522611 25912 25931 TCAAATCTGTGTGACCTTGG 69 3716 522612 25913 25932 TTCAAATCTGTGTGACCTTG 75 3717 522613 25914 25933 ATTCAAATCTGTGTGACCTT 79 3718 522614 25916 25935 GGATTCAAATCTGTGTGACC 75 3719 522615 25917 25936 GGGATTCAAATCTGTGTGAC 54 3720 522616 25918 25937 AGGGATTCAAATCTGTGTGA 57 3721 522617 25919 25938 CAGGGATTCAAATCTGTGTG 63 3722 522618 25951 25970 GGGAAAGGCACAGGCTTTGG 81 3723 522619 25952 25971 TGGGAAAGGCACAGGCTTTG 70 3724 522620 25954 25973 TGTGGGAAAGGCACAGGCTT 66 3725 522621 26006 26025 AGGCAAAGTAACATACCTGC 87 3726 522622 26007 26026 AAGGCAAAGTAACATACCTG 76 3727 522623 26008 26027 TAAGGCAAAGTAACATACCT 62 3728 522624 26009 26028 TTAAGGCAAAGTAACATACC 69 3729 522625 26011 26030 CCTTAAGGCAAAGTAACATA 73 3730 522626 26012 26031 ACCTTAAGGCAAAGTAACAT 45 3731 522627 26166 26185 ATTTCTTATCTTCAATCCTC 84 3732 522628 26167 26186 CATTTCTTATCTTCAATCCT 64 3733 522629 26168 26187 TCATTTCTTATCTTCAATCC 62 3734 522630 26169 26188 CTCATTTCTTATCTTCAATC 81 3735 522631 26171 26190 CTCTCATTTCTTATCTTCAA 86 3736 522632 26172 26191 GCTCTCATTTCTTATCTTCA 90 3737 522633 26173 26192 TGCTCTCATTTCTTATCTTC 78

3738 522634 26174 26193 GTGCTCTCATTTCTTATCTT 80 3739 522635 26176 26195 AGGTGCTCTCATTTCTTATC 78 3740 522636 26177 26196 CAGGTGCTCTCATTTCTTAT 73 3741 522637 26178 26197 CCAGGTGCTCTCATTTCTTA 74 3742 522638 26179 26198 GCCAGGTGCTCTCATTTCTT 84 3743 522639 26201 26220 GCCCATCATGACCAGGCCCT 60 3744 522640 26202 26221 GGCCCATCATGACCAGGCCC 21 3745 522641 26203 26222 GGGCCCATCATGACCAGGCC 41 3746 522642 26301 26320 CAGGCCTCGGGAAGCACCTG 71 3747 522643 26302 26321 GCAGGCCTCGGGAAGCACCT 86 3748 522644 26303 26322 AGCAGGCCTCGGGAAGCACC 77 3749 522645 26304 26323 GAGCAGGCCTCGGGAAGCAC 80 3750 522646 26354 26373 CAGAAAAAGCAACCAAAGCA 31 3751 522647 26355 26374 GCAGAAAAAGCAACCAAAGC 31 3752 522648 26356 26375 TGCAGAAAAAGCAACCAAAG 34 3753 522649 26357 26376 CTGCAGAAAAAGCAACCAAA 52 3754 522650 26359 26378 ACCTGCAGAAAAAGCAACCA 32 3755 522651 26360 26379 GACCTGCAGAAAAAGCAACC 24 3756 522652 26361 26380 AGACCTGCAGAAAAAGCAAC 17 3757 522653 26362 26381 CAGACCTGCAGAAAAAGCAA 23 3758 522654 26386 26405 ACCTAAGGAGTGAGGGTATC 46 3759 522655 26387 26406 AACCTAAGGAGTGAGGGTAT 58 3760 522656 26388 26407 CAACCTAAGGAGTGAGGGTA 59 3761 522657 26389 26408 GCAACCTAAGGAGTGAGGGT 73 3762 522658 27639 27658 TGCTTGCCAGGTGAGCTTCT 76 3763 522659 27640 27659 TTGCTTGCCAGGTGAGCTTC 65 3764 522660 27641 27660 TTTGCTTGCCAGGTGAGCTT 65 3765 522661 27642 27661 CTTTGCTTGCCAGGTGAGCT 77 3766 522662 27644 27663 GTCTTTGCTTGCCAGGTGAG 74 3767 522663 27645 27664 GGTCTTTGCTTGCCAGGTGA 75 3768 522664 27767 27786 CCTTGTATTAAATACTCTAA 59 3769 522665 27768 27787 ACCTTGTATTAAATACTCTA 57 3770 522666 27769 27788 CACCTTGTATTAAATACTCT 78 3771 522667 27770 27789 GCACCTTGTATTAAATACTC 83 3772 522668 27772 27791 ATGCACCTTGTATTAAATAC 69 3773 522669 27773 27792 AATGCACCTTGTATTAAATA 41 3774 522670 27774 27793 CAATGCACCTTGTATTAAAT 59 3775 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 83 425

TABLE-US-00055 TABLE 55 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ ID SEQ ID NO: 2 NO: 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 522671 27775 27794 CCAATGCACCTTGTATTAAA 83 3776 522672 27777 27796 ATCCAATGCACCTTGTATTA 85 3777 522673 27778 27797 AATCCAATGCACCTTGTATT 77 3778 522674 27779 27798 AAATCCAATGCACCTTGTAT 80 3779 522675 27780 27799 GAAATCCAATGCACCTTGTA 85 3780 522676 27782 27801 TTGAAATCCAATGCACCTTG 86 3781 522677 27783 27802 TTTGAAATCCAATGCACCTT 87 3782 522678 27784 27803 CTTTGAAATCCAATGCACCT 82 3783 522679 27785 27804 CCTTTGAAATCCAATGCACC 88 3784 522680 27787 27806 TTCCTTTGAAATCCAATGCA 82 3785 522681 27788 27807 TTTCCTTTGAAATCCAATGC 82 3786 522682 27789 27808 GTTTCCTTTGAAATCCAATG 87 3787 522683 27790 27809 AGTTTCCTTTGAAATCCAAT 82 3788 522684 27850 27869 CTGTTACTACATGTCTAATC 61 3789 522685 27851 27870 CCTGTTACTACATGTCTAAT 51 3790 522686 27852 27871 ACCTGTTACTACATGTCTAA 68 3791 522687 27853 27872 CACCTGTTACTACATGTCTA 82 3792 522688 27855 27874 GGCACCTGTTACTACATGTC 90 3793 522689 27856 27875 AGGCACCTGTTACTACATGT 88 3794 522690 27857 27876 AAGGCACCTGTTACTACATG 82 3795 522691 27858 27877 AAAGGCACCTGTTACTACAT 80 3796 522692 28027 28046 ATCACAGAATTATCAGCAGT 84 3797 522693 28028 28047 AATCACAGAATTATCAGCAG 79 3798 522694 28029 28048 AAATCACAGAATTATCAGCA 86 3799 522695 28030 28049 TAAATCACAGAATTATCAGC 64 3800 522696 28128 28147 CACACAATATCCATGGACTT 80 3801 522697 28129 28148 ACACACAATATCCATGGACT 88 3802 522698 28130 28149

GACACACAATATCCATGGAC 89 3803 522699 28131 28150 TGACACACAATATCCATGGA 75 3804 522700 28133 28152 CCTGACACACAATATCCATG 80 3805 522701 28134 28153 GCCTGACACACAATATCCAT 79 3806 522702 28135 28154 AGCCTGACACACAATATCCA 72 3807 522703 28460 28479 CATGAGAAGCTTAGAAGGCC 59 3808 522704 28462 28481 CTCATGAGAAGCTTAGAAGG 63 3809 522705 28463 28482 GCTCATGAGAAGCTTAGAAG 83 3810 522706 28465 28484 GAGCTCATGAGAAGCTTAGA 70 3811 522707 28466 28485 TGAGCTCATGAGAAGCTTAG 60 3812 522708 28467 28486 CTGAGCTCATGAGAAGCTTA 69 3813 522709 28468 28487 GCTGAGCTCATGAGAAGCTT 80 3814 522710 28470 28489 TTGCTGAGCTCATGAGAAGC 74 3815 522711 28471 28490 GTTGCTGAGCTCATGAGAAG 66 3816 522712 28472 28491 TGTTGCTGAGCTCATGAGAA 60 3817 522713 28473 28492 CTGTTGCTGAGCTCATGAGA 65 3818 522714 28475 28494 CACTGTTGCTGAGCTCATGA 73 3819 522715 28476 28495 CCACTGTTGCTGAGCTCATG 86 3820 522716 28477 28496 ACCACTGTTGCTGAGCTCAT 86 3821 522717 28478 28497 AACCACTGTTGCTGAGCTCA 87 3822 522718 28480 28499 AAAACCACTGTTGCTGAGCT 80 3823 522719 28481 28500 AAAAACCACTGTTGCTGAGC 83 3824 522720 28482 28501 TAAAAACCACTGTTGCTGAG 69 3825 522721 28483 28502 GTAAAAACCACTGTTGCTGA 69 3826 522722 28516 28535 TCTGGTTCCATGCTCTTAGG 70 3827 522723 28517 28536 CTCTGGTTCCATGCTCTTAG 74 3828 522724 28518 28537 GCTCTGGTTCCATGCTCTTA 78 3829 522725 28519 28538 GGCTCTGGTTCCATGCTCTT 75 3830 522726 28521 28540 CAGGCTCTGGTTCCATGCTC 73 3831 522727 28522 28541 ACAGGCTCTGGTTCCATGCT 78 3832 522728 28523 28542 AACAGGCTCTGGTTCCATGC 78 3833 522729 28581 28600 TGAACTAAATCCACCAAATG 22 3834 522730 28582 28601 ATGAACTAAATCCACCAAAT 46 3835 522731 28583 28602 GATGAACTAAATCCACCAAA 46 3836 522732 28584 28603 GGATGAACTAAATCCACCAA 67 3837 522733 28586 28605 AAGGATGAACTAAATCCACC 65 3838 522734 28587 28606 AAAGGATGAACTAAATCCAC 43 3839 522735 28588 28607 AAAAGGATGAACTAAATCCA 33 3840 522736 28589 28608 CAAAAGGATGAACTAAATCC 16 3841 522737 28803 28822 AGAGACTGAATTCTAGCCTG 59 3842 522738 28804 28823 CAGAGACTGAATTCTAGCCT 67 3843 522739 28805 28824 CCAGAGACTGAATTCTAGCC 67 3844 522740 28806 28825 TCCAGAGACTGAATTCTAGC 77 3845 522741 28808 28827 TATCCAGAGACTGAATTCTA 52 3846 522742 28809 28828 GTATCCAGAGACTGAATTCT 46 3847 522743 28810 28829 GGTATCCAGAGACTGAATTC 50 3848 522744 28811 28830 GGGTATCCAGAGACTGAATT 65 3849 522745 29011 29030 GTTTAGGCTATGGTCTGAGC 90 3850 522746 29012 29031 GGTTTAGGCTATGGTCTGAG 78 3851 522747 29013 29032 AGGTTTAGGCTATGGTCTGA 71 3852 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 84 425

TABLE-US-00056 TABLE 56 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ SEQ ID ID NO: NO: 2 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 522748 29014 29033 GAGGTTTAGGCTATGGTCTG 71 3853 522749 29016 29035 ATGAGGTTTAGGCTATGGTC 67 3854 522750 29045 29064 GGATGCTCCAGGTGGGCCAG 59 3855 522751 29046 29065 TGGATGCTCCAGGTGGGCCA 52 3856 522752 29047 29066 GTGGATGCTCCAGGTGGGCC 57 3857 522753 29048 29067 GGTGGATGCTCCAGGTGGGC 52 3858 522754 29140 29159 GTCTGCCTGATTCCCTTTCT 75 3859 522755 29141 29160 AGTCTGCCTGATTCCCTTTC 58 3860 522756 29142 29161 CAGTCTGCCTGATTCCCTTT 59 3861 522757 29143 29162 GCAGTCTGCCTGATTCCCTT 85 3862 522758 29145 29164 CAGCAGTCTGCCTGATTCCC 79 3863 522759 29146 29165 TCAGCAGTCTGCCTGATTCC 72 3864 522760 29147 29166 TTCAGCAGTCTGCCTGATTC 61 3865 522761 29148 29167 GTTCAGCAGTCTGCCTGATT 71 3866 522762 29150 29169 CTGTTCAGCAGTCTGCCTGA 55 3867 522763 29151 29170 ACTGTTCAGCAGTCTGCCTG 62 3868 522764 29152 29171 TACTGTTCAGCAGTCTGCCT 64 3869 522765 29153 29172 TTACTGTTCAGCAGTCTGCC 61 3870 522766 29155 29174 ACTTACTGTTCAGCAGTCTG 73 3871 522767 29156 29175 TACTTACTGTTCAGCAGTCT 75 3872 522768 29157 29176

ATACTTACTGTTCAGCAGTC 74 3873 522769 29158 29177 CATACTTACTGTTCAGCAGT 74 3874 522770 29160 29179 GTCATACTTACTGTTCAGCA 89 3875 522771 29161 29180 AGTCATACTTACTGTTCAGC 74 3876 522772 29162 29181 AAGTCATACTTACTGTTCAG 39 3877 522773 29163 29182 AAAGTCATACTTACTGTTCA 43 3878 522774 29194 29213 TGGTGAATAGCTATGTCTAA 54 3879 522775 29195 29214 TTGGTGAATAGCTATGTCTA 58 3880 522776 29196 29215 CTTGGTGAATAGCTATGTCT 71 3881 522777 29197 29216 GCTTGGTGAATAGCTATGTC 77 3882 522778 29199 29218 TAGCTTGGTGAATAGCTATG 70 3883 522779 29200 29219 GTAGCTTGGTGAATAGCTAT 50 3884 522780 29201 29220 GGTAGCTTGGTGAATAGCTA 71 3885 522781 29243 29262 AGCCTACAAGAGCCTGTTAA 39 3886 522782 29244 29263 CAGCCTACAAGAGCCTGTTA 67 3887 522783 29245 29264 GCAGCCTACAAGAGCCTGTT 82 3888 522784 29246 29265 TGCAGCCTACAAGAGCCTGT 82 3889 522785 29248 29267 TGTGCAGCCTACAAGAGCCT 64 3890 522786 29249 29268 ATGTGCAGCCTACAAGAGCC 59 3891 522787 29250 29269 CATGTGCAGCCTACAAGAGC 39 3892 522788 29251 29270 GCATGTGCAGCCTACAAGAG 64 3893 522789 29253 29272 AAGCATGTGCAGCCTACAAG 70 3894 522790 29254 29273 GAAGCATGTGCAGCCTACAA 66 3895 522791 29255 29274 GGAAGCATGTGCAGCCTACA 78 3896 522792 29256 29275 GGGAAGCATGTGCAGCCTAC 69 3897 522793 29258 29277 TAGGGAAGCATGTGCAGCCT 74 3898 522794 29259 29278 CTAGGGAAGCATGTGCAGCC 72 3899 522795 29260 29279 TCTAGGGAAGCATGTGCAGC 65 3900 522796 29261 29280 TTCTAGGGAAGCATGTGCAG 61 3901 522797 29263 29282 GTTTCTAGGGAAGCATGTGC 72 3902 522798 29264 29283 AGTTTCTAGGGAAGCATGTG 40 3903 522799 29265 29284 AAGTTTCTAGGGAAGCATGT 46 3904 522800 29266 29285 CAAGTTTCTAGGGAAGCATG 42 3905 522801 29362 29381 TACAACTCTCTTGTTGGGAG 70 3906 522802 29363 29382 GTACAACTCTCTTGTTGGGA 73 3907 522803 29364 29383 GGTACAACTCTCTTGTTGGG 63 3908 522804 29365 29384 GGGTACAACTCTCTTGTTGG 57 3909 522805 29367 29386 CAGGGTACAACTCTCTTGTT 64 3910 522806 29368 29387 ACAGGGTACAACTCTCTTGT 57 3911 522807 29369 29388 AACAGGGTACAACTCTCTTG 85 3912 522808 29370 29389 AAACAGGGTACAACTCTCTT 63 3913 522809 29372 29391 AAAAACAGGGTACAACTCTC 62 3914 522810 29373 29392 TAAAAACAGGGTACAACTCT 32 3915 522811 29374 29393 CTAAAAACAGGGTACAACTC 21 3916 522812 29375 29394 GCTAAAAACAGGGTACAACT 31 3917 522813 29476 29495 TGTAGACTCTCCATGAGCCT 65 3918 522814 29477 29496 ATGTAGACTCTCCATGAGCC 45 3919 522815 29478 29497 GATGTAGACTCTCCATGAGC 25 3920 522816 29479 29498 GGATGTAGACTCTCCATGAG 42 3921 522817 30968 30987 TAAGCTAGGCACACCCAGGG 43 3922 522818 30969 30988 CTAAGCTAGGCACACCCAGG 23 3923 522819 30970 30989 ACTAAGCTAGGCACACCCAG 33 3924 522820 30971 30990 CACTAAGCTAGGCACACCCA 48 3925 522821 30973 30992 GGCACTAAGCTAGGCACACC 78 3926 522822 30975 30994 GTGGCACTAAGCTAGGCACA 69 3927 522823 30976 30995 TGTGGCACTAAGCTAGGCAC 66 3928 522824 30978 30997 ACTGTGGCACTAAGCTAGGC 69 3929 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 85 425

TABLE-US-00057 TABLE 57 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ SEQ ID ID NO: NO: 2 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 522825 30979 30998 TACTGTGGCACTAAGCTAGG 53 3930 522826 30980 30999 TTACTGTGGCACTAAGCTAG 36 3931 522827 30981 31000 TTTACTGTGGCACTAAGCTA 40 3932 522828 30983 31002 TGTTTACTGTGGCACTAAGC 59 3933 522829 30984 31003 GTGTTTACTGTGGCACTAAG 68 3934 522830 30985 31004 AGTGTTTACTGTGGCACTAA 67 3935 522831 30986 31005 GAGTGTTTACTGTGGCACTA 72 3936 522832 31770 31789 ATGGATACTCAGAAGAGCAG 34 3937 522833 31771 31790 GATGGATACTCAGAAGAGCA 46 3938 522834 31793 31812 ACCTCAGGGCTGGCCCAAAG 47 3939 522835 31794 31813 GACCTCAGGGCTGGCCCAAA 67 3940 522836 31795 31814

GGACCTCAGGGCTGGCCCAA 66 3941 522837 31796 31815
AGGACCTCAGGGCTGGCCCA 68 3942 522838 31798 31817
TCAGGACCTCAGGGCTGGCC 78 3943 522839 31799 31818
GTCAGGACCTCAGGGCTGGC 66 3944 522840 31800 31819
TGTCAGGACCTCAGGGCTGG 53 3945 522841 31801 31820 CTGTCAGGACCTCAGGGCTG 49 3946 522842 31933 31952 GAGTTATCCTCAATTCACCA 74 3947 522843 31934 31953 AGAGTTATCCTCAATTCACC 59 3948 522844 31936 31955 CCAGAGTTATCCTCAATTCA 71 3949 522845 31938 31957 TGCCAGAGTTATCCTCAATT 52 3950 522846 31939 31958 CTGCCAGAGTTATCCTCAAT 62 3951 522847 31940 31959 CCTGCCAGAGTTATCCTCAA 64 3952 522848 31941 31960 TCCTGCCAGAGTTATCCTCA 68 3953 522849 31943 31962 GATCCTGCCAGAGTTATCCT 69 3954 522850 31944 31963 GGATCCTGCCAGAGTTATCC 62 3955 522851 31945 31964 AGGATCCTGCCAGAGTTATC 45 3956 522852 31946 31965 CAGGATCCTGCCAGAGTTAT 60 3957 522853 32116 32135 CAGCAAATATACACCACCCT 70 3958 522854 32117 32136 ACAGCAAATATACACCACCC 66 3959 522855 32118 32137 AACAGCAAATATACACCACC 56 3960 522856 32119 32138 CAACAGCAAATATACACCAC 66 3961 522857 32121 32140 CACAACAGCAAATATACACC 70 3962 522858 32122 32141 TCACAACAGCAAATATACAC 55 3963 522859 32123 32142 CTCACAACAGCAAATATACA 41 3964 522860 32124 32143 ACTCACAACAGCAAATATAC 48 3965 522861 32126 32145 TGACTCACAACAGCAAATAT 21 3966 522862 32127 32146 CTGACTCACAACAGCAAATA 33 3967 522863 32128 32147 ACTGACTCACAACAGCAAAT 44 3968 522864 32129 32148 CACTGACTCACAACAGCAAA 54 3969 522865 32131 32150
GTCACTGACTCACAACAGCA 85 3970 522866 32132 32151 AGTCACTGACTCACAACAGC 80 3971 522867 32133 32152 CAGTCACTGACTCACAACAG 62 3972 522868 32134 32153 TCAGTCACTGACTCACAACA 62 3973 522869 32136 32155 TCTCAGTCACTGACTCACAA 71 3974 522870 32137 32156 ATCTCAGTCACTGACTCACA 77 3975 522871 32138 32157 GATCTCAGTCACTGACTCAC 75 3976 522872 32139 32158 GGATCTCAGTCACTGACTCA 73 3977 522873 32173 32192 GGTTACCCAGCCACTGCCCC 65 3978 522874 32174 32193 GGGTTACCCAGCCACTGCCC 54 3979 522875 32175 32194 AGGGTTACCCAGCCACTGCC 61 3980 522876 32176 32195 CAGGGTTACCCAGCCACTGC 65 3981 522877 32178 32197 TGCAGGGTTACCCAGCCACT 74 3982 522878 32179 32198 ATGCAGGGTTACCCAGCCAC 73 3983 522879 32180 32199 GATGCAGGGTTACCCAGCCA 67 3984 522880 32181 32200 GGATGCAGGGTTACCCAGCC 69 3985 522881 32183 32202 AAGGATGCAGGGTTACCCAG 58 3986 522882 32184 32203 GAAGGATGCAGGGTTACCCA 58 3987 522883 32185 32204 TGAAGGATGCAGGGTTACCC 51 3988 522884 32186 32205 GTGAAGGATGCAGGGTTACC 43 3989 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 80 425 522885 33409 33428 ATCTTCATTACATGATTACT 59 3990 522886 33410 33429 GATCTTCATTACATGATTAC 58 3991 522887 33411 33430 AGATCTTCATTACATGATTA 69 3992 522888 33412 33431 CAGATCTTCATTACATGATT 81 3993 522889 33414 33433 GGCAGATCTTCATTACATGA 83 3994 522890 33669 33688 ATGGACAGGTGATTCTAAGC 69 3995 522891 33670 33689 GATGGACAGGTGATTCTAAG 56 3996 522892 33672 33691 CAGATGGACAGGTGATTCTA 62 3997 522893 33674 33693 GGCAGATGGACAGGTGATTC 67 3998 522894 33675 33694 AGGCAGATGGACAGGTGATT 76 3999 522895 33676 33695 GAGGCAGATGGACAGGTGAT 69 4000 522896 33677 33696 TGAGGCAGATGGACAGGTGA 50 4001 522897 33730 33749 GTGACCTTGGGATAGTCCCT 88 4002 522898 33731 33750 TGTGACCTTGGGATAGTCCC 76 4003 522899 33732 33751 TTGTGACCTTGGGATAGTCC 69 4004 522900 33733 33752 TTTGTGACCTTGGGATAGTC 52 4005 522901 33857 33876 GAGCAAAGGCTTGCTTAAGT 60 4006

TABLE-US-00058 TABLE 58 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 2 SEQ SEQ ID ID NO: NO: 2 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 413433 32431 32450 GCCTGGACAAGTCCTGCCCA

77 425 522902 33858 33877 AGAGCAAAGGCTTGCTTAAG 40 4007 522903 33859 33878 AAGAGCAAAGGCTTGCTTAA 41 4008 522904 33860 33879 GAAGAGCAAAGGCTTGCTTA 67 4009 522905 33862 33881 CAGAAGAGCAAAGGCTTGCT 74 4010 522906 33863 33882 ACAGAAGAGCAAAGGCTTGC 61 4011 522907 33864 33883 CACAGAAGAGCAAAGGCTTG 33 4012 522908 33865 33884 CCACAGAAGAGCAAAGGCTT 58 4013 522909 33867 33886 ACCCACAGAAGAGCAAAGGC 63 4014 522910 33868 33887 GACCCACAGAAGAGCAAAGG 40 4015 522911 33869 33888 AGACCCACAGAAGAGCAAAG 37 4016 522912 33870 33889 CAGACCCACAGAAGAGCAAA 37 4017 522913 34212 34231 ATGCCCTTGCTGTAGGACCC 84 4018 522914 34213 34232 AATGCCCTTGCTGTAGGACC 69 4019 522915 34214 34233 CAATGCCCTTGCTGTAGGAC 59 4020 522916 34215 34234 TCAATGCCCTTGCTGTAGGA 56 4021 522917 34740 34759 TTAGCTCTCAGTGAAGCTGG 76 4022 522918 34869 34888 AACAGCAGCAGTCTAATCCA 49 4023 522919 34870 34889 AAACAGCAGCAGTCTAATCC 50 4024 522920 34871 34890 GAAACAGCAGCAGTCTAATC 30 4025 522921 34872 34891 GGAAACAGCAGCAGTCTAAT 33 4026 522922 34899 34918 AACTCTCAGCTACTCCCCAA 46 4027 522923 34900 34919 CAACTCTCAGCTACTCCCCA 56 4028 522924 34901 34920 CCAACTCTCAGCTACTCCCC 75 4029 522925 34902 34921 ACCAACTCTCAGCTACTCCC 75 4030 522926 34932 34951 GCAAACAGATTAAAGTTGCT 55 4031 522927 34933 34952 GGCAAACAGATTAAAGTTGC 63 4032 522928 35274 35293 CACCTGAGTGCTGCAAAGTG 47 4033 522929 35275 35294 CCACCTGAGTGCTGCAAAGT 55 4034 522930 35276 35295 CCCACCTGAGTGCTGCAAAG 58 4035 522931 35277 35296 TCCCACCTGAGTGCTGCAAA 60 4036 522932 35279 35298 ACTCCCACCTGAGTGCTGCA 77 4037 522933 35280 35299 CACTCCCACCTGAGTGCTGC 62 4038 522934 35281 35300 ACACTCCCACCTGAGTGCTG 64 4039 522935 35282 35301 GACACTCCCACCTGAGTGCT 69 4040 522936 35319 35338 TTTTTGGCTGCCCTGCCTCA 35 4041 522937 35320 35339 CTTTTTGGCTGCCCTGCCTC 56 4042 522938 35321 35340 TCTTTTTGGCTGCCCTGCCT 58 4043 522939 35322 35341 GTCTTTTTGGCTGCCCTGCC 74 4044 522940 35324 35343 TGGTCTTTTTGGCTGCCCTG 72 4045 522941 35325 35344 TTGGTCTTTTTGGCTGCCCT 79 4046 522942 35326 35345 GTTGGTCTTTTTGGCTGCCC 81 4047 522943 35327 35346 CGTTGGTCTTTTTGGCTGCC 79 4048 522944 35407 35426 CTTATCTTTGTCCACATATC 57 4049 522945 35408 35427 CCTTATCTTTGTCCACATAT 66 4050 522946 35409 35428 ACCTTATCTTTGTCCACATA 60 4051 522947 35410 35429 TACCTTATCTTTGTCCACAT 75 4052 522948 35412 35431 AATACCTTATCTTTGTCCAC 67 4053 522949 35413 35432 AAATACCTTATCTTTGTCCA 61 4054 522950 35414 35433 TAAATACCTTATCTTTGTCC 71 4055 522951 35415 35434 ATAAATACCTTATCTTTGTC 16 4056 522952 35417 35436 CAATAAATACCTTATCTTTG 22 4057 522953 35418 35437 TCAATAAATACCTTATCTTT 3 4058 522954 35419 35438 TTCAATAAATACCTTATCTT 12 4059 522955 35420 35439 CTTCAATAAATACCTTATCT 47 4060 522956 35422 35441 TGCTTCAATAAATACCTTAT 65 4061 522957 35423 35442 ATGCTTCAATAAATACCTTA 55 4062 522958 35424 35443 AATGCTTCAATAAATACCTT 42 4063 522959 35425 35444 TAATGCTTCAATAAATACCT 42 4064 522960 35432 35451 TTAGAAGTAATGCTTCAATA 37 4065 522961 35433 35452 CTTAGAAGTAATGCTTCAAT 44 4066 522962 35434 35453 TCTTAGAAGTAATGCTTCAA 44 4067 522963 35435 35454 CTCTTAGAAGTAATGCTTCA 59 4068 522964 35437 35456 CCCTCTTAGAAGTAATGCTT 80 4069 522965 35438 35457 TCCCTCTTAGAAGTAATGCT 73 4070 522966 35439 35458 TTCCCTCTTAGAAGTAATGC 59 4071 522967 35440 35459 TTTCCCTCTTAGAAGTAATG 48 4072 522968 37835 37854 GCTAATGATACAAGGTTGTT 57 4073 522969 37836 37855 AGCTAATGATACAAGGTTGT 54 4074 522970 37838 37857 GCAGCTAATGATACAAGGTT 65 4075 522971 37840 37859 ATGCAGCTAATGATACAAGG 57 4076 522972 37841 37860 AATGCAGCTAATGATACAAG 33 4077 522973 37842 37861 AAATGCAGCTAATGATACAA

24 4078 522974 37843 37862 AAAATGCAGCTAATGATACA 22 4079 522975 39666 39685 AAGAGTTACCTCCTCCCTGG 30 4080 522976 39667 39686 CAAGAGTTACCTCCTCCCTG 36 4081 522977 39668 39687 GCAAGAGTTACCTCCTCCCT 62 4082 522978 39669 39688 TGCAAGAGTTACCTCCTCCC 61 4083

TABLE-US-00059 TABLE 59 Inhibition of DGAT2 mRNA by 3-10-4 MOE gapmers targeting SEQ ID NO: 2 SEQ SEQ ID ID NO: NO: 2 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 525386 10047 10063 CCACAGGGCCCTTTCCA 49 4141 525387 10048 10064 GCCACAGGGCCCTTTCC 53 4142 525388 10215 10231 GTTCTCGAATATCCCTA 67 4143 525389 10216 10232 AGTTCTCGAATATCCCT 55 4144 525390 10217 10233 GAGTTCTCGAATATCCC 52 4145 525391 10218 10234 GGAGTTCTCGAATATCC 60 4146 525392 10219 10235 AGGAGTTCTCGAATATC 34 4147 525393 10220 10236 GAGGAGTTCTCGAATAT 41 4148 525394 10490 10506 GGTCCTCTCCGCTGCCT 62 4149 525395 10491 10507 GGGTCCTCTCCGCTGCC 70 4150 525396 10492 10508 AGGGTCCTCTCCGCTGC 57 4151 525397 10529 10545 TTCACTGAGGACCTCAG 29 4152 525398 10530 10546 ATTCACTGAGGACCTCA 44 4153 525399 10531 10547 GATTCACTGAGGACCTC 66 4154 525400 10532 10548 CGATTCACTGAGGACCT 48 4155 525401 10533 10549 GCGATTCACTGAGGACC 83 4156 525402 10534 10550 CGCGATTCACTGAGGAC 71 4157 525403 10535 10551 GCGCGATTCACTGAGGA 68 4158 525404 10536 10552 TGCGCGATTCACTGAGG 65 4159 525405 10537 10553 CTGCGCGATTCACTGAG 60 4160 525406 10538 10554 TCTGCGCGATTCACTGA 45 4161 525407 10650 10666 TCCTCCCACTTCAGTTT 28 4162 525408 10651 10667 TTCCTCCCACTTCAGTT 18 4163 525409 10652 10668 CTTCCTCCCACTTCAGT 30 4164 525410 10654 10670 TGCTTCCTCCCACTTCA 30 4165 525411 10655 10671 ATGCTTCCTCCCACTTC 37 4166 525412 10656 10672 CATGCTTCCTCCCACTT 32 4167 525413 10657 10673 GCATGCTTCCTCCCACT 59 4168 525414 10658 10674 GGCATGCTTCCTCCCAC 69 4169 525415 10659 10675 AGGCATGCTTCCTCCCA 71 4170 525416 10660 10676 TAGGCATGCTTCCTCCC 62 4171 525417 10661 10677 TTAGGCATGCTTCCTCC 53 4172 525418 10662 10678 CTTAGGCATGCTTCCTC 53 4173 525419 10663 10679 ACTTAGGCATGCTTCCT 63 4174 525420 10664 10680 AACTTAGGCATGCTTCC 62 4175 525421 10665 10681 AAACTTAGGCATGCTTC 50 4176 525422 10666 10682 AAAACTTAGGCATGCTT 58 4177 525423 10667 10683 GAAAACTTAGGCATGCT 68 4178 525424 10668 10684 GGAAAACTTAGGCATGC 65 4179 525425 10669 10685 AGGAAAACTTAGGCATG 41 4180 525426 10670 10686 AAGGAAAACTTAGGCAT 15 4181 525427 10671 10687 TAAGGAAAACTTAGGCA 15 4182 525428 10672 10688 CTAAGGAAAACTTAGGC 12 4183 525429 10728 10744 GATGACTTCCCAGGGTC 32 4184 525430 10729 10745 TGATGACTTCCCAGGGT 41 4185 525431 10730 10746 GTGATGACTTCCCAGGG 78 4186 525432 10731 10747 TGTGATGACTTCCCAGG 51 4187 525433 10732 10748 CTGTGATGACTTCCCAG 41 4188 525434 10733 10749 GCTGTGATGACTTCCCA 63 4189 525435 10734 10750 AGCTGTGATGACTTCCC 61 4190 525436 10735 10751 CAGCTGTGATGACTTCC 60 4191 525437 10736 10752 ACAGCTGTGATGACTTC 31 4192 525438 10737 10753 GACAGCTGTGATGACTT 0 4193 525439 10816 10832 TGTCAGAGAGGCTCAGC 59 4194 525440 10817 10833 ATGTCAGAGAGGCTCAG 41 4195 525441 10818 10834 CATGTCAGAGAGGCTCA 60 4196 525442 10819 10835 CCATGTCAGAGAGGCTC 73 4197 525443 10820 10836 TCCATGTCAGAGAGGCT 75 4198 525444 10821 10837 ATCCATGTCAGAGAGGC 64 4199 525445 10822 10838 AATCCATGTCAGAGAGG 44 4200 525446 10823 10839 AAATCCATGTCAGAGAG 10 4201 525447 10940 10956 TAGTCGATTTACCAGAA 31 4202 525448 10941 10957 ATAGTCGATTTACCAGA 31 4203 525449 10942 10958 GATAGTCGATTTACCAG 43 4204 525450 10943 10959 GGATAGTCGATTTACCA 62 4205 525451 10944 10960

TGGATAGTCGATTTACC 48 4206 525452 10945 10961 TTGGATAGTCGATTTAC 36 4207 525453 10946 10962 TTTGGATAGTCGATTTA 16 4208 525454 11074 11090 CGATGTTACATTAAGGG 26 4209 525455 11075 11091 TCGATGTTACATTAAGG 24 4210 525456 11076 11092 CTCGATGTTACATTAAG 11 4211 496041 31122 31138 CACAGCGATGAGCCAGC 52 1096 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 69 425 525765 36860 36876 ATTATTCTAAAACTCAA 11 4212 525766 36861 36877 GATTATTCTAAAACTCA 6 4213 525767 36862 36878 TGATTATTCTAAAACTC 7 4214 525768 37022 37038 GGGCAGTCACCATTTGC 14 4215 525769 37023 37039 TGGGCAGTCACCATTTG 12 4216 525770 37024 37040 TTGGGCAGTCACCATTT 7 4217 TABLE-US-00060 TABLE 60 Inhibition of DGAT2 mRNA by 3-10-4 MOE gapmers targeting SEQ ID NO: 2 SEQ SEQ ID ID NO: NO: 2 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 525457 11077 11093 CCTCGATGTTACATTAA 44 4218 525458 11127 11143 TATTGCAACCACTAGGA 36 4219 525459 11128 11144 TTATTGCAACCACTAGG 41 4220 525460 11129 11145 GTTATTGCAACCACTAG 49 4221 525461 11130 11146 GGTTATTGCAACCACTA 61 4222 525462 11158 11174 AGTTAAAGTGTGGTACA 4 4223 525463 11159 11175 CAGTTAAAGTGTGGTAC 37 4224 525464 11160 11176 ACAGTTAAAGTGTGGTA 14 4225 525465 11171 11187 ATGCCTAGGTCACAGTT 55 4226 525466 11172 11188 AATGCCTAGGTCACAGT 48 4227 525467 11173 11189 CAATGCCTAGGTCACAG 44 4228 525468 11174 11190 CCAATGCCTAGGTCACA 73 4229 525469 11175 11191 GCCAATGCCTAGGTCAC 89 4230 525470 11176 11192 TGCCAATGCCTAGGTCA 82 4231 525471 11177 11193 ATGCCAATGCCTAGGTC 79 4232 525472 11178 11194 AATGCCAATGCCTAGGT 76 4233 525473 11179 11195 CAATGCCAATGCCTAGG 69 4234 525474 11180 11196 GCAATGCCAATGCCTAG 78 4235 525475 11199 11215 AGCGATAATCACACAAG 54 4236 525476 11200 11216 CAGCGATAATCACACAA 58 4237 525477 11201 11217 ACAGCGATAATCACACA 72 4238 525478 11202 11218 CACAGCGATAATCACAC 49 4239 525479 11203 11219 CCACAGCGATAATCACA 79 4240 525480 11204 11220 ACCACAGCGATAATCAC 76 4241 525481 11205 11221 AACCACAGCGATAATCA 40 4242 525482 14503 14519 GGTCTGCCACTCTCTAC 49 4243 525483 14504 14520 GGGTCTGCCACTCTCTA 47 4244 525484 14505 14521 AGGGTCTGCCACTCTCT 64 4245 525485 14648 14664 AGTCATCTTGTGTACAT 60 4246 525486 14649 14665 CAGTCATCTTGTGTACA 49 4247 525487 14650 14666 ACAGTCATCTTGTGTAC 52 4248 525488 14663 14679 CAGATCACTCTGCACAG 62 4249 525489 14664 14680 TCAGATCACTCTGCACA 65 4250 525490 14665 14681 CTCAGATCACTCTGCAC 61 4251 525491 14667 14683 TGCTCAGATCACTCTGC 60 4252 525492 14668 14684 TTGCTCAGATCACTCTG 51 4253 525493 14669 14685 ATTGCTCAGATCACTCT 54 4254 525494 14745 14761 ACCTCTTCCAGGAAGAC 41 4255 525495 14746 14762 CACCTCTTCCAGGAAGA 44 4256 525496 14747 14763 TCACCTCTTCCAGGAAG 28 4257 525497 14755 14771 ACTGAATGTCACCTCTT 57 4258 525498 14756 14772 GACTGAATGTCACCTCT 62 4259 525499 14757 14773 GGACTGAATGTCACCTC 71 4260 525500 14758 14774 CGGACTGAATGTCACCT 74 4261 525501 14759 14775 CCGGACTGAATGTCACC 80 4262 525502 14760 14776 TCCGGACTGAATGTCAC 39 4263 525503 14770 14786 TCTTTCCAGATCCGGAC 53 4264 525504 14771 14787 ATCTTTCCAGATCCGGA 62 4265 525505 14772 14788 CATCTTTCCAGATCCGG 70 4266 525506 14773 14789 TCATCTTTCCAGATCCG 71 4267 525507 14774 14790 TTCATCTTTCCAGATCC 63 4268 525508 14775 14791 ATTCATCTTTCCAGATC 23 4269 525509 14996 15012 TCTCAAGTCTTCCTCCT 33 4270 525510 14997 15013 CTCTCAAGTCTTCCTCC 46 4271 525511 14998 15014 GCTCTCAAGTCTTCCTC 65 4272 525512 14999 15015 AGCTCTCAAGTCTTCCT 72 4273 525513 15000 15016 GAGCTCTCAAGTCTTCC 73 4274 525514 15001 15017 TGAGCTCTCAAGTCTTC 64 4275

525515 15250 15266 ATAATCTGCACAGGTTC 72 4276 525516 15251 15267 CATAATCTGCACAGGTT 67 4277 525517 15252 15268 CCATAATCTGCACAGGT 69 4278 525518 15253 15269 ACCATAATCTGCACAGG 68 4279 525519 15254 15270 CACCATAATCTGCACAG 49 4280 525520 15255 15271 GCACCATAATCTGCACA 60 4281 525521 18156 18172 TTAATCCAGGATTGTCA 55 4282 525522 18157 18173 TTTAATCCAGGATTGTC 57 4283 525523 18158 18174 CTTTAATCCAGGATTGT 26 4284 525524 18161 18177 TAGCTTTAATCCAGGAT 42 4285 525525 18162 18178 TTAGCTTTAATCCAGGA 28 4286 525526 18163 18179 CTTAGCTTTAATCCAGG 45 4287 525527 18164 18180 CCTTAGCTTTAATCCAG 27 4288 525528 18165 18181 TCCTTAGCTTTAATCCA 60 4289 525529 18166 18182 CTCCTTAGCTTTAATCC 23 4290 525530 18167 18183 CCTCCTTAGCTTTAATC 32 4291 525531 18168 18184 TCCTCCTTAGCTTTAAT 28 4292 525532 18174 18190 TCAGTGTCCTCCTTAGC 39 4293 525533 18175 18191 CTCAGTGTCCTCCTTAG 32 4294 496041 31122 31138 CACAGCGATGAGCCAGC 64 1096 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 78 425

TABLE-US-00061 TABLE 61 Inhibition of DGAT2 mRNA by 3-10-4 MOE gapmers targeting SEQ ID NO: 2 SEQ SEQ ID ID NO: NO: 2 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 525534 18176 18192 ACTCAGTGTCCTCCTTA 35 4295 525535 18177 18193 GACTCAGTGTCCTCCTT 64 4296 525536 18178 18194 GGACTCAGTGTCCTCCT 73 4297 525537 18180 18196 TGGGACTCAGTGTCCTC 62 4298 525538 18181 18197 CTGGGACTCAGTGTCCT 59 4299 525539 18182 18198 CCTGGGACTCAGTGTCC 63 4300 525540 18308 18324 CTGTTTGATGGGTAAAA 20 4301 525541 18309 18325 CCTGTTTGATGGGTAAA 62 4302 525542 18310 18326 TCCTGTTTGATGGGTAA 68 4303 525543 18311 18327 CTCCTGTTTGATGGGTA 39 4304 525544 18312 18328 TCTCCTGTTTGATGGGT 73 4305 525545 18313 18329 CTCTCCTGTTTGATGGG 47 4306 525546 18314 18330 CCTCTCCTGTTTGATGG 43 4307 525547 18321 18337 TCGGTGTCCTCTCCTGT 63 4308 525548 18322 18338 CTCGGTGTCCTCTCCTG 64 4309 525549 18323 18339 CCTCGGTGTCCTCTCCT 60 4310 525550 18324 18340 GCCTCGGTGTCCTCTCC 68 4311 525551 18325 18341 AGCCTCGGTGTCCTCTC 78 4312 525552 18326 18342 AAGCCTCGGTGTCCTCT 73 4313 525553 18327 18343 TAAGCCTCGGTGTCCTC 70 4314 525554 18328 18344 GTAAGCCTCGGTGTCCT 75 4315 525555 18329 18345 AGTAAGCCTCGGTGTCC 64 4316 525556 18417 18433 CCAGATTGTTGTGGGAA 66 4317 525557 18418 18434 ACCAGATTGTTGTGGGA 68 4318 525558 18419 18435 CACCAGATTGTTGTGGG 63 4319 525559 18432 18448 CTAATACCTACCTCACC 35 4320 525560 18433 18449 GCTAATACCTACCTCAC 25 4321 525561 18434 18450 GGCTAATACCTACCTCA 34 4322 525562 18450 18466 CTCATCTATACAGTGGG 58 4323 525563 18451 18467 CCTCATCTATACAGTGG 51 4324 525564 18452 18468 TCCTCATCTATACAGTG 28 4325 525565 18581 18597 ACTGGCATCTGGCAGGG 69 4326 525566 18582 18598 TACTGGCATCTGGCAGG 52 4327 525567 18583 18599 ATACTGGCATCTGGCAG 31 4328 525568 18591 18607 TCACCTCCATACTGGCA 21 4329 525569 18592 18608 CTCACCTCCATACTGGC 20 4330 525570 18593 18609 CCTCACCTCCATACTGG 6 4331 525571 18828 18844 ATGCTTAACTTCTGCTG 48 4332 525572 18829 18845 GATGCTTAACTTCTGCT 46 4333 525573 18830 18846 GGATGCTTAACTTCTGC 53 4334 525574 18831 18847 AGGATGCTTAACTTCTG 40 4335 525575 18832 18848 TAGGATGCTTAACTTCT 40 4336 525576 18833 18849 TTAGGATGCTTAACTTC 20 4337 525577 18869 18885 GGCCTGGATGCCCAAGT 65 4338 20779 20795 525578 18870 18886 AGGCCTGGATGCCCAAG 71 4339 20780 20796 525579 18871 18887 TAGGCCTGGATGCCCAA 66 4340 20781 20797 525580 18900 18916 CAGATGATGTCCTGCCT 25 4341 525581 18901 18917 GCAGATGATGTCCTGCC 55 4342

525582 18902 18918 AGCAGATGATGTCCTGC 38 4343 525583 19018 19034 TAGTCAAAGGTGGCTTC 44 4344 525584 19019 19035 GTAGTCAAAGGTGGCTT 57 4345 525585 19020 19036 AGTAGTCAAAGGTGGCT 55 4346 525589 22548 22564 AACAAGTGGGAAATGCA 17 4347 525590 22549 22565 CAACAAGTGGGAAATGC 19 4348 525591 22550 22566 CCAACAAGTGGGAAATG 23 4349 525592 22771 22787 AGCTTATTAGGTGTCTT 42 4350 525593 22772 22788 AAGCTTATTAGGTGTCT 53 4351 525594 22773 22789 TAAGCTTATTAGGTGTC 44 4352 525595 22774 22790 CTAAGCTTATTAGGTGT 36 4353 525596 22886 22902 CCCCCATGCAGCTTGGA 29 4354 525597 22887 22903 GCCCCCATGCAGCTTGG 43 4355 525598 22888 22904 AGCCCCCATGCAGCTTG 50 4356 525599 23096 23112 CCAAACACTCAGGTAGG 49 4357 525600 23097 23113 TCCAAACACTCAGGTAG 41 4358 525601 23098 23114 GTCCAAACACTCAGGTA 54 4359 525602 23100 23116 CAGTCCAAACACTCAGG 47 4360 525603 23101 23117 TCAGTCCAAACACTCAG 57 4361 525604 23102 23118 TTCAGTCCAAACACTCA 49 4362 525605 23239 23255 GAACAGCAGCATCAGCA 47 4363 525606 23240 23256 GGAACAGCAGCATCAGC 67 4364 525607 23241 23257 GGGAACAGCAGCATCAG 40 4365 525608 23242 23258 TGGGAACAGCAGCATCA 44 4366 525609 23243 23259 CTGGGAACAGCAGCATC 69 4367 525610 23244 23260 TCTGGGAACAGCAGCAT 65 4368 525586 36855 36871 TCTAAAACTCAAATCCA 0 4369 525587 36856 36872 TTCTAAAACTCAAATCC 0 4370 525588 36857 36873 ATTCTAAAACTCAAATC 0 4371 496041 31122 31138 CACAGCGATGAGCCAGC 64 1096 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 78 425 TABLE-US-00062 TABLE 62 Inhibition of DGAT2 mRNA by 3-10-4 MOE gapmers targeting SEQ ID NO: 2 SEQ SEQ ID ID NO: NO: 2 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 525611 23245 23261 GTCTGGGAACAGCAGCA 65 4372 525612 23246 23262 GGTCTGGGAACAGCAGC 83 4373 525613 23247 23263 TGGTCTGGGAACAGCAG 42 4374 525614 23427 23443 ACAAGTCATCTTCCAGC 16 4375 525615 23428 23444 GACAAGTCATCTTCCAG 24 4376 525616 23429 23445 GGACAAGTCATCTTCCA 37 4377 525617 23551 23567 ATCCTTGAGGGTCTCAT 55 4378 525618 23552 23568 TATCCTTGAGGGTCTCA 55 4379 525619 23553 23569 TTATCCTTGAGGGTCTC 65 4380 525620 23554 23570 CTTATCCTTGAGGGTCT 46 4381 525621 26400 26416 CCTGGATGGCAACCTAA 42 4382 525622 26401 26417 GCCTGGATGGCAACCTA 33 4383 525623 26402 26418 GGCCTGGATGGCAACCT 43 4384 525624 26484 26500 CTGCTATGCTGAGAGCA 44 4385 525625 26485 26501 CCTGCTATGCTGAGAGC 42 4386 525626 26486 26502 ACCTGCTATGCTGAGAG 8 4387 525627 26516 26532 GTTCATCTGCCTTGACG 44 4388 525628 26517 26533 GGTTCATCTGCCTTGAC 56 4389 525629 26518 26534 AGGTTCATCTGCCTTGA 33 4390 525630 26519 26535 CAGGTTCATCTGCCTTG 55 4391 525631 26520 26536 GCAGGTTCATCTGCCTT 75 4392 525632 26521 26537 AGCAGGTTCATCTGCCT 62 4393 525633 26536 26552 TCTGTGATGCTCTGGAG 33 4394 525634 26537 26553 CTCTGTGATGCTCTGGA 43 4395 525635 26538 26554 ACTCTGTGATGCTCTGG 46 4396 525636 26539 26555 CACTCTGTGATGCTCTG 39 4397 525637 26627 26643 ACATGGTAAGTCCTGAT 38 4398 525638 26628 26644 GACATGGTAAGTCCTGA 55 4399 525639 26629 26645 TGACATGGTAAGTCCTG 58 4400 525640 26630 26646 CTGACATGGTAAGTCCT 55 4401 525641 26631 26647 ACTGACATGGTAAGTCC 35 4402 525642 26632 26648 CACTGACATGGTAAGTC 48 4403 525643 26783 26799 TCTGCCATTTAATGAGC 19 4404 525644 26784 26800 CTCTGCCATTTAATGAG 22 4405 525645 26785 26801 ACTCTGCCATTTAATGA 12 4406 525646 26788 26804 CCAACTCTGCCATTTAA 43 4407 525647 26789 26805 CCCAACTCTGCCATTTA 47 4408 525648 26790 26806 TCCCAACTCTGCCATTT 35 4409 525649 26926 26942 AGGGTTCATGGATCCCC 69 4410 525650 26927 26943 AAGGGTTCATGGATCCC 67 4411

525651 26928 26944 TAAGGGTTCATGGATCC 44 4412 525652 27251 27267 CTGGAGTCCCTGGGTTC 45 4413 525653 27252 27268 GCTGGAGTCCCTGGGTT 30 4414 525654 27253 27269 GGCTGGAGTCCCTGGGT 31 4415 525655 27352 27368 AGCCAGGCCCAGACCCA 32 4416 525656 27353 27369 CAGCCAGGCCCAGACCC 22 4417 525657 27354 27370 CCAGCCAGGCCCAGACC 11 4418 525658 27382 27398 GGTGTTCTACAAGCTGC 61 4419 525659 27383 27399 TGGTGTTCTACAAGCTG 50 4420 525660 27384 27400 CTGGTGTTCTACAAGCT 49 4421 525661 27387 27403 GAGCTGGTGTTCTACAA 34 4422 525662 27388 27404 TGAGCTGGTGTTCTACA 47 4423 525663 27389 27405 GTGAGCTGGTGTTCTAC 55 4424 525664 27437 27453 GATCAATTATTAACCTA 14 4425 525665 27438 27454 TGATCAATTATTAACCT 0 4426 525666 27439 27455 CTGATCAATTATTAACC 5 4427 525667 29678 29694 GCAGGTACTCATGTTTG 50 4428 525668 29679 29695 AGCAGGTACTCATGTTT 39 4429 525669 29680 29696 CAGCAGGTACTCATGTT 31 4430 525670 29782 29798 CCAATATCATACTATCT 5 4431 525671 29783 29799 CCCAATATCATACTATC 13 4432 525672 29784 29800 CCCCAATATCATACTAT 40 4433 525673 29828 29844 AACCTTTAAGGCCTATC 16 4434 525674 29829 29845 CAACCTTTAAGGCCTAT 23 4435 525675 29830 29846 CCAACCTTTAAGGCCTA 35 4436 525676 29831 29847 CCCAACCTTTAAGGCCT 40 4437 525677 29839 29855 ATTTTTTACCCAACCTT 30 4438 525678 29840 29856 CATTTTTTACCCAACCT 24 4439 525679 29841 29857 CCATTTTTTACCCAACC 50 4440 525680 29842 29858 TCCATTTTTTACCCAAC 40 4441 525681 29843 29859 TTCCATTTTTTACCCAA 24 4442 525682 30483 30499 GCTTTAGAAATTCACCA 45 4443 525683 30484 30500 GGCTTTAGAAATTCACC 68 4444 525684 30485 30501 AGGCTTTAGAAATTCAC 39 4445 496041 31122 31138 CACAGCGATGAGCCAGC 58 1096 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 76 425 525685 32431 32447 TGGACAAGTCCTGCCCA 47 4446 525686 32432 32448 CTGGACAAGTCCTGCCC 58 4447 525687 32433 32449 CCTGGACAAGTCCTGCC 60 4448 TABLE-US-00063 TABLE 63 Inhibition of DGAT2 mRNA by 3-10-4 MOE gapmers targeting SEQ ID NO: 2 SEQ SEQ ID ID NO: NO: 2 2 % SEQ ISIS Start Stop inhi- ID NO Site Site Sequence bition NO 496041 31122 31138 CACAGCGATGAGCCAGC 44 1096 413433 32431 32450 GCCTGGACAAGTCCTGCCCA 74 425 525688 32434 32450 GCCTGGACAAGTCCTGC 77 4449 525689 32435 32451 AGCCTGGACAAGTCCTG 50 4450 525690 32447 32463 ACTAGACTATGCAGCCT 62 4451 525691 32448 32464 TACTAGACTATGCAGCC 55 4452 525692 32449 32465 ATACTAGACTATGCAGC 37 4453 525693 32450 32466 CATACTAGACTATGCAG 17 4454 525694 32451 32467 TCATACTAGACTATGCA 20 4455 525695 32452 32468 ATCATACTAGACTATGC 20 4456 525696 32453 32469 CATCATACTAGACTATG 13 4457 525697 32454 32470 CCATCATACTAGACTAT 32 4458 525698 32455 32471 GCCATCATACTAGACTA 58 4459 525699 32456 32472 TGCCATCATACTAGACT 51 4460 525700 32457 32473 TTGCCATCATACTAGAC 40 4461 525701 32458 32474 GTTGCCATCATACTAGA 51 4462 525702 32460 32476 ATGTTGCCATCATACTA 52 4463 525703 32461 32477 AATGTTGCCATCATACT 28 4464 525704 32462 32478 CAATGTTGCCATCATAC 40 4465 525705 32463 32479 GCAATGTTGCCATCATA 73 4466 525706 32464 32480 TGCAATGTTGCCATCAT 62 4467 525707 32465 32481 TTGCAATGTTGCCATCA 61 4468 525708 32466 32482 GTTGCAATGTTGCCATC 79 4469 525709 32467 32483 GGTTGCAATGTTGCCAT 69 4470 525710 32468 32484 TGGTTGCAATGTTGCCA 66 4471 525711 32469 32485 GTGGTTGCAATGTTGCC 73 4472 525712 32470 32486 GGTGGTTGCAATGTTGC 45 4473 525713 32476 32492 CTGGATGGTGGTTGCAA 20 4474 525714 32477 32493 CCTGGATGGTGGTTGCA 59 4475 525715 32478 32494 GCCTGGATGGTGGTTGC 58 4476 525716 32479 32495 AGCCTGGATGGTGGTTG 34 4477 525717 32619 32635 GAGATCTCCAGCAGCAA 52 4478 525718 32620 32636

TGAGATCTCCAGCAGCA 57 4479 525719 32621 32637 CTGAGATCTCCAGCAGC 51 4480 525720 32622 32638 ACTGAGATCTCCAGCAG 16 4481 525721 32645 32661 GGAGTGACAGGGCAGGA 42 4482 525722 32646 32662 TGGAGTGACAGGGCAGG 42 4483 525723 32647 32663 ATGGAGTGACAGGGCAG 35 4484 525724 32752 32768 AAGTCTATCAGGATGCA 43 4485 525725 32753 32769 AAAGTCTATCAGGATGC 37 4486 525726 32754 32770 CAAAGTCTATCAGGATG 20 4487 525727 32755 32771 ACAAAGTCTATCAGGAT 0 4488 525728 32757 32773 TGACAAAGTCTATCAGG 25 4489 525729 32758 32774 GTGACAAAGTCTATCAG 51 4490 525730 32759 32775 AGTGACAAAGTCTATCA 28 4491 525731 33180 33196 CCTAGTGCCCCAGGGAG 29 4492 525732 33181 33197 TCCTAGTGCCCCAGGGA 25 4493 525733 33182 33198 GTCCTAGTGCCCCAGGG 74 4494 525734 33183 33199 TGTCCTAGTGCCCCAGG 53 4495 525735 36248 36264 GGTATTCTGCTCATAAA 43 4496 525736 36249 36265 GGGTATTCTGCTCATAA 50 4497 525737 36250 36266 AGGGTATTCTGCTCATA 65 4498 525738 36251 36267 AAGGGTATTCTGCTCAT 55 4499 525739 36252 36268 TAAGGGTATTCTGCTCA 58 4500 525740 36253 36269 GTAAGGGTATTCTGCTC 64 4501 525741 36254 36270 AGTAAGGGTATTCTGCT 60 4502 525742 36263 36279 GAGACAATGAGTAAGGG 25 4503 525743 36264 36280 AGAGACAATGAGTAAGG 19 4504 525744 36265 36281 GAGAGACAATGAGTAAG 11 4505 525745 36522 36538 AAGCGACAGCCCTGTGC 17 4506 525746 36523 36539 CAAGCGACAGCCCTGTG 16 4507 525747 36524 36540 ACAAGCGACAGCCCTGT 28 4508 525748 36652 36668 AATGAGACAGGCAGCCC 40 4509 525749 36653 36669 GAATGAGACAGGCAGCC 35 4510 525750 36654 36670 AGAATGAGACAGGCAGC 37 4511 525751 36679 36695 CTCTGTGGGCTCCTCCA 25 4512 525752 36680 36696 GCTCTGTGGGCTCCTCC 57 4513 525753 36681 36697 TGCTCTGTGGGCTCCTC 46 4514 525754 36682 36698 GTGCTCTGTGGGCTCCT 76 4515 525755 36683 36699 TGTGCTCTGTGGGCTCC 59 4516 525756 36686 36702 CCCTGTGCTCTGTGGGC 30 4517 525757 36687 36703 GCCCTGTGCTCTGTGGG 32 4518 525758 36688 36704 GGCCCTGTGCTCTGTGG 40 4519 525759 36738 36754 CCACTGTCAGTCTCTCC 40 4520 525760 36739 36755 CCCACTGTCAGTCTCTC 46 4521 525761 36740 36756 CCCCACTGTCAGTCTCT 29 4522 525762 36758 36774 CTTGGCTGCAAGCTCTG 22 4523 525763 36759 36775 CCTTGGCTGCAAGCTCT 37 4524 525764 36760 36776 GCCTTGGCTGCAAGCTC 30 4525

Example 5: Antisense Inhibition of Human DGAT2 in HepG2 Cells by MOE Gapmers

[0460] Antisense oligonucleotides were designed targeting a diacylglycerol acyltransferase 2 (DGAT2) nucleic acid and were tested for their effects on DGAT2 mRNA in vitro. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. The results for each experiment are presented in separate tables shown below. Cultured HepG2 cells at a density of 20,000 cells per well were transfected using electroporation with 1,000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ISIS 413433, which consistently demonstrated higher potency than any of the previously disclosed oligonucleotides in the studies above was included in this study as a benchmark oligonucleotide. Antisense oligonucleotides that demonstrated about the same or greater potency than ISIS 413433 were therefore considered for further experimentation.

[0461] The newly designed chimeric antisense oligonucleotides in the Tables below were designed as 5-10-5 MOE gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment comprises of ten 2′-deoxynucleosides and is flanked by wing segments on the 5′ direction and the 3′ direction comprising five nucleosides each. Each nucleoside in the 5′ wing segment and each nucleoside in the 3′ wing segment has a 2′-MOE modification. The internucleoside linkages throughout each gapmer are phosphorothioate (P═S) linkages. All cytosine residues throughout each gapmer are 5-methylcytosines. “Start site” indicates the 5′-most nucleoside to which the gapmer is targeted in the human gene sequence. “Stop site” indicates the 3′-most nucleoside to which the gapmer is targeted human gene sequence. Each gapmer listed in the Tables below is targeted to the human DGAT2 genomic sequence, designated herein as SEQ ID NO: 2 (Ref Seq No. NT 033927.5 truncated from nucleotides 5669186 to 5712008). ‘n/a’ indicates that the antisense oligonucleotide does not target that particular gene sequence with 100% complementarity. In case the sequence alignment for a target gene in a particular table is not shown, it is understood that none of the oligonucleotides presented in that table align with 100% complementarity with that target gene.

TABLE-US-00064 TABLE 64 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ Sequence SEQ SEQ ID ID ID ID NO: NO: NO: NO: 1 1 % 2 2 SEQ ISIS Start Stop inhi- Start Stop ID NO Site Site bition Site Site NO 472445 1000 1019 CTGCATTCTTGCC 70 38153 38172 4527 AGGCATG 472446 1001 1020 ACTGCATTCTTGC 76 38154 38173 4528 CAGGCAT 472447 1003 1022 TGACTGCATTCTT 82 38156 38175 4529 GCCAGGC 472448 1004 1023 GTGACTGCATTCT 76 38157 38176 4530 TGCCAGG 472449 1006 1025 GGGTGACTGCATT 42 38159 38178 4531 CTTGCCA 472450 1007 1026 AGGGTGACTGCAT 81 38160 38179 4532 TCTTGCC 472451 1008 1027 CAGGGTGACTGCA 71 38161 38180 4533 TTCTTGC 472452 1010 1029 CGCAGGGTGACTG 69 38163 38182 4534 CATTCTT 472453 1011 1030 CCGCAGGGTGACT 78 38164 38183 4535 GCATTCT 472454 1013 1032 TTCCGCAGGGTGA 53 38166 38185 4536 CTGCATT 472455 1014 1033 GTTCCGCAGGGTG 76 38167 38186 4537 ACTGCAT 472456 1015 1034 GGTTCCGCAGGGT 86 38168 38187 4538 GACTGCA 472457 1017 1036 GCGGTTCCGCAGG 77 38170 38189 4539 GTGACTG 472458 1018 1037 TGCGGTTCCGCAG 81 38171 38190 4540 GGTGACT 472459 1020 1039 CTTGCGGTTCCGC 60 38173 38192 4541 AGGGTGA 472460 1021 1040 CCTTGCGGTTCCG 55 38174 38193 4542 CAGGGTG 472461 1023 1042 GCCCTTGCGGTTC 23 38176 38195 4543 CGCAGGG 472462 1024 1043 AGCCCTTGCGGTT 61 38177 38196 4544 CCGCAGG 472463 1026 1045 AAAGCCCTTGCGG 76 38179 38198 4545 TTCCGCA 472464 1027 1046 CAAAGCCCTTGCG 74 38180 38199 4546 GTTCCGC 472465 1029 1048 CACAAAGCCCTTG 53 38182 38201 4547 CGGTTCC 472466 1030 1049 TCACAAAGCCCTT 23 38183 38202 4548 GCGGTTC 472467 1032 1051 TTTCACAAAGCCC 0 38185 38204 4549 TTGCGGT 472468 1033 1052 GTTTCACAAAGCC 46 38186 38205 4550 CTTGCGG 472469 1035 1054 CAGTTTCACAAAG 51 38188 38207 4551 CCCTTGC 472470 1036 1055 CCAGTTTCACAAA 55 38189 38208 4552 GCCCTTG 472471 1038 1057 GGCCAGTTTCACA 37 38191 38210 4553 AAGCCCT 472472 1039 1058 GGGCCAGTTTCAC 0 38192 38211 4554 AAAGCCC 472473 1041 1060 CAGGGCCAGTTTC 34 38194 38213 4555 ACAAAGC 472474 1042 1061 GCAGGGCCAGTTT 63 38195 38214 4556 CACAAAG 472475 1089 1108 TTCATTCTCTCCA 66 39136 39155 4557 AAGGAGT 472476 1090 1109 CTTCATTCTCTCC 75 39137 39156 4558 AAAGGAG 472477 1092 1111 CACTTCATTCTCT 78 39139 39158 4559 CCAAAGG 472478 1093 1112 ACACTTCATTCTC 84 39140 39159 4560 TCCAAAG 472479 1094 1113 TACACTTCATTCT 62 39141 39160 4561 CTCCAAA 472480 1096 1115 TGTACACTTCATT 95 39143 39162 4562 CTCTCCA 472481 1097 1116 TTGTACACTTCAT 89 39144 39163 4563 TCTCTCC 472482 1099 1118 GCTTGTACACTTC 91 39146 39165 4564 ATTCTCT 472483 1100 1119 TGCTTGTACACTT 75 39147 39166 4565 CATTCTC 472484 1102 1121 CCTGCTTGTACAC 90 39149 39168 4566 TTCATTC 472485 1103 1122 ACCTGCTTGTACA 86 39150 39169 4567 CTTCATT 472486 1104 1123 CACCTGCTTGTAC 95 39151 39170 4568 ACTTCAT 472487 1106 1125 ATCACCTGCTTGT 90 39153 39172 4569 ACACTTC 472488 1107 1126 GATCACCTGCTTG 86 39154 39173 4570 TACACTT 472489 1109 1128 AAGATCACCTGCT

69 39156 39175 4571 TGTACAC 472490 1110 1129 GAAGATCACCTGC 63 39157 39176 4572 TTGTACA 472491 1112 1131 TCGAAGATCACCT 72 39159 39178 4573 GCTTGTA 472492 1113 1132 CTCGAAGATCACC 51 39160 39179 4574 TGCTTGT 472493 1115 1134 TCCTCGAAGATCA 71 39162 39181 4575 CCTGCTT 472494 1116 1135 CTCCTCGAAGATC 70 39163 39182 4576 ACCTGCT 472495 1118 1137 CCCTCCTCGAAGA 69 39165 39184 4577 TCACCTG 472496 1119 1138 GCCCTCCTCGAAG 87 39166 39185 4578 ATCACCT 472497 1121 1140 GAGCCCTCCTCGA 54 39168 39187 4579 AGATCAC 472498 1122 1141 GGAGCCCTCCTCG 77 39169 39188 4580 AAGATCA 472499 1124 1143 CAGGAGCCCTCCT 65 39171 39190 4581 CGAAGAT 472500 1125 1144 CCAGGAGCCCTCC 72 39172 39191 4582 TCGAAGA 472501 1127 1146 CCCCAGGAGCCCT 60 39174 39193 4583 CCTCGAA 472502 1128 1147 GCCCCAGGAGCCC 48 39175 39194 4584 TCCTCGA 472503 1130 1149 CGGCCCCAGGAGC 21 39177 39196 4585 CCTCCTC 472504 1131 1150 TCGGCCCCAGGAG 26 39178 39197 4586 CCCTCCT 472505 1132 1151 ATCGGCCCCAGGA 1 39179 39198 4587 GCCCTCC 472506 1134 1153 CCATCGGCCCCAG 70 39181 39200 4588 GAGCCCT 472507 1135 1154 CCCATCGGCCCCA 74 39182 39201 4589 GGAGCCC 472508 1137 1156 GACCCATCGGCCC 44 39184 39203 4590 CAGGAGC 472509 1138 1157 GGACCCATCGGCC 45 39185 39204 4591 CCAGGAG 472510 1158 1177 GTATTTCTGGAAC 71 39205 39224 4592 TTCTTCT 472511 1159 1178 TGTATTTCTGGAA 63 39206 39225 4593 CTTCTTC 472512 1161 1180 AATGTATTTCTGG 60 39208 39227 4594 AACTTCT 472513 1162 1181 CAATGTATTTCTG 36 39209 39228 4595 GAACTTC 472514 1164 1183 ACCAATGTATTTC 72 39211 39230 4596 TGGAACT 472515 1165 1184 AACCAATGTATTT 68 39212 39231 4597 CTGGAAC 472516 1166 1185 AAACCAATGTATT 50 39213 39232 4598 TCTGGAA 472517 1167 1186 GAAACCAATGTAT 65 39214 39233 4599 TTCTGGA 472518 1169 1188 GCGAAACCAATGT 76 39216 39235 4600 ATTTCTG 472519 1170 1189 GGCGAAACCAATG 75 39217 39236 4601 TATTTCT 472520 1209 1228 GTCGGAGGAGAAG 43 39256 39275 4602 AGGCCTC 472444 998 1017 GCATTCTTGCCAG 89 38151 38170 4526 GCATGGA 413433 n/a n/a GCCTGGACAAGTC 87 32431 32450 425 CTGCCCA

TABLE-US-00065 TABLE 65 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID ID ID NO: NO: NO: NO: 1 1 % 2 2 SEQ ISIS Start Stop inhi- Start Stop ID NO Site Site Sequence bition Site Site NO 472588 1558 1577 TTCTTAAAAAAGA 22 41529 41548 4603 CCTAACA 472589 1560 1579 CCTTCTTAAAAAA 70 41531 41550 4604 GACCTAA 472590 1561 1580 TCCTTCTTAAAAA 65 41532 41551 4605 AGACCTA 472591 1562 1581 TTCCTTCTTAAAA 58 41533 41552 4606 AAGACCT 472592 1564 1583 TTTTCCTTCTTAA 14 41535 41554 4607 AAAAGAC 472593 1565 1584 TTTTTCCTTCTTA 9 41536 41555 4608 AAAAAGA 472594 1566 1585 CTTTTTCCTTCTT 15 41537 41556 4609 AAAAAAG 472595 1568 1587 GACTTTTTCCTTC 36 41539 41558 4610 TTAAAAA 472596 1569 1588 TGACTTTTTCCTT 36 41540 41559 4611 CTTAAAA 472597 1570 1589 CTGACTTTTTCCT 75 41541 41560 4612 TCTTAAA 472598 1571 1590 ACTGACTTTTTCC 77 41542 41561 4613 TTCTTAA 472599 1615 1634 CACCACCTAGAAC 70 41586 41605 4614 AGGGCAA 472600 1617 1636 GCCACCACCTAGA 44 41588 41607 4615 ACAGGGC 472601 1618 1637 AGCCACCACCTAG 70 41589 41608 4616 AACAGGG 472602 1619 1638 TAGCCACCACCTA 46 41590 41609 4617 GAACAGG 472603 1621 1640 TTTAGCCACCACC 51 41592 41611 4618 TAGAACA 472604 1622 1641 ATTTAGCCACCAC 40 41593 41612 4619 CTAGAAC 472605 1623 1642 GATTTAGCCACCA 63 41594 41613 4620 CCTAGAA 472606 1624 1643 AGATTTAGCCACC 54 41595 41614 4621 ACCTAGA 472607 1626 1645 CCAGATTTAGCCA 74 41597 41616 4622 CCACCTA 472608 1627 1646 CCCAGATTTAGCC 88 41598 41617 4623 ACCACCT 472609 1628 1647 GCCCAGATTTAGC 78 41599 41618 4624 CACCACC 472610 1629 1648 GGCCCAGATTTAG 83 41600 41619 4625 CCACCAC 472611 1631 1650 TAGGCCCAGATTT 51 41602 41621 4626 AGCCACC 472612 1632 1651 TTAGGCCCAGATT 14 41603 41622 4627

TAGCCAC 472613 1633 1652 ATTAGGCCCAGAT 45 41604 41623 4628 TTAGCCA 472614 1634 1653 GATTAGGCCCAGA 44 41605 41624 4629 TTTAGCC 472615 1636 1655 CAGATTAGGCCCA 16 41607 41626 4630 GATTTAG 472616 1637 1656 CCAGATTAGGCCC 40 41608 41627 4631 AGATTTA 472617 1638 1657 CCCAGATTAGGCC 40 41609 41628 4632 CAGATTT 472618 1639 1658 ACCCAGATTAGGC 50 41610 41629 4633 CCAGATT 472619 1641 1660 CCACCCAGATTAG 71 41612 41631 4634 GCCCAGA 472620 1642 1661 GCCACCCAGATTA 73 41613 41632 4635 GGCCCAG 472621 1643 1662 AGCCACCCAGATT 65 41614 41633 4636 AGGCCCA 472622 1644 1663 GAGCCACCCAGAT 26 41615 41634 4637 TAGGCCC 472623 1646 1665 CTGAGCCACCCAG 23 41617 41636 4638 ATTAGGC 472624 1647 1666 GCTGAGCCACCCA 24 41618 41637 4639 GATTAGG 472625 1648 1667 AGCTGAGCCACCC 2 41619 41638 4640 AGATTAG 472626 1649 1668 TAGCTGAGCCACC 31 41620 41639 4641 CAGATTA 472627 1651 1670 GTTAGCTGAGCCA 48 41622 41641 4642 CCCAGAT 472628 1652 1671 GGTTAGCTGAGCC 57 41623 41642 4643 ACCCAGA 472629 1654 1673 GAGGTTAGCTGAG 66 41625 41644 4644 CCACCCA 472630 1655 1674 AGAGGTTAGCTGA 51 41626 41645 4645 GCCACCC 472631 1677 1696 GTCACTTCAGGAA 54 41648 41667 4646 GGGAAGA 472632 1678 1697 TGTCACTTCAGGA 58 41649 41668 4647 AGGGAAG 472633 1749 1768 AAAAGTGAATCAT 23 41720 41739 4648 CTAACTG 472634 1750 1769 AAAAAGTGAATCA 0 41721 41740 4649 TCTAACT 472635 1751 1770 CAAAAAGTGAATC 2 41722 41741 4650 ATCTAAC 472636 1752 1771 GCAAAAAGTGAAT 55 41723 41742 4651 CATCTAA 472637 1754 1773 GGGCAAAAAGTGA 79 41725 41744 4652 ATCATCT 472638 1790 1809 CTTGTATGAGAAG 58 41761 41780 4653 TGGCTTT 472639 1791 1810 GCTTGTATGAGAA 49 41762 41781 4654 GTGGCTT 472640 1793 1812 GGGCTTGTATGAG 67 41764 41783 4655 AAGTGGC 472641 1838 1857 CCTGCAGTTTCAG 64 41809 41828 4656 GACTAGA 472642 1839 1858 TCCTGCAGTTTCA 50 41810 41829 4657 GGACTAG 472643 1841 1860 GGTCCTGCAGTTT 11 41812 41831 4658 CAGGACT 472644 1842 1861 TGGTCCTGCAGTT 20 41813 41832 4659 TCAGGAC 472645 1843 1862 CTGGTCCTGCAGT 41 41814 41833 4660 TTCAGGA 472646 1845 1864 AACTGGTCCTGCA 50 41816 41835 4661 GTTTCAG 472647 1846 1865 AAACTGGTCCTGC 56 41817 41836 4662 AGTTTCA 472648 1848 1867 AGAAACTGGTCCT 40 41819 41838 4663 GCAGTTT 472649 1849 1868 GAGAAACTGGTCC 24 41820 41839 4664 TGCAGTT 472650 1850 1869 AGAGAAACTGGTC 29 41821 41840 4665 CTGCAGT 472651 1851 1870 CAGAGAAACTGGT 35 41822 41841 4666 CCTGCAG 472652 1853 1872 GGCAGAGAAACTG 84 41824 41843 4667 GTCCTGC 472653 1854 1873 TGGCAGAGAAACT 71 41825 41844 4668 GGTCCTG 472654 1855 1874 TTGGCAGAGAAAC 63 41826 41845 4669 TGGTCCT 472655 1856 1875 CTTGGCAGAGAAA 62 41827 41846 4670 CTGGTCC 472656 1858 1877 CCCTTGGCAGAGA 49 41829 41848 4671 AACTGGT 472657 1859 1878 CCCCTTGGCAGAG 66 41830 41849 4672 AAACTGG 472658 1861 1880 CTCCCCTTGGCAG 58 41832 41851 4673 AGAAACT 472659 1862 1881 CCTCCCCTTGGCA 54 41833 41852 4674 GAGAAAC 472660 1863 1882 TCCTCCCCTTGGC 49 41834 41853 4675 AGAGAAA 472661 1865 1884 ACTCCTCCCCTTG 47 41836 41855 4676 GCAGAGA 472662 1866 1885 AACTCCTCCCCTT 39 41837 41856 4677 GGCAGAG 472663 1867 1886 CAACTCCTCCCCT 0 41838 41857 4678 TGGCAGA 472664 1869 1888 TCCAACTCCTCCC 29 41840 41859 4679 CTTGGCA 413433 n/a n/a GCCTGGACAAGTC 85 32431 32450 425 CTGCCCA

TABLE-US-00066 TABLE 66 Inhibition of DGAT2 mRNA by 5-10-5 MOE gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ SEQ SEQ ID ID ID ID NO NO: NO: NO: 1 1 % 2 2 SEQ ISIS Start Stop inhi- Start Stop ID NO Site Site Sequence bition Site Site NO 472445 1000 1019 CTGCATTCTTGCC 53 38153 38172 4527 AGGCATG 472446 1001 1020 ACTGCATTCTTGC 60 38154 38173 4528 CAGGCAT 472447 1003 1022 TGACTGCATTCTT 52 38156 38175 4529 GCCAGGC 472448 1004 1023 GTGACTGCATTCT

60 38157 38176 4530 TGCCAGG 472449 1006 1025 GGGTGACTGCATT 0 38159 38178 4531 CTTGCCA 472450 1007 1026 AGGGTGACTGCAT 62 38160 38179 4532 TCTTGCC 472451 1008 1027 CAGGGTGACTGCA 43 38161 38180 4533 TTCTTGC 472452 1010 1029 CGCAGGGTGACTG 25 38163 38182 4534 CATTCTT 472453 1011 1030 CCGCAGGGTGACT 45 38164 38183 4535 GCATTCT 472454 1013 1032 TTCCGCAGGGTGA 16 38166 38185 4536 CTGCATT 472455 1014 1033 GTTCCGCAGGGTG 29 38167 38186 4537 ACTGCAT 472456 1015 1034 GGTTCCGCAGGGT 62 38168 38187 4538 GACTGCA 472457 1017 1036 GCGGTTCCGCAGG 33 38170 38189 4539 GTGACTG 472458 1018 1037 TGCGGTTCCGCAG 57 38171 38190 4540 GGTGACT 472459 1020 1039 CTTGCGGTTCCGC 39 38173 38192 4541 AGGGTGA 472460 1021 1040 CCTTGCGGTTCCG 25 38174 38193 4542 CAGGGTG 472461 1023 1042 GCCCTTGCGGTTC 9 38176 38195 4543 CGCAGGG 472462 1024 1043 AGCCCTTGCGGTT 32 38177 38196 4544 CCGCAGG 472463 1026 1045 AAAGCCCTTGCGG 46 38179 38198 4545 TTCCGCA 472464 1027 1046 CAAAGCCCTTGCG 48 38180 38199 4546 GTTCCGC 472465 1029 1048 CACAAAGCCCTTG 8 38182 38201 4547 CGGTTCC 472466 1030 1049 TCACAAAGCCCTT 0 38183 38202 4548 GCGGTTC 472467 1032 1051 TTTCACAAAGCCC 0 38185 38204 4549 TTGCGGT 472468 1033 1052 GTTTCACAAAGCC 18 38186 38205 4550 CTTGCGG 472469 1035 1054 CAGTTTCACAAAG 0 38188 38207 4551 CCCTTGC 472470 1036 1055 CCAGTTTCACAAA 23 38189 38208 4552 GCCCTTG 472471 1038 1057 GGCCAGTTTCACA 2 38191 38210 4553 AAGCCCT 472472 1039 1058 GGGCCAGTTTCAC 0 38192 38211 4554 AAAGCCC 472473 1041 1060 CAGGGCCAGTTTC 0 38194 38213 4555 ACAAAGC 472474 1042 1061 GCAGGGCCAGTTT 10 38195 38214 4556 CACAAAG 472475 1089 1108 TTCATTCTCTCCA 47 39136 39155 4557 AAGGAGT 472476 1090 1109 CTTCATTCTCTCC 65 39137 39156 4558 AAAGGAG 472477 1092 1111 CACTTCATTCTCT 51 39139 39158 4559 CCAAAGG 472478 1093 1112 ACACTTCATTCTC 68 39140 39159 4560 TCCAAAG 472479 1094 1113 TACACTTCATTCT 41 39141 39160 4561 CTCCAAA 472480 1096 1115 TGTACACTTCATT 85 39143 39162 4562 CTCTCCA 472481 1097 1116 TTGTACACTTCAT 72 39144 39163 4563 TCTCTCC 472482 1099 1118 GCTTGTACACTTC 80 39146 39165 4564 ATTCTCT 472483 1100 1119 TGCTTGTACACTT 64 39147 39166 4565 CATTCTC 472484 1102 1121 CCTGCTTGTACAC 75 39149 39168 4566 TTCATTC 472485 1103 1122 ACCTGCTTGTACA 67 39150 39169 4567 CTTCATT 472486 1104 1123 CACCTGCTTGTAC 88 39151 39170 4568 ACTTCAT 472487 1106 1125 ATCACCTGCTTGT 59 39153 39172 4569 ACACTTC 472488 1107 1126 GATCACCTGCTTG 70 39154 39173 4570 TACACTT 472489 1109 1128 AAGATCACCTGCT 28 39156 39175 4571 TGTACAC 472490 1110 1129 GAAGATCACCTGC 53 39157 39176 4572 TTGTACA 472491 1112 1131 TCGAAGATCACCT 52 39159 39178 4573 GCTTGTA 472492 1113 1132 CTCGAAGATCACC 49 39160 39179 4574 TGCTTGT 472493 1115 1134 TCCTCGAAGATCA 46 39162 39181 4575 CCTGCTT 472494 1116 1135 CTCCTCGAAGATC 55 39163 39182 4576 ACCTGCT 472495 1118 1137 CCCTCCTCGAAGA 54 39165 39184 4577 TCACCTG 472496 1119 1138 GCCCTCCTCGAAG 74 39166 39185 4578 ATCACCT 472497 1121 1140 GAGCCCTCCTCGA 54 39168 39187 4579 AGATCAC 472498 1122 1141 GGAGCCCTCCTCG 68 39169 39188 4580 AAGATCA 472499 1124 1143 CAGGAGCCCTCCT 37 39171 39190 4581 CGAAGAT 472500 1125 1144 CCAGGAGCCCTCC 41 39172 39191 4582 TCGAAGA 472501 1127 1146 CCCCAGGAGCCCT 34 39174 39193 4583 CCTCGAA 472502 1128 1147 GCCCCAGGAGCCC 17 39175 39194 4584 TCCTCGA 472503 1130 1149 CGGCCCCAGGAGC 0 39177 39196 4585 CCTCCTC 472504 1131 1150 TCGGCCCCAGGAG 16 39178 39197 4586 CCCTCCT 472505 1132 1151 ATCGGCCCCAGGA 0 39179 39198 4587 GCCCTCC 472506 1134 1153 CCATCGGCCCCAG 64 39181 39200 4588 GAGCCCT 472507 1135 1154 CCCATCGGCCCCA 55 39182 39201 4589 GGAGCCC 472508 1137 1156 GACCCATCGGCCC 34 39184 39203 4590 CAGGAGC 472509 1138 1157 GGACCCATCGGCC 50 39185 39204 4591 CCAGGAG 472510 1158 1177 GTATTTCTGGAAC

64 39205 39224 4592 TTCTTCT 472511 1159 1178 TGTATTTCTGGAA 49 39206 39225 4593 CTTCTTC 472512 1161 1180 AATGTATTTCTGG 40 39208 39227 4594 AACTTCT 472513 1162 1181 CAATGTATTTCTG 4 39209 39228 4595 GAACTTC 472514 1164 1183 ACCAATGTATTTC 41 39211 39230 4596 TGGAACT 472515 1165 1184 AACCAATGTATTT 41 39212 39231 4597 CTGGAAC 472516 1166 1185 AAACCAATGTATT 34 39213 39232 4598 TCTGGAA 472517 1167 1186 GAAACCAATGTAT 51 39214 39233 4599 TTCTGGA 472518 1169 1188 GCGAAACCAATGT 54 39216 39235 4600 ATTTCTG 472519 1170 1189 GGCGAAACCAATG 59 39217 39236 4601 TATTTCT 472520 1209 1228 GTCGGAGGAGAAG 17 39256 39275 4602 AGGCCTC 472444 998 1017 GCATTCTTGCCAG 77 38151 38170 4526 GCATGGA 413433 n/a n/a GCCTGGACAAGTC 76 32431 32450 425 CTGCCCA

Example 6: Dose-Dependent Antisense Inhibition of Human DGAT2 in HepG2 Cells by MOE Gapmers

[0462] Gapmers from the studies described in the Examples above exhibiting significant in vitro inhibition of DGAT2 mRNA were selected and tested at various doses in HepG2 cells. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. Previously disclosed oligonucleotides, ISIS 21316, ISIS 217317, ISIS 217328, ISIS 369185, ISIS 366714, ISIS 366730, ISIS 366746, and ISIS 369241 from earlier published application, WO 2005/019418, were included in the assay. The results for each experiment are presented in separate tables shown below. Cells were plated at a density of 10,000 cells per well and transfected using Lipofectin reagent with 6.25 nM, 12.5 nM, 25.0 nM, 50 nM, 100 nM, or 200.0 nM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2367 was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited. ‘n.d.’ indicates that the inhibition levels were not recorded.

[0463] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells. The results also demonstrate that several newly designed oligonucleotides had greater efficacy than the previously disclosed oligonucleotides.

TABLE-US-00067 TABLE 67

ISIS No	6.25 nM	12.5 nM	25.0 nM	50.0 nM	100.0 nM	200.0 nM	IC.sub.50 (nM)
217316	7	7	27	47	68	63	55
217317	3	12	32	50	75	84	44
369185	17	12	34	63	78	81	42
381726	4	30	44	56	76	83	37
411874	15	31	27	54	66	66	40
411899	5	22	31	57	74	87	39
411901	0	6	24	46	66	65	58
411905	6	30	29	58	73	76	34
411912	8	24	31	56	79	89	35
411941	11	22	30	59	77	82	35

TABLE-US-00068 TABLE 68

ISIS No	6.25 nM	12.5 nM	25.0 nM	50.0 nM	100.0 nM	200.0 nM	IC.sub.50 (nM)
366714	7	8	42	67	79	82	33
366730	0	18	33	72	84	91	43
411944	2	14	46	65	82	93	30
411945	14	21	51	68	86	n.d.	24
411950	6	5	28	66	84	n.d.	29
413176	7	16	24	58	78	88	38
413192	0	6	30	62	80	80	57
413200	2	18	20	60	78	77	36
413213	9	12	34	55	84	90	45
413226	3	22	13	36	72	87	41

TABLE-US-00069 TABLE 69

ISIS No	6.25 nM	12.5 nM	25.0 nM	50.0 nM	100.0 nM	200.0 nM	IC.sub.50 (nM)
366746	5	7	21	32	62	77	66
369241	0	4	10	35	56	63	106
413198	10	16	24	53	72	73	51
413214	1	17	43	61	81	77	34
413232	8	23	33	60	78	82	35
413236	0	12	3	52	71	72	47
413253	6	7	27	59	72	75	64
413258	0	12	27	46	78	77	62
413266	0	0	21	48	76	75	52
413284	4	0	22	45	57	65	78

TABLE-US-00070 TABLE 70

ISIS No	6.25 nM	12.5 nM	25.0 nM	50.0 nM	100.0 nM	200.0 nM	IC.sub.50 (nM)
217328	4	15	31	50	64	72	62
413243	0	10	28	50	69	77	66
413351	4	16	16	39	56	69	51
413356	16	12	23	51	63	69	34
413364	0	0	21	41	58	53	52
413399	0	3	23	35	60	58	78
413413	1	0	30	50	71	77	64
413422	0	0	8	41	63	70	62
413433	5	5	20	55	77	82	47
413441	2	5	26	50	70	79	106
413446	0	0	10	41	63	78	35

TABLE-US-00071 TABLE 71 6.25 12.5 25.0 50.0 100.0 200.0 IC.sub.50 ISIS No nM nM nM nM nM nM (nM) 217317 0 0 11 34 64 84 77 366730 0 15 33 61 87 91 45 381726 1 11 31 51 71 83 48 411899 6 0 25 39 70 89 55 411912 0 0 13 58 77 87 40 411944 7 21 32 67 76 87 42 413176 0 12 20 54 89 88 46 413232 16 10 30 60 79 90 31 413433 15 17 37 71 84 87 24 413446 0 0 44 73 77 86 47

Example 7: Dose-Dependent Antisense Inhibition of Human DGAT2 in HepG2 Cells by MOE Gapmers

[0464] Gapmers from the studies described in the Examples above exhibiting significant in vitro inhibition of DGAT2 mRNA were selected and tested at various doses in HepG2 cells. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. Previously disclosed oligonucleotides from earlier published application, WO 2005/019418, were included in the assay. The results for each experiment are presented in separate tables shown below.

Study 1

[0465] Cells were plated at a density of 10,000 cells per well and transfected using Lipofectin reagent with 18.75 nM, 37.5 nM, 75.0 nM, or 150.0 nM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.

[0466] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE-US-00072 TABLE 72 18.75 37.5 75.0 150.0 IC.sub.50 ISIS No Motif nM nM nM nM (nM) 217328 5-10-5 27 44 62 73 47 366730 5-10-5 27 36 61 77 48 411944 5-10-5 19 38 48 69 67 411945 5-10-5 14 26 45 61 93 413253 5-10-5 24 34 50 68 73 413433 5-10-5 28 39 56 68 58 423440 5-10-5 18 39 53 70 67 423441 5-10-5 23 36 54 66 63 423444 5-10-5 15 32 53 69 71 423447 5-10-5 24 36 55 71 60 423463 5-10-5 23 39 59 73 56 423489 3-14-3 21 32 55 73 60 423490 3-14-3 27 32 51 60 76 423498 3-14-3 19 38 56 77 59 423499 5-10-5 23 37 59 77 53 423523 3-14-3 25 41 60 73 56 423524 3-14-3 24 37 62 75 54 423601 2-13-5 28 38 58 75 51

TABLE-US-00073 TABLE 73 18.75 37.5 75.0 150.0 IC.sub.50 ISIS No Motif nM nM nM nM (nM) 217317 5-10-5 4 23 39 61 104 217328 5-10-5 20 33 54 67 73 411899 5-10-5 11 34 51 66 74 413214 5-10-5 8 32 52 68 73 413232 5-10-5 17 26 50 70 71 423452 5-10-5 12 26 45 65 88 423453 5-10-5 15 33 46 67 76 423458 5-10-5 14 31 49 70 74 423459 5-10-5 17 29 61 73 64 423464 5-10-5 19 33 52 64 74 423507 3-14-3 15 32 52 68 73 423508 3-14-3 14 37 48 69 73 423526 3-14-3 13 23 41 58 109 423527 3-14-3 13 28 51 58 102 423565 2-13-5 17 27 48 69 80 423567 2-13-5 0 29 53 74 79 423585 2-13-5 12 25 42 64 94 423595 2-13-5 12 26 47 69 81 423606 2-13-5 13 27 55 71 68

[0467] The same gapmers were also tested at various doses in HepG2 cells using electroporation. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 2.5 μM, 5.0 μM, 10.0 μM, or 20.0 μM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.

[0468] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE-US-00074 TABLE 74 2.5 5.0 10.0 20.0 IC.sub.50 ISIS No Motif μM μM μM μM (μM)

217328 5-10-5 55 70 85 94 2 366730 5-10-5 38 51 78 89 5 411944 5-10-5 30 32 48 73 10 411945 5-10-5 31 26 48 69 11 413253 5-10-5 39 51 77 84 4 413433 5-10-5 72 81 82 85 <2.5 423440 5-10-5 30 54 65 91 5 423441 5-10-5 41 47 60 90 6 423444 5-10-5 19 46 70 81 6 423447 5-10-5 25 49 76 79 5 423463 5-10-5 59 79 85 86 2 423489 3-14-3 33 56 72 92 5 423490 3-14-3 17 25 53 82 9 423498 3-14-3 30 41 59 88 7 423499 5-10-5 27 26 63 78 10 423523 3-14-3 45 48 81 89 5 423524 3-14-3 35 59 84 92 4 423601 2-13-5 44 55 87 90 4

TABLE-US-00075 TABLE 75 18.75 37.5 75.0 150.0 IC.sub.50 ISIS No Motif nM nM nM nM (nM) 217317 5-10-5 16 33 63 65 7 217328 5-10-5 32 56 79 91 4 411899 5-10-5 28 38 74 87 5 413214 5-10-5 37 46 65 91 6 413232 5-10-5 18 20 43 78 11 423452 5-10-5 14 47 73 85 6 423453 5-10-5 52 70 83 93 2 423458 5-10-5 28 31 65 87 8 423459 5-10-5 24 33 57 88 8 423464 5-10-5 68 81 86 88 1 423507 3-14-3 29 42 71 89 5 423508 3-14-3 16 38 53 83 7 423526 3-14-3 48 73 81 90 3 423527 3-14-3 27 51 68 85 6 423565 2-13-5 13 35 66 84 7 423567 2-13-5 27 42 61 91 7 423585 2-13-5 24 28 67 91 7 423595 2-13-5 12 25 50 78 10 423606 2-13-5 42 59 74 85 3

Study 2

[0469] Cells were plated at a density of 35,000 cells per well and transfected using electroporation with 0.22 μM, 0.67 μM, 2.0 μM, 6.0 μM, or 18.0 μM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2977_MGB (forward sequence ACTGGGCGGGCTTCATATT, designated herein as SEQ ID NO: 10; reverse sequence CCCGGAGTAGGCGGCTAT, designated herein as SEQ ID NO: 11; probe sequence AGCCATGAAGACCC, designated herein as SEQ ID NO: 12) was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.

[0470] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE-US-00076 TABLE 76 0.22 0.67 2.0 6.0 18.0 IC.sub.50 ISIS No Motif μM μM μM μM μM (μM) 217328 5-10-5 0 17 52 83 96 2.1 366730 5-10-5 0 0 27 72 93 3.9 413433 5-10-5 0 10 58 84 93 2.2 423453 5-10-5 14 21 53 86 94 1.9 423459 5-10-5 0 18 36 68 92 3.0 423463 5-10-5 16 15 50 76 80 2.7 423464 5-10-5 0 29 56 86 92 1.6 423489 3-14-3 0 0 37 63 91 3.8 423498 3-14-3 0 0 21 56 82 5.3 423499 3-14-3 0 6 26 62 88 4.1 423523 3-14-3 0 0 34 69 95 4.2 423524 3-14-3 0 6 15 62 90 4.4 423526 3-14-3 5 7 45 87 94 2.6 423601 2-13-5 0 0 31 65 89 4.3 423606 2-13-5 58 19 43 72 94 2.6

Study 3

[0471] Cells were plated at a density of 10,000 cells per well and transfected using Lipofectin reagent with 12.5 nM, 25.0 nM, 50.0 nM, 100.0 nM, or 200.0 nM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.

[0472] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE-US-00077 TABLE 77 0.22 0.67 2.0 6.0 18.0 IC.sub.50 ISIS No Motif μM μM μM μM μM (μM) 217328 5-10-5 25 31 58 71 85 38 366730 5-10-5 21 34 53 74 90 44 413433 5-10-5 23 40 63 75 86 37 423453 5-10-5 0 2 33 61 72 90 423459 5-10-5 16 22 44 65 79 64 423463 5-10-5 6 24 46 63 82 62 423464 5-10-5 20 28 45 64 79 60 423489 3-14-3 15 28 57 75 84 42 423498 3-14-3 8 21

52 71 85 49 423499 3-14-3 13 23 43 65 84 62 423523 3-14-3 25 38 58 72 86 38 423524 3-14-3 16 32 56 66 83 52 423526 3-14-3 28 33 51 54 68 71 423601 2-13-5 13 30 54 71 84 49 423606 2-13-5 12 36 52 69 80 47

Study 4

[0473] Cells were plated at a density of 10,000 cells per well and transfected using Lipofectin reagent with 12.5 nM, 25.0 nM, 50.0 nM, 100.0 nM, or 200.0 nM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.

[0474] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE-US-00078 TABLE 78 0.22 0.67 2.0 6.0 18.0 IC.sub.50 ISIS No Motif μM μM μM μM μM (μM) 217328 5-10-5 42 54 60 88 95 2 366730 5-10-5 26 44 61 85 95 3 413433 5-10-5 55 69 76 83 87 <1.25 423453 5-10-5 40 62 78 91 95 2 423459 5-10-5 19 25 47 79 92 5 423463 5-10-5 51 52 83 86 90 <1.25 423464 5-10-5 42 67 82 90 89 2 423489 3-14-3 8 19 40 77 96 5 423498 3-14-3 5 1 20 51 89 9 423499 3-14-3 23 24 35 71 95 5 423523 3-14-3 19 42 67 87 93 3 423524 3-14-3 38 53 71 90 95 2 423526 3-14-3 33 59 78 91 93 2 423601 2-13-5 22 45 72 88 93 3 423606 2-13-5 30 46 64 83 86 3

Study 5

[0475] Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 12.5 μM, 25.0 μM, 50.0 μM, 100.0 μM, or 200.0 μM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB (forward sequence AACTGGCCCTGCGTCATG, designated herein as SEQ ID NO: 7; reverse sequence CTTGTACACTTCATTCTCTCCAAAGG; designated herein as SEQ ID NO: 8; probe sequence CTGACCTGGTTCCC, designated herein as SEQ ID NO: 9) was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.

[0476] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE-US-00079 TABLE 79 12.5 25 50 100 200 IC.sub.50 ISIS No Motif μM μM μM μM μM (μM) 217316 5-10-5 0 0 17 43 76 10.9 217318 5-10-5 0 0 10 48 45 18.4 217328 5-10-5 12 22 40 69 82 6.0 217376 5-10-5 0 0 51 50 70 9.8 381728 5-10-5 0 3 17 37 58 15.8 411896 5-10-5 5 16 52 73 87 5.5 411899 5-10-5 0 0 15 42 54 15.7 411900 5-10-5 0 0 18 38 76 11.5 411950 5-10-5 5 13 21 49 70 10.7 411951 5-10-5 0 10 31 52 76 8.9 413433 5-10-5 0 37 58 75 82 5.6 423453 5-10-5 16 21 41 73 86 5.5 423463 5-10-5 26 33 62 81 85 3.5 423464 5-10-5 21 43 58 82 84 3.4 423606 2-13-5 30 43 53 76 80 3.6

Study 6

[0477] Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.625 μM, 1.25 μM, 2.5 μM, 5.0 μM, 10 μM, or 20 μM, concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB was used to measure mRNA levels.

DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.
[0478] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.
TABLE-US-00080 TABLE 80 0.625 1.25 2.5 5.0 10.0 20.0 IC.sub.50 ISIS No Motif μM μM μM μM μM μM (μM) 217317 5-10-5 11 18 39 59 79 94 3.5 217328 5-10-5 13 45 58 87 84 95 1.9 217376 5-10-5 10 9 4 28 12 11 >20 217376 5-10-5 18 15 12 23 9 0 >20 366730 5-10-5 14 23 55 74 86 97 2.6 411897 5-10-5 30 28 28 57 73 94 3.3 411901 5-10-5 19 22 30 58 76 93 4.0 411902 5-10-5 11 6 19 42 72 83 5.6 411947 5-10-5 14 14 10 35 66 87 7.1 411948 5-10-5 25 21 34 50 57 80 5.4 411950 5-10-5 7 24 41 56 74 83 3.9 411950 5-10-5 13 14 34 46 79 88 4.6 411951 5-10-5 28 18 22 43 83 92 5.4 411955 5-10-5 11 26 52 71 96 97 2.5 411958 5-10-5 19 17 21 60 81 92 4.7 413433 5-10-5 39 62 82 87 88 89 0.9 423463 5-10-5 15 36 76 87 92 90 1.7 423463 5-10-5 31 64 85 90 91 89 1.0 423489 3-14-3 21 17 39 73 89 98 3.2 423606 2-13-5 18 36 50 85 87 92 2.2
Example 8: Dose-Dependent Antisense Inhibition of Human DGAT2 in HepG2 Cells by MOE Gapmers
[0479] Gapmers from the studies described in the Examples above exhibiting significant in vitro inhibition of DGAT2 mRNA were selected and tested at various doses in HepG2 cells. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. The results for each experiment are presented in separate tables shown below. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.625 μM, 1.25 μM, 2.50 μM, 5.00 μM, or 10.00 μM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.
[0480] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.
TABLE-US-00081 TABLE 81 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 483817 47 71 82 89 90 <0.6 483825 40 56 78 85 85 0.8 483828 45 69 75 70 76 <0.6 483832 42 59 67 78 72 0.8 483834 26 38 64 86 90 1.6 483852 47 60 69 81 85 0.7 483866 76 77 82 86 86 <0.6 483869 39 69 74 72 80 0.6 483873 46 71 78 82 87 <0.6
TABLE-US-00082 TABLE 82 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 483811 29 34 58 87 87 1.7 483816 21 14 34 71 87 3.0 483831 34 37 55 81 83 1.7 483835 28 47 56 82 89 1.6 483840 4 17 49 75 81 2.9 483870 13 31 52 84 88 2.2 483889 21 0 56 81 85 2.7 483895 43 21 74 88 92 2.0 483898 46 47 81 89 86 0.8 483908 29 16 73 84 88 1.9 483910 31 39 73 79 76 1.5 483923 24 0 62 81 89 3.4 483952 63 67 86 91 89 <0.6
TABLE-US-00083 TABLE 83 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 483968 39 55 76 89 88 0.9 483969 29 57 70 82 89 1.2 483970 16 48 62 71 85 1.9 483972 17 52 69 78 78 1.7 483984 53 74 85 83 78 <0.6 483986 15 37 59 71 83 2.2 483987 9 0 76 78 91 1.5 483993 42 58 69 69 78 0.8 483996 20 46 53 63 74 2.3 483997 20 61 72 82 82 1.3 484004 27 57 65 74 83 1.4 484006 22 47 65 72 86 1.7 484010 27 0 51 78 86 2.0 484017 28 51 66 79 88 1.4
TABLE-US-00084 TABLE 84 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No UM UM μM μM μM (μM) 413433 11 53 74 85 83 1.6 484012 19 36 49 76 88 2.2 484020 26 49 66 70 78 1.6 484041 12 45 51 68 80 2.4 484049 35 62 68 82 81 0.9 484094 20 37 52 62 74 2.7 484099 32 44 77 84 89 1.3 484111 22 50 65 54 80 2.0 484114 29 47 69 68 72 1.6 484118 10 42 60 74 84 2.1 484182 0 57 59

79 74 2.3
TABLE-US-00085 TABLE 85 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 33 46 65 81 87 1.4 484109 17 1 46 65 83 3.3 484138 9 15 17 28 53 >10 484139 14 21 60 75 90 2.3 484152 25 46 75 89 91 1.4 484175 10 15 21 35 43 >10 484183 3 15 20 34 55 >10 484203 23 35 54 70 75 2.3 484209 21 36 50 74 88 2.2 484215 42 45 59 84 91 1.2 484217 23 36 56 82 93 1.9 484231 34 0 72 79 82 1 423463 35 53 73 86 87 1.1
TABLE-US-00086 TABLE 86 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 37 55 72 84 88 1 423464 41 62 79 92 90 0.7 484268 34 57 73 81 84 1 484269 24 47 64 67 75 1.8 484270 29 50 65 88 91 1.3 484271 43 66 81 88 84 0.6 484273 42 62 73 79 78 0.7 484283 26 0 61 85 93 1.6 484284 27 38 54 76 81 1.9 484292 31 31 53 81 91 1.8 484293 26 36 61 78 87 1.8 484327 24 48 48 67 79 2.1 484342 21 34 48 67 88 2.4 484387 21 0 21 40 58 9.6 484396 15 13 18 24 44 >10
TABLE-US-00087 TABLE 87 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 30 57 80 81 86 1.1 484345 17 28 68 60 79 2.4 484354 20 28 75 60 79 2.2 484366 25 33 37 62 82 2.8 484368 16 39 57 73 89 2.1 484369 22 18 57 27 35 >10 484372 3 28 37 69 76 3.3 484376 12 0 52 64 67 3.1 484379 0 19 21 37 75 5.6 484380 0 26 34 47 71 4.7 484395 7 41 61 29 46 >10 484397 0 7 52 16 34 >10 484406 17 6 53 39 53 7.5 484408 4 0 16 24 38 >10 484409 0 6 0 41 59 9.0
TABLE-US-00088 TABLE 88 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 22 51 68 72 88 1.6 472316 15 45 45 82 89 2.1 472317 34 30 43 77 91 2.0 472318 22 38 52 68 85 2.2 472444 22 27 41 39 74 4.4 472447 0 18 16 18 69 9.0 472456 10 10 48 51 68 4.3 472480 7 0 41 59 87 3.0 472481 29 29 37 62 84 2.7 472482 16 29 42 67 85 2.7 472484 23 19 51 72 87 2.4 472486 28 40 59 83 90 1.7 472488 27 31 36 53 64 4.4 472496 0 0 37 51 76 3.8 484410 0 40 19 24 39 >10
TABLE-US-00089 TABLE 89 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 37 51 70 84 88 1.1 472548 15 30 25 54 78 3.9 472552 15 28 33 51 66 4.7 472570 29 46 56 83 90 1.6 472571 29 34 53 70 86 2 411955 0 41 46 68 85 2.2 472608 28 26 56 59 72 2.7 472610 18 0 51 70 79 2.6 472637 24 37 39 53 72 3.4 472652 18 43 55 71 81 2.1 472725 35 46 61 77 86 1.4 472733 39 34 41 63 74 2.5 472738 38 53 65 81 85 1.1 472793 0 0 51 67 75 2 472811 0 44 65 78 77 1.4
TABLE-US-00090 TABLE 90 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 0 47 72 84 88 2 472316 21 42 69 86 94 2 484152 20 57 79 90 93 1 484215 33 39 70 91 95 1 484217 19 18 49 89 94 2 484231 4 52 76 83 91 2 423463 33 54 79 91 91 1 423464 0 48 78 83 88 3 484268 0 3 0 13 50 >10 484270 7 36 49 80 83 2 484271 16 59 76 86 91 1 484283 10 35 56 87 92 2 484369 25 1 0 5 33 >10
TABLE-US-00091 TABLE 91 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 28 60 69 84 92 1 483811 0 19 53 77 85 3 483817 47 69 85 88 94 <0.6 483825 18 50 80 82 87 1 483834 16 37 72 89 87 2 483835 1 29 56 80 85 2 483866 55 79 88 83 90 <0.6 483873 27 75 87 86 91 1 483898 49 72 82 89 89 <0.6 483908 35 45 76 89 88 1 483913 0 42 76 86 89 2
TABLE-US-00092 TABLE 92 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 47 68 84 90 88 <0.6 423453 36 66 81 95 96 0.8 472158 56 51 75 91 96 0.6 472725 43 59 63 78 80 0.8 472738 38 58 65 77 81 1 483895 35 53 76 92 93 1.0 483923 26 46 76 88 92 1.3 483952 60 82 88 93 92 <0.6 483968 46 70 82 91 93 <0.6 483984 39 67 87 88 88 0.6 483987 40 60 75 90 95 0.9 483997 50 63 73 90 92 0.6 484099 56 75 88 93 95 <0.6
TABLE-US-00093 TABLE 93 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 26 53 75 86 91 1.3 495429 8 42 68 86 92 1.9 495449 19 53 74 87 86 1.4 495450 30 58 74 87 93 1.1 495470 0 36 59 82 92 2.2 495481 15 39 75 84 86 1.7 495484 19 50 73 87 87 1.5 495488 5 32 45 67 86 2.8 495489 13 36 54 81 91 2.1 495490 20 42 66 83 88 1.7 495492 5 38 64 81 90 2.1 495495 29 32 62 78 90 1.8 495496 21 44 62 79 92 1.7 495497 20 42 59 69 88 2.0 495498 21 54 68 81 90 1.5

TABLE-US-00094 TABLE 94 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 35 42 73 86 90 1.3 495442 6 29 43 68 80 3.0 495451 21 31 51 76 87 2.2 495453 21 34 64 80 84 1.9 495454 23 33 53 80 86 2.1 495463 4 28 53 74 85 2.6 495464 21 41 65 84 84 1.7 495466 12 35 57 76 84 2.2 495467 17 28 44 69 89 2.6 495469 17 24 53 81 83 2.4 495471 29 43 58 83 84 1.6 495472 4 38 63 78 80 2.3 495482 7 32 52 76 91 2.4 495485 8 29 55 76 76 2.6 495491 25 42 64 85 92 1.6

TABLE-US-00095 TABLE 95 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 40 48 75 86 86 1 495430 0 5 11 47 67 6.2 495441 17 17 55 77 87 2.5 495443 28 51 59 74 84 1.6 495448 13 28 32 61 79 3.4 495452 13 51 47 71 78 2.3 495462 46 47 51 76 84 1.2 495465 2 20 35 68 78 3.5 495468 22 20 46 56 84 3 495473 11 24 48 69 75 3 495479 19 29 54 76 85 2.2 495483 15 9 47 77 88 2.7 495516 49 76 89 90 93 <0.6 495562 20 55 70 88 92 1.4 495576 51 69 88 93 94 <0.6

TABLE-US-00096 TABLE 96 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 34 51 71 86 89 1.1 495505 34 58 80 87 91 1.0 495517 19 55 78 85 91 1.3 495518 24 51 72 87 90 1.4 495519 33 46 70 86 90 1.3 495520 63 84 89 90 91 <0.6 495522 45 32 54 82 89 1.5 495523 14 46 66 86 91 1.7 495524 12 39 60 84 93 1.9 495554 35 64 75 89 90 0.9 495555 42 49 72 87 91 1 495563 35 41 66 84 91 1.4 495571 21 41 67 87 93 1.6 495575 33 43 62 82 84 1.5 495577 33 56 71 85 91 1.1

TABLE-US-00097 TABLE 97 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 39 52 66 81 87 1.1 495570 27 50 55 78 89 1.6 495650 0 38 58 65 85 2.7 495561 24 54 71 87 92 1.3 495618 0 35 50 79 87 2.5 495540 11 35 48 74 88 2.4 495649 0 24 43 76 82 3.1 495515 22 45 69 83 91 1.6 495599 25 23 42 73 86 2.5 495620 27 54 63 85 91 1.4 495606 16 27 56 80 90 2.2 495604 31 48 77 83 90 1.2 495609 52 66 80 84 90 <0.6 495619 29 42 63 73 90 1.6 495621 0 28 31 69 84 3.3

TABLE-US-00098 TABLE 98 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 19 24 66 81 88 2.1 495591 7 24 43 66 82 3.0 495744 25 41 51 73 84 2.0 495685 34 58 73 84 83 1.0 495738 26 41 61 81 83 1.7 495707 33 46 61 76 72 1.5 495837 30 33 67 90 94 1.6 495752 14 30 68 86 91 1.9 495839 15 67 83 90 91 1.2 495736 28 41 65 83 85 1.6 495840 39 59 78 92 93 0.8 495753 43 48 69 87 97 1.0 495878 34 62 78 92 96 0.9 495749 21 23 52 84 89 2.2 495756 24 29 58 80 89 2.0

TABLE-US-00099 TABLE 99 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 28 60 70 83 84 1.1 495853 39 51 78 92 96 1 495857 41 49 67 87 91 1.1 495829 41 68 81 90 90 0.6 495818 12 38 57 77 93 2.1 495842 40 67 80 88 91 0.7 495849 37 53 66 79 82 1.1 495836 16 54 63 80 88 1.6 495875 22 58 63 80 93 1.4 495841 43 66 76 87 90 0.6 495819 13 38 54 74 90 2.2 495825 36 52 69 88 93 1.1 495832 20 50 65 78 85 1.6 495809 16 37 46 78 85 2.3 495876 19 39 65 86 95 1.7

TABLE-US-00100 TABLE 100 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 46 61 82 86 86 0.6 496037 23 38 57 74 93 1.9 496038 10 31 42 67 93 2.6 496039 9 19 35 67 88 3.1 496040 4 38 45 70 88 2.6 496041 21 30 50 84 95 2.1 496043 2 26 58 80 87 2.4 495826 7 44 72 82 85 1.9 495868 27 38 75 83 85 1.5 495901 18 46 69 79 83 1.7 495902 20 48 66 82 85 1.6 495955 3 3 54 66 75 3.5 495992 26 43 74 86 92 1.4 496100 20 30 56 71 85 2.3 496104 37 35 48 72 81 1.9

TABLE-US-00101 TABLE 101 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 33 49 70 75 90 1.3 501838 32 52 67 73 71 1.3 501849 39 63 76 85 93 0.8 501850 29 59 72 88 92 1.1 501852 43 63 75 80 87 0.7 501855 57 70 87 90 92 <0.6 501861 39 63 80 90 93 0.8 501871 28 53 76 84 93 1.2 501883 33 45 62 73 82 1.5 501884 41 59 75 86 92 0.8 501886 42 63 71 78 84 0.7 501890 32 49 61 72 79 1.5 501932 38 49 68 80 85 1.1 501944 30 43 69 80 87 1.4 501950 32 49 70 72 91 1.3

TABLE-US-00102 TABLE 102 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 35 53 69 86 91 1.1 501903 32 39 62 80 90 1.6 501916 29 42 61 78 88 1.6 501933 37

51 66 75 77 1.2 501934 26 41 52 69 82 2 501946 15 34 57 76 90 2.1 501947 35 54 74 87 93 1.1 501951 36 54 75 85 93 1 501957 27 41 67 76 86 1.7 501959 40 52 72 81 87 1 501960 35 52 67 82 90 1.2 501966 34 42 64 75 84 1.5 501968 29 49 73 84 91 1.3 501977 19 28 49 74 88 2.3 502040 24 38 54 73 88 2

TABLE-US-00103 TABLE 103 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 37 53 72 83 39 1.1 501981 21 43 62 79 90 1.7 501982 28 53 64 70 82 1.5 501983 29 41 57 75 84 1.8 502013 24 42 60 74 80 1.9 502015 19 39 57 63 71 2.5 502025 22 48 59 78 89 1.7 502026 21 39 56 77 86 2 502037 15 13 38 56 87 3.4 502038 3 14 52 55 77 3.5 502045 45 47 63 75 85 1.1 502046 21 37 50 71 81 2.3 502056 44 67 82 86 41 0.6 502083 37 55 76 89 93 1 502119 37 60 77 88 92 0.9

TABLE-US-00104 TABLE 104 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 40 58 77 88 91 0.8 502075 30 47 69 82 91 1.4 502098 37 59 74 88 94 0.9 502099 43 59 76 88 94 0.8 502154 64 72 84 88 90 <0.6 502155 34 54 72 82 87 1.1 502156 17 38 57 83 87 2 502163 41 60 82 92 93 0.7 502164 41 73 78 89 90 0.6 502175 33 58 74 87 90 1 502179 61 74 81 87 87 <0.6 502189 33 52 79 91 94 1.1 502191 53 67 84 92 92 <0.6 502194 48 64 83 87 92 <0.6 502228 34 47 67 80 89 1.3

TABLE-US-00105 TABLE 105 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 24 28 51 68 83 2.4 495756 49 71 87 88 94 <0.6 507692 48 70 83 88 93 <0.6 507693 48 68 86 91 93 <0.6 507694 41 61 81 88 93 0.7 507695 36 53 72 84 93 1.1 507696 51 74 83 89 94 <0.6 507710 48 72 71 90 91 <0.6 507716 42 54 83 91 95 0.8 507717 44 65 79 86 90 0.6 502314 25 38 61 79 93 1.7 502319 47 66 80 92 94 0.6 502322 45 69 81 89 86 <0.6 502331 34 45 53 71 89 1.6 502334 52 68 77 89 93 <0.6

TABLE-US-00106 TABLE 106 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 30 49 73 87 90 1.2 522366 32 53 68 80 84 1.2 522373 39 62 81 87 91 0.8 522374 39 50 71 85 91 1.1 522375 23 49 73 87 91 1.4 522383 46 64 79 83 85 <0.6 522435 39 62 76 86 88 0.8 522437 33 62 78 87 90 0.9 522444 37 69 81 88 92 0.7 522445 62 54 68 85 91 <0.6 522450 29 46 69 86 86 1.4 522473 19 33 66 78 86 2 522484 63 72 81 80 81 <0.6 522485 37 43 70 82 90 1.3 522501 38 66 79 84 86 0.7

TABLE-US-00107 TABLE 107 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 20 48 75 84 89 1.5 522440 31 51 69 79 85 1.3 522465 26 45 68 81 89 1.5 522474 17 40 65 78 87 1.9 522495 15 28 69 82 85 2 522509 24 38 61 88 87 1.7 522529 21 38 68 58 87 2 522553 15 29 61 81 90 2.1 522579 23 43 58 83 89 1.7 522580 50 36 68 84 93 1.1 522581 16 21 59 82 92 2.2 522582 23 48 59 86 91 1.6 522587 21 52 69 90 95 1.4 522588 19 41 65 88 94 1.7 522589 38 57 80 96 92 0.9

TABLE-US-00108 TABLE 108 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 34 49 77 79 91 1.2 522550 16 45 58 80 89 1.9 522554 26 39 60 71 87 1.9 522555 38 50 66 74 81 1.2 522556 33 39 59 76 85 1.7 522609 41 67 87 92 84 0.6 522610 21 41 65 71 93 1.8 522621 35 54 71 85 94 1.1 522627 18 9 57 81 95 2.3 522631 26 34 70 81 93 1.7 522632 40 55 82 91 95 0.8 522638 8 33 63 74 91 2.2 522643 64 53 69 82 93 <0.6 522667 23 55 75 86 91 1.3 522688 45 57 80 88 94 0.7

TABLE-US-00109 TABLE 109 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 33 51 73 80 86 1.2 522672 20 20 57 89 93 2.1 522675 20 29 63 78 94 2 522676 6 43 62 88 93 1.9 522677 6 39 63 83 96 2 522679 17 35 62 82 92 1.9 522682 61 57 69 86 90 <0.6 522689 29 57 78 87 93 1.1 522694 28 33 57 76 92 1.8 522697 19 31 68 86 95 1.8 522698 20 54 72 85 93 1.4 522715 12 48 68 80 88 1.8 522716 25 35 67 84 90 1.7 522717 27 56 74 90 88 1.2 522745 42 62 85 90 92 0.7

TABLE-US-00110 TABLE 110 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 34 52 78 84 91 1.1 522671 36 43 72 85 92 1.2 522678 35 43 65 83 92 1.3 522680 25 50 63 82 86 1.5 522681 26 57 69 86 90 1.3 522683 24 33 65 77 88 1.8 522687 36 41 68 82 93 1.3 522690 23 51 67 84 93 1.5 522692 33 38 59 77 92 1.6 522705 35 48 66 73 86 1.3 522719 29 38 62

80 92 1.7 522757 29 41 60 79 93 1.6 522770 33 47 78 91 94 1.1 522784 20 33 54 71 90 2.2 522807 20 35 61 78 92 1.9

TABLE-US-00111 TABLE 111 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 32 54 75 84 89 1.1 522758 26 49 66 87 94 1.4 522783 32 33 59 77 89 1.8 522838 16 39 64 75 82 2.0 522865 16 34 58 80 92 2.0 522866 16 25 45 66 83 2.8 522870 18 38 46 79 95 2.1 522871 13 35 57 80 91 2.1 522888 27 52 71 85 89 1.3 522889 36 59 75 81 80 0.9 522894 31 42 54 80 92 1.6 522897 29 53 76 88 92 1.2 522898 18 34 63 77 87 2.0 522913 40 54 79 90 90 0.9 522942 27 57 77 85 88 1.1

TABLE-US-00112 TABLE 112 ISIS 0.625 1.25 2.50 5.00 10.00 IC.sub.50 No μM μM μM μM μM (μM) 413433 20 46 71 82 89 1.6 334177 9 30 60 74 92 2.3 522905 9 24 50 76 94 2.5 522917 17 35 58 82 93 2.0 522924 7 36 59 79 95 2.1 522925 15 36 64 85 95 1.8 522932 9 29 53 80 92 2.3 522941 23 44 65 85 91 1.6 522943 13 44 75 81 88 1.7 522947 45 30 62 78 94 1.4 522964 27 48 71 85 95 1.4 495876 24 33 58 86 94 1.8 495877 16 25 52 77 87 2.4 495878 37 69 86 91 96 0.7 523002 34 54 80 89 93 1.0

Example 9: Dose-Dependent Antisense Inhibition of Human DGAT2 in HepG2 Cells by MOE Gapmers

[0481] Gapmers from the studies described in the Examples above exhibiting significant in vitro inhibition of DGAT2 mRNA were selected and tested at various doses in HepG2 cells. The antisense oligonucleotides were tested in a series of experiments that had similar culture conditions. The results for each experiment are presented in separate tables shown below. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.3125 μM, 0.625 μM, 1.25 μM, 2.50 μM, 5.00 μM, or 10.00 μM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.

[0482] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE-US-00113 TABLE 113 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 0 41 65 75 81 87 1.3 484157 24 52 78 87 95 94 0.6 484127 10 39 71 86 93 94 0.9 484181 49 57 76 79 89 90 0.3 484137 29 41 62 79 89 91 0.8 484148 0 30 72 77 91 93 1.2 484169 18 31 48 68 87 88 1.3 484170 0 51 58 75 94 90 1.3 484133 36 63 50 58 85 91 1.0 484140 31 60 65 73 87 92 0.6 484129 48 62 73 87 91 93 0.3 484141 40 33 55 64 85 96 0.9

TABLE-US-00114 TABLE 114 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 36 38 57 64 83 87 0.9 484158 8 6 39 61 86 91 1.9 484377 0 2 15 33 78 94 2.8 484130 18 41 51 79 91 96 1.0 484336 26 24 45 67 90 92 1.3 484167 0 0 33 72 89 96 2.3 484344 17 18 29 48 75 83 2.2 484156 0 5 29 47 76 81 2.9 484391 0 0 42 66 84 96 2.0 484353 0 16 48 72 91 99 1.6 484350 0 37 49 79 89 87 1.5 484357 15 23 39 69 91 94 1.5 484343 25 28 58 80 90 93 1.0

TABLE-US-00115 TABLE 115 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 34 34 65 83 86 85 0.8 472155 21 14 20 35 54 73 4.1 472156 11 23 41 42 66 84 2.3 472158 27 36 62 77 87 95 0.9 472175 4 25 30 60 69 89 2.1 472178 0 0 6 30 54 82 4.2 423453 32 31 57 77 90 94 0.9 380132 17 9 0 30 21 54 >10 472433 0 0 0 0 18 62 >10 217328 29 28 45 65 76 83 1.4

TABLE-US-00116 TABLE 116 0.3125 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM 0.625 μM μM μM μM (μM) 413433 36 55 70 81 92 93 <0.6 501094 43 60 76 85 90 90 <0.6 501097 23 39 56 74 81 82 1.1 501098 56 70 83 92 90 91 <0.6 501100 51 62 82 91 92 90 <0.6 501103 58 65 77 86 93

92 <0.6 501118 26 50 54 72 85 88 0.9 501122 28 40 59 69 85 84 1.0 501127 38 61 75 81 89 88 <0.6 501165 34 24 59 70 85 82 1.1 501171 38 57 64 80 88 89 <0.6 501214 36 50 68 84 83 85 <0.6

TABLE-US-00117 TABLE 117 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 17 38 61 84 88 91 1.0 500998 37 57 74 84 90 92 0.4 501020 29 53 73 83 89 92 0.6 501033 9 27 45 66 76 80 1.7 501034 14 42 56 75 83 90 1.1 501035 39 61 77 86 86 86 0.3 501040 17 25 40 71 86 91 1.4 501041 49 66 80 81 85 83 <0.3 501062 24 29 50 66 81 88 1.3 501064 45 51 73 84 85 85 0.4 501093 12 34 55 72 76 78 1.4 501111 14 26 43 65 69 78 1.9

TABLE-US-00118 TABLE 118 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 41 47 65 81 91 94 0.6 500841 32 38 67 75 83 84 0.8 500844 35 51 77 74 77 63 0.4 500852 35 38 63 79 88 89 0.8 500859 58 39 52 85 91 96 0.5 500867 21 54 46 88 90 93 0.8 500913 42 65 58 89 91 90 0.4 500942 32 31 50 86 92 95 0.9 500966 53 75 78 84 88 89 <0.3 500989 32 69 74 90 87 86 0.3 501018 13 58 72 86 87 82 0.7 501067 7 63 45 77 88 94 1.0

TABLE-US-00119 TABLE 119 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 0 39 44 70 77 92 1.7 501385 0 9 30 25 37 63 7.0 501387 0 0 0 14 27 75 6.8 501404 28 37 48 59 61 83 1.5 501412 0 23 24 16 51 75 4.6 501427 3 18 50 41 68 92 2.1 501430 21 7 47 53 74 90 1.9 501442 35 39 65 72 92 94 0.8 501443 36 50 34 48 62 75 1.6 501447 21 9 49 56 74 90 1.8 501448 23 25 48 60 69 88 1.6 501450 27 36 28 31 52 77 3.4

TABLE-US-00120 TABLE 120 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 0 13 31 51 80 86 2.1 501391 0 9 15 7 46 67 8.6 501393 0 0 0 50 64 81 3.1 501398 0 5 0 0 41 58 >10 501405 0 31 33 57 73 80 2.0 501429 5 10 0 2 40 62 >10 501435 20 15 34 50 79 89 1.9 501438 12 39 30 64 83 89 1.6 501440 42 21 41 57 81 92 1.3 501451 0 0 19 39 67 84 3.5 501452 0 7 14 38 54 85 3.6 501454 12 4 1 19 65 82 4.2

TABLE-US-00121 TABLE 121 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 0 26 56 75 82 92 1.2 501388 0 15 28 63 86 88 1.8 501389 0 0 0 58 69 84 2.1 501390 0 33 0 0 73 86 2.0 501392 0 19 57 60 75 81 2.2 501396 0 13 0 53 81 83 2.3 501411 0 0 0 31 45 78 5.5 501428 14 0 14 38 74 81 3.2 501436 0 0 41 61 80 78 2.2 501449 0 0 35 56 71 82 2.7 501455 0 0 19 30 61 75 4.1 501457 18 0 23 39 63 87 3.1

TABLE-US-00122 TABLE 122 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 19 33 46 69 80 91 1.3 501154 24 22 33 42 74 81 2.2 501183 18 42 54 71 71 80 1.2 501199 23 22 44 68 80 79 1.5 501200 8 13 40 65 78 87 1.9 501212 39 32 24 52 65 70 2.3 501213 12 34 40 69 75 79 1.6 501224 36 33 53 65 74 81 1.1 501270 12 9 42 59 67 76 2.3 501287 2 24 33 46 71 80 2.5 501322 8 9 6 31 54 77 4.4 501345 10 17 21 50 55 81 3.0

TABLE-US-00123 TABLE 123 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 0 11 17 33 60 69 4.2 525395 18 18 15 47 49 69 4.4 525401 10 28 33 58 68 68 2.5 525402 0 9 23 43 62 73 3.5 525414 2 11 29 36 56 56 5.2 525415 22 0 21 39 49 73 4.7 525431 6 19 36 49 67 66 2.9 525442 13 0 23 44 56 81 3.4 525443 4 6 24 61 65 78 2.7 525469 28 38 50 69 79 85 1.1 525470 3 17 23 41 70 82 2.8 525501 0 5 34 31 58 79 3.5

TABLE-US-00124 TABLE 124 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 22 35 48 62 76 86 1.4 525468 11 30 21 61 72 83 2.1 525471 29 47 62 70 77 84 0.8 525472 23 35 54 70 81 81 1.2 525474 22 38 54 69 79 78 1.2 525479 25 26 35 63 75 84 1.6 525480 0 21 29 50 75 75 2.7 525500 4 23 39 60 80 87 1.8 525513 32 21 45 55 77 86 1.6 525551 15 31 38 8 84 90 2.3 525552 11 29 49 56 78 87 1.6 525554 16 7 31 62 68 79 2.3

TABLE-US-00125 TABLE 125 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 28 21 44 50 68 79 1.9 525544 0 3 22 41 52 63 4.8 525612 26 36 56 72 76 79 1.1 525631 0 8 26 54 77 82 2.7 525649 0 0 10 46 64 76 3.7 496041 14 9 8 31 53 65 5.7 525688 5 25 43 68 82 89 1.6 525705 27 30 48 69 82 90 1.2 525708 5 26 54 69 84 89 1.5 525711 14 28 49 67 81 88 1.4 525733 21 42 60 76 82 93 1.0 525754 0 17 34 55 79 85 2.2

Example 10: Final Confirmation of Antisense Inhibition of Human DGAT2 in HepG2 Cells by MOE Gapmers

[0483] Gapmers from the studies described in the Examples above exhibiting significant in vitro

inhibition of DGAT2 mRNA were selected and tested at various doses in HepG2 cells. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.3125 μM, 0.625 μM, 1.25 μM, 2.50 M, 5.00 μM, or 10.00 μM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.
[0484] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.
TABLE-US-00126 TABLE 126 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 19 26 52 72 86 92 1.3 483817 18 31 60 74 86 90 1.1 483898 40 41 68 77 89 90 0.6 483908 21 47 63 83 89 92 0.8 484152 18 25 47 63 84 91 1.4 484181 42 45 73 81 87 86 0.5 484215 27 37 47 69 80 93 1.1 484231 16 33 58 71 83 89 1.2 472316 0 5 40 50 84 92 2.3 423463 17 34 55 80 90 93 1.1 484271 26 29 62 83 89 92 1.0 484283 7 12 30 50 79 90 2.2
TABLE-US-00127 TABLE 127 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 3 25 51 69 89 92 1.5 472158 18 25 26 44 71 80 2.4 483874 30 42 62 82 90 93 0.8 483910 20 30 49 74 85 90 1.2 483952 36 62 80 88 92 93 0.4 483968 8 35 57 73 86 91 1.2 483984 32 34 62 78 88 91 0.9 483987 12 10 36 61 80 92 1.9 483997 0 7 33 57 77 89 2.4 484085 25 24 53 72 87 89 1.2 484099 29 44 67 82 91 94 0.7 423453 19 24 46 62 88 95 1.4
TABLE-US-00128 TABLE 128 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 22 19 46 61 83 90 1.5 484127 16 18 54 74 90 95 1.3 484129 33 39 64 79 90 89 0.8 484130 30 29 52 66 85 92 1.1 484140 0 16 37 63 86 87 1.9 484133 1 2 44 71 83 93 1.8 484137 0 27 64 81 92 92 1.3 484148 16 37 67 79 91 94 1.0 484157 0 31 70 86 92 93 0.9 484170 23 40 57 77 90 94 1.0 484181 9 18 63 83 76 88 1.4 484343 18 21 40 72 88 91 1.4
TABLE-US-00129 TABLE 129 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 0 23 30 57 71 83 2.4 501849 22 33 46 65 79 87 1.4 501850 12 19 40 52 76 83 2.0 501852 17 25 47 63 68 77 1.8 501855 24 37 59 78 82 84 1.0 501861 15 23 45 61 77 84 1.7 501871 14 19 39 57 71 85 2.0 501884 14 25 42 57 74 83 1.9 501886 30 40 47 58 70 73 1.4 501932 8 24 28 55 68 78 2.4 501944 6 31 0 44 67 82 3.3 501950 0 14 19 55 79 85 2.4
TABLE-US-00130 TABLE 130 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 31 43 53 74 80 84 0.9 502154 29 46 68 81 86 91 0.7 502163 22 33 55 71 89 97 1.1 502164 20 32 71 72 85 91 1.0 502179 28 50 67 84 81 90 0.7 502191 29 47 64 74 94 97 0.7 502194 24 23 62 78 90 90 1.1 502314 25 14 21 44 66 85 2.6 502319 20 32 56 78 86 92 1.1 502322 17 35 49 68 79 83 1.4 502331 11 22 11 45 60 82 3.1 502334 35 40 59 69 82 92 0.8
TABLE-US-00131 TABLE 131 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 13 16 25 38 59 75 3.5 495702 29 37 49 65 78 80 1.2 495718 21 34 28 56 71 77 2.0 495756 23 40 56 76 87 95 1.0 507692 20 33 56 68 89 91 1.1 507693 28 38 56 79 89 94 0.9 507694 19 30 45 72 87 94 1.3 507695 28 32 44 70 84 92 1.2 507696 18 37 59 78 88 90 1.0 507710 27 39 52 73 91 92 1.0 507716 32 51 52 68 88 93 0.8 507717 31 49 44 66 84 90 1.0
TABLE-US-00132 TABLE 132 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 0 43 64 80 84 90 1.2 501947 7 23 42 65 79 89 1.7 501951 13 25 45 68 85 92 1.5 501959 8 29 38 66 74 90 1.7 501960 14 22 41 68 75 88 1.7 501968 17 33 44 59 88 87 1.4 502038 0 7 8 36 58 84 3.7 502045 13 38 53 67 81 88 1.3 502056 0 31 66 83 88 91 1.2 502098 9 35 48 72 86 93 1.3 502099 9 31 52 74 82 92 1.3 502119 1 45 61 79 89 95 1.1
TABLE-US-00133 TABLE 133 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM uM μM μM (μM) 413433 0 43 59 69 84 90 1.1 495697 15 32 50 66 78 79 1.5 495705 25 37 53 59 76 79 1.3 495719 21 30 44 64 76 84 1.5 501388 13 34 44 62 71 87 1.6 501389 11 44 47 54 80 91

1.4 501392 0 25 42 51 75 87 1.9 501396 19 31 52 66 76 87 1.4 501428 0 0 33 46 63 85 2.8 501436 4 28 23 57 76 88 2.1 501449 19 21 42 59 75 92 1.7 501457 7 29 35 50 75 92 1.9

TABLE-US-00134 TABLE 134 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 13 35 53 73 83 87 1.3 522373 19 34 54 72 83 84 1.2 522374 13 31 54 65 79 85 1.4 522383 23 44 62 72 76 81 1.0 522435 25 39 55 73 81 80 1.1 522437 26 35 56 70 81 85 1.1 522440 14 33 50 67 77 82 1.5 522444 24 35 57 78 83 89 1.0 522445 25 45 57 69 83 86 1.0 522465 14 15 40 55 75 82 2.0 522484 37 0 72 75 76 74 1.3 522501 25 45 55 72 78 76 1.0

TABLE-US-00135 TABLE 135 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 25 36 50 79 83 87 1.1 522580 27 34 55 62 86 85 1.1 522587 35 40 56 73 88 92 0.8 522589 27 40 56 71 82 91 1.0 522609 37 48 70 82 94 93 0.5 522621 20 35 55 74 86 91 1.1 522632 31 52 66 80 90 93 0.6 522643 16 28 48 67 78 87 1.5 522688 16 38 57 75 76 89 1.2 522689 29 34 58 78 84 91 0.9 522717 18 24 50 72 82 88 1.4 522745 22 35 57 75 89 92 1.0

TABLE-US-00136 TABLE 136 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 23 32 45 74 78 88 1.3 522682 26 34 63 68 74 81 1.1 522770 8 26 50 65 83 91 1.5 522889 25 40 59 70 77 78 1.1 522897 14 22 45 69 83 89 1.5 522913 19 30 57 73 86 90 1.2 522941 8 18 41 56 73 82 2.1 522942 20 30 49 67 83 86 1.3 522947 14 0 40 58 80 91 2.0 522964 9 20 51 63 78 89 1.7 495878 16 36 62 80 90 93 1.0 523002 21 32 51 67 73 84 1.4

TABLE-US-00137 TABLE 137 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 40 35 66 81 87 89 0.7 525401 17 58 64 79 85 86 0.8 525402 30 41 49 74 87 90 0.9 525431 31 29 64 82 86 84 0.9 525442 35 39 59 76 90 87 0.8 525443 39 53 64 76 88 89 0.5 525469 32 68 85 93 92 93 0.3 525470 39 34 59 74 91 89 0.8 525471 27 34 59 84 86 0 0.9 525474 32 47 69 80 93 87 0.6 525479 25 33 51 60 75 75 1.5 525501 31 29 56 82 91 88 1.0

TABLE-US-00138 TABLE 138 0.3125 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM μM (μM) 413433 22 41 54 69 81 88 1.1 525472 26 31 50 69 84 82 1.2 525500 11 33 52 68 83 83 1.4 525513 16 20 48 64 79 87 1.6 525552 27 30 57 70 83 90 1.1 525612 0 44 65 76 83 82 0.3 525688 11 40 60 75 88 97 1.1 525705 25 41 65 76 90 95 0.8 525708 36 38 64 75 93 96 0.7 525711 24 53 67 80 90 94 0.7 525733 27 70 73 86 91 95 0.4 525754 23 23 48 70 85 89 1.3

Example 11: Final Confirmation of Antisense Inhibition of Human DGAT2 in HepG2 Cells by MOE Gapmers

[0485] Gapmers from the studies described in the Examples above exhibiting significant in vitro inhibition of DGAT2 mRNA were selected and tested at various doses in HepG2 cells. Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.625 μM, 1.25 μM, 2.50 μM, 5.00 μM, or 10.00 μM concentrations of antisense oligonucleotide, as specified in the Tables below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and DGAT2 mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS2988_MGB was used to measure mRNA levels. DGAT2 mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of DGAT2, relative to untreated control cells. ‘0’ indicates that the mRNA levels were not inhibited.

[0486] The half maximal inhibitory concentration (IC.sub.50) of each oligonucleotide is also presented. DGAT2 mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE-US-00139 TABLE 139 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 48 60 77 85 86 0.6 495450 42 50 73 83 83 1.0 495516 64 78 87 84 84 <0.6 495520 70 76 87 83 82 <0.6 495554 26 61 75 76 85 1.1 495555 30 72 80 87 84 0.8 495576 61 73 86 88 88 <0.6 495577 44 64 75 85 80 0.6 495609 64 69 83 86 89 <0.6 495685 0 53 75 82 85 2.0 495707 9 41 72 75 66 2.2 495736 16 61 80 85 81 1.3 495749 16 33 76 90 93 1.7 495752 0 66 75 74 93 1.7 495753 0 59 79 91 92 1.8

TABLE-US-00140 TABLE 140 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 54 70 87 91 93 <0.6 495738 55 57 81 92 92 <0.6 495744 37 59 78 88 90 0.9 495756

52 61 80 90 94 <0.6 495825 34 69 89 94 97 0.7 495829 38 79 88 90 91 <0.6 495837 52 67 93 93 96 <0.6 495839 66 89 91 93 93 <0.6 495840 64 84 94 96 94 <0.6 495841 51 82 90 92 93 <0.6 495842 54 73 88 90 92 <0.6 495849 63 69 82 85 91 <0.6 495853 72 77 91 96 94 <0.6 495857 52 61 88 91 92 <0.6 495878 54 77 93 97 96 <0.6

TABLE-US-00141 TABLE 141 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 38 63 73 85 88 0.8 501171 38 53 69 83 84 1.0 501183 51 70 79 87 89 <0.6 501199 44 68 88 91 87 <0.6 501213 25 57 82 93 96 1.1 501214 35 56 72 88 92 1.0 501224 48 68 83 90 90 <0.6 501405 27 51 72 90 96 1.3 501430 10 26 65 90 94 2.0 501435 35 41 71 92 94 1.3 501438 27 71 83 92 92 0.9 501440 24 40 66 90 97 1.6 501442 58 79 91 96 94 <0.6 501443 32 29 57 89 96 1.7 501448 27 46 72 91 93 1.4

TABLE-US-00142 TABLE 142 0.625 1.25 2.50 5.00 10.00 IC.sub.50 ISIS No μM μM μM μM μM (μM) 413433 43 58 82 91 94 0.8 501127 46 68 81 91 92 <0.6 501103 34 70 85 93 91 0.7 501100 54 75 88 93 94 <0.6 501098 55 81 91 91 89 <0.6 501094 62 70 87 90 92 <0.6 501064 52 67 81 88 89 <0.6 501041 53 71 82 86 85 <0.6 501035 57 77 87 88 90 <0.6 501020 47 63 86 94 94 0.6 500998 47 61 86 91 95 0.6 500989 55 75 83 86 85 <0.6 500966 71 82 85 87 87 <0.6 500913 29 46 74 85 88 1.3 500859 6 44 75 87 90 1.8

Example 12: Tolerability of Antisense Oligonucleotides Targeting Human DGAT2 in CD1 Mice

[0487] CD1® mice (Charles River, MA) are a multipurpose mice model, frequently utilized for safety and efficacy testing. The mice were treated with ISIS antisense oligonucleotides selected from studies described above and evaluated for changes in the levels of various plasma chemistry markers. Liver and kidney tissue from the treated mice were microscopically reviewed; no toxicity or tissue injury was seen in any treated animal.

Treatment

[0488] Groups of five male CD1 mice each were injected subcutaneously twice a week for 6 weeks with 100 mg/kg (200 mg/kg/week) of ISIS oligonucleotide. One group of ten male CD1 mice was injected subcutaneously twice a week for 6 weeks with PBS. Mice were euthanized 48 hours after the last dose, and organs and plasma were harvested for further analysis.

Plasma Chemistry Markers

[0489] To evaluate the effect of ISIS oligonucleotides on liver and kidney function, plasma levels of transaminases, bilirubin, albumin, and BUN were measured on day 27 using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, NY). The results are presented in the Table below. ISIS oligonucleotides that caused changes in the levels of any of the liver or kidney function markers outside the expected range for antisense oligonucleotides were excluded in further studies.

TABLE-US-00143 TABLE 143 Plasma chemistry markers in CD1 mice ALT AST BUN Bilirubin (IU/L) (IU/L) (mg/dL) (mg/dL) PBS 33 50 30 0.3 ISIS 483984 193 177 30 0.3 ISIS 484099 69 77 23 0.2 ISIS 484129 43 46 24 0.2 ISIS 484157 136 105 26 0.2 ISIS 495576 36 53 30 0.3 ISIS 495609 102 112 28 0.3 ISIS 495753 68 76 28 0.2 ISIS 495756 154 123 26 0.3 ISIS 501849 45 58 26 0.2 ISIS 501850 294 306 26 0.3 ISIS 501855 62 56 28 0.2 ISIS 501861 100 97 30 0.3 ISIS 502322 42 46 28 0.2 ISIS 507696 99 132 30 0.2 ISIS 507710 50 68 27 0.3 ISIS 507716 3112 3302 19 7.8

Body and Organ Weights

[0490] Body weights of the mice were measured on day 35 and are presented in the Table below. Liver, spleen and kidney weights were measured at the end of the study (day 41), and are also presented in the Table below. ISIS oligonucleotides that caused any changes in body or organ weights outside the expected range for antisense oligonucleotides were excluded from further studies. ‘n/a’ indicates that the particular endpoint was not measured in that group of mice.

TABLE-US-00144 TABLE 144 Body and organ weights of CD1 mice (g) Body weight Kidney Liver Spleen PBS 41.2 0.7 2.0 0.1 ISIS 483984 41.0 n/a n/a n/a ISIS 484129 41.0 0.6 2.1 0.2 ISIS 484157 38.8 n/a n/a n/a ISIS 495576 41.4 0.6 2.1 0.2 ISIS 495609 42.5 0.6 2.1 0.2 ISIS 495753

	40.0	0.5	2.2	0.2
ISIS 495756	42.0	0.7	2.6	0.2
ISIS 501849	42.3	0.6	2.4	0.2
ISIS 501850	41.7	0.7	2.5	0.2
ISIS 501855	39.9	n/a	n/a	n/a
ISIS 501861	40.3	0.6	2.3	0.2
ISIS 502322	41.1	0.7	2.2	0.2
ISIS 507696	38.9	0.6	2.3	0.2
ISIS 507710	42.7	0.6	2.5	0.2

Example 13: Tolerability of Antisense Oligonucleotides Targeting Human DGAT2 in CD1 Mice

[0491] CD1® mice were treated with ISIS antisense oligonucleotides selected from studies described above and evaluated for changes in the levels of various plasma chemistry markers.

Treatment

[0492] Groups of five male CD1 mice each were injected subcutaneously twice a week for 6 weeks with 50 mg/kg of ISIS 483817, ISIS 483910, ISIS 484085, ISIS 484127, ISIS 484130, ISIS 484137, ISIS 484215, ISIS 484231, ISIS 484271, ISIS 501871, ISIS 502098, ISIS 502119, ISIS 502154, ISIS 502163, ISIS 502164, ISIS 502194, ISIS 525443, ISIS 525474, ISIS 525552, ISIS 525612, ISIS 525688, ISIS 525705, ISIS 525708, ISIS 525711, and ISIS 525733. One group of ten male CD1 mice was injected subcutaneously twice a week for 6 weeks with PBS. Mice were euthanized 48 hours after the last dose, and organs and plasma were harvested for further analysis.

Plasma Chemistry Markers

[0493] To evaluate the effect of ISIS oligonucleotides on liver and kidney function, plasma levels of transaminases, bilirubin, albumin, and BUN were measured on day 26 using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, NY). The results are presented in the Table below. ISIS oligonucleotides that caused changes in the levels of any of the liver or kidney function markers outside the expected range for antisense oligonucleotides were excluded in further studies.

TABLE-US-00145 TABLE 145 Plasma chemistry markers in CD1 mice

	ALT (IU/L)	AST (IU/L)	BUN (mg/dL)	Bilirubin (mg/dL)
PBS	26	55	29	0.6
ISIS 483817	40	54	27	0.3
ISIS 483910	244	155	28	0.6
ISIS 484085	46	58	27	0.4
ISIS 484127	100	147	25	0.4
ISIS 484130	151	91	29	0.3
ISIS 484137	31	49	26	0.3
ISIS 484215	42	49	25	0.2
ISIS 484231	31	45	26	0.2
ISIS 484271	2847	3267	26	2.9
ISIS 501871	108	96	23	0.2
ISIS 502098	60	62	24	0.3
ISIS 502119	402	370	26	0.3
ISIS 502154	293	294	28	0.3
ISIS 502163	45	60	23	0.4
ISIS 502164	81	75	25	0.4
ISIS 502194	32	43	24	0.4
ISIS 525443	93	101	24	0.3
ISIS 525474	49	94	24	0.6
ISIS 525552	1374	874	25	0.3
ISIS 525612	78	54	27	0.4
ISIS 525705	76	117	31	0.3
ISIS 525708	59	79	29	0.3
ISIS 525711	1670	787	26	0.3

Body and Organ Weights

[0494] Body weights of the mice were measured on day 36 and are presented in the Table below. Liver, spleen and kidney weights were measured at the end of the study, and are also presented in the Table below. ISIS oligonucleotides that caused any changes in body or organ weights outside the expected range for antisense oligonucleotides were excluded from further studies. 'n/a' indicates that the particular endpoint was not measured in that group of mice.

TABLE-US-00146 TABLE 146 Body and organ weights of CD1 mice (g)

	Body weight	Kidney	Liver	Spleen
PBS	39.4	0.7	2.0	0.1
ISIS 483817	38.8	0.6	2.3	0.2
ISIS 483910	39.2	0.6	2.3	0.2
ISIS 484085	37.9	0.6	2.3	0.1
ISIS 484127	35.3	n/a	n/a	n/a
ISIS 484130	41.5	n/a	n/a	n/a
ISIS 484137	39.6	0.7	2.3	0.1
ISIS 484215	38.5	0.6	2.4	0.2
ISIS 484231	38.3	0.6	2.1	0.1
ISIS 501871	38.7	0.6	2.3	0.2
ISIS 502098	38.8	0.6	2.5	0.2
ISIS 502163	39.1	0.6	2.0	0.2
ISIS 502164	37.7	0.6	2.1	0.2
ISIS 502194	40.5	0.6	1.9	0.1
ISIS 525443	39.5	0.6	2.0	0.2
ISIS 525474	37.0	0.6	1.8	0.1
ISIS 525612	37.8	0.6	2.0	0.1
ISIS 525705	36.5	0.7	2.0	0.4
ISIS 525708	38.7	0.7	2.0	0.3

Example 14: Tolerability of Antisense Oligonucleotides Targeting Human DGAT2 in Sprague-Dawley Rats

[0495] Sprague-Dawley rats are a multipurpose model used for safety and efficacy evaluations. The rats were treated with ISIS antisense oligonucleotides from the study described above and evaluated for changes in the levels of various plasma chemistry markers.

Treatment

[0496] Male Sprague-Dawley rats were maintained on a 12-hour light/dark cycle and fed ad libitum with Purina normal rat chow, diet 5001. Groups of 4 Sprague-Dawley rats each were injected subcutaneously twice a week for 8 weeks with 50 mg/kg of ISIS 484129, ISIS 495576, ISIS 495609, ISIS 495753, ISIS 495756, ISIS 501849, ISIS 501850, ISIS 501861, ISIS 502322, ISIS 507696, and ISIS 507710. Forty eight hours after the last dose, rats were euthanized and organs and plasma were harvested for further analysis.

Liver Function

[0497] To evaluate the effect of ISIS oligonucleotides on hepatic function, plasma levels of transaminases were measured on day 56 using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, NY). Plasma levels of ALT and AST were measured and the results are presented in the Table below expressed in IU/L. Plasma levels of Bilirubin were also measured using the same clinical chemistry analyzer and the results are also presented in the Table below. ISIS oligonucleotides that caused changes in the levels of any markers of liver function outside the expected range for antisense oligonucleotides were excluded in further studies.

TABLE-US-00147 TABLE 147 Liver function markers in Sprague-Dawley rats ALT AST Bilirubin (IU/L) TU/L) (mg/dL) PBS 49 70 0.16 ISIS 484129 133 130 0.18 ISIS 495576 99 148 0.17 ISIS 495609 406 269 0.32 ISIS 495753 968 1258 1.83 ISIS 495756 165 206 0.24 ISIS 501849 57 103 0.16 ISIS 501850 1218 1916 6.55 ISIS 501861 61 113 0.14 ISIS 502322 79 84 0.14

Kidney Function

[0498] To evaluate the effect of ISIS oligonucleotides on kidney function, urine levels of creatinine and total protein were measured on day 56 using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, NY). Results are presented in the Table below, expressed in mg/dL.

TABLE-US-00148 TABLE 148 Kidney function markers (mg/dL) in Sprague-Dawley rats Total Creatinine protein PBS 146 114 ISIS 495756 63 978 ISIS 501849 82 445 ISIS 501861 67 309 ISIS 501850 52 268 ISIS 502322 84 507 ISIS 495576 108 587 ISIS 495609 38 264 ISIS 495753 73 411 ISIS 484129 66 312

Body and Organ Weights

[0499] Body weights of the rat were measured on day 50 and are presented in the Table below. Liver, heart, spleen and kidney weights were measured at the end of the study, and are also presented in the Table below. ISIS oligonucleotides that caused any changes in body or organ weights outside the expected range for antisense oligonucleotides were excluded from further studies. 'n/a' indicates that the particular endpoint was not measured in that group of mice.

TABLE-US-00149 TABLE 149 Body and organ weights of Sprague-Dawley rats (g) Body weight Heart Kidney Liver Spleen PBS 486 1.7 3.5 14.2 0.7 ISIS 495756 374 1.1 3.5 15.5 2.2 ISIS 501849 404 1.2 3.3 13.7 1.9 ISIS 501861 390 1.1 3.6 16.0 2.7 ISIS 501850 329 1.3 4.1 16.3 4.5 ISIS 502322 424 1.3 3.4 15.6 1.7 ISIS 495576 461 1.3 3.5 17.3 2.1 ISIS 495609 383 1.4 3.8 18.9 3.8 ISIS 495753 384 1.2 3.4 16.3 3.6 ISIS 484129 415 1.3 3.1 14.9 1.5

Example 15: Tolerability of Antisense Oligonucleotides Targeting Human DGAT2 in Sprague-Dawley Rats

[0500] Sprague-Dawley rats were treated with ISIS antisense oligonucleotides from the studies described above and evaluated for changes in the levels of various plasma chemistry markers.

Treatment

[0501] Male Sprague-Dawley rats were maintained on a 12-hour light/dark cycle and fed ad libitum with Purina normal rat chow, diet 5001. Groups of 4 Sprague-Dawley rats each were injected subcutaneously twice a week for 8 weeks with 50 mg/kg of ISIS 483817, ISIS 483910, ISIS 484085, ISIS 484137, ISIS 484215, ISIS 484231, ISIS 501871, ISIS 502098, ISIS 502163, ISIS 502164, ISIS 502194, ISIS 525443, ISIS 525474, ISIS 525612, and ISIS 525708. Forty eight hours after the last dose, rats were euthanized and organs and plasma were harvested for further analysis.

Liver Function

[0502] To evaluate the effect of ISIS oligonucleotides on hepatic function, plasma levels of transaminases were measured on day 53 using an automated clinical chemistry analyzer (Hitachi

Olympus AU400e, Melville, NY). Plasma levels of ALT and AST were measured and the results are presented in the Table below expressed in IU/L. Plasma levels of Bilirubin were also measured using the same clinical chemistry analyzer and the results are also presented in the Table below. ISIS oligonucleotides that caused changes in the levels of any markers of liver function outside the expected range for antisense oligonucleotides were excluded in further studies.

TABLE-US-00150 TABLE 150 Liver function markers in Sprague-Dawley rats ALT AST Bilirubin (IU/L) (IU/L) (mg/dL) PBS 53 83 0.21 ISIS 483817 81 127 0.13 ISIS 483910 283 428 3.38 ISIS 484085 131 208 0.27 ISIS 484137 63 98 0.16 ISIS 484215 50 86 0.13 ISIS 484231 75 106 0.17 ISIS 501871 45 73 0.10 ISIS 502098 57 156 0.12 ISIS 502163 85 177 0.21 ISIS 502164 67 94 0.15 ISIS 502194 54 82 0.15 ISIS 525443 50 82 0.13 ISIS 525474 118 136 0.22 ISIS 525612 313 314 0.18 ISIS 525708 65 117 0.16

Kidney Function

[0503] To evaluate the effect of ISIS oligonucleotides on kidney function, plasma levels of creatinine and total protein were measured on day 53 using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, NY). Results are presented in the Table below, expressed in mg/dL. Urine levels of creatinine and total protein were also measured on day 53 using the same automated clinical chemistry analyzer. Results are presented in the Table below, expressed in mg/dL.

TABLE-US-00151 TABLE 151 Kidney function markers in the plasma (mg/dL) in Sprague-Dawley rats BUN Creatinine PBS 20 0.44 ISIS 483817 22 0.40 ISIS 483910 24 0.42 ISIS 484085 19 0.37 ISIS 484137 22 0.43 ISIS 484215 19 0.36 ISIS 484231 21 0.39 ISIS 501871 23 0.31 ISIS 502098 25 0.42 ISIS 502163 21 0.37 ISIS 502164 23 0.43 ISIS 502194 21 0.44 ISIS 525443 24 0.51 ISIS 525474 22 0.39 ISIS 525612 18 0.38 ISIS 525708 31 0.53

TABLE-US-00152 TABLE 152 Kidney function markers in the urine (mg/dL) in Sprague-Dawley rats Creatinine Total protein PBS 227 193 ISIS 483817 95 2129 ISIS 483910 60 1030 ISIS 484085 87 868 ISIS 484137 84 517 ISIS 484215 146 1115 ISIS 484231 89 652 ISIS 501871 52 3426 ISIS 502098 64 550 ISIS 502163 73 522 ISIS 502164 100 554 ISIS 502194 95 410 ISIS 525443 73 595 ISIS 525474 79 1547 ISIS 525612 94 453 ISIS 525708 41 2043

Body and Organ Weights

[0504] Body weights of the rat were measured on day 49 and are presented in the Table below. Liver, heart, spleen and kidney weights were measured at the end of the study, and are also presented in the Table below. ISIS oligonucleotides that caused any changes in body or organ weights outside the expected range for antisense oligonucleotides were excluded from further studies. ‘n/a’ indicates that the particular endpoint was not measured in that group of mice.

TABLE-US-00153 TABLE 153 Body and organ weights of Sprague-Dawley rats (g) Body weight Heart Kidney Liver Spleen PBS 485 1.6 3.5 14.5 0.9 ISIS 483817 362 1.2 4.2 17.0 2.3 ISIS 483910 358 1.0 3.2 19.0 3.0 ISIS 484085 348 1.1 2.9 15.3 1.6 ISIS 484137 353 1.1 3.0 14.2 1.6 ISIS 484215 391 1.2 3.7 16.6 2.0 ISIS 484231 386 1.1 3.2 16.6 2.5 ISIS 501871 322 1.1 3.5 17.4 1.8 ISIS 502098 315 1.2 2.9 15.8 2.3 ISIS 502163 326 1.0 3.5 14.5 3.4 ISIS 502164 381 1.2 2.8 15.4 2.2 ISIS 502194 439 1.3 3.4 18.6 2.0 ISIS 525443 469 1.5 3.7 22.1 1.7 ISIS 525474 445 1.7 3.8 19.2 2.2 ISIS 525612 427 1.5 3.1 13.1 1.4 ISIS 525708 338 1.0 3.6 15.9 2.3

Example 16: Effect of ISIS Antisense Oligonucleotides Targeting Human DGAT2 in Cynomolgus Monkeys

[0505] Cynomolgus monkeys were treated with ISIS antisense oligonucleotides selected from studies described above. Antisense oligonucleotide efficacy and tolerability were evaluated.

[0506] At the time this study was undertaken, the cynomolgus monkey genomic sequence was not available in the National Center for Biotechnology Information (NCBI) database; therefore, cross-reactivity with the cynomolgus monkey gene sequence could not be confirmed. Instead, the sequences of the ISIS antisense oligonucleotides used in the cynomolgus monkeys was compared to a rhesus monkey sequence for homology. It is expected that ISIS oligonucleotides with

homology to the rhesus monkey sequence are fully cross-reactive with the cynomolgus monkey sequence as well. The human antisense oligonucleotides tested are cross-reactive with the rhesus genomic sequence (Ref Seq No. NW_001100387.1 truncated from nucleotides 1232000 to 1268000, designated herein as SEQ ID NO: 3). The greater the complementarity between the human oligonucleotide and the rhesus monkey sequence, the more likely the human oligonucleotide can cross-react with the rhesus monkey sequence. The start and stop sites of each oligonucleotide to SEQ ID NO: 3 is presented in the Table below. “Start site” indicates the 5′-most nucleotide to which the gapmer is targeted in the rhesus monkey gene sequence.
TABLE-US-00154 TABLE 154 Antisense oligonucleotides complementary to SEQ ID NO: 3 Unmod- Mod- ified ified SEQ SEQ ISIS Start Stop ID ID No Site Site Sequence Motif NO NO 484085 14986 15005 GTCTGGGAACAG 5-10-5 1371 N/A CAGCATCA 484129 18207 18226 GCACTGACATGG 5-10-5 1415 N/A TAAGTCCT 484137 18355 18374 TGCCATTTAATG 5-10-5 1423 4680 AGCTTCAC 495576 7054 7073 CACCATAATCTG 5-10-5 1849 N/A CACAGGTT 501861 19517 19536 TCACAGAATTAT 5-10-5 2959 N/A CAGCAGTA 502194 27298 27317 CCTCTTAGAAGT 5-10-5 3292 N/A AATGCTTC 525443 2716 2732 TCCATGTCAGAG 3-10-4 4198 N/A AGGCT 525612 14990 15006 GGTCTGGGAACA 3-10-4 4373 N/A GCAGC
Treatment
[0507] Prior to the study, the monkeys were kept in quarantine for a 30-week period, during which the animals were observed daily for general health. The monkeys were 2-4 years old and weighed between 2 and 4 kg. Eight groups of 5 randomly assigned male cynomolgus monkeys each were injected subcutaneously with ISIS oligonucleotide or PBS using a stainless steel dosing needle and syringe of appropriate size into the intracapsular region and outer thigh of the monkeys. The monkeys were dosed 3 times a week for the first week (days 1, 3, and 5) as loading doses, and subsequently twice a week for weeks 2-13, with 20 mg/kg of ISIS oligonucleotide. A control group of 6 cynomolgus monkeys was injected with 0.9% sterile saline subcutaneously three times a week for the first week (days 1, 3, and 5), and subsequently once a week for weeks 2-13.
[0508] During the study period, the monkeys were observed for signs of mortality, clinical observations, and body weight, qualitative food consumption, ophthalmoscopic and electro cardiac examination, clinical and anatomic pathology. Scheduled euthanasia of the animals was conducted on day 93 for animals assigned to terminal necroscopy and on Day 182 for animals assigned to the recovery necroscopy. A full panel of tissues were taken, processed to slides and examined microscopically for histopathology. The protocols described in the Example were approved by the Institutional Animal Care and Use Committee (IACUC).
Hepatic Target Reduction
Target Gene RNA Analysis
[0509] On day 93, RNA was extracted from liver tissue for real-time PCR analysis of DGAT2 using primer probe set mkDGAT2 (forward sequence CCGCAAGGGCTTTGTGAA, designated herein as SEQ ID NO: 13, reverse sequence TTCTCTCCAAAGGAGTACATGGG, designated herein as SEQ ID NO: 14, probe sequence CCTGCGCCATGGAGCCGAC, designated herein as SEQ ID NO: 15). Results are presented as percent inhibition of DGAT2 mRNA, relative to PBS control, normalized to the house keeping gene CyclophilinA. Similar results were obtained on normalization with RIBOGREEN®. As shown in the Table below, treatment with ISIS antisense oligonucleotides resulted in significant reduction of DGAT2 mRNA in comparison to the sterile saline control. Specifically, treatment with ISIS 484137 and ISIS 501861 resulted in the significant reduction of DGAT2 mRNA expression.
[0510] Inhibition levels with select oligonucleotides were also measured with the human primer probe set RTS2977_MGB. As presented in the Table below, treatment with ISIS oligonucleotides significantly reduced DGAT2 levels. Specifically, treatment with ISIS 484137 and ISIS 501861 resulted in the significant reduction of DGAT2 mRNA expression.

TABLE-US-00155 TABLE 155 Percent Inhibition of DGAT2 mRNA in the cynomolgus monkey liver relative to the saline control (mkDGAT2 primer probe set) ISIS No RIBOGREEN CyclophilinA 484085 20 21 484129 54 52 484137 71 69 495576 70 66 501861 89 88 502194 26 29 525443 49 48 525612 35 43

TABLE-US-00156 TABLE 156 Percent Inhibition of DGAT2 mRNA in the cynomolgus monkey liver relative to the saline control (RTS2977_MGB primer probe set) ISIS No RIBOGREEN CyclophilinA 501861 82 79 484137 63 58 525443 22 17

Secondary Lipid Gene RNA Analysis

[0511] Gene expression analysis of secondary lipid genes, DGAT1, ACC1, ACC2, FAS, and SCD12 was also performed. The results are presented in the Table below, expressed as % expression of each gene compared to the PBS control. As presented in the Table below, treatment with ISIS oligonucleotides significantly reduced lipogenic gene levels. Specifically, treatment with ISIS 484137 resulted in the significant reduction of mRNA expression.

TABLE-US-00157 TABLE 157 % lipid gene expression DGAT1 ACC1 ACC2 FAS SCD-1 sterile saline 100 100 100 100 100 ISIS 484129 85 95 95 122 64 ISIS 495576 82 66 59 65 40 ISIS 501861 140 69 67 51 22 ISIS 484085 118 122 86 281 123 ISIS 484137 81 99 77 115 51 ISIS 502194 103 91 81 114 63 ISIS 525443 118 87 75 57 25 ISIS 525612 170 106 96 83 42

Tolerability Studies

Body Weight Measurements

[0512] To evaluate the effect of ISIS oligonucleotides on the overall health of the animals, body weights were measured at regularly and are presented in the Table below. The results indicate that effect of treatment with antisense oligonucleotides on body weights was within the expected range for antisense oligonucleotides. Specifically, treatment with ISIS 484137 was well tolerated in terms of the body weights of the monkeys.

TABLE-US-00158 TABLE 158 Body weights in the cynomolgus monkey Day 1 Day 15 Day 29 Day 36 Day 50 Day 64 Day 78 Day 85 sterile saline 2734 2795 2775 2719 2743 2739 2807 2779 ISIS 484085 2759 2785 2822 2762 2815 2812 2910 2911 ISIS 484129 2780 2889 2906 2852 2918 2937 3043 3029 ISIS 484137 2756 2793 2847 2763 2850 2823 2906 2900 ISIS 495576 2748 2865 2905 2823 2869 2908 2996 2979 ISIS 501861 2702 2775 2797 2750 2775 2828 2893 2860 ISIS 502194 2817 2886 2902 2840 2926 2929 3038 3045 ISIS 525443 2790 2835 2814 2767 2790 2772 2843 2859 ISIS 525612 2921 2962 2995 2937 3006 3020 3093 3166

Liver Function

[0513] To evaluate the effect of ISIS oligonucleotides on hepatic function, blood samples were collected from all the study groups. The blood samples were collected via femoral venipuncture on day 93, 48 hrs post-dosing. The monkeys were fasted overnight prior to blood collection. Blood was collected in tubes without any anticoagulant, and kept at room temperature for a minimum of 30 min for serum separation. The tubes were then centrifuged to obtain serum. Levels of various liver function markers were measured using a Toshiba 200FR NEO chemistry analyzer (Toshiba Co., Japan). Serum levels of ALT and AST were measured and the results are presented in the Table below, expressed in IU/L. Bilirubin, a liver function marker, was similarly measured and is presented in the Table below, expressed in mg/dL. The results indicate that antisense oligonucleotides had no effect on liver function outside the expected range for antisense oligonucleotides. Specifically, treatment with ISIS 484137 was well tolerated in terms of the liver function in monkeys.

TABLE-US-00159 TABLE 159 Liver function markers in cynomolgus monkey serum ALT AST Bilirubin (IU/L) (IU/L) (mg/dL) sterile saline 40 42 0.20 ISIS 484085 57 52 0.15 ISIS 484129 43 43 0.16 ISIS 484137 52 57 0.16 ISIS 495576 56 45 0.15 ISIS 501861 70 67 0.16 ISIS 502194 88 54 0.16 ISIS 525443 56 45 0.23 ISIS 525612 38 37 0.23

Kidney Function

[0514] To evaluate the effect of ISIS oligonucleotides on kidney function, blood samples were collected from all the study groups. The blood samples were collected via femoral venipuncture on day 93, 48 hrs post-dosing. The monkeys were fasted overnight prior to blood collection. Blood was collected in tubes without any anticoagulant, and kept at room temperature for a minimum of 30 min for serum separation. The tubes were then centrifuged to obtain serum. Levels of BUN and creatinine were measured using a Toshiba 200FR NEO chemistry analyzer (Toshiba Co., Japan). Results are presented in the Table below, expressed in mg/dL.

[0515] The plasma chemistry data indicate that most of the ISIS oligonucleotides did not have any effect on the kidney function outside the expected range for antisense oligonucleotides. Specifically, treatment with ISIS 484137 was well tolerated in terms of the kidney function of the monkeys.

TABLE-US-00160 TABLE 160 BUN and creatinine levels (mg/dL) in cynomolgus monkeys BUN Creatinine sterile saline 25 0.87 ISIS 484085 26 0.94 ISIS 484129 21 0.81 ISIS 484137 24 0.94 ISIS 495576 22 0.92 ISIS 501861 25 0.90 ISIS 502194 27 0.86 ISIS 525443 26 1.08 ISIS 525612 23 0.98

Example 17: Effect of ISIS 484137 Targeting Human DGAT2 in Cynomolgus Monkeys

[0516] Cynomolgus monkeys were treated with ISIS 484137 to evaluate the safety of this antisense oligonucleotide over a 13-week dosing period followed by a 13 week recovery period. The protocols described in the Example were approved by the Testing Facility's Institutional Animal Care and Use Committee (IACUC).

Treatment

[0517] The monkeys were 2-4 years old and weighed between 2.3 and 3.7 kg. Five groups of 6-10 randomly assigned cynomolgus monkeys each were injected subcutaneously with ISIS oligonucleotide or 0.9% sterile saline using a stainless steel dosing needle and syringe of appropriate size into one of four delineated sites on the backs of the monkeys. The monkeys were dosed on Days 1, 3, 5, and 7, and then once weekly thereafter for a total of 13 weeks at dose levels of 4 mg/kg, 8 mg/kg, 12 mg/kg, or 40 mg/kg of ISIS 484137. A control group of 10 cynomolgus monkeys was injected with 0.9% sterile saline subcutaneously using the same regimen. Toxicokinetics were assessed in a sixth group of 14 animals (7 of each sex) at the 8 mg/kg dose level. Recovery was assessed in the controls and at the 12 mg/kg and 40 mg/kg dose levels of ISIS 484137 (2 per sex).

[0518] During the study period, the monkeys were observed for signs of mortality, clinical observations, and body weight, qualitative food consumption, ophthalmoscopic and electrocardiographic examination, clinical and anatomic pathology. Scheduled euthanasia of the animals was conducted on day 93 for animals assigned to terminal necroscopy and on Day 182 for animals assigned to the recovery necroscopy. A full panel of tissues were taken, processed to slides and examined microscopically for histopathology.

Body and Organ Weight Measurements

[0519] To evaluate the effect of ISIS 484137 on the overall health of the animals, body weights were measured at regularly and are presented in the Table below. Organ weights were measured after euthanisia. ‘n.d.’ indicates that there is no data for that timepoint. The results indicate that effect of treatment with ISIS 484137 on body and organ weights was within the expected range for antisense oligonucleotides. Specifically, treatment with ISIS 484137 was well tolerated in terms of the body and organ weights of the monkeys.

TABLE-US-00161 TABLE 161 Body weights (g) of cynomolgus monkeys Dose Day (mg/kg) Day 93 182 sterile saline — 3035 3075 ISIS 484137 4 2848 n.d. 8 2850 n.d. 12 3125 3325 40 3027 2875

TABLE-US-00162 TABLE 162 Organ weights (g) of cynomolgus monkeys on day 93 Dose Adrenal (mg/kg) Brain gland Heart Kidney Liver Spleen Thyroid sterile saline — 67 0.46 10.8 12.6 60.0 4.1 0.4 ISIS 484137 4 67 0.52 9.7 13.2 60.8 3.9 0.3 8 67 0.51 10.9 14.2 73.7 3.6 0.4 12 69 0.56 10.1 14.0 65.1 3.2 0.3 40 69 0.59 10.2 15.8 79.2 6.2 0.4

Plasma Homeostasis

[0520] To evaluate the potential for ISIS 484137 to produce complement activation, plasma samples were obtained once prior to treatment initiation on Day-7, then following the first dose on Day 1 at 4, 8 and 24 hours, and finally following the last dose on Day 91 predose and at 4 hours for determination of complement split product Bb using ELISA. Serum samples were also obtained for measurement of Complement C.sub.5 on Day-7, on Day 1 (at 24 hours post dose), on Day 91 (predose and at 24 hours post dose) and for recovery animals on Days 121 and 182 using an automated clinical chemistry analyzer. Additionally, blood for coagulation analyses was taken on Day 1 (4 hours post dose) and on Day 91 (predose, 1, 4, 8, and 24 hours post dose) and evaluated using an automated coagulation analyzer.

[0521] At the 40 mg/kg dose level, indications of alternative pathway activation were observed as acute, transient increases in Complement Bb (up to 7-fold over baseline) at 4 hours post dose on Days 1 and 91. Complement Bb then returned to baseline levels at 24 hours on Day 1 or to near baseline levels on Day 91. There were no significant changes in Complement C5 observed during the study. Mild, transient prolongations of APTT (up to 29%), relative to controls were observed at 4 to 8 hours post dose and then diminished by 24 or 48 hours post dose. The acute transient increases in Complement Bb and APTT observed following treatment with ISIS 484137 were typical of those commonly seen oligonucleotide-treated monkeys.

Liver Function

[0522] To evaluate the effect of ISIS 484137 on hepatic function, blood samples were collected from all the study groups on Days 44, 93, 121, and 182 for determination of liver transaminases and liver functions. Serum levels of ALT and AST were measured and the results are presented in the Table below, expressed in IU/L. Bilirubin, a liver function marker, was similarly measured and the results are presented in the Table below, expressed in mg/dL. ISIS 484137 had no adverse effect on liver function outside the expected range for antisense oligonucleotides. ‘n.d.’ indicates that there is no data for that timepoint. The results indicate that treatment with ISIS 484137 was well tolerated in terms of the liver function in monkeys.

TABLE-US-00163 TABLE 163 ALT levels (IU/L) in cynomolgus monkey serum Dose (mg/kg) Day 44 Day 93 Day 182 sterile saline — 43 36 36 ISIS 484137 4 31 38 n.d. 8 33 31 n.d. 12 32 40 27 40 41 62 42

TABLE-US-00164 TABLE 164 AST levels (IU/L) in cynomolgus monkey serum Dose (mg/kg) Day 44 Day 93 Day 121 sterile saline — 54 65 41 ISIS 484137 4 33 53 n.d. 8 50 53 n.d. 12 29 71 31 40 49 90 44

TABLE-US-00165 TABLE 165 Total bilirubin levels (mg/dL) in cynomolgus monkey serum Dose (mg/kg) Day 44 Day 93 Day 121 Day 182 sterile saline — 0.20 0.25 0.23 0.20 ISIS 484137 4 0.17 0.20 n.d. n.d. 8 0.20 0.20 n.d. n.d. 12 0.20 0.35 0.20 0.20 40 0.20 0.20 0.18 0.23

Kidney Function

[0523] To evaluate the effect of ISIS 484137 on kidney function, blood samples were collected from all the study groups on days 44, 93, 121, and 182. Results are presented in the Table below, expressed in mg/dL.

[0524] The serum chemistry data indicate that ISIS 484137 did not have any effect on the kidney function outside the expected range for antisense oligonucleotides. Similar results were found with urine samples of the monkeys. Treatment with ISIS 484137 was therefore well tolerated in terms of the kidney function of the monkeys.

TABLE-US-00166 TABLE 166 Albumin levels (g/dL) in cynomolgus monkey serum Dose (mg/kg) Day 44 Day 93 sterile saline — 4.2 4.1 ISIS 484137 4 4.2 4.1 8 4.3 3.9 12 3.8 4.0 40 3.8 3.8

TABLE-US-00167 TABLE 167 BUN levels (mg/dL) in cynomolgus monkey serum Dose mg/kg Day 44 Day 93 Day 121 sterile saline — 23.7 22.5 23.0 ISIS 484137 4 18.0 17.7 n.d. 8 20.0 21.0 n.d. 12 22.0 22.0 21.0 40 21.3 20.0 24.0

TABLE-US-00168 TABLE 168 Creatinine levels (mg/dL) in cynomolgus monkey serum Dose (mg/kg) Day 44 Day 93 Day 121 Day 182 sterile saline — 128 81 120 93 ISIS 484137 4 129 138 n.d. n.d. 8 69 79 n.d. n.d. 12 183 153 79 84 40 129 85 74 72

Plasma Toxicokinetics

[0525] Following subcutaneous injection in monkeys, ISIS 484137 was quickly absorbed into the systemic circulation with a median T.sub.max (time to reach plasma C.sub.max) ranging from 1 to 4 hours. Peak exposure (C.sub.max) and total exposure (AUC.sub.0-48 hr) were dose-dependent on Day 1, Day 7, and Day 91. Similar AUC (area under the curve) values were observed between Day 1 (after a single dose) and Day 91 (after 16 doses). These results indicate a lack of plasma accumulation of ISIS 484137 following multiple doses. Mean clearance values (CL/F.sub.0-48 hr) following subcutaneous administration at all dose levels following single or multiple administrations ranged from 31.3 to 75.8 mL/hr/kg and appeared to decrease with increasing dose. The post-distribution plasma elimination half-life (t.sub.1/2λz) ranged from 9.42 days to 20.2 days following the 13 weeks of treatment. A review of ISIS 484137 plasma toxicokinetic parameters revealed no gender difference. The Table below presents the results. ‘n.d.’ means ‘not determined’.

TABLE-US-00169 TABLE 169 Plasma toxicokinetic summary in monkeys for ISIS 484137 Number AUC.sub.0-48 hr Dose of C.sub.max (hr* t.sub.1/2λz (mg/kg) Day animals (μg/mL) T.sub.max (hr) μg/mL) (days) 4 1 6 15 1.5 59 n.d. 91 6 7 3.0 53 n.d. 8 1 6 30 1.5 138 n.d. 7 12 28 2.0 137 9.42 91 6 19 2.0 127 n.d. 12 1 10 43 1.0 263 n.d. 91 10 24 4.0 261 15.1 40 1 10 113 1.0 1030 n.d. 91 10 92 1.0 1300 20.2

Tissue Toxicokinetics

[0526] There was a dose-dependent increase in kidney cortex and liver concentrations over the 10-fold increase in dose of ISIS 484137. Based on the mean tissue ISIS 484137 concentrations at the end of treatment and after recovery, the estimated tissue half-lives were 16.4 to 21.3 days and 17.8 to 22.8 days in kidney and liver, respectively, similar to the tissue half-lives determined following 8 mg/kg ISIS 484137. These findings are also consistent with the estimated terminal elimination half-life values in plasma. The Table below presents the results. ‘n.d.’ means ‘not determined’.

[0527] The tissue half-life (approximately 2-3 weeks) observed supports an infrequent clinical dosing regimen.

TABLE-US-00170 TABLE 170 Tissue toxicokinetic summary in monkeys for ISIS 484137 Intact ISIS 484137 (μg/g) Day 3 (2 days Day 93 (2 days Day 182 (91 days Dose after 1.sup.st dose) after last dose) after last dose) (mg/kg) Kidney Liver Kidney Liver Kidney Liver 4 n.d. n.d. 610 221 n.d. n.d. 8 594 84 889 406 n.d. n.d. 12 n.d. n.d. 1320 586 30 18 40 n.d. n.d. 4220 993 233 66

Pro-Inflammatory Effects

[0528] None of the inflammatory marker levels were changed beyond the expected range for antisense oligonucleotides. Therefore, treatment with ISIS 484137 did not cause any adverse inflammation and was well tolerated in the monkeys.

[0529] In summary, 13 weeks treatment with ISIS 484137 was clinically well tolerated at doses up to 40 mg/kg/week. There was no mortality during the study and there were no treatment-related effects in clinical findings, body weights, food consumption/appetence, ophthalmoscopic and electrocardiographic examinations, hematology, urinalysis, or complement C3 levels during the study. Overall, the results of the study indicate that ISIS 484137 is the most potent and well tolerated compound of those tested for inhibiting DGAT2 and is an important candidate for the treatment of metabolic diseases, such as NAFLD and NASH.

Example 18: Double-Blind, Placebo-Controlled, Dose-Escalation Phase I Study

[0530] A double-blind, placebo-controlled, dose-escalation Phase I study to assess the safety, tolerability, pharmacokinetics and pharmacodynamics of single and multiple doses of ISIS 484137 administered subcutaneously to healthy overweight volunteers is conducted in a Study Center.

Treatment Protocol

[0531] Approximately 48 subjects are planned to be enrolled in the study: 16 subjects in the single-dose cohorts and 32 subjects in the multiple-dose cohorts. Subjects are healthy males or females aged 18-65, inclusive, who have given written consent and are able to comply with all the study requirements. BMI of the volunteers is 29.0-38.0 kg/m.sup.2 inclusive. Any volunteers with clinically significant abnormalities in medical history are excluded from the study. ISIS 484137 solution or placebo (0.9% sterile saline) is prepared by an unblinded pharmacist or qualified delegate shortly before use, using aseptic technique. Study staff, who are blinded to the identity of the drug, administer the drug to the subjects.

[0532] There are 4 single-dose cohorts (Cohort A-D) with 4 subjects per cohort randomized 3:1 of ISIS 484137: placebo. The length of each subject's participation is approximately 8 weeks, which includes a 4-week screening period, a single dose, and a 4-week post-treatment evaluation period. Subjects receive a single dose of ISIS 484137 by subcutaneous administration. The dose of the antisense oligonucleotide to each cohort is presented in the Table below.

TABLE-US-00171 TABLE 171 Single-dose cohort dosing regimen of ISIS 484137 Total dose Cohort (mg) A 50 B 100 C 200 D 400

[0533] There are 4 multiple-dose cohorts (Cohort AA-DD) with 8 subjects per cohort randomized 3:1 of ISIS 484137: placebo. The length of each subject's participation is approximately 23 weeks, which includes a 4-week screening period, a 6-week treatment period, and a 13-week post-treatment evaluation period. Subjects receive 3 doses of ISIS 484137 by subcutaneous administration during the first week (Days 1, 3, and 5) and subsequently receive one subcutaneous dose once a week for the next 5 weeks (Days 8, 15, 22, 29, and 36) for a total of 8 doses. Subjects have follow-up visits at the Study Center on Days 37, 43, 50, 64, 78, 92, 106, and 127.

TABLE-US-00172 TABLE 172 Multiple-dose cohort dosing regimen of ISIS 484137 Dose per Total administration dose Cohort (mg) mg AA 100 800 BB 200 1600 CC 300 2400 DD 400 3200

[0534] Blood and urine samples are collected regularly throughout the study for safety, pharmacokinetic, and pharmacodynamics analysis. The safety and tolerability of ISIS 484137 is assessed by determining the incidence, severity, and dose-relationship of adverse events, vital signs, physical examination, ECG findings, and clinical laboratory parameters. Safety results in subjects dosed with ISIS 484137 are compared with those in subjects dosed with placebo.

Claims

1. A compound comprising a modified oligonucleotide 8 to 80 linked nucleosides in length having a nucleobase comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NO:16-1422, 1426-4679.

2. The compound of claim 1, wherein the modified oligonucleotide has a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NO:16-1422, 1426-4679.

3. The compound of claim 1, wherein the modified oligonucleotide has a nucleobase sequence consisting of any one of SEQ ID NO: 16-1422, 1426-4679.

4. The compound of claim 1, wherein the modified oligonucleotide has a nucleobase sequence comprising any one of SEQ ID NO: 1371, 1415, 1849, 2959, 3292, 4198, and 4373, wherein the modified oligonucleotide comprises a gap segment consisting of linked deoxynucleosides; a 5′ wing segment consisting of linked nucleosides; and a 3′ wing segment consisting of linked nucleosides; wherein the gap segment is positioned between the 5′ wing segment and the 3′ wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

5. The compound of claim 1, wherein the oligonucleotide is at least 80% complementary to SEQ ID Nos: 1 or 2.

6. The compound of claim 1, wherein the modified oligonucleotide comprises at least one modified internucleoside linkage, at least one modified sugar, or at least one modified nucleobase.

7. The compound of claim 6, wherein the modified internucleoside linkage is a phosphorothioate internucleoside linkage.

8. The compound of claim 6, wherein the modified sugar is a bicyclic sugar.
9. The compound of claim 8, wherein the bicyclic sugar is selected from the group consisting of: 4′—(CH.sub.2)—O-2′ (LNA); 4′—(CH.sub.2).sub.2—O-2′ (ENA); and 4′-CH(CH.sub.3)—O-2′ (cEt).
10. The compound of claim 6, wherein the modified sugar is 2′-O-methoxyethyl.
11. The compound of claim 6, wherein the modified nucleobase is a 5-methylcytosine.
12. The compound of claim 1, wherein the compound is double-stranded.
13. The compound of claim 1, wherein the compound comprises ribonucleotides.
14. The compound of claim 1, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides.
15. The compound of claim 1, further comprising a conjugated GalNAc moiety.
16. A composition comprising the compound claim 1, or salt thereof, and a pharmaceutically acceptable carrier.
17. A method of treating, preventing, or ameliorating a disease associated with DGAT2 in an individual comprising administering to the individual the compound of claim 1, or salt thereof, thereby treating, preventing, or ameliorating the disease.
18. The method of claim 17, wherein the disease is NAFLD, NASH, lipodystrophy, or partial lipodystrophy.
19. A method of inhibiting expression of DGAT2 in a cell comprising contacting the cell with the compound of claim 1, thereby inhibiting expression of DGAT2 in the cell.
20. A method of reducing or inhibiting triglyceride synthesis, lipid synthesis and insulin resistance in the liver or white adipose tissue of an individual having, or at risk of having, a disease associated with DGAT2 comprising administering the compound of claim 1 to the individual, thereby reducing or inhibiting triglyceride synthesis, lipid synthesis and insulin resistance in the liver or white adipose tissue of the individual.